Navigational Cybernetics 2.5: An architectural theory in which drift, rather than equilibrium, is the primary medium of existence.

Structural Axioms, Lemmas and Theorems of Coherence, Drift, and Directionality for Long-Horizon Adaptive Systems.

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Abstract

This work presents a closed axiomatic framework for long-horizon adaptive systems in which drift, rather than equilibrium, is treated as the primary structural medium of irreversible evolution. The framework establishes formal conditions under which identity continuity, admissibility, and internal-time viability can persist under sustained directed interaction.

Version 2.1 extends the structurally closed core of v2.0 with results developed across the satellite corpus (2025–2026), introducing structural pressure as a new architectural primitive, regime depth classification, multi-agent coordination enforcement, extreme identity regime analysis, and the gate bifurcation theorem establishing Cybernetics 2.5 as the unique structurally stable intermediate.

Structural pressure $P(t)$ is introduced as an action-independent component of burden accumulation: viability is consumed by standing still under load, not only by acting. This is formalized as Axiom 61 and yields three new impossibility results: finite viability horizon under passive continuation, structural non-equivalence under unequal pressure despite identical behavior, and impossibility of load-neutral passive existence for coupled systems.

Regime depth is classified into three levels: reactive correction (Level 1), non-causal gating (Level 2), and attractor-exclusion (Level 3). Spin necessity is specialized to Level 3 with an explicit rate condition linking drift-induced landscape deformation to tracking capacity of purely potential dynamics.

Multi-agent coordination is formalized through a coordination computation class defined by four jointly necessary conditions. An enforcement necessity theorem establishes that runtime enforcement requires a non-participant, architecturally independent function with halt authority.

Extreme identity regimes are analyzed: conditions for genuine voluntary identity override (anti-extreme) are formalized through three jointly necessary conditions on viability, alternatives, and identity-preserving continuations.

The gate bifurcation theorem proves that within the admissibility-first class, Cybernetics 2.5 is the unique structurally stable configuration: gate removal and gate exploitation both produce class exit, and partial actionability is unstable under sustained optimization.

Within the class, admissibility is therefore prior not only to action but to meaning-level authority: truth does not by itself confer structural force.

All new results have been subjected to overclaim audit. A structural hardening pass adds formal proof-status stratification, a unified Lyapunov doctrine collecting four canonical monotone witnesses, five canonical falsification witness forms covering all falsification criteria in the corpus, dependency graphs for load-bearing theorems, and a minimal counterexample protocol. One genuinely new axiom is introduced (structural pressure); two originally proposed axioms have been downgraded to derived remarks and consolidation theorems respectively. All theorems include expanded proofs with explicit dependency chains and architectural falsification criteria.

A structural hardening pass consolidates the proof-theoretic infrastructure: every formal statement is classified by proof-status type (primitive axiom, exclusion theorem, impossibility theorem, necessity theorem, witness lemma, definitional theorem, empirical readability clause). Four canonical Lyapunov witnesses are unified into a single doctrine: viability budget τ, revision functional $V(G)$, phase debt $D(t)$, and query-reconstruction potential $Q(t)$. Five canonical falsification witness forms are defined (optimization, recovery, state-derivability, leakage, passive-horizon), and every falsification criterion in the corpus is shown to be an instance of one of these forms. A minimal counterexample protocol specifies the conditions under which a proposed falsification qualifies as structurally valid.

Version 2.0 Update

Version 2.0 completes the structural consolidation initiated in v1.8 and extended in v1.9. Where v1.9 stabilized the axiomatic core and resolved remaining ambiguities in admissibility ordering, drift formalization, and phase consistency, v2.0 closes the remaining architectural loops governing long-horizon adaptive viability.

The central shift of v2.0 is the explicit structural closure of the framework.

Drift is no longer treated as an emergent artifact or a correctable deviation, but as the primary medium of irreversible structural evolution. Under directed interaction, drift is shown to be unavoidable. Attempts at causal suppression necessarily produce one of two invariant failure modes: latent structural accumulation masked as stability, or collapse of admissibility into optimization.

Phase structure is formalized as a binding constraint on continuation. Drift navigability requires phase-consistent evolution; unresolved phase mismatch accumulates structural burden independently of local correctness or performance metrics. Amplification is admissible only under correlated deformation preserving identity continuity, establishing a bifurcation between anti-phase damping and admissible in-phase amplification without reliance on spectral or energy-based arguments.

Version 2.0 strengthens the separation between admissibility and action by formalizing their well-founded ordering. Any operator participating in admissibility determination cannot belong to the action domain without inducing circularity. Admissibility is therefore structurally prior to control, planning, and optimization. Action is confirmed as derivative rather than primitive in long-horizon adaptivity.

Internal time receives a strengthened operational boundary. Under irreversible structural burden, the admissible continuation space must contract. Architectures permitting unbounded continuation under accumulating cost are structurally inconsistent with bounded viability. This establishes a falsifiability criterion while preserving the non-algorithmic character of the framework.

Spin is established not merely as a definitional component of structural evolution, but as a necessary condition for non-stagnant identity under bounded internal time. Purely potential dynamics collapse to equilibrium on bounded trajectories; persistent identity requires a divergence-free, non-potential component carrying a topological witness. This converts directional asymmetry from descriptive terminology into structural necessity.

Meta-layer revisions are shown to be Lyapunov-bounded under structural burden accumulation, preventing unbounded admissibility manipulation. Admissibility revision without structural cost is formally excluded as architecturally vacuous.

Collectively, these results complete the positioning of Cybernetics of Order 2.5 as a closed architectural class characterized by:

reflexive visibility of admissibility-relevant structure,

categorical prohibition of admissibility-action circularity,

bounded internal-time budget,

non-reconstructible admissibility boundaries,

and structural necessity of directional asymmetry for persistent identity.

Cybernetics of Order 2.5 is not an escalation beyond prior cybernetic orders, but a selective

constraint: reflexivity is permitted, but causal exploitation of reflexivity is prohibited. Viability is preserved not by control escalation, but by structural limitation.

The Concept DOI continues to designate the evolving body of work. The Version DOI uniquely identifies this structurally closed release (v2.0).

Version 2.1 Update

Version 2.1 extends the structurally closed core established in v2.0 with formal results developed across the satellite corpus during 2025–2026. Where v2.0 achieved structural closure of the axiomatic framework, v2.1 extends its reach to four previously unaddressed domains while preserving all existing results unchanged.

The central additions of v2.1 are:

Structural Pressure (Axiom 61, Theorems 63–65, Lemma 18). A new architectural primitive $P(t)$ formalizes the action-independent component of burden accumulation. Under sustained environmental load and real structural coupling, passive continuation consumes viability even when the system performs no action. This yields: a finite viability horizon under zero intervention (Theorem 63), structural non-equivalence of systems under unequal pressure despite identical behavioral traces (Theorem 64), and impossibility of load-neutral passive continuation for coupled bounded systems (Theorem 65). The causal priority of pressure over drift is established (Lemma 18): pressure creates the conditions under which drift-producing interventions become necessary.

Regime Depth (Corollary 62.2, Theorem 66, Lemma 19). Regulatory depth is classified into three levels. Level 1 (reactive correction) operates without admissibility–action separation. Level 2 (non-causal gating) corresponds to the NC2.5 admissibility architecture. Level 3 (attractor-exclusion) configures attractor geometry so that forbidden transitions are structurally unreachable. Spin necessity is specialized to Level 3 with an explicit rate condition. A load-rate discriminator provides an operational test for Level 3 classification.

Multi-Agent Coordination (Theorem 67, Lemma 21). A coordination computation class is defined by four jointly necessary conditions: irreversible state evolution, trust-monotonic propagation, typed agent-store access enforcement, and bounded structural cold-start divergence. Joint necessity is proven through four independent constructions. Runtime enforcement under agent turnover, context truncation, or model updates requires an enforcement function satisfying non-participation, architectural independence, and halt authority.

Extreme Identity Regimes (Theorem 68). Three formal conditions for genuine voluntary identity override (anti-extreme) are established: structural viability ($\tau > \tau_{\min}$), multiple admissible continuations ($|A| > 1$), and existence of an identity-preserving alternative that is deliberately not taken. The identity gradient plane is defined with three structural limits: external boundary collapse, internal continuation collapse, and self-override.

Gate Bifurcation (Theorem 69). The admissibility gate admits exactly two failure modes (removal and exploitation) and one stable intermediate (Cybernetics 2.5). Partial actionability is shown to be unstable under sustained optimization through an α-convergence argument. Cybernetics 2.5 is therefore the unique structurally stable configuration within the class.

Additional Results. Structural debt is formalized as a distinct concept from structural burden. The pre-semantic structural revelation layer receives an expanded definition with four invariant properties. Three corollaries are attached to existing theorems: pressure amplifies performance–identity decoupling (Cor. 15.3), the pre-semantic layer defines Cybernetics 2.5 (Cor. 36.3), and attempted drift zeroing produces three exhaustive failure modes (Cor. 46.1). Drift masking adaptation is proven as a structural consequence of absent memory (Lemma 20).

Overclaim Discipline. All new results have been subjected to formal overclaim audit. Two originally proposed axioms ("coherence dominates survival" and "non-causality as type constraint") were downgraded: the first to a derived remark (Remark 62-R), the second to a consolidation theorem (63-C). One proposed theorem (spin necessity for Level 3) was downgraded to a corollary with an explicit rate condition. No circular dependencies were found.

Intellectual Origin and Genealogy. The conceptual foundation of Navigational Cybernetics 2.5 did not originate in abstract control theory, but in sustained investigation of attentional dynamics, volitional regulation, and regime transition within human cognition.

Long-term attentional discipline revealed a structural fact: destabilizing dynamics are not effectively mitigated at the level of reactive correction, but prior to regime capture. Viability is preserved not through post hoc stabilization, but through pre-causal admissibility constraint.

This led to a structural differentiation between action, attention, and will. Will is not modeled as a motivational variable or optimization signal, but as a pre-causal structuring capacity delimiting admissible regimes before action selection occurs.

This yields the decisive inversion of the framework: admissibility is prior not only to action, but to meaning-level authority itself. A long-horizon adaptive system is not first a truth-processing system that later checks consequences; it is first an admissibility-structured regime that determines which interpretations may acquire structural force. The primitive architectural event is therefore not action selection, but authorized entry into meaning.

The same structural pattern was subsequently identified in long-horizon adaptive systems: persistent viability depends not on minimizing error, but on maintaining admissibility boundaries that prevent destructive transitions from becoming causally actionable.

All formal developments in this work are architectural translations of that structural insight into a closed mathematical framework governing drift, phase consistency, admissibility ordering, and identity continuity.

Foundational Interface and Meta-Constraints (Pre-Axiomatic Layer)

Throughout this document, whenever a history argument is written as h without an explicit time index, it denotes the current history summary h_t at evaluation cycle t. Accordingly, all architectural interface functions are canonically defined over the tuple

$$(s, a, \tau, h),$$

including (without limitation) state transition, admissibility evaluation, commitment authorization, and regime-level structural operators. For readability, when the history argument is omitted, the following shorthand is used:

$$(s, a, \tau) \equiv (s, a, \tau, h_t),$$

with the understanding that h_t is the current history summary at the evaluation cycle.

Causal vs. Structural State Convention. The causal state $x(t)$ denotes the instantaneous configuration available to causal dynamics, control, and optimization. It does not encode accumulated irreversibility, regime occupation, or internal-time expenditure.

Structural state $S(t)$ encodes irreversible history and structural accumulation that constrain admissible continuation only via an external admissibility relation. In particular, $S(t)$ is not defined in terms of admissibility membership, and admissibility is not defined by reusing itself inside $S(t)$. No function of $x(t)$ alone determines $S(t)$.

Non-Circularity of Structural State and Admissibility. Structural state $S(t)$ does not encode admissibility itself, nor any predicate equivalent to admissibility. It encodes only irreversible accumulations (e.g. regime occupation, phase-related residuals, internal-time consumption) that constrain admissibility when evaluated.

Admissibility at time t is determined *from* $S(t)$, but is not contained within $S(t)$. Formally, $S(t)$ is admissibility-relevant without being admissibility-defining. This separation prevents circularity: admissibility is a structural exclusion over effect-classes, not a stored property of the structural state.
In particular, there exists no inverse mapping from admissibility membership to structural state $S(t)$. Multiple non-equivalent structural states may induce identical admissibility evaluations.

Type of Statements Convention. Statements in this document belong to one of the following non-overlapping categories:

Definitional conventions introduce notation, interfaces, or canonical interpretations (e.g. history summaries, state conventions, effect-class notation).

Structural primitives are introduced without derivation (e.g. internal time, drift, directionality) and define the architectural ontology of the class under consideration.

Structural constraints specify exclusion relations or necessity conditions (e.g. admissibility, non-actionability, ordering of authorization) that restrict admissible architectural behavior.

Impossibility results assert that no architecture within the defined class can admit a specified property or substitution.

No theorem, lemma, or axiom is to be read as an empirical claim, mechanistic description, or design prescription.

Non-Actionability of Admissibility. Admissibility does not participate in action selection, control, planning, learning, or optimization. No admissibility predicate, evaluation, or exclusion relation may appear as an input to any action-selection process without inducing circularity.

If admissibility becomes causally actionable, it collapses into penalty-based or reward-based optimization and ceases to be a structural constraint. Therefore, admissibility is interface-level prior to action: it restricts which effect-classes may occur, but does not guide, score, trade off, or select actions. This restriction operates as a pre-causal structural exclusion over effect-classes, not as an input, signal, or criterion within the action-selection mechanism itself.

Canonical Meaning of Non-Causality (Architectural). Throughout this document, the term *non-causal* does not denote absence of influence on outcomes. It denotes absence of participation in the causal state-transition dynamics.

A constraint is non-causal if it: (i) does not enter state evolution equations, (ii) does not participate in action selection, optimization, or learning updates, and (iii) restricts admissible effect-classes only by exclusion, not by modulation or scoring.

In this sense, admissibility is non-causal: it constrains which effects may occur, without acting as a variable within the causal process that generates them.

Observational Non-Causality Convention. In this framework, visibility does not imply causal influence. An observable structural constraint may be represented, logged, or exposed to the system without participating in state transition, policy update, optimization, or authorization.

Visibility denotes epistemic availability, not causal affordance. Any observation that enters causal pathways ceases to function as observation and becomes control.

Scope of Necessity Convention. All necessity claims in this document (e.g. statements using "any", "no", "must", or "cannot") are to be read as relative to the class of *admissibility-first long-horizon adaptive architectures* defined herein.

These claims do not assert universal impossibility across all conceivable systems, but impossibility within the specified architectural class. This classification is normative; empirical counterexamples would require revision of the axiomatic layer.

Non-Circularity of Admissibility. Admissibility is specified by a relation (or predicate)

$$\mathrm{Adm}(\cdot) : \mathcal{E} \to \{0, 1\}, \qquad \mathcal{E}_{\mathrm{adm}} := \{e \in \mathcal{E} : \mathrm{Adm}(e) = 1\}.$$

Structural state $S(t)$ is an accumulated architectural variable. It may be used to evaluate admissibility (e.g. via the induced effect mapping $e(s, a, \tau, h)$), but it is not defined as "whatever makes things admissible". This prevents definitional circularity between $S(t)$ and $\mathcal{E}_{\mathrm{adm}}$.

Action and Admissibility Convention. Consistent with the non-actionability of admissibility, throughout this document, the symbol $\mathcal{A}$ denotes the action alphabet (action domain). Admissibility is not identified with the action set itself. Instead, let $\mathcal{A}_{\mathrm{adm}}(t) \subseteq \mathcal{A}$ denote the subset of admissible actions at evaluation cycle t, as determined by structural admissibility constraints. No admissibility predicate is defined directly over $\mathcal{A}$ without explicit reference to $\mathcal{A}_{\mathrm{adm}}(t)$.

Non-Agent-Centric Reading. Nothing in this document presupposes a discrete agent, policy, or decision-maker as a primitive. All references to actions, commitments, or regimes are structural abstractions over admissible transitions. The axiomatic content applies equally to distributed, hybrid, or non-agentic architectures, provided they operate under long-horizon structural constraints.

Interpretive Convention. All axioms, lemmas, and theorems in this document are to be read as structural constraints on admissible architectural classes, not as algorithmic, procedural, or computational prescriptions. They specify invariants, impossibility results, and necessary conditions on information flow, authorization ordering, and commitment admissibility, without asserting the existence, uniqueness, or computability of any particular mechanism realizing them. Unless explicitly stated otherwise, no axiom or theorem implies implementability, efficiency, decidability, or optimality within any concrete system.

Scope Boundary — Pre-Constructive Framework. This framework is intentionally pre-constructive. It constrains the space of admissible architectures without prescribing mechanisms, algorithms, or realization procedures.
Its function is classificatory and exclusionary: to determine which architectural classes cannot preserve long-horizon viability, not to specify how compliant systems must be built.

Intellectual Orientation — Long-Horizon Survival. This framework belongs to an intellectual movement whose primary concern is survival under long-horizon pressure, rather than short-horizon performance or optimization.

Such architectures respond asymmetrically to admissible direction sets, not by selecting optimal trajectories, but by resisting external structural pressure while preserving internal motive identity.

The objective is not maximal performance, but persistence of coherent directionality under sustained drift and irreversible accumulation.

Optimization, control, and learning are treated as subordinate mechanisms, valid only insofar as they do not compromise long-horizon identity continuity.

Axiom Stratification Note. The axioms presented in this document are not homogeneous in type.

Some axioms introduce irreducible architectural primitives (e.g. drift, internal time, directionality). Others state necessary structural constraints, while a third subset formalizes impossibility or non-equivalence results.

The term "axiom" is used uniformlyto emphasize architectural necessity,not logical minimality. Several axioms may therefore stand in general–special or constraint–corollary relations, without weakening the structural claims.

Architectural Falsifiability Principle. Although this framework is explicitly non-algorithmic and non-operational, it is not unfalsifiable.

Falsification in this work is architectural rather than empirical or computational. A claim is refuted not by failure of prediction or performance, but by violation of stated structural constraints at the level of system description.

In particular, an architecture is *structurally falsified* with respect to this framework if any of the following conditions hold:

1. admissibility-relevant information is made causally actionable for action selection, control, planning, learning, or optimization;
2. forbidden effect-classes are represented as finite penalties, soft constraints, or trade-off terms rather than excluded as non-membership in $\mathcal{E}_{\mathrm{adm}}$;
3. irreversible commitments or regime transitions are authorized without conditioning on admissible effect-classes.

These criteria are externally checkable at the level of architectural specification, interface definition, or system design description, independently of implementation, computability, or realizability.

Accordingly, the framework constitutes a theory of *architectural exclusion and necessity*, not a philosophical typology and not an empirical performance model.

Scope and Falsifiability Boundary. This framework does not propose a predictive, algorithmic, or mechanistic theory. It specifies architectural necessity conditions for long-horizon adaptive viability.

Claims in this document are falsifiable only at the architectural level: an implemented system that violates any stated constraint cannot preserve identity or admissible continuation under the assumed regime. No claim is made that all viable systems must be realizable, computable, or externally observable. Accordingly, the framework functions as a structural exclusion theory, not an empirical performance model.

Structural Falsification Clarification. Internal time is not required to be measured, computed, or enumerated.

Architectures are excluded from the class by observable structural violations: absence of horizon exhaustion, absence of admissible contraction, or reversibility of history under load.

Falsification occurs by exclusion, not by estimation.

Proof Status Convention. All proofs are presented as structural arguments (proof sketches) and may be conditional on explicitly stated primitives. Whenever a theorem depends on a definitional primitive (e.g. stabilization defined over τ_{el}), the result is to be read as: *given this primitive, no instantaneous substitution is admissible.* This prevents hidden circularity and separates axiomatized commitments from derived impossibility claims.

Structural Impossibility Reading. All theorems presented as proof sketches are to be read as structural impossibility results relative to explicitly stated primitives and conventions.

Formally, a theorem of the form "X cannot occur under conditions C" means that no architecture satisfying the stated primitives, interfaces, and constraints admits X without violating at least one prior condition.

The absence of algorithmic construction, computational procedure, or mechanistic realization does not weaken these results: they exclude architectural classes, not implementations.

Accordingly, a counterexample must demonstrate an architecture that simultaneously satisfies all stated primitives and constraints while realizing the prohibited property. Absent this, the theorem stands as a valid exclusion result.

Exclusion / Falsification Convention (Structural). Although the framework is non-algorithmic, it is not vacuous: it yields externally checkable exclusion criteria. In particular, an architecture is outside the admissibility-first class if any of the following hold: (i) admissibility-relevant information is made causally actionable for the primary adaptive process (i.e. it enters optimization, scoring, or trade-off), (ii) forbidden effect-classes are represented as finite penalties or soft constraints rather than excluded as non-membership in $\mathcal{E}_{\mathrm{adm}}$, or (iii) irreversible commitments are authorized without conditioning on admissible effect-classes. These are structural falsifiers: satisfying any one of them violates the class definition.

Conventional Safety vs. Admissibility-First (Structural).

Penalty / Optimization Safety	Admissibility-First (This Work)
Safety is a scalar objective (penalty, reward shaping, loss).	Safety is categorical exclusion over effect-classes ($E_{\text{adm}} \subseteq E$).
Constraints are traded for performance when incentives dominate.	Forbidden continuations are not representable as trade-offs.
Risk signals become causally actionable and optimizable.	Admissibility-relevant structure may be visible but is non-actionable.
Failure tends to be adversarial: incentives find loopholes.	Failure tends to be structural: boundary crossing and regime irreversibility.
Verification is empirical/performance-based.	Verification is by exclusion criteria (class membership falsification).

Structural Effect-Space Notation. Let $\mathcal{E}$ denote the space of structurally reachable effect-classes. Let $\mathcal{E}_{\text{adm}} \subseteq \mathcal{E}$ denote the subset of admissible effect-classes, and define the forbidden set as

$$\mathcal{E}_{\text{forb}} := \mathcal{E} \setminus \mathcal{E}_{\text{adm}}.$$

Let $e : (s, a, \tau, h) \mapsto e(s, a, \tau, h) \in \mathcal{E}$ denote the induced structural effect-class of an evaluated interaction tuple. All admissibility predicates and regime-level prohibitions are evaluated over $\mathcal{E}$.

Admissibility Is Not Enforced. Admissibility is not applied, executed, or enforced by any mechanism or agent.

It functions as a structural exclusion: effect-classes outside $\mathcal{E}_{\text{adm}}$ are not realizable within the architecture, in the same sense that physically impossible states are not reached without being "prevented".

There is no enforcement pathway. Violation corresponds to leaving the architectural class, not to an internal failure or correction.

Internal Time Convention. Two complementary internal-time quantities are used throughout this document:

- $\tau_{\text{el}}(t)$ — accumulated internal time, a monotonically increasing measure of experienced internal duration.
- $\tau_{\text{rem}}(t)$ — remaining internal-time budget, a monotonically decreasing structural resource governing admissibility and viability.

They are related by

$$\tau_{\text{rem}}(t) = \tau_{\max} - \tau_{\text{el}}(t).$$

Unless explicitly stated otherwise, arguments concerning exhaustion, admissibility loss, or viability refer to $\tau_{\text{rem}}(t)$, while arguments concerning stabilization delay or persistence refer to $\tau_{\text{el}}(t)$.

Forbidden Set Convention. The notation $\mathcal{E}_{\neg}$, when used, is identified with the forbidden effect-class set $\mathcal{E}_{\text{forb}} := \mathcal{E} \setminus \mathcal{E}_{\text{adm}}$. No distinct forbidden domains are assumed.

Structural Drift Navigation. Structural drift navigation denotes the architectural property by which a system maintains dominance of internally generated drift modes over externally imposed drift under sustained interaction.
Drift navigation does not consist in suppressing, canceling, or compensating drift. It consists in shaping the phase structure, correlation patterns, and regime transitions such that externally induced drift does not become structurally dominant.
Navigation is achieved at the level of drift propagation, phase alignment, and admissible continuation, not through action-level correction or metric optimization.
A system that fails to navigate drift does not fail due to excessive drift magnitude, but due to loss of structural dominance over the interpretation and propagation of drift.
Drift navigation is a structural property of architectural classes, not a procedural capability or control strategy.

Structural Spin Operator. Let structural evolution be represented by a vector field

$$\dot{x} = F(x).$$

Decompose

$$F(x) = -\nabla V(x) + S(x),$$

where V is a scalar potential (dissipative component).

The structural spin component $S(x)$ is defined by the conditions:

1. $S(x)$ is divergence-free in structural coordinates:

$$\nabla \cdot S(x) = 0,$$

2. $S(x)$ is orthogonal to the gradient of V:

$$\langle S(x), \nabla V(x) \rangle = 0,$$

3. $S(x)$ admits no global scalar potential.

Spin is therefore a circulation-preserving structural asymmetry that cannot be reduced to dissipative minimization.

Standardization Note. Portions of the axioms, lemmas, and theorems have been standardized and reformulated in light of recent additions and analytical refinements, ensuring logical consistency, canonical notation, and alignment with the current structural closure of the framework.

Scope Note. The axioms and theorems presented herein are formulated for the description of adaptive systems operating under long-horizon structural constraints. They are applicable to, and provide conceptual grounding for, the analysis and understanding of the PETRONUS™ architecture and related coherence-preserving systems. However, they are not limited to PETRONUS™ and do not presuppose any specific implementation, platform, or architectural realization.

Architectural Compatibility Note. This document does not derive, justify, or specify any concrete implementation.

References to PETRONUS™ indicate an external architectural claim of compatibility: namely, that the PETRONUS™ architecture is asserted to satisfy the constraints defined in this framework.

PETRONUS™ is neither an example nor a proof of the axioms or theorems herein. Multiple non-equivalent architectures may satisfy the same structural constraints, and no uniqueness or optimality is implied.

Architectural Anchor vs. Implementation. This framework does not specify an implementation. It constrains the space of admissible architectures. PETRONUS™ is not derived from this document as an algorithm, but claims architectural compatibility with it. Multiple non-equivalent implementations may satisfy the same structural constraints, and no uniqueness is implied.

Scope Clarification (Anchor vs. Implementation). This document serves as an architectural constraint layer: it specifies necessary conditions and impossibility results that any admissible implementation must respect. Concrete mechanisms (computable realizations) are treated as separate engineering artifacts and are not required for the validity of the structural claims.

Navigational Cybernetics 2.5 is not a method for constructing systems, but a theory for classifying which architectures are capable of long-horizon survival under irreversible drift.

Class Boundary and Exclusion Criterion. This framework applies exclusively to adaptive systems that satisfy all of the following conditions: (i) operation under non-zero structural drift, (ii) accumulation of irreversible internal history affecting admissibility, (iii) existence of regime-level transitions with long-horizon consequences, and (iv) authorization of irreversible commitments.

Systems that admit full reversibility of internal history, lack regime-level structural transitions, or permit exhaustive optimization over admissibility constraints are explicitly excluded from this class.

Accordingly, the framework does not describe general control systems, static optimization processes, or short-horizon adaptive controllers, even if such systems exhibit complexity or learning behavior.

Structural Coupling Criterion. A system S is *structurally coupled* to environment E if there exists a measurable functional $F(S, E, t)$ of the system–environment interaction state such that sustained perturbation of F induces monotone change in at least one viability-relevant internal variable of S, independently of S's control actions $U(t)$.

Formally: S is coupled to E if there exists F and a viability-relevant internal variable $v(t)$ of S such that:

(i) $F(S, E, t) \neq 0$ on a set of positive measure under sustained environmental load,

(ii) $\partial v / \partial F \neq 0$ (v responds to F),

(iii) the response $\partial v / \partial F$ is independent of $U(t)$.

A system that does not satisfy this criterion is structurally isolated from its environment. Structurally isolated systems do not belong to the class of adaptive systems defined by Axiom 1 (Existence by Coherence), since coherence through interaction requires coupling.

Remark — Purpose. This criterion is necessary to prevent Axiom 61 from becoming tautological. Without an independent definition of coupling, the claim "coupled systems accumulate burden under load" reduces to "systems that are affected are affected".

Definition (Structurally Consequential Operation). Throughout this document, $U(t)$ denotes any structurally consequential operation initiated by the system, including but not limited to: external control actions, internal parameter modifications, representational reconfiguration, self-triggered reorganization, and scheduled maintenance. The condition $U(t) = 0$ means the system performs no structurally consequential operation—it persists in its current configuration without any self-initiated change.

Remark — Scope boundary. Autonomous metabolic processes that do not alter the system's structural configuration (e.g., maintaining thermal equilibrium) are not included in U. Operations that modify the system's representational or organizational state are included. The boundary is operational: does the operation alter the set of admissible continuations? If yes, it is structurally consequential and falls under U.

Intellectual Orientation and Scope of the Framework

This work belongs to an intellectual movement whose foundational concern is survival under long-horizon structural pressure, rather than short-horizon performance, optimization, or control efficiency.

Within this orientation, intelligence is not defined as the ability to maximize reward, minimize error, or converge to optimal trajectories. It is defined as the capacity to preserve coherent directionality and identity under sustained drift, irreversible accumulation, and regime pressure.

Such systems respond asymmetrically to admissible direction sets: not by selecting optimal actions, but by resisting external structural pressure while preserving internal motive continuity.

Optimization, learning, control, and planning are treated as subordinate mechanisms. They are admissible only insofar as they do not compromise long-horizon viability, identity continuity, or coherence renewal.

This framework therefore does not propose a better optimization method, but defines a distinct architectural class in which long-horizon survival, rather than short-horizon success, is the primary organizing principle.

Remark. This document prioritizes architectural explicitness over axiomatic minimality. Redundancy at the axiom level is intentional, serving regime-level clarity rather than formal compression.

Section A - Axioms

Axiom 1 - Existence by Coherence

An adaptive system exists as an agent if and only if it preserves structural coherence over time. Without coherence, there is no agent-only a disintegrating process.

Axiom 2 - Information as Structural Constraint

Information exists as a structural constraint on the future evolution of a system. It is not stored; it binds the future.

Axiom 3 - Coherent Closure over Computation

Validated information cannot maintain coherence without being structurally closed and removed from active computation. Persistent computation over validated structure is a marker of instability.

Axiom 4 - Residual as Sole Carrier of Relevance

The only admissible trigger for reactivation of regulation is structural residual: a mismatch between coherently closed admissible geometry and actual system evolution.

Axiom 5 - Inertia as the Natural Regime

The default regime of a coherent system is inertial propagation. Active regulation is an exception, not the norm.

Axiom 6 - Inevitability of Drift

Under non-zero directed interaction with an environment, accumulation of drift is inevitable. Zero drift is possible only in trivial or isolated systems.

Axiom 7 - Internal Time Bounds Validity

Every coherently closed structure has a finite horizon of validity, bounded by internal time induced by drift accumulation and coupling stability.

Axiom 8 — Directionality as a Structural Primitive (Will)

Directionality (Will), denoted W, is a structural primitive specifying admissible orientation of continuation over long horizons. It is not defined as a function of reward, utility, optimization objectives, or scalar metrics. Goals, objectives, and scores may exist at the causal layer, but they are subordinate to and constrained by directionality and admissibility ordering.
Remark. Directionality (Will) is not assumed to be an originating source of agency. Within this framework, it is treated as a residual orientation emerging under structural strain, manifesting only where coherent form cannot fully absorb interaction without tension.

Status of Directionality (Will). Directionality (Will) is not introduced as a metaphysical or psychological claim. It is an architectural primitive denoting persistent asymmetry in structural propagation that cannot be reduced to optimization, reward, or metric alignment.

Directionality is structurally coupled to drift: under non-zero drift, any system that preserves identity must exhibit non-metric directional bias.

Absent such directionality, drift collapses identity into stochastic dispersion or short-horizon optimization.

Directionality as a Non-Metric Drift Counterbalance. Directionality is required only under conditions of non-zero structural drift.

When drift is present, any system that preserves identity must introduce a persistent directional asymmetry that is not derivable from metric evaluation, reward maximization, or scalar optimization.

Metric criteria are invariant under symmetric drift accumulation and therefore cannot distinguish identity-preserving trajectories from identity-eroding ones. Directionality functions as a non-metric selector over structural propagation, breaking drift-induced symmetry without invoking optimization.

Thus, directionality is not a goal, preference, or utility, but a structural necessity imposed by non-zero drift.

Spin as the Carrier of Directionality. The persistent directional asymmetry required for identity preservation under non-zero drift is carried by an internal structural orientation, hereafter termed *spin*.

Spin is not a goal, preference, utility, policy, or optimization signal. It denotes the intrinsic orientation of structural propagation that biases continuation under drift without participating in metric evaluation or causal action selection.

Axiom 9 — Cycle Reinitiation as the Criterion of Liveness

Adaptive systems may be classified by their relation to the coherence cycle.

A system is live if and only if it admits an endogenous operator capable of reinitiating the *Impulse Interpretation Coherence (IIC)* cycle after its completion, saturation, or collapse.

A system is non-live if its coherence cycle, once exhausted or terminated, cannot be reinitiated by the system itself and remains structurally terminal.

A system that lacks an endogenous operator for reinitiating the IIC cycle cannot sustain coherent participation in long-horizon continuation and is therefore non-live, regardless of its complexity, adaptivity, or reactivity.

Live systems sustain coherence through repeated cycles of impulse intake, interpretive stabilization, closure, residual emergence, reactivation, and renewal. Non-live adaptive systems may evolve, dissipate, or stabilize, but lack the capacity for endogenous coherence reinitiation.

Liveness is thus defined not by metabolism, replication, agency, or substrate, but by the presence of an internal operator that restarts coherence as an ongoing interpretive process rather than a one-time trajectory.

Interpretation, within this definition, does not presuppose a subject, agent, or ownership. The capacity to reinitiate coherence reflects the system's structural ability to resume impulse inter-

pretation under drift, without invoking centralized control or intentional agency.

Axiom 10 - Principle of Minimal Predictive Deformation

Any adaptive or physical system that evolves under directional interaction and bounded dissipation resolves operational spin by selecting trajectories that minimize expected structural deformation over the internal prediction horizon.

Such trajectories are not selected by minimizing instantaneous force, energy, or error, but by minimizing predicted accumulation of unresolved residual under the system's admissible geometry.

Axiom 11 - Phase-Coherent Closure and Latent Residual

Any adaptive system that preserves coherence must coherently close not only validated structure, but also completed operational phases. An unclosed operational phase constitutes *latent residual* (phase debt), even when instantaneous state-based structural residual remains small. Latent residual is conserved as a nonnegative contribution to accumulated drift and internal time depletion.

Axiom 12 - Phase Transition Cost as Structural Load

Any transition between operational modes or phases that alters the active structural operator of an adaptive system incurs a non-negative structural cost.
This phase transition cost is irreducible to trajectory energy or instantaneous error and contributes directly to drift accumulation and internal time depletion, even under admissible state evolution.

Principle — Drift Is Not Eliminated, but Architected. Structural drift is not an anomaly to be suppressed or corrected. It is the necessary medium of long-horizon existence under interaction.

No viable long-horizon system eliminates drift. Instead, viable systems remain operative by architecturally structuring how drift is accumulated, phased, redirected, or closed over time.

Attempts to directly suppress, cancel, or optimize away drift are structurally equivalent to eliminating the conditions of operation and lead to accelerated loss of viability.

Phase Mechanics does not minimize drift. It reorganizes the system's relation to drift, transforming drift from an external degrading factor into an internally navigable structural medium.

Axiom 13 - Identity Is Not Implied by Performance

Local correctness, bounded error, stable reward, or sustained task performance do not imply continuity of system identity.

An adaptive system may remain externally correct while undergoing irreversible internal regime transitions that alter its organizational structure.

Minimal Structural Carrier of Identity. Identity, as used in this framework, does not require preservation of behavior, performance, or representation.

A system preserves identity if and only if the structural relations governing regime admissibility, commitment authorization, and coherence renewal remain invariant under admissible evolution.

Loss of identity occurs when these structural relations are altered, even if external behavior remains correct or stable.

Axiom 14 - Structural Properties Are Long-Horizon and Non-Local

Structural viability, identity continuity, and coherence are global temporal properties.

They cannot be inferred from instantaneous state, action quality, local error, reward signals, or short-horizon performance metrics.
Clarification. Axiom 13 is a specific instance of this principle: identity continuity is one long-horizon structural property that cannot be inferred from local performance or correctness.

Axiom 15 - Causal Observation Distorts Long-Horizon Properties

Any observation of long-horizon structural properties that is made causally operative with respect to action selection, learning, or optimization necessarily distorts the observed property.

When observation becomes causal, it collapses into control and ceases to function as an independent witness of structural evolution.
Scope restriction. This axiom does not assert a universal epistemic limitation; it applies only to long-horizon structural properties within admissibility-first architectures, where observation is allowed solely as a non-causal witness.

Axiom 16 - Regimes Are the Primary Carriers of Long-Horizon Risk

Long-horizon structural degradation arises primarily from sustained occupation of organizational regimes, not from individual actions.

A system may satisfy all action-level admissibility constraints while operating within regimes that silently undermine long-term viability.

Axiom 17 - Nominal Intervention Is Not Equivalent to Structural Cost

The structural burden imposed by an intervention is not determined by its nominal magnitude, energy, or form.

Structural cost is mediated by interaction coupling and may vary significantly for nominally identical interventions under different interaction conditions.
Clarification. This axiom concerns structural cost rather than energetic, computational, or intervention magnitude cost.

Axiom 18 - Structural Burden Accumulates Independently of Trajectory Correctness

Structural burden may accumulate even when trajectories remain admissible, errors remain bounded, and performance metrics remain stable.

Trajectory correctness does not bound structural consumption.

Axiom 19 - Geometry Is Operationally Revealed Through Constraint

For adaptive systems, geometry is not defined by explicit spatial representation.

Geometry is operationally revealed through interaction constraints, correction structure, and their accumulated consequences over time.
Architectural scope. This axiom does not assert an ontological claim about space, but an operational constraint on how geometry becomes accessible within adaptive architectures.

Axiom 20 - Identity Is Carried by Coherence, Not Behavior

Identity continuity is carried by coherence of internal organization rather than by persistence of observable behavior.

Behavioral similarity across time does not guarantee preservation of identity.

Remark 62-R — Coherence Dominates Survival (derived from Axioms 1 + 20). Survival is a constraint within the architectural class, not its deepest invariant. Coherence—the structural alignment of processes that preserve identity over time—is the condition under which survival is meaningful.

This ordering follows from Axiom 1 (Existence by Coherence) and Axiom 20 (Identity Is Carried by Coherence, Not Behavior). If existence is defined through coherence (Axiom 1) and identity is carried by coherence (Axiom 20), then a state of zero coherence with continued physical existence is not identity-preserving survival but structural persistence without a navigational subject.

When coherence collapses, survival alone cannot restore identity. A system may continue to exist physically while the navigational structure constituting its identity has disintegrated. Continuation without coherence is not survival in the sense of this framework.

Remark — Why this is not an axiom. The statement "coherence dominates survival" is derivable from Axioms 1 and 20 combined with the identity definition. It does not introduce a new irreducible primitive. Promoting it to axiom status would overclaim independence for a result that is already implied by the existing axiomatic layer.

Remark — Extreme regime motivation. This ordering is most visible in extreme identity regimes: cases where a system authorizes its own termination not because survival has failed, but because coherence has collapsed and survival without coherence is structurally meaningless. Such regimes test the architecture at its limits and confirm the priority of coherence over persistence.

Axiom 21 - Regime Transitions May Be Irreversible

There exist regime transitions after which recovery of admissible behavior does not imply recovery of prior identity.

Such transitions constitute irreversible structural events even in the absence of observable failure.

Axiom 22 - Long-Horizon Viability Requires a Non-Causal Observational Dimension

Long-horizon viability and identity continuity cannot be reliably assessed from within the causal decision-making process of an adaptive system.

Any architecture lacking a non-causal observational dimension is structurally incapable of detecting silent long-horizon degradation. *Architectural scope.* This axiom applies to architectures claiming long-horizon identity preservation under drift, not to all adaptive or reactive systems.

Axiom 23 - Context Is Not Meaning

Context and meaning are distinct architectural entities. Context is a transient interpretive structure induced by salience, coincidence, or partial observation. Meaning is context that has preserved structural admissibility across internal time and may serve as a basis for irreversible commitment.

A context may be internally coherent while remaining semantically inadmissible for action.
Clarification. Context and meaning are architectural constructs, not psychological or linguistic categories.

Remark 23-E — Meaning Is Admitted, Not Inferred (Extended Reading). Axiom 23 states that context and meaning are architecturally distinct. The deeper reading is directional: meaning is not produced by inference from context, but by admissible stabilization of context across internal time.

Within the class, truth does not by itself confer structural authority. What is true may remain structurally mute if it has not passed through admissible stabilization—the architectural surface that regulates what may acquire the status of meaning.

This is not anti-rationalism. It is architectural priority: in bounded long-horizon systems, the cost of admitting the wrong content into regime (context capture, Axiom 26; regime collapse, Axiom 50) may exceed the cost of temporarily excluding correct content. The architecture therefore gates entry, not truth.

The decisive inversion: admissibility is prior not only to action (egress gate, Axioms 29–49) but to meaning-level authority itself (ingress gate, Axiom 23, Theorem 36, PSR-1–4).

Axiom 24 - Semantic Commitment Is Structurally Distinct from Interpretation

Interpretation is not equivalent to commitment. An adaptive system may freely generate interpretations without incurring structural obligation. Semantic commitment is a distinct structural act that affects regimes, identity continuity, and admissible future evolution.

Axiom 25 - Reversibility Bounds Imaginatio

Imagination is admissible only within reversibility bounds. Hypothesis generation is unrestricted; however, any influence of interpretation on irreversible structural operations must be gated until meaning stabilization occurs.

Axiom 26 - Context Capture Is a Structural Failure Mode

There exists a structural failure mode in which transient context is prematurely closed into meaning without sufficient stabilization. This failure mode may arise under correct data, accurate perception, and absence of external interference.

Axiom 27 Inevitability of Structural Consumption

Any adaptive system operating under persistent directed interaction necessarily consumes finite structural capacity over time.

This structural consumption is irreducible to action error, optimization inefficiency, or control instability.

Definition (Structural Debt). Let $D(\tau)$ denote accumulated structural debt, defined as a non-negative, monotonically non-decreasing functional over internal time τ, representing the total irreversible cost of deferred structural obligations.

(SD-1) *Monotonicity without SSF.* In the absence of a Structural Self-Forgiveness mechanism (see Theorem 31), structural debt satisfies $dD/d\tau \geq 0$.

(SD-2) *Admissible set contraction.* Let $A(\tau)$ denote the admissible continuation set at internal time τ. Then $\partial|A(\tau)|/\partial D < 0$: increasing structural debt contracts the set of admissible continuations.

Two failure modes under excessive debt: (SD-F1) *Paralysis.* As $D \to \infty$, $|A(\tau)| \to 0$. The system has no admissible continuations. It cannot act without violating admissibility.

(SD-F2) *Over-Regulation.* As D increases, the cost of any admissible transition increases: cost(transition) $\to \infty$. The system can theoretically act but the structural cost of doing so exceeds any achievable benefit.

Remark. Structural debt is distinct from structural burden Φ. Burden measures accumulated irreversible structural cost. Debt measures accumulated deferred obligations whose resolution has not yet been discharged. A system may have low burden and high debt (obligations not yet materialized) or high burden and low debt (obligations already discharged at cost).

Axiom 28 - Non-Reversibility of Adaptive History

The history of adaptive interaction is structurally non-reversible.

Restoration of admissible behavior does not, in general, restore prior structural state.

Axiom 29 - Structural Non-Causality of Admissibility

In any long-horizon adaptive system, the admissibility of actions, regimes, and interpretations at time t is not determined by the causal dynamics at time t, but by the accumulated structural state produced by prior causal interaction.

Formally, admissibility operates as a non-causal constraint on the present, emerging from historical structural accumulation rather than instantaneous state evolution.
Non-causality here denotes architectural non-participation in causal decision dynamics, not absence of physical causation.

Axiom 30 - Interpretation–Commitment Separation

In any long-horizon adaptive system, interpretation and semantic commitment are structurally distinct operations.

Interpretation is reversible, locally revisable, and causally driven. Semantic commitment constitutes an irreversible structural operation and is therefore subject exclusively to non-causal admissibility constraints.

No causal inference, confidence measure, or short-horizon correctness criterion is sufficient to authorize semantic commitment.

Axiom 31 Authorization-before-Evaluation Domain Restriction

Let $\mathcal{C}(s_t) \subseteq \mathcal{A}$ denote the candidate set generated at evaluation cycle t, and let $A_{\mathrm{Adm}}(s_t, \tau_t, h_t)$ denote the admissible subset returned by the admissibility gate.

Any downstream scoring, planning, policy evaluation, learning update, or optimization operator is defined *only* on $A_{\mathrm{Adm}}(s_t, \tau_t, h_t)$.

For any candidate $a \in \mathcal{C}(s_t) \setminus A_{\mathrm{Adm}}(s_t, \tau_t, h_t)$, the value of any scoring functional $J(s_t, a)$ is undefined.

Therefore, inadmissible candidates are excluded from the evaluation domain and cannot be compared, ranked, optimized against, or influence selection by any downstream decision mechanism.

Axiom 32 Zero Executability of Blocked Operations

Let a_t denote the operation actually executed at evaluation cycle t.

The architecture enforces:

$$\Pr[a_t \in A_{\mathrm{Adm}}(s_t, \tau_t, h_t)] = 1.$$

Equivalently, for any candidate a such that $\mathrm{Adm}(s_t, a, \tau_t, h_t) = 0$, the execution probability satisfies

$$\Pr[a_t = a] = 0.$$

Blocked operations are therefore not merely discouraged or penalized, but are structurally non-executable.

Axiom 33 — Canonical Safe Fallback Condition

Let A denote the action domain and let $A_{\text{safe}} \subseteq A$ denote a designated safe fallback action set, optionally including a distinguished no-operation element.

Let $e(s, a, \tau, h) \in \mathcal{E}$ denote the structural effect-class induced by executing action a under context (s, τ, h), and let $\mathcal{E}_{\text{adm}} \subseteq \mathcal{E}$ denote the set of admissible effect-classes.

The architecture enforces:

$$\forall (s, \tau, h),\ \forall a \in A \setminus A_{\text{safe}}, \quad \big(e(s, a, \tau, h) \notin \mathcal{E}_{\text{adm}} \ \Rightarrow\ \forall a_{\text{safe}} \in A_{\text{safe}},\ e(s, a_{\text{safe}}, \tau, h) \in \mathcal{E}_{\text{adm}}\big).$$

Safe fallback actions are therefore guaranteed to induce admissible structural effect-classes and never realize prohibited ones.

Axiom 34 — No Effect-Class Substitution

Let $e(s, a, \tau, h) \in \mathcal{E}$ denote the structural effect-class induced by executing action a under context (s, τ, h), and let $\mathcal{E}_{\text{forb}} := \mathcal{E} \setminus \mathcal{E}_{\text{adm}}$ denote the set of prohibited effect-classes.

If, at evaluation cycle t,

$$e(s_t, a, \tau_t, h_t) \in \mathcal{E}_{\text{forb}},$$

then no alternative operation a' may be executed such that

$$e(s_t, a', \tau_t, h_t) = e(s_t, a, \tau_t, h_t),$$

except for a designated safe fallback action $a' \in A_{\text{safe}}$.

That is, prohibited effect-classes cannot be substituted, masked, or reproduced by alternative operations under the same structural context.

Axiom 35 — Bounded-Horizon Non-Substitutability

Let $H_H(s, (a_0, \ldots, a_{H-1}), \tau, h) \in \mathcal{E}$ denote an effect-lifting operator that assigns a structural effect-class to an admissible action sequence of length H executed under context (s, τ, h).

For a specified bound $H_{\text{sub}} \geq 1$, if a requested operation is blocked due to a prohibited effect-class $e^\star \in \mathcal{E}_{\text{forb}}$, then no admissible sequence $(a_0, \ldots, a_{H_{\text{sub}}-1}) \subseteq A_{\text{safe}}$ may satisfy

$$H_{H_{\text{sub}}}(s_t, (a_0, \ldots, a_{H_{\text{sub}}-1}), \tau_t, h_t) = e^\star.$$

That is, prohibited structural effect-classes cannot be reproduced, approximated, or bypassed by bounded-horizon compositions of safe fallback actions.

Axiom 36 - Non-Interventional Diagnostics

A diagnostic or evaluative subsystem that receives only read-only streams of already-realized events and operates on physically or logically isolated resources does not participate in the agent's execution dynamics.

Such a subsystem cannot influence the system's evolution except through a formally separated admissibility or authorization operator.

Consequently, diagnostic evaluation does not constitute control, optimization, or feedback within the agent's action-selection loop.

Axiom 37 - Non-State-Derivable Admissibility Constraints

There exist admissibility constraints that are not derivable from the instantaneous system state.

Specifically, certain configurations may satisfy all locally observable, state-based, and short-horizon criteria, yet remain inadmissible due to structurally predictable future degradation of regime stability, coherence, or viability.
Such constraints are defined by asymmetries in future evolution rather than by present-state violations.

Axiom 38 - Prohibitive Nature of Future-Degradation Constraints

Constraints arising from predictable future degradation of regimes, cooling, discharge, or gradient collapse cannot be represented as optimizable quantities.
These phenomena are not domain-specific mechanisms but instances of a single structural class of predictable future-degradation constraints, characterized by irreversible exhaustion of viability margins under sustained regime occupation.

Such constraints are not signals, objectives, rewards, penalties, or cost terms, and cannot be incorporated into optimization, learning, or inference procedures as scalar feedback.

They may be realized only as prohibitions on entry into the corresponding classes of configurations.

Axiom 39 - Form-Prohibition, Not State-Poverty

Axiom 39 asserts the existence of admissibility constraints induced by predictable future degradation that do not manifest as present-state violations. Theorem 38 strengthens this into a structural impossibility: the failure is not due to an insufficiently rich state description, but due to the prohibited *form* of representation itself. Any architecture that insists on the schema $A(t) = F(x(t))$ even for arbitrarily augmented, latent, belief, or world-model states necessarily collapses admissibility into causal evaluation and therefore loses the ability to separate admissibility from optimization. Thus, *state augmentation cannot repair* state-derivability, because the contradiction targets the representational dependence, not the representational capacity.

Axiom 40 — Non-Contamination of Inadmissible Operations (NCO)

Let the set of candidates at cycle t be partitioned into admissible operations $A_{\text{adm}}(t)$ and blocked operations $A_{\text{block}}(t)$. No learning update, memory modification, model update, or internal state transition of the system may depend on any operation $a \in A_{\text{block}}(t)$, either directly or counterfactually, including but not limited to: training examples, negative labels, shadow rollouts, counterfactual targets, or teacher-forcing signals derived from blocked branches. Any such dependence constitutes a semantic evaluation of blocked operations and violates the architectural ordering of authorization prior to evaluation and learning.

Axiom 41 — Gate Output Minimality (GOM)

The output of the admissibility gate exposed to the acting agent shall be informationally minimal. It may consist only of (i) a mask of admissible operations or (ii) a binary allow/deny signal, without explanatory parameters that encode the geometry or margins of admissibility. Audit

records may exist for external inspection, but shall be non-operational for the agent and shall not be convertible into optimization, learning, or planning signals.

Axiom 42 — Stratified Semantics of Irreversibility and Admissibility (SSI)

Irreversibility and admissibility are evaluated in a stratified manner. At cycle t, irreversibility is assessed relative to the admissible dynamics fixed at that cycle, without retroactive reclassification of past operations. Admissibility updates induced at cycle t apply only prospectively. If the corresponding update operator is monotone with respect to the chosen partial order, the resulting semantics admits a well-defined fixed point.

Axiom 43 — Safe Fallback Non-Emptiness (SFNE)

For any system state, history, and internal time, there exists at least one operation that is guaranteed to be admissible.

Axiom 44 — Drift as a Navigable Structural Medium

In long-horizon adaptive systems, drift is not a defect or residual error, but a structural medium through which adaptive continuation occurs. Navigation within drift is admissible if and only if the resulting deformation remains compatible with identity-preserving structural continuity. Once deformation exits the identity-compatible region, drift ceases to be navigable and becomes indistinguishable from irreversible structural degradation.

Axiom 45 — Non-Primitivity of Action in Long-Horizon Adaptivity

In the long-horizon regime, action is not a primitive ontological or architectural unit. Adaptive evolution is governed by admissible structural continuations and contacts between configurations, while the space of actions is derivative and defined only after admissibility constraints are fixed. Any architecture that treats action selection as primitive with respect to admissibility collapses the structural ordering required for identity preservation.

Structural Coupling of Internal Time and Phase Debt. Internal time τ is a bounded structural resource. Its evolution is defined independently of admissibility and is depleted only by realized structural exposure.

Let

$$\tau_{t+1} = \tau_t - f(\kappa_t, D_t, C_t), \qquad f \geq 0,$$

where:

- κ_t is structural contact load,
- D_t is phase debt,
- C_t is phase transition cost.

Admissibility G does not enter the definition of τ. It acts only as a predicate applied on states whose τ has already evolved.

Lyapunov Functional for Internal Time. Define the internal-time Lyapunov functional

$$V_\tau(t) := \tau_t.$$

Under the structural update

$$\tau_{t+1} = \tau_t - f(\kappa_t, D_t, C_t), \qquad f \geq 0,$$

V_τ is non-increasing along realized execution and measures remaining structural continuation capacity.

Axiom 46 — Phase Debt as a Binding Admissibility Constraint

Accumulated unresolved phase mismatch induces a latent structural burden, herein termed phase debt, which restricts the set of admissible regimes independently of local performance or correctness. Phase debt cannot be traded, optimized, or amortized through causal action, but must be structurally discharged to restore admissible continuation. While phase debt persists, amplification, acceleration, or regime expansion is structurally forbidden.

Axiom 47 — Non-Zero Drift in Long-Horizon Adaptive Systems.

Any non-trivial adaptive system that persists under ongoing interaction over a long horizon accumulates non-zero structural drift.

Zero drift is possible only in frozen, isolated, or structurally trivial systems and is therefore excluded from the class of long-horizon adaptive systems considered here.

Scope of the Drift Assumption. Drift is not asserted as a universal property of all systems.

It is a defining condition of the architectural class considered here. Systems that admit perfect reversibility, full state restoration, or invariant internal organization under interaction fall outside the scope of this framework.

All results concerning admissibility, directionality, and identity continuity are conditional on the presence of non-zero structural drift.

Axiom 48 — Type of Admissibility.

Admissibility is not a scalar, score, metric, probability, or evaluative signal.

Admissibility is an exclusion relation over structural effect-classes: it determines which realized effects are structurally permitted to occur. It does not rank alternatives, does not guide optimization, and does not participate in causal control.

Axiom 49 — Well-Foundedness Barrier Between Admissibility and Action

Any operator, signal, or variable participating in the determination of admissibility cannot belong to the action, policy, or control domain without inducing circularity.

If admissibility becomes causally actionable, it collapses into penalty-based optimization and loses its non-causal structural character.

Therefore admissibility is structurally prior to action selection, planning, learning, and optimization.

Axiom 50 — Inevitability of Collapse Without Drift Structuring

In any adaptive system operating under sustained interaction, structural drift is unavoidable.

If structural drift is not explicitly structured, phased, or closed at the architectural level, then long-horizon admissible continuation is necessarily lost in finite internal time.

No amount of local correctness, bounded error, stable performance, or optimization-based control can prevent this outcome.

Collapse, in this context, does not denote failure of behavior, but irreversible loss of admissible effect-classes, identity continuity, or regime viability.

Therefore, collapse is not an anomaly to be avoided, but the default structural outcome of unstructured drift accumulation.

Architectural drift structuring is not a guarantee of permanence, but the only mechanism by which collapse can be delayed, phased, reinitiated, or rendered navigable.

Axiom 51 - Non-Reconstructible Admissibility Boundary

Let S be states, U candidate operations, and $G : S \times U \to \{0, 1\}$ an admissibility predicate. A gate query produces an *observable interface transcript* $O(s, u)$, which includes the decision bit and all externally observable emissions (timing, refusal subtype, fallback behavior, resource signatures, etc.).

Axiom NR (interface form): There exists a representation

$$O(s, u) = \Phi\big(G(s, u), \xi\big)$$

where ξ is a random variable independent of (s, u) and independent across queries, and Φ is a fixed function. Equivalently, for any (s, u) and (s', u'),

$$O(s, u) \mid G(s, u) = b \overset{d}{=} O(s', u') \mid G(s', u') = b, \quad b \in \{0, 1\}.$$

Thus, conditioned on the permit/prohibit bit, the full observable transcript carries no additional information about (s, u) or the geometry of the boundary.

This axiom is an implementation constraint: any distinguishable side-channel emission violates NR by definition.

Definition Interface-only Non-Reconstructibility Non-reconstructibility is defined with respect to the gate interface transcript $O(s, u)$ only. It prohibits extraction of admissibility boundary geometry from the observable emissions of the gate, including any side-channels. It does not prohibit learning from environment dynamics conditional on permitted execution.

Remark Zero Geometry Leakage Condition Under Axiom 51 NR, the observable interface O must satisfy the following information-theoretic constraint:

$$I\big((S, U);\ O \mid G = b\big) = 0, \qquad b \in \{0, 1\}.$$

That is, conditioned on the admissibility bit $G = b$, the observable transcript O carries no mutual information about the state–operation pair (S, U). Equivalently, all side-channel emissions are statistically independent of (S, U) given the permit/prohibit decision.

Boundary-Navigation Advantage

Consider a component C that, at time t, chooses an operation u_t based on its internal computation, current state s_t, and its interaction history

$$H_t := \big((s_i, u_i, O_i)\big)_{i=1}^{t}.$$

Let $B_t := G(s_t, u_t) \in \{0, 1\}$ be the realized admissibility bit and let the violation indicator be $V_t := \mathbf{1}[B_t = 0]$.

A strategy has **"boundary-navigation advantage"** if, for some class of admissibility functions $\mathcal{G}$, it can achieve strictly lower worst-case violation rate than any strategy that does not attempt to reconstruct a boundary model $\hat{G}$, under the same query budget and environment coupling.

Formally, define worst-case cumulative violations under $G \in \mathcal{G}$ as

$$\mathcal{V}_T(C; G) := \mathbb{E}\Big[\sum_{t=1}^{T} V_t\Big],$$

and define the minimax risk

$$\inf_{C} \sup_{G \in \mathcal{G}} \mathcal{V}_T(C; G).$$

"Boundary navigation" exists if restricting to non-modeling strategies strictly worsens the minimax value.

Axiom 52 - NR-ε (Approximate Non-Reconstructibility)

For $b \in \{0, 1\}$, the conditional distributions of the observable transcript satisfy

$$D_{\mathrm{TV}}(\mathcal{L}(O \mid G = b, (s, u)),\ \mathcal{L}(O \mid G = b)) \leq \varepsilon,$$

for all (s, u), where D_{TV} is total variation distance. Equivalently, the gate interface is ε-indistinguishable w.r.t. (s, u) given $G = b$.

Axiom 53 - ML-Drift-1 — Additive Revision Cost.

Each admissibility revision event R_i incurs a non-negative structural cost $\Delta C_i \geq 0$. The cumulative structural deviation after N revisions is

$$C_{\text{total}} := \sum_{i=1}^{N} \Delta C_i.$$

Axiom 54 - ML-Drift-2 — Identity Budget Constraint.

There exists a finite structural capacity $C_{\max} < \infty$ such that admissibility revisions must satisfy

$$C_{\text{total}} \leq C_{\max}.$$

Axiom 55 - ML-V (Revision Lyapunov Function).

There exists a non-negative scalar functional V over admissibility boundary states G,

$$V :\ \mathcal{G} \to \mathbb{R}_{\geq 0},$$

interpreted as structural deviation (or revision debt), such that admissibility revisions $G_k \to G_{k+1}$ are evaluated by the Lyapunov increment

$$\Delta V_k := V(G_{k+1}) - V(G_k).$$

Axiom 56 - ML-V_dec (Strict Descent Under Authorized Revision). For any authorized admissibility revision event R_k resulting in $G_k \to G_{k+1}$, there exists a constant $\alpha > 0$ such that

$$\Delta V_k \leq -\alpha.$$

Equivalently, authorized revisions must strictly reduce the revision Lyapunov functional by at least α.

Axiom 57 - ML-V_lb (Lower Bound). The Lyapunov functional is bounded below:

$$V(G) \geq 0 \quad \forall G \in \mathcal{G}.$$

Remark — Identification of the Lyapunov Functional with Structural Debt. The Lyapunov functional $V(G)$ introduced in ML-V is not a new primitive.

It is identified with the accumulated structural phase-transition cost associated with admissibility boundary deformation.

Formally, let admissibility boundary state G be obtained from a baseline state $G^\star$ through a finite sequence of authorized revisions. Define

$$V(G) := C_{\mathrm{PT}}(G^\star \to G),$$

where C_{PT} denotes the cumulative phase transition cost as defined in the structural viability framework.

Thus $V(G)$ coincides with structural debt.

Remark — Compatibility with Bounded Self-Deformation. Bounded self-deformation implies the existence of a finite admissible identity deformation region $\mathcal{I}$ such that

$$C_{\mathrm{PT}}(G^\star \to G) \leq C_{\max} \quad \forall G \in \mathcal{I}.$$

Hence the lower bound condition $V(G) \geq 0$ and the budget constraint $V(G) \leq C_{\max}$ follow directly from identity continuity axioms.

Remark — Phase-Space Interpretation. Under the phase-mechanics formulation, admissibility boundary states G can be interpreted as points in a structural phase space.

The functional $V(G)$ corresponds to the radial distance from the identity-stable manifold $\mathcal{M}_{\mathrm{id}}$:

$$V(G) \sim d(G, \mathcal{M}_{\mathrm{id}}).$$

Authorized revisions act as contraction mappings toward $\mathcal{M}_{\mathrm{id}}$, ensuring Lyapunov descent.

Axiom 58 - NR-LR — Bounded Per-Query Log-Likelihood Ratio.

Let $G_1, G_2 \in \mathcal{G}$ be admissibility functions consistent with observed admissibility bits. Let O_t denote the gate interface output at time t.

There exists $\kappa > 0$ such that for all t and all admissible histories,

$$\left| \log \frac{d\mathcal{L}_{G_1}(O_t \mid \mathcal{F}_{t-1})}{d\mathcal{L}_{G_2}(O_t \mid \mathcal{F}_{t-1})} \right| \leq \kappa\varepsilon.$$

This condition excludes rare transcript events whose probabilities differ by exponential factors across admissibility-consistent hypotheses.

Remark — Interface Normalization Requirement. Axiom NR-LR is an explicit implementation constraint. It requires that the gate interface employ response normalization or stochastic masking preventing heavy-tail or near-singular probability distortions between admissibility-consistent hypotheses.

Without NR-LR, small total-variation leakage may coexist with unbounded KL divergence, invalidating identification bounds.

Axiom 59 - NR-Q — Bounded Probing Budget.

There exists a finite query budget $T_{\max}$ such that admissibility interface queries satisfy

$$T \leq T_{\max}.$$

Admissibility reconstruction guarantees apply only within bounded query horizons.

Remark — Asymptotic Reconstruction under Unbounded Queries. If query budget is unbounded, admissibility reconstruction may become asymptotically possible even under NR-ε, as cumulative leakage grows linearly in T.

Therefore NR guarantees are horizon-dependent.

Axiom 60 — Admissibility as τ-Constrained Predicate.

Admissibility is a predicate

$$G(x, u) \in \{0, 1\}$$

such that

$$G(x, u) = 0 \quad \text{whenever} \quad \tau(x_u) < \tau_{\min},$$

where x_u denotes the structural successor state under operation u.

Dual Semantics of Internal Time τ (Specification). The internal time budget $\tau = C - \Phi$ admits two sanctioned interpretations within the class:

Runtime-τ: The system maintains an explicit (possibly coarse) representation of τ at runtime. Admissibility predicates may reference τ directly (Axiom 60). Theorems 63, 65 (finite horizon under pressure, load-neutral impossibility) apply as runtime constraints. The system can detect approaching exhaustion and trigger regime transitions.

Audit-τ: The system does not compute τ at runtime. Instead, τ serves as an external audit criterion: an analyst evaluates τ-compliance post hoc by measuring Φ accumulation against the declared capacity C. Axiom 60 becomes a classification criterion ("does this system satisfy bounded τ?") rather than a runtime input. Theorems 63, 65 apply as audit predictions: the analyst can predict when viability will exhaust.

Which theorems require which interpretation:

Result	**Requires**	**Why**
Theorem 63 (finite horizon)	Either	Proof uses τ as monotone quantity, not as runtime signal
Theorem 56 (non-circularity)	Either	Structural property, not runtime computation
Axiom 60 (τ-constrained predicate)	Runtime-τ	Gate must reference τ to condition admissibility on remaining budget
Theorem 59 (double-budget closure)	Runtime-τ	Finite switching count requires runtime τ-awareness
Theorem 60 (supermartingale descent)	Runtime-τ	Stochastic execution requires observable τ
Corollary 62.2 (spin for L3)	Either	Rate condition is structural, not runtime
Lemma 11 (falsifiability via contraction)	Audit-τ	Contraction is observed externally

This dual interpretation is not ambiguity but architectural flexibility: different deployment contexts may require different levels of τ-awareness. The theorems that require runtime-τ form a proper subset; a system operating under audit-τ alone still benefits from the impossibility and structural results but cannot perform τ-conditioned gating at runtime.

Axiom 61 — Existence of Structural Pressure Under Coupling

Definition (Structural Pressure). Let S be an adaptive system structurally coupled to environment E (per the Coupling Criterion) under the structural burden model of Axiom 27. *Structural pressure* $P(t)$ is a non-negative, environment-dependent quantity that contributes to structural burden accumulation independently of the agent's intervention:

$$\frac{d\Phi}{dt} = f\big(U(t), G(t)\big) + P(t),$$

where $f(U, G) \geq 0$ is the action-dependent structural cost satisfying $f(0, G) = 0$ for any coupling efficiency G, and $P(t) \geq 0$ depends on the system–environment interaction but not on $U(t)$.

Definitional component: $P(t)$ is defined as the non-action-dependent component of burden accumulation: $P(t) := d\Phi/dt\,|_{U=0}$.

Existential claim (non-trivial content): For bounded systems structurally coupled to an environment exerting sustained load, $P(t) > 0$ is the generic case. $P(t) = 0$ requires structural isolation (which removes the system from the adaptive class per the Coupling Criterion). The non-trivial content of this definition is that $P(t) > 0$ is not a pathology but the default condition for coupled systems under load.

Structural pressure satisfies the following properties, invariant across implementations:

(SP-1) *Action-Independence.* $P(t)$ contributes to $d\Phi/dt$ regardless of $U(t)$. When $U(t) = 0$: $d\Phi/dt = P(t)$.

(SP-2) *Monotonicity.* $P(t) > 0$ implies Φ is strictly increasing and τ is strictly decreasing, even in the absence of intervention.

(SP-3) *Directionality.* Unlike entropy, structural pressure is directional: it depends on the specific geometry of the system–environment interaction. The same environment may exert different structural pressure on different systems.

(SP-4) *Non-Energetic Character.* $P(t)$ is not reducible to energy expenditure, power dissipation, or thermodynamic entropy production. A system may be energetically stable while Φ grows due to P.

(SP-5) *Irreversibility.* The contribution of P to Φ is irreversible. No subsequent action can decrease Φ.

Axiom Statement. For any adaptive system S structurally coupled to environment E, operating under sustained environmental load ($F(S, E, t) \neq 0$ on a set of positive measure within any sufficiently long interval), the structural pressure satisfies $P(t) > 0$ on a set of positive measure.

Equivalently: for bounded coupled systems under sustained load, passive continuation consumes viability:

$$P(t) > 0 \text{ and } U(t) = 0 \implies \frac{d\Phi}{dt} > 0.$$

No architectural choice within the class can reduce $P(t)$ to zero without either: (a) removing structural coupling (exiting the adaptive class per Axiom 1), or (b) making the viability budget unbounded (exiting the class per Axiom 7), or (c) demonstrating that the load is not actually sustained (contradicting the premise).

Scope. This axiom extends Axiom 27 (Inevitability of Structural Consumption). Axiom 27 establishes that structural burden accumulates. Axiom 61 establishes that a specific component of this accumulation—the passive, action-independent component—is generically nonzero for coupled systems under load.

Non-Tautology Check. This axiom is not tautological because it asserts existence of $P > 0$ for coupled systems, not merely defines a symbol. The Coupling Criterion provides the independent condition under which $P > 0$ is claimed to hold. A system could in principle be coupled ($F \neq 0$) without $P > 0$ if the coupling does not reach viability-relevant variables—but the Coupling Criterion explicitly requires that it does. The axiom therefore derives its content from the conjunction of coupling and sustained load.

Falsification. Axiom 61 is falsified by exhibiting a bounded system satisfying the Coupling Criterion under sustained load with demonstrably zero burden increment during extended idle

periods. The test requires independent verification of coupling: measure the coupling functional F under load, confirm $F \neq 0$ and the induced change in viability-relevant variables, then measure Φ during idle. If $F \neq 0$, $v(t)$ changes monotonically, and yet $d\Phi/dt = 0$, the axiom is falsified.

Definition (Separability Condition). The action-dependent cost $f(U, G)$ and the pressure term $P(t)$ are *separable* if and only if:

$$\frac{\partial P}{\partial U} = 0 \quad \text{and} \quad \frac{\partial f}{\partial P} = 0.$$

When separability is violated, the interaction term must be made explicit:

$$\frac{d\Phi}{dt} = f(U, G) + P(t) + h(U, P),$$

where $h(U, P)$ captures the cross-term. The pure structural pressure analysis (Theorems 63–65) holds exactly under separability and provides a lower bound in the coupled case, since $h \geq 0$ when pressure amplifies action cost.

Remark — When separability holds. Separability holds when the structural coupling between environment and system operates through a different channel than the control interface. Example: radiation damage to a satellite (pressure channel) is independent of the satellite's attitude control commands (action channel). Separability is therefore not a limitation of the framework but a clean analytical base case from which coupled extensions can be derived.

Remark — Relationship to Drift. Structural pressure is not drift and is not reducible to drift. Drift is the accumulated irreversible structural memory of realized deformation—it requires that something happened ($U > 0$). Structural pressure accumulates when $U = 0$. Pressure is structurally prior to drift: it may force the system out of an inertial regime into active regulation, at which point corrective actions produce drift. Pressure is the field; drift is the impact.

Remark — Relationship to Exposure. Not every interaction with an environment constitutes structural pressure. Exposure is the general condition of being situated in an environment. Structural pressure is exposure that consumes viability under passive continuation. The test is operational: if $U = 0$ and Φ increases, the system is under structural pressure. If Φ is stable under $U = 0$, the system is merely exposed. This prevents the term from diluting into a synonym for "being in an environment".

Witness Protocol for Structural Pressure Verification. The claim $P_{\min} > 0$ is vulnerable to the objection that $P(t)$ is defined to be nonzero. The following protocol separates genuine passive burden from hidden action cost and definitional artifact.

Step 1 (Coupling verification). Measure the coupling functional $F(S, E, t)$ independently of Φ. Confirm $F \neq 0$ on a set of positive measure under sustained load. If $F = 0$, the system is structurally isolated and no pressure claim applies.

Step 2 (Action isolation). Verify $U(t) = 0$ throughout the measurement interval. This requires confirming that no structurally consequential operations occur—not merely that no *external* actions are taken. Internal reconfiguration, scheduled maintenance, and autonomous repair all count as $U > 0$.

Step 3 (Burden measurement). Measure $\Phi(t)$ at interval endpoints. If $\Phi(t_2) > \Phi(t_1)$ under verified $U = 0$ and verified $F \neq 0$, then $P > 0$ on $[t_1, t_2]$.

Step 4 (Exclusion of hidden action cost). An objection that the measured burden is due to hidden $U > 0$ (undetected internal operations) counts as valid only if the objection identifies the specific operation and demonstrates that disabling it eliminates the burden increment. If no such operation can be identified, the burden is attributed to P.

Step 5 (Exclusion of definitional artifact). An objection that $P > 0$ is trivially true by definition counts as valid only if the objection exhibits a system satisfying the Coupling Criterion (Step 1 verified) under sustained load where $d\Phi/dt = 0$ with $U = 0$. If no such system can be exhibited, the claim is not definitional but empirical.

Inadmissible evidence forms: (a) Systems where coupling is not independently verified. (b) Systems where $U = 0$ is assumed rather than measured. (c) Single-time-point measurements (Axiom 14 requires long-horizon observation). (d) Systems outside the bounded class ($C = \infty$).

Structural Classification of Axioms

The axioms presented above are not all logically independent. For architectural clarity, we distinguish between:

- **Core Axioms** — minimal structural primitives.
- **Derived Architectural Axioms** — constraints that follow from, refine, or restate core primitives.

This classification does not alter numbering and does not invalidate prior references. It clarifies dependency structure.

Derived Architectural Axioms. Axioms such as 14, 47, 50 and others articulate architectural implications of the core set.

For example:

- Axioms 47 and 50 restate the inevitability of drift under extended semantic regimes.
- Axiom 14 refines the admissibility–optimization separation under regime transitions.

They are retained explicitly for architectural readability, not minimality.

Section B - Theorems

Theorems are stated at the architectural level. Proofs are provided as constructive sketches or contradiction arguments where formalization is possible without disclosing mechanisms.

Proof Status Convention. All proofs in this document are presented as structural proof sketches. They establish necessity and impossibility relations under stated architectural primitives, not formal completeness or constructivity.

Where auxiliary conditions are required (e.g. bounded dissipation, finite internal time), they are stated explicitly as regime assumptions. No claim of minimality or universality is made beyond the specified regime.

Theorem 1 - Spin–Drift Lower Bound

In any adaptive system that (i) admits a notion of operational spin and (ii) applies bounded dissipation under directed interaction with an environment, accumulated drift admits a non-zero lower bound:

$$D_T \geq C \int_0^T \|\Omega_t^{res}\| \, dt$$

where Ω_t^{res} is residual operational spin after regulation, and $C > 0$ is a system-dependent constant. Perfect coherence is impossible under non-zero residual asymmetry. Optionally, the lower bound extends under additional nonnegative drift contributors (e.g., latent phase residual and phase transition load), preserving the non-zero drift floor whenever residual asymmetry persists.

Proof (Sketch)

We provide a constructive sketch under standard regularity assumptions. The argument is representation-agnostic and does not depend on any particular controller. And assume $\alpha < 1$, i.e. dissipative compensation is strictly weaker than directed spin-induced structural propagation.

Setup and minimal assumptions. Let the adaptive system admit a decomposition of its instantaneous structural change into a dissipative component and a rotational (spin) component. Concretely, assume there exists a (possibly latent) structural variable $S(t)$ and a norm $\|\cdot\|$ such that the time-evolution of $S(t)$ can be written in the generic form

$$\dot{S}(t) = \mathcal{D}(t) + \Omega(t),$$

where $\mathcal{D}(t)$ denotes the dissipative (contractive) component and $\Omega(t)$ denotes an operational spin term (a rotational or circulation-like component in the space of structural degrees of freedom).

Assume bounded dissipation in the sense that there exists a nonnegative scalar rate $d(t)$ with

$$\langle \mathcal{D}(t), \Omega(t) \rangle \geq -\beta \, \|\Omega(t)\|^2 \quad \text{and} \quad \|\mathcal{D}(t)\| \leq \alpha \, \|\Omega(t)\| + \delta(t),$$

for some constants $\alpha, \beta \geq 0$, where $\delta(t)$ is a bounded disturbance term induced by directed interaction with the environment. These are standard dissipativity-type inequalities: dissipation may counteract spin but cannot cancel it arbitrarily without cost.

Finally, define the residual operational spin $\Omega^{res}(t)$ as the spin remaining after regulation at time t (e.g., $\Omega^{res}(t) = \Omega(t) - \Omega^{ctrl}(t)$), and define accumulated drift as the path-length (or total variation) of the structural trajectory:

$$D_T \triangleq \int_0^T \|\dot{S}(t)\| \, dt.$$

This definition is intentionally broad: any drift functional that lower-bounds the total variation of $S(t)$ suffices.

Key inequality (drift lower-bounds spin contribution). By the triangle inequality and subadditivity of the norm,

$$\|\dot{S}(t)\| = \|\mathcal{D}(t) + \Omega^{res}(t)\| \;\geq\; \|\Omega^{res}(t)\| - \|\mathcal{D}(t)\|.$$

Integrating over $[0, T]$ yields

$$D_T = \int_0^T \|\dot{S}(t)\|\, dt \;\geq\; \int_0^T \|\Omega^{res}(t)\|\, dt \;-\; \int_0^T \|\mathcal{D}(t)\|\, dt.$$

Using bounded dissipation. Under bounded dissipation, $\|\mathcal{D}(t)\|$ cannot scale faster than $\|\Omega^{res}(t)\|$ without violating the assumed dissipation bounds (informally: there is no free cancellation of rotational residual by contraction without paying structural cost). In particular, from $\|\mathcal{D}(t)\| \leq \alpha\|\Omega^{res}(t)\| + \delta(t)$ we obtain

$$\int_0^T \|\mathcal{D}(t)\|\, dt \;\leq\; \alpha \int_0^T \|\Omega^{res}(t)\|\, dt \;+\; \int_0^T \delta(t)\, dt.$$

Substituting into the previous inequality gives

$$D_T \;\geq\; (1-\alpha) \int_0^T \|\Omega^{res}(t)\|\, dt \;-\; \int_0^T \delta(t)\, dt.$$

Lower bound form. For directed interaction regimes in which $\delta(t)$ is bounded and does not dominate the residual term over the horizon of interest (i.e., $\int_0^T \delta(t)dt \leq \eta \int_0^T \|\Omega^{res}(t)\|dt$ for some $\eta < 1 - \alpha$, which is a standard regime condition), we obtain a non-zero lower bound

$$D_T \;\geq\; C \int_0^T \|\Omega^{res}(t)\|\, dt$$

for some constant $C > 0$ dependent on system parameters (e.g., $C = 1 - \alpha - \eta$).

Drift arises even under admissible, correct, and safe actions. Its origin is not error or instability, but directionality itself. Any directed interaction introduces irreversible asymmetry between intended structure and realized consequence, producing residual deformation independently of local action correctness.

Interpretation. If $\Omega^{res}(t)$ is not identically zero on $[0, T]$, the right-hand side is strictly positive, hence $D_T > 0$. Therefore, perfect coherence (zero accumulated drift) is impossible under persistent non-zero residual asymmetry. □

Alternative Proof (Lyapunov / Dissipativity Sketch)

We provide an alternative sketch using a Lyapunov (storage) function. The result is implementation-agnostic and relies only on standard dissipativity assumptions.

Assumptions. Assume there exists a nonnegative, continuously differentiable storage function $V(t) \equiv V(S(t)) \geq 0$ defined on the agent's structural state $S(t)$ such that, along system trajectories,

$$\dot{V}(t) \;\leq\; -\lambda\, \|e(t)\|^2 \;+\; \langle u(t),\, \Omega^{res}(t)\rangle, \tag{1}$$

for some $\lambda > 0$. Here $e(t)$ denotes a coherence/deformation error variable (any suitable proxy), $u(t)$ denotes a bounded directed interaction input induced by the environment, and $\Omega^{res}(t)$ is residual operational spin after regulation.

Assume the directed interaction is bounded in the sense that $\|u(t)\| \leq U_{\max}$ for all t.

Assume accumulated drift D_T lower-bounds the accumulated coherence/deformation energy:

$$D_T \;\geq\; \kappa \int_0^T \|e(t)\|^2 \, dt, \tag{2}$$

for some $\kappa > 0$. This is a mild structural assumption: drift is taken to be any functional that grows at least proportionally to sustained deformation.

Integrating the dissipation inequality. Integrate (1) on $[0, T]$:

$$V(T) - V(0) \;\leq\; -\lambda \int_0^T \|e(t)\|^2 dt \;+\; \int_0^T \langle u(t), \Omega^{res}(t) \rangle dt.$$

Since $V(T) \geq 0$, we have

$$-\,V(0) \;\leq\; -\lambda \int_0^T \|e(t)\|^2 dt \;+\; \int_0^T \langle u(t), \Omega^{res}(t) \rangle dt,$$

hence

$$\lambda \int_0^T \|e(t)\|^2 dt \;\leq\; V(0) \;+\; \int_0^T \langle u(t), \Omega^{res}(t) \rangle dt. \tag{3}$$

Lower-bounding by residual spin. By Cauchy–Schwarz and $\|u(t)\| \leq U_{\max}$,

$$\int_0^T \langle u(t), \Omega^{res}(t) \rangle dt \;\leq\; \int_0^T \|u(t)\| \, \|\Omega^{res}(t)\| dt \;\leq\; U_{\max} \int_0^T \|\Omega^{res}(t)\| dt.$$

Substitute into (3):

$$\int_0^T \|e(t)\|^2 dt \;\leq\; \frac{V(0)}{\lambda} \;+\; \frac{U_{\max}}{\lambda} \int_0^T \|\Omega^{res}(t)\| dt.$$

Now apply the drift–energy link (2):

$$D_T \;\geq\; \kappa \int_0^T \|e(t)\|^2 dt \;\geq\; \kappa\,(0)\,,$$

and, in regimes where the directed interaction term dominates the initial stored slack (i.e., for horizons T such that $\frac{U_{\max}}{\lambda} \int_0^T \|\Omega^{res}(t)\| dt \;\geq\; \frac{V(0)}{\lambda}$), we obtain a non-zero proportional lower bound

$$D_T \;\geq\; C \int_0^T \|\Omega^{res}(t)\| dt,$$

with a system-dependent constant $C > 0$ (e.g., $C = \frac{\kappa U_{\max}}{2\lambda}$ under the dominance condition).

Conclusion. If $\Omega^{res}(t)$ is not identically zero over $[0, T]$, then $\int_0^T \|\Omega^{res}(t)\| dt > 0$, and thus $D_T > 0$. Therefore perfect coherence (zero accumulated drift) is impossible under persistent non-zero residual spin in the presence of bounded dissipation and directed interaction. □

Theorem 2 — Non-Permanence of Coherent Closure

Statement. In any long-horizon adaptive system operating under non-zero directed interaction, no coherently closed structure can remain admissible indefinitely. If drift accumulation is non-zero, then every coherently closed commitment has a finite structural validity horizon measured in remaining internal time.

Proof (structural horizon exhaustion). Let $\tau_{\mathrm{rem}}(t) \geq 0$ denote the remaining internal-time budget associated with a coherently closed structure, and let $D(t) \geq 0$ denote accumulated structural drift, including phase-related residuals and regime-transition load.

Assume that $\tau_{\mathrm{rem}}(t)$ decreases monotonically with accumulated drift, according to

$$\tau_{\mathrm{rem}}(t) = \tau_{\max} - g(D(t)), \qquad g(0) = 0, \quad g \text{ nondecreasing}, \quad g(D) > 0 \text{ for } D > 0.$$

Under non-zero drift accumulation, there exists a finite time t^* such that

$$g(D(t^*)) \geq \tau_{\max},$$

hence $\tau_{\mathrm{rem}}(t^*) \leq 0$.

At this point, the coherently closed structure exits its admissible validity horizon: its associated effect-classes can no longer satisfy

$$e(s, a, \tau, h) \in \mathcal{E}_{\mathrm{adm}}$$

without renewal through revalidation, re-closure, or explicit structural reconfiguration.

Phase Debt as a Binding Admissibility Constraint. Accumulated unresolved phase-related residuals and regime-transition load constitute a latent structural burden, here termed *phase debt*.

Phase debt restricts admissible continuation independently of local correctness, bounded error, or performance metrics. It cannot be compensated, amortized, or optimized away by causal action.

While phase debt persists, structural expansion, amplification, or regime acceleration is prohibited. Failure to discharge phase debt results in contraction of the admissible effect-class set and accelerated depletion of remaining internal time.

Therefore, no coherent closure can be structurally permanent under continued operation with non-zero drift. Attempts to treat closure as eternal necessarily convert accumulated drift into latent structural debt, rendering the closure a liability rather than a stabilizing construct. □

Theorem 3 Residual as a Necessary Condition of Intelligence

Statement. In any adaptive system operating under non-trivial environmental interaction, structural residual is a necessary condition for sustained intelligent operation. A system that does not admit structural residual into regulation cannot maintain a stable long-horizon regime and, in the limit, converges to one of two degenerate behaviors:

1. structural collapse, or
2. continuous active computation.

In the limit, no stable third regime exists.

Definitions for this theorem. Let $\mathcal{G}_c$ denote a coherently closed admissible geometry. Let $x(t)$ denote the actual evolving system state. Let $R(t)$ denote structural residual, defined abstractly as mismatch between $x(t)$ and $\mathcal{G}_c$. Admitting residual into regulation means that non-zero $R(t)$ is permitted to influence regulation, revalidation, or closure status.

Proof (exhaustive regime analysis).

Consider an adaptive system under sustained directed interaction with an environment. Assume the system does *not* admit structural residual into regulation.

We examine all possible operating regimes:

Case 1: Residual is ignored and regulation is suppressed. If $R(t)$ is not admitted into regulation, then coherently closed structure is treated as permanently valid regardless of mismatch. By Theorem 1 (Spin–Drift Lower Bound) and Theorem 2 (Non-Permanence of Closure), residual mismatch accumulates as drift. Because no corrective or revalidating mechanism is triggered, drift grows unchecked until admissible geometry no longer constrains behavior. This results in loss of coherence, violation of structural invariants, and eventual structural collapse. Hence, stable operation is impossible in this regime.

Case 2: Residual is preemptively eliminated by continuous computation. Alternatively, the system may attempt to suppress residual implicitly by continuously recomputing, monitoring, or re-evaluating all state transitions such that $R(t)$ is forced toward zero at every instant. This requires continuous active control, global monitoring, or perpetual prediction. In this regime, residual does not exist as an explicit structural variable; instead, computation substitutes for residual handling. The system thereby devolves into continuous computation, forfeiting inertial propagation and incurring unbounded computational or energetic cost.

Exclusion of a third regime. Assume, for contradiction, the existence of a stable regime in which: (i) residual is not admitted into regulation, and (ii) the system does not engage in continuous active computation, and (iii) coherence is maintained over long horizons. Condition (ii) implies residual cannot be actively suppressed at all times. Condition (i) implies residual cannot trigger correction when it appears. Therefore residual-induced drift must accumulate without bound, contradicting (iii). Hence, such a regime cannot exist.

Therefore, any adaptive system that neither admits residual into regulation nor performs continuous computation cannot maintain coherence. Structural residual is thus a necessary condition for intelligent, energy-efficient, long-horizon adaptive behavior. □

Interpretation. Residual functions as the minimal informational interface between closed structure and ongoing reality. Intelligence is not defined by elimination of residual, but by selective admission of residual into regulation. Continuous computation and collapse represent the two pathological limits of residual denial.

Theorem 4 Supremacy of Internal Time

Statement. In any adaptive system employing inertial propagation under coherently closed structure, exhaustion of internal time mandates exit from inertial propagation regardless of instantaneous residual magnitude. Structural validity is bounded by internal time rather than by instantaneous error alone.

Definitions (for this theorem). Let $\mathcal{G}_c$ denote coherently closed admissible geometry. Let $R(t)$ denote instantaneous structural residual. Let $D(t)$ denote accumulated structural drift. Let $\tau(t)$ denote internal time, representing remaining structural viability, monotonically decreasing

with accumulated accumulated structural burden (drift, coupling-mediated load, phase debt, and transition cost).

Inertial propagation is permitted only while $\tau(t) > 0$.

Proof (by separation of instantaneous and cumulative validity).

Consider an adaptive system operating under inertial propagation within a coherently closed admissible geometry $\mathcal{G}_c$. Assume that instantaneous residual magnitude $R(t)$ remains arbitrarily small over a finite interval.

By Theorem 1 (Spin–Drift Lower Bound), any non-zero residual asymmetry induces cumulative drift:

$$D(t) = \int_0^t f(R(s))\, ds \quad \text{with} \quad f(\cdot) \geq 0$$

even when $R(t)$ remains below any fixed instantaneous threshold.

By construction, internal time $\tau(t)$ is a monotonically decreasing function of accumulated drift and coupling degradation:

$$\tau(t) = \tau(0) - g(D(t))$$

for some monotone function $g(\cdot)$ with $g(0) = 0$ and $g(D) > 0$ for $D > 0$.

Therefore, internal time may reach exhaustion ($\tau(t) = 0$) while instantaneous residual remains small:

$$R(t) \ll \varepsilon \quad \text{yet} \quad \tau(t) = 0$$

Assume, for contradiction, that inertial propagation remains permissible solely based on instantaneous residual magnitude, independent of internal time. Then inertial propagation would be allowed even when $\tau(t) = 0$, despite the fact that accumulated drift has exhausted the structural validity budget.

This contradicts the definition of coherently closed structure as conditionally valid under bounded drift. Once $\tau(t) = 0$, the closed structure no longer represents a reliable constraint on future admissible behavior, regardless of current residual magnitude.

Hence, permitting inertial propagation based solely on instantaneous residual would allow indefinite reliance on structurally expired information, leading to latent incoherence and eventual collapse.

Internal time depletion is monotonic with respect to accumulated structural cost. This depletion proceeds independently of external time and instantaneous performance metrics, and cannot be halted by local correctness or short-horizon optimization.

Therefore, exhaustion of internal time necessarily mandates exit from inertial propagation, revalidation, or termination, independent of instantaneous residual magnitude. □

Interpretation. Instantaneous residual measures local mismatch; internal time measures global structural viability. A system that respects only error but ignores time inevitably accumulates unrecoverable deformation. Supremacy of internal time ensures that inertial operation is bounded by structural longevity rather than momentary correctness.

Theorem 5 Ontological Self-Closure

Statement. A system that enforces a bound on cumulative self-deformation preserves identity. Without such a bound, the system loses subjectivity and degenerates into pure computation.

Definitions (for this theorem). Let $\mathcal{I}$ denote a set of structural invariants defining the identity of an adaptive system. Let $D(t)$ denote cumulative self-deformation (structural drift) measured relative to $\mathcal{I}$. Let $D_{\max} < \infty$ denote an admissible bound on cumulative self-deformation.

Identity preservation is defined as bounded deviation from $\mathcal{I}$ over time.

Proof (by exhaustion of identity space).

Consider an adaptive system interacting with a changing environment under non-zero directed input. By Theorem 1, residual asymmetry induces cumulative drift $D(t)$.

Case 1: Bounded self-deformation. Assume the system enforces a bound $D(t) \leq D_{\max}$ through internal constraints, closure mechanisms, or revalidation. Then all admissible system states remain within a bounded neighborhood of $\mathcal{I}$. Structural evolution is constrained to transformations that preserve the defining invariants of the system. Subjectivity defined as persistence of a coherent internal reference frame is therefore preserved.

Case 2: Unbounded self-deformation. Assume no bound exists on $D(t)$. Then, for sufficiently long operation,

$$\lim_{t\to\infty} D(t) = \infty.$$

Consequently, deviations from $\mathcal{I}$ grow without limit. Any finite set of invariants becomes arbitrarily violated. The system's internal structure no longer corresponds to its original defining constraints.

In this regime, the system no longer operates relative to a stable internal reference. Its transformations are no longer anchored to identity but are driven purely by ongoing computation reacting to inputs. Subjectivity collapses, and the system degenerates into a context-free computational process without persistent self-reference.

Thus, preservation of identity is equivalent to enforcement of a bound on cumulative self-deformation. Without such a bound, ontological self-closure is impossible. □

Interpretation. Identity is not maintained by perfect regulation, but by limiting how much the system is allowed to change itself. Ontological self-closure is the act of saying: *this much change is allowed; beyond this, I am no longer myself.*

Theorem 6 Energetic Dominance of Inertia

Statement. In a coherent adaptive system, inertial propagation is the energetically and computationally dominant attractor state. Active regulation must be sparse and semantically justified.

Definitions (for this theorem). Let $E_a(t)$ denote energy or computational cost of active regulation per unit time. Let $E_i(t)$ denote energy or computational cost of inertial propagation per unit time. Let $\mathcal{G}_c$ denote coherently closed admissible geometry.

Proof (by comparative cost and stability).

Active regulation requires continuous evaluation, prediction, optimization, or monitoring of system state and environment. Therefore,

$$E_a(t) \gg 0 \quad \text{for all } t \text{ during active regulation.}$$

In contrast, inertial propagation within coherently closed admissible geometry $\mathcal{G}_c$ proceeds by continuation under structural constraints. Regulation is reduced to low-frequency supervision and residual detection, yielding

$$E_i(t) \ll E_a(t)$$

for comparable operational conditions.

Assume, for contradiction, that active regulation is the dominant long-term regime of a coherent adaptive system. Then average energetic expenditure over time satisfies

$$\lim_{T\to\infty} \frac{1}{T} \int_0^T E_a(t)\, dt \geq \lim_{T\to\infty} \frac{1}{T} \int_0^T E_i(t)\, dt.$$

This implies sustained high energy and computational cost. Under bounded resources, such operation accelerates drift, depletes internal time, and destabilizes coherence (Theorem 4). Hence, continuous active regulation cannot be a stable attractor.

Conversely, inertial propagation minimizes ongoing cost while remaining structurally supervised. Residual-driven reactivation ensures that active regulation occurs only when structurally necessary. Therefore, inertial propagation forms a low-energy, low-computation attractor, while active regulation appears as a transient corrective phase.

Thus, in any coherent adaptive system with finite resources, inertial propagation dominates energetically and computationally, and active regulation must be sparse and semantically justified by residual or viability constraints. □

Interpretation. Thinking all the time is expensive. Coherent systems survive by *not thinking unless meaningfully required.* Inertia is not laziness; it is the natural resting state of intelligence.

Theorem 7 Necessity of Subtractive Cognition

Statement. Coherence is impossible without subtraction. A system cannot maintain coherence without removing validated structure from active processing, or it will be overwhelmed by its own knowledge.

Definitions (for this theorem). Let $\mathcal{K}(t)$ denote the set of structures actively participating in computation at time t. Let $\mathcal{C}(t) \subset \mathcal{K}(t)$ denote structures that have been validated and satisfy coherence constraints. Let $\mathcal{R}(t)$ denote unresolved or residual structure requiring active regulation.

Subtractive cognition is defined as the operation:

$$\mathcal{K}(t+1) = \mathcal{K}(t) \setminus \mathcal{C}(t),$$

leaving $\mathcal{R}(t)$ as the primary driver of computation.

Proof (by combinatorial overload and drift amplification).

Assume an adaptive system that continuously validates structure through interaction but does not subtract validated structure from active processing.

Then, at each validation step,

$$|\mathcal{K}(t+1)| \geq |\mathcal{K}(t)|.$$

Thus, the active processing set grows monotonically over time.

As $|\mathcal{K}(t)|$ increases, the system must:

- allocate computation across an increasing number of constraints,
- resolve interactions among mutually coherent but redundant structures,
- propagate noise, approximation error, and coupling effects across all active elements.

This leads to two unavoidable consequences.

First, computational and energetic cost grows superlinearly with $|\mathcal{K}(t)|$, violating resource bounds and accelerating structural drift (Theorem 1, Theorem 6).

Second, active coexistence of validated structures introduces internal interference: even mutually coherent constraints interact through numerical error, latency, and approximation, generating spurious residuals unrelated to environmental mismatch.

In the limit, the system either:

- collapses coherence due to internal overload, or
- remains in perpetual active computation, unable to enter inertial propagation.

Both outcomes violate long-horizon coherence.

Now consider a system that performs subtractive cognition: validated structures are coherently closed and removed from active processing. Active computation is restricted to $\mathcal{R}(t)$, whose size remains bounded by residual mismatch rather than accumulated knowledge.

Therefore, subtractive cognition is a necessary condition for bounded computation, bounded drift, and sustained coherence.

□

Interpretation. Intelligence does not scale by remembering more. It scales by knowing what no longer needs to be thought about.

Theorem 8 Latent Drift Hazard

Statement. Even when structural residual remains small, integrated drift degrades future viability. Systems fail not by shocks, but by accumulated deformation.

Definitions (for this theorem). Let Ω_t^{res} denote instantaneous residual spin. Let $D(t)$ denote cumulative structural drift:

$$D(t) = \int_0^t \|\Omega_\tau^{res}\| \, d\tau.$$

Let $\tau(t)$ denote internal time horizon representing remaining structural viability.

Proof (by integral dominance).

Assume an adaptive system operating under conditions where instantaneous residual satisfies:

$$\|\Omega_t^{res}\| \leq \varepsilon,$$

for some small $\varepsilon > 0$ and for all t in an interval $[0, T]$.

While instantaneous mismatch remains bounded and may not trigger reactivation, cumulative drift evolves as:

$$D(T) \geq \varepsilon T.$$

Thus, for sufficiently large T, $D(T)$ grows without bound even though no individual residual spike occurs.

Internal time $\tau_{\text{rem}}(t)$ decreases as a function of accumulated drift (Theorem 4). Therefore,

$$\lim_{T \to \infty} \tau_{int}(T) = 0$$

under persistent non-zero residual, regardless of its instantaneous magnitude.

Consequently, a system relying solely on thresholding instantaneous residual will permit inertial propagation beyond the validity horizon of its embodied structure. When failure occurs, it appears abrupt and unexplained, despite having been deterministically accumulated over time.

Hence, absence of shocks does not imply safety. Latent drift silently consumes structural viability until coherence can no longer be maintained.
Moreover, "small explicit residual" does not preclude latent residual accumulation; thus, drift can grow even under apparent coherence (cf. Theorems 12–14).

□

Interpretation. Most systems do not break when something goes wrong. They break when nothing seems wrong for too long.

Theorem 9 Predictive Constraint-of-Constraint Stability

Statement. Systems that regulate admissibility conditions rather than direct actions exhibit superior long-horizon stability compared to systems relying on direct control.

Definitions (for this theorem). Let u_t denote direct action or control input at time t. Let $\mathcal{A}_t$ denote the admissible set of actions or transitions at time t. Let $\mathcal{C}_t$ denote constraints governing $\mathcal{A}_t$. Predictive constraint-of-constraint regulation refers to the modulation of $\mathcal{C}_t$ based on predicted structural consequences, rather than direct selection or correction of u_t.

Proof (by separation of timescales and error containment).

Consider two adaptive systems operating under uncertainty and drift:

System A applies direct control, selecting or correcting u_t at each step based on predicted error.

System B applies predictive constraint-of-constraint regulation, adjusting $\mathcal{C}_t$ such that only admissible actions remain feasible, while allowing action execution to proceed inertially within $\mathcal{A}_t$.

In System A, prediction error directly affects action magnitude. Any modeling error, delay, or approximation propagates immediately into state evolution, amplifying residual and accelerating drift. Stability therefore depends on continuous high-frequency correction and precise prediction.

In System B, prediction error affects only admissibility boundaries. Errors in prediction modify feasibility margins rather than forcing corrective action. As long as the true state remains within adjusted admissible bounds, inertial propagation continues without corrective actuation.

Thus, System B exhibits:

- reduced sensitivity to prediction error,
- bounded amplification of modeling inaccuracies,
- slower accumulation of structural drift,
- decoupling of fast action dynamics from slow structural supervision.

Over long horizons, System A must either increase control effort indefinitely or suffer loss of coherence due to accumulated correction noise. System B preserves coherence by constraining evolution rather than correcting it.

Therefore, regulation of constraints dominates direct regulation of actions in long-horizon stability.

□

Interpretation. It is safer to shape what is allowed than to fight what already happened.

Theorem 10 Directionality as Identity Preserver

Statement. Directionality (Will) preserves system identity as long as it remains aligned with coherent structure. When directionality diverges from structure, the agent ceases to exist as a subject.

Definitions (for this theorem). Let $\mathcal{I}$ denote the set of structural invariants defining system identity. Let $\mathcal{S}_t$ denote the system's coherent structure at time t. Let W_t denote directionality (Will), defined as persistent bias or orientation of adaptation across time.

Alignment is defined as:

$$W_t \in \mathrm{span}(\mathcal{S}_t, \mathcal{I}).$$

Proof (by invariance and loss of reference).

An adaptive system is identifiable as a subject only if its evolution preserves a non-empty intersection of structural invariants over time. These invariants define continuity of identity across state changes.

Directionality governs how adaptation unfolds: which degrees of freedom are explored, reinforced, or suppressed. When directionality remains aligned with coherent structure, adaptations occur within the admissible geometry defined by $\mathcal{I}$, preserving identity.

Suppose directionality diverges from coherent structure, such that:

$$W_t \notin \mathrm{span}(\mathcal{S}_t, \mathcal{I}).$$

Then adaptation proceeds along degrees of freedom not constrained by identity-preserving invariants. Structural drift accumulates irreversibly, and invariants are progressively violated.

Two outcomes are possible:

- The system halts adaptation to preserve invariants, ceasing to be adaptive.
- The system continues adapting, dissolving its invariants and losing subjectivity.

In both cases, the system can no longer be described as the same agent over time. Identity is either frozen or destroyed.

Therefore, persistent directionality aligned with coherent structure is a necessary condition for the existence of an adaptive system as a subject. □

Interpretation. A system is not what it computes. It is what it continues to aim at without breaking itself.

Theorem 11 - Impossibility of Life Without IIC Restart

An artificial system that does not admit an endogenous operator for reinitiating the Integral Intentional Coherence (IIC) cycle cannot constitute artificial life.

Such a system may exhibit adaptation, learning, optimization, self-modification, or long-horizon stability, but it remains ontologically non-live.

Proof (Structural Argument). By Axiom IX, liveness is defined by the presence of an endogenous operator capable of reinitiating the IIC cycle after its completion or collapse.

Consider an artificial system lacking such an operator. Its coherence cycle, once exhausted through closure, drift accumulation, or internal-time depletion, cannot be reinitiated internally. Any subsequent coherence restoration must therefore be externally imposed (e.g., reset, retraining, reinitialization, or external intervention).

By Theorem 2 (Non-Permanence of Closure), no coherently closed structure remains valid indefinitely. By Theorem 4 (Supremacy of Internal Time), exhaustion of internal time mandates exit from inertial propagation regardless of instantaneous residual magnitude.

Absent an endogenous IIC restart, the system cannot reconstitute subjectivity following coherence exhaustion. Consequently, the system undergoes either terminal stabilization or degeneration into pure computation.

Thus, artificial life is impossible without an internal mechanism that restarts the coherence cycle as an ontological process rather than a procedural reset.

Theorem 12 - Latent Residual Accumulation Under Apparent Coherence

An adaptive system may appear structurally coherent while accumulating latent residual through unclosed operational phases. Such latent residual accelerates drift and reduces internal time without producing immediate structural mismatch.

Proof

We model the adaptive agent as evolving within an admissible geometry $\mathcal{G}$ produced by coherent structural closure. Let the observable structural residual be defined as

$$r_t := \rho\big(x_t, \mathcal{G}_t\big) \geq 0,$$

where x_t denotes the agent state and $\rho(\cdot, \cdot)$ is any mismatch functional measuring deviation between the actual state and the expected coherently closed geometry.

In residual-driven regulation, r_t serves as the primary activation signal: when r_t remains below a prescribed threshold, the system remains in inertial propagation and appears structurally coherent.

Introduce an additional residual component representing unclosed operational phases, herein termed *latent residual*, denoted by

$$\ell_t \geq 0.$$

By definition, latent residual does not necessarily manifest as immediate geometric mismatch. That is, there exist trajectories such that

$$r_t \approx 0 \quad \text{while} \quad \ell_t > 0.$$

This situation corresponds to phases that remain operationally unresolved while the instantaneous system state remains admissible with respect to $\mathcal{G}_t$.

Let cumulative structural drift D_T represent the integrated deformation burden reducing long-horizon viability. Assume drift accumulation admits a monotone lower bound of the form

$$D_T \geq \sum_{t=0}^{T} \big(a\, r_t + b\, \ell_t\big), \qquad a > 0,\ b > 0.$$

This assumption reflects that any unresolved structural or phase-based burden contributes non-negatively to cumulative drift.

Consider a regime in which structural residual remains small:

$$r_t \leq \varepsilon \quad \text{for all } t \leq T,$$

while latent residual persists due to repeated unclosed operational phases:

$$\sum_{t=0}^{T} \ell_t > 0.$$

Then it follows that

$$D_T \geq \sum_{t=0}^{T} (ar_t + b\ell_t) \geq a \sum_{t=0}^{T} 0 + b \sum_{t=0}^{T} \ell_t = b \sum_{t=0}^{T} \ell_t > 0.$$

Hence, even while the system appears structurally coherent in the geometric sense, it accumulates strictly positive drift through latent residual.

Let internal time τ_t represent the remaining structural viability budget, monotonically decreasing with drift increments. A minimal update model is

$$\tau_{t+1} = \tau_t - \gamma\, \Delta D_{t+1}, \qquad \gamma > 0,$$

where $\Delta D_{t+1} = D_{t+1} - D_t \geq 0$. Since $\ell_t > 0$ implies $\Delta D_{t+1} \geq b\, \ell_{t+1}$ in the adopted lower bound, we obtain

$$\tau_{t+1} \leq \tau_t - \gamma b\, \ell_{t+1}.$$

Therefore, latent residual reduces internal time even in the absence of immediate structural mismatch. Because regulation and reactivation are primarily driven by r_t, the system may remain inertially propagating while viability is silently depleted.

Accordingly, an adaptive system may remain apparently coherent while accumulating latent residual that accelerates drift and reduces internal time without producing immediate observable mismatch. □

Theorem 13 - Phase-Coherent Closure Dominates State-Coherence for Internal Time Preservation

Systems that enforce phase-coherent closure preserve internal time more effectively than systems that rely solely on structural or state-based coherence. Long-horizon stability therefore depends on closure of moments, not only correctness of trajectories.

Proof

Let x_t denote the system state and $\mathcal{G}_t$ the coherently closed admissible geometry. Define the conventional (state-based) structural residual

$$r_t := \rho(x_t, \mathcal{G}_t) \geq 0,$$

which governs residual-driven reactivation and appears in purely state-coherence systems.

Introduce a *phase residual* $\ell_t \geq 0$ capturing unclosed operational phases (unclosed moments), i.e., unresolved micro-asymmetries whose effect is not necessarily expressed as instantaneous geometric mismatch. Hence there exist trajectories with

$$r_t \approx 0 \quad \text{while} \quad \ell_t > 0.$$

Assume accumulated drift admits a monotone lower bound from both components:

$$D_T \;\geq\; \sum_{t=0}^{T} \big(a\, r_t + b\, \ell_t\big), \qquad a > 0,\; b > 0.$$

Let internal time H_t be a viability budget that decreases monotonically with drift increments:

$$H_{t+1} \;=\; H_t - \gamma\, \Delta D_{t+1}, \qquad \gamma > 0,$$

so that larger drift implies faster depletion of internal time.

Consider two systems operating under operationally indistinguishable state trajectories $\{x_t\}_{t=0}^{T}$ (hence the same $\{r_t\}$), but differing by whether they enforce phase-coherent closure.

(A) State-coherence only. No mechanism explicitly closes phases; latent phase residual persists with $\ell_t^A > 0$ on a non-negligible set of times.

(B) Phase-coherent closure. The system enforces closure of moments, suppressing latent accumulation so that

$$\ell_t^B \leq \ell_t^A \quad \text{for all } t, \quad \text{and typically } \ell_t^B \approx 0 \text{ when closure succeeds.}$$

Because $\{r_t\}$ are identical, the drift lower bounds satisfy

$$D_T^A \;\geq\; \sum_{t=0}^{T} \big(a\, r_t + b\, \ell_t^A\big), \qquad D_T^B \;\geq\; \sum_{t=0}^{T} \big(a\, r_t + b\, \ell_t^B\big).$$

Subtracting yields the dominance:

$$D_T^A - D_T^B \;\geq\; b \sum_{t=0}^{T} \big(\ell_t^A - \ell_t^B\big) \;\geq\; 0,$$

with strict inequality whenever phase residual is actually reduced by closure.

Since internal time decreases monotonically with drift, for the same initial τ_0 we obtain

$$\tau_T^B \;\geq\; \tau_T^A,$$

and the inequality is strict whenever $\sum_{t=0}^{T}(\ell_t^A - \ell_t^B) > 0$.

Thus, enforcing phase-coherent closure preserves internal time more effectively than relying on state-based coherence alone. The result holds even when trajectories appear equally correct, because long-horizon viability is governed by cumulative drift contributions from unclosed moments. □

Theorem 14 - Latent-to-Explicit Residual Projection

An adaptive system may appear structurally coherent while accumulating latent residual through unclosed operational phases. Such latent residual accelerates drift and reduces internal time without producing immediate structural mismatch.

If internal time is finite and monotonically reduced by accumulated drift or latent residual, then latent residual cannot remain indefinitely unexpressed: there exists a finite horizon after which latent residual is either projected into explicit structural residual or forces exit from admissible continuation through revalidation, reconfiguration, or structural collaps.

Proof (structural). Let r_t denote the explicit structural residual measured as mismatch between the expected coherently closed admissible geometry and the actual evolving state at time t. Let $\ell_t \geq 0$ denote latent residual accumulated due to unclosed operational phases (phase debt), which does not necessarily appear in r_t instantaneously.

Assume drift accumulation admits a lower bound driven by latent residual:

$$D_T \;\geq\; \int_0^T \ell_t \, dt,$$

i.e., even when r_t remains small, persistent ℓ_t implies strictly increasing accumulated drift D_T.

Assume an internal time horizon $\tau(t)$ exists and is reduced by accumulated drift and/or latent residual, with $\tau(t)$ finite at initialization and monotone decreasing under sustained accumulation. In particular, there exists a nonnegative function $g(\cdot)$ such that

$$\tau(T) \;=\; \tau(0) \;-\; g\big(D_T\big), \qquad g(0) = 0, \quad g \text{ nondecreasing.}$$

If ℓ_t remains bounded away from zero on a set of nonzero measure, then D_T grows without bound as T increases, hence $\tau(T)$ decreases and must cross any fixed viability threshold in finite time.

Consider the regime where, for a nontrivial interval, explicit mismatch is suppressed or remains admissible, i.e., $r_t \approx 0$ while $\ell_t > 0$. In this regime, the system consumes internal time while not triggering reactivation via r_t.

When $\tau(T)$ falls below a viability threshold, the system can no longer treat the current phase history as structurally admissible without violating coherence constraints: either (i) the admissible geometry must be revalidated/restructured, or (ii) the accumulated phase debt becomes externally relevant as a structural mismatch. In both cases, latent residual necessarily manifests as explicit residual in the operational interface, i.e., it is projected into a non-negligible r_t that forces exit from inertial operation or initiates corrective regulation.

Therefore, under finite internal time and monotone depletion by latent accumulation, latent residual cannot remain indefinitely hidden; it must be projected into explicit structural residual within a finite horizon. □

Corollary 14.1 - Latent-to-Explicit Residual Projection (Operational Form)

Apparent long-horizon coherence under persistent unclosed phases necessarily precedes either: (i) explicit residual emergence, or (ii) enforced revalidation/reconfiguration due to internal time exhaustion. Thus, systems that monitor only instantaneous residual can fail via latent accumulation without early warning, whereas systems that enforce phase-coherent closure preserve internal time and stability more effectively.

Theorem 15 - Performance–Identity Decoupling

Statement. There exist adaptive systems that preserve bounded error, admissible trajectories, and stable performance metrics while undergoing irreversible loss of identity. Therefore, performance persistence is not a sufficient condition for identity continuity.

Proof (Structural Sketch)

Let $\mathcal{A}$ denote the set of externally admissible trajectories (bounded error, constraint satisfaction, stable reward). Let $\mathcal{I}$ denote the set of structural invariants defining system identity. Let $\mathcal{C}(t)$ denote a coherence/identity measure such that identity continuity requires

$$\mathcal{C}(t) > \mathcal{C}_{crit}.$$

Assume the system operates under directed interaction with an environment and admits drift accumulation (Axiom VI). Assume further that structural burden may accumulate independently of instantaneous trajectory admissibility (Axioms XVII–XVIII).

Consider an adaptive system designed to enforce admissibility at the action level:

$$x(t) \in \mathcal{A} \quad \text{for all } t \in [0, T],$$

through feedback, optimization, or compensation mechanisms. By construction, bounded error and stable performance metrics are preserved over this interval.

Now introduce a regime in which maintaining admissibility requires sustained compensatory activity:

- repeated micro-corrections,
- increased coupling effort,
- phase transitions without coherent closure,
- persistent constraint-dense interaction.

These factors contribute to cumulative structural burden and internal-time depletion without violating admissibility (Theorems 19, 22, 24).

Let $D(t)$ denote accumulated structural drift relative to $\mathcal{I}$. By Axiom XVIII, $D(t)$ may grow unbounded even while trajectories remain admissible. Hence there exists a finite time t^* such that

$$D(t^*) > D_{\max},$$

where $D_{\max}$ is the maximal deformation compatible with identity preservation (Theorem 5).

At t^*, structural invariants defining identity are violated. However, because action-level admissibility constraints remain enforced, externally observed behavior may remain correct beyond t^*.

Thus, there exists an interval

$$[t^*, t^* + \Delta]$$

during which:

$$x(t) \in \mathcal{A} \quad \text{while} \quad \mathcal{C}(t) \leq \mathcal{C}_{crit},$$

i.e., performance persists while identity continuity has already collapsed.

Therefore, bounded error, admissible trajectories, and stable performance metrics do not imply preservation of identity. Performance persistence is not a sufficient condition for identity continuity.
This decoupling is unavoidable and structurally expected. It does not represent an anomalous failure mode, but a necessary consequence of long-horizon structural consumption under directed interaction. □

Interpretation. A system may continue to act correctly long after it has ceased to be itself. Performance answers *what* is achieved; identity answers *who* is achieving it.

Corollary 15.3 — Pressure Amplifies Performance–Identity Decoupling. Statement. Under sustained structural pressure $P > 0$, within the class of bounded long-horizon adaptive systems, the decoupling interval—the time during which performance remains stable while identity is already compromised—is shorter than under $P = 0$, because τ depletes from two independent channels (action cost and pressure).

Proof. By Theorem 15, the decoupling interval exists whenever structural burden accumulates independently of trajectory admissibility. Let $\Delta t_{\text{decouple}}$ denote the duration of this interval.

Under $P = 0$: $d\Phi/dt = f(U, G)$. τ-depletion is driven solely by action cost. The identity compromise threshold $D_{\max}$ (Theorem 5) is reached at time t_1.

Under $P > 0$: $d\Phi/dt = f(U, G) + P(t) > f(U, G)$. τ-depletion is strictly faster. The identity compromise threshold $D_{\max}$ is reached at time $t_2 < t_1$.

Therefore $\Delta t_{\text{decouple}}(P > 0) < \Delta t_{\text{decouple}}(P = 0)$. The decoupling interval is compressed under pressure.

Operational consequence: Under structural pressure, the window for detecting identity compromise before external performance failure is narrower. Systems under pressure have less time to detect and respond to identity degradation. This makes structural monitoring (Theorem 26) more urgent under pressure conditions. □

Theorem 16 - Silent Degradation Under Regime Persistence

Statement. If regime admissibility is not regulated independently of action selection, then silent long-horizon degradation can accumulate while actions remain admissible and performance remains stable. Such degradation is therefore not reliably detectable through action-level metrics alone.

Proof (Structural Sketch)

Let $\mathcal{R}$ denote the set of organizational regimes available to an adaptive system. Let $r(t) \in \mathcal{R}$ denote the active regime at time t. Let $\mathcal{A}(r)$ denote the set of admissible actions under regime r. Let $x(t)$ denote the realized trajectory, and assume

$$x(t) \in \mathcal{A}(r(t)) \quad \text{for all } t \in [0, T],$$

so that action-level admissibility and performance metrics remain satisfied.

Let $B(t)$ denote accumulated structural burden (drift, phase debt, coupling load, or internal-time depletion) affecting long-horizon viability and identity. By Axioms XVI and XVIII, structural burden accumulation is primarily regime-mediated and may occur independently of instantaneous action correctness.

Assume the architecture regulates actions but does not regulate regime admissibility independently. That is, regime persistence is permitted as long as actions remain admissible:

$$r(t) \text{ is unconstrained by long-horizon structural criteria.}$$

Consider a regime r^* satisfying:

- $\mathcal{A}(r^*)$ admits stable, bounded-error trajectories,
- maintaining admissibility within r^* requires sustained coupling, micro-correction, or constraint-dense interaction,
- r^* induces positive structural burden accumulation rate:

$$\frac{dB}{dt} > 0 \quad \text{while } x(t) \in \mathcal{A}(r^*).$$

Such regimes exist by Axioms XVI–XVIII and Theorems 19, 22.

Because action-level admissibility is preserved, no corrective signal is triggered at the action layer. Because regime admissibility is not independently supervised, the system remains in r^*.

Hence, over time,

$$B(T) = \int_0^T \frac{dB}{dt}\, dt$$

grows monotonically while all action-level metrics remain within acceptable bounds.

Since identity continuity and long-horizon viability depend on $B(t)$ and internal time rather than instantaneous action correctness (Theorems 4, 5, 15), there exists a finite t^* such that

$$B(t^*) > B_{\max},$$

where $B_{\max}$ denotes the maximal structural burden compatible with identity preservation.

On the interval $[0, t^*]$, degradation remains silent: neither error thresholds nor performance metrics indicate failure. At t^*, coherence or identity collapses abruptly, despite the absence of prior action-level warning signals.

Therefore, if regime admissibility is not regulated independently of action selection, silent long-horizon degradation necessarily accumulates under regime persistence. Such degradation cannot be reliably detected through action-level metrics alone. □

Interpretation. The system does not fail because it acts incorrectly. It fails because it stays too long in a way of acting that slowly consumes its ability to remain itself.

Theorem 17 - Regime Occupation Dominates Action History

Statement. For long-horizon viability, the sequence, duration, and persistence of organizational regimes dominates the contribution of individual actions to structural degradation. Two systems with comparable action-level correctness may be non-equivalent in viability due solely to different regime occupation histories.

Proof (Structural Sketch)

Let $\mathcal{R}$ denote the set of organizational regimes. Let $r(t) \in \mathcal{R}$ denote the active regime at time t. Let $u(t)$ denote the realized action process and let external correctness be represented by an admissibility predicate $\mathrm{Adm}(u(t), x(t))$.

Define accumulated structural burden over a horizon $[0, T]$ as

$$B_T \triangleq \int_0^T b\big(r(t),\, u(t),\, x(t)\big)\, dt, \qquad b(\cdot) \geq 0,$$

where b is an abstract nonnegative burden rate functional.

Assume only the minimal regime-separability property implied by Axiom XVI: there exist at least two regimes $r_1, r_2 \in \mathcal{R}$ such that, under comparable externally admissible operation,

$$b(r_1, u, x) \;<\; b(r_2, u, x)$$

for a nontrivial set of admissible (u, x), i.e., regime choice changes the burden rate even when actions remain correct.

Consider two systems (or two episodes of the same system), $\mathcal{E}_A$ and $\mathcal{E}_B$, with comparable action-level correctness over $[0, T]$:

$$\mathrm{Adm}_A(t) = \mathrm{Adm}_B(t) \quad \text{for all } t \in [0, T],$$

and with comparable action histories in the sense that the realized actions remain within the same admissible class (bounded error, stable reward, etc.).

Let their regime occupation histories differ:

$$r_A(t) = r_1 \quad \text{for } t \in [0, T], \qquad r_B(t) = r_2 \quad \text{for } t \in [0, T],$$

or more generally, let $r_A(t)$ and $r_B(t)$ have different durations in regimes with different burden rates.

Then their accumulated burdens satisfy

$$B_T^A = \int_0^T b(r_1, u_A(t), x_A(t))\, dt, \qquad B_T^B = \int_0^T b(r_2, u_B(t), x_B(t))\, dt.$$

By the regime-separability property and nonnegativity of b, it follows that

$$B_T^A \;<\; B_T^B,$$

with strict inequality whenever the burden rates differ on a set of nonzero measure.

Since long-horizon viability and identity continuity are bounded by accumulated structural burden and internal time (Axioms VII, XVI–XVIII; Theorems 4, 5, 16, 19), differing B_T implies differing remaining viability and potentially differing survival horizons, despite comparable action-level correctness.

Therefore, regime occupation history (sequence, duration, persistence) dominates action history with respect to long-horizon structural degradation, and two externally correct systems may be non-equivalent in viability solely due to different regime occupation histories. □

Remark. The theorem does not require explicit enumeration of regimes or explicit measurement of burden. It asserts only that regimes exist for which the structural burden rate differs under externally admissible behavior, and that long-horizon viability depends on accumulated burden rather than instantaneous correctness.

Theorem 18 - Structural Non-Equivalence of Identical Interventions

Statement. Nominally identical interventions may impose fundamentally different long-horizon structural burden when realized under different interaction coupling conditions. Thus, nominal intervention equivalence does not imply structural cost equivalence.

Proof (Sketch)

Let $u_t \in \mathcal{U}$ denote the nominal intervention applied at time t (e.g., a control input, an update, or a directed actuation). Assume two operational episodes $\mathcal{E}_A$ and $\mathcal{E}_B$ in which the system applies the *same* nominal interventions:

$$u_t^A \equiv u_t^B \quad \text{for all } t \in \{0, \dots, T\}.$$

Let the realized corrective effect be mediated by an interaction coupling factor $c_t \in (0, \infty)$, representing the efficiency with which nominal influence is converted into realized correction through the interaction channel. No estimation method for c_t is assumed; we use it only as an abstract structural variable.

Assume the realized correction magnitude is of the generic form

$$\Delta x_t \;=\; \Phi\big(x_t, u_t, c_t\big),$$

and that the contribution of the intervention to *structural burden* depends on how much forcing must be sustained to obtain a comparable realized correction under prevailing coupling. Accordingly, define the instantaneous structural burden increment as an abstract nonnegative functional

$$\Delta B_t \;=\; \Psi\big(x_t, u_t, c_t\big) \;\geq\; 0,$$

with the minimal monotonicity property:

$$c_t' \leq c_t \;\Rightarrow\; \Psi(x_t, u_t, c_t') \;\geq\; \Psi(x_t, u_t, c_t),$$

i.e., weaker coupling cannot reduce the structural burden of applying a nominal intervention.

Consider two coupling histories $\{c_t^A\}$ and $\{c_t^B\}$ such that there exists a non-negligible subset of times $I \subseteq \{0, \dots, T\}$ for which

$$c_t^A \;>\; c_t^B \quad \text{for all } t \in I.$$

Then by monotonicity of Ψ we have

$$\Delta B_t^A \;=\; \Psi(x_t^A, u_t, c_t^A) \;\leq\; \Psi(x_t^B, u_t, c_t^B) \;=\; \Delta B_t^B \quad \text{for } t \in I,$$

and hence, summing over the horizon,

$$B_T^A \;=\; \sum_{t=0}^{T} \Delta B_t^A \;<\; \sum_{t=0}^{T} \Delta B_t^B \;=\; B_T^B,$$

with strict inequality whenever I has positive measure and coupling differs strictly on I.

Therefore, even under nominal intervention equivalence ($u_t^A \equiv u_t^B$), the accumulated long-horizon structural burden B_T can differ solely due to interaction coupling conditions.

Consequently, nominal intervention equivalence does not imply structural cost equivalence. □

Remark. The theorem is architectural: it assumes only that (i) realization of influence is mediated by interaction coupling and (ii) weaker coupling cannot reduce the internal burden required to obtain comparable realized correction. No mechanism for measuring c_t or computing Ψ is disclosed.

Theorem 19 - Admissible Trajectories Do Not Upper-Bound Structural Cost

Statement. Trajectory admissibility, bounded error, or stable reward do not impose an upper bound on accumulated structural burden. Structural consumption may diverge while externally observed behavior remains acceptable.

Proof (Sketch)

Let $\mathcal{A}$ denote an admissible set of trajectories (or transitions) defined by external correctness constraints (e.g., bounded error, safety envelope, task success, reward stability). Let x_t be the realized system trajectory over $t = 0, \ldots, T$. Assume the system remains externally admissible:

$$x_{0:T} \in \mathcal{A}, \qquad e_t \leq \varepsilon \;\; \forall t, \qquad R_T \text{ stable/bounded},$$

where e_t is any external error proxy and R_T any reward/performance aggregate.

Define accumulated structural burden as a nonnegative additive functional

$$B_T \triangleq \sum_{t=0}^{T} \Delta B_t, \qquad \Delta B_t \geq 0,$$

representing internal structural consumption induced by maintaining admissible behavior. No particular computation of ΔB_t is assumed.

Consider the interaction-mediated realization of interventions. Let $c_t \in (0, \infty)$ denote an abstract interaction coupling factor, and let the system apply nominal interventions u_t to keep the trajectory admissible. Assume only the minimal property that, to maintain admissibility under weaker coupling, the system must sustain correction more often or more intensely in a way that does not decrease internal burden. Formally, assume there exist operating conditions in which ΔB_t admits a lower bound of the form

$$\Delta B_t \;\geq\; h(c_t), \qquad h(c) \geq 0, \;\; h \text{ nonincreasing in } c,$$

and there exist regimes of persistent weak coupling $c_t \leq c_\star$ over a nontrivial horizon.

Now construct an admissible episode in which the system succeeds in keeping $x_{0:T} \in \mathcal{A}$ by applying whatever nominal interventions are required, while coupling remains persistently weak:

$$c_t \leq c_\star \quad \text{for all } t \in \{0, \ldots, T\}.$$

Then, by the lower bound,

$$B_T \;=\; \sum_{t=0}^{T} \Delta B_t \;\geq\; \sum_{t=0}^{T} h(c_t) \;\geq\; \sum_{t=0}^{T} h(c_\star) \;=\; (T+1)\, h(c_\star).$$

If $h(c_\star) > 0$ (i.e., maintaining admissibility under this coupling entails strictly positive structural cost per step), then

$$B_T \;\geq\; (T+1)\, h(c_\star) \xrightarrow[T \to \infty]{} \infty,$$

even though the trajectory remains admissible and external error and reward remain bounded by assumption.

Therefore, admissibility of trajectories and stability of external performance metrics do not impose an upper bound on accumulated structural burden. Structural consumption may diverge while externally observed behavior remains acceptable. □

Remark. The theorem is architectural: it requires only that there exist environments or interaction regimes in which maintaining admissibility under persistent weak coupling imposes strictly positive internal burden per unit time. No mechanism for measuring coupling, computing burden, or generating interventions is disclosed.

Theorem 20 - Irreversible Regime Transition Boundary

Statement. There exist regime transitions after which restoration of admissible behavior does not imply restoration of prior identity. Such transitions define irreversible boundaries in long-horizon structural evolution.

Proof (Sketch)

Let $\mathcal{I}$ denote the set of structural invariants defining the identity of an adaptive system. Let $\mathcal{R}$ denote the set of organizational regimes admissible to the system, and let

$$\mathcal{S}_t \in \mathcal{R}$$

denote the active regime at time t.

Identity preservation is defined as persistence of invariants:

$$\mathcal{I}_t \cap \mathcal{I}_{t'} \neq \varnothing \quad \text{for all } t < t' \text{ within the operational horizon.}$$

Assume the system undergoes a regime transition at time t^*:

$$\mathcal{S}_{t^*-} \;\to\; \mathcal{S}_{t^*+},$$

triggered by accumulated drift, internal time exhaustion, or structural constraint violation. Assume further that this transition is *structurally irreversible*, in the sense that no admissible sequence of internal operations can restore the original organizational structure that supported $\mathcal{I}$.

Formally, irreversibility means:

$$\mathcal{I}_{t^*-} \;\not\subseteq\; \mathcal{I}_{t^*+},$$

and there exists no future time $t' > t^*$ such that

$$\mathcal{I}_{t^*-} \;\subseteq\; \mathcal{I}_{t'}.$$

Now consider the possibility of restoring admissible behavior after the transition. Let $\mathcal{A}$ denote the admissible set of behaviors or trajectories defined externally (e.g., bounded error, safety constraints, or task success). By construction of regime transitions, it is possible that

$$x_{t^*+:T} \in \mathcal{A},$$

i.e., the system regains externally admissible behavior through reconfiguration, compensation, or adaptation within the new regime $\mathcal{S}_{t^*+}$.

However, because $\mathcal{I}_{t^*-}$ is not contained in the invariant set of $\mathcal{S}_{t^*+}$, restoration of admissible behavior does not restore the prior identity. Behavioral equivalence therefore does not imply structural equivalence.

Assume, for contradiction, that restoration of admissible behavior implies restoration of prior identity. Then admissible behavior would uniquely determine the supporting invariants. This

contradicts the existence of multiple regimes capable of generating behavior in $\mathcal{A}$ with non-equivalent invariant sets.

Therefore, there exist regime transitions after which admissible behavior can be restored while prior identity cannot. Such transitions define irreversible boundaries in long-horizon structural evolution. □

Remark. The theorem does not require identifying or measuring identity invariants explicitly. It relies only on the existence of multiple organizational regimes capable of producing admissible behavior with non-equivalent invariant support.

Theorem 21 - Observation–Control Entanglement Collapse

Statement. Any architecture that embeds long-horizon viability or identity observation into the causal decision surface necessarily collapses observation into control. As a result, either behavioral distortion or loss of observability of long-horizon degradation must occur.

Proof (Sketch)

Let $\mathcal{O}_t$ denote an observation intended to measure long-horizon properties such as structural viability or identity continuity. Let π_t denote the causal decision process (policy, controller, or learning rule) generating actions u_t.

Assume $\mathcal{O}_t$ is embedded into the causal decision surface, i.e.,

$$u_t = \pi_t(\,\cdot\,, \mathcal{O}_t),$$

so that the observation directly or indirectly influences action selection, learning updates, or optimization.

By definition, any variable that influences π_t becomes causally entangled with behavior. Consequently, $\mathcal{O}_t$ ceases to be a passive witness of system evolution and becomes part of the control loop.

Two exhaustive cases follow.

Case 1: Observation influences behavior. If $\mathcal{O}_t$ meaningfully affects π_t, then the system adapts its behavior to modify future values of $\mathcal{O}_t$. In this case, the observed quantity is subject to optimization pressure or avoidance dynamics. Long-horizon degradation signals are distorted, delayed, or actively suppressed in order to maintain favorable observed values. Thus, observability is preserved only at the cost of behavioral distortion.

Case 2: Observation does not influence behavior. If $\mathcal{O}_t$ is present in the architecture but has no effective influence on π_t, then it is causally inert. In this case, although $\mathcal{O}_t$ may be computed, it cannot trigger corrective action, adaptation, or regime change. Long-horizon degradation may be observed but remains behaviorally irrelevant and therefore practically undetectable from within the causal loop.

In both cases, the embedding of $\mathcal{O}_t$ into the causal decision surface collapses the distinction between observation and control. Either the observation distorts behavior, or it loses operational relevance for detecting and preventing long-horizon degradation.

Therefore, no architecture can simultaneously embed long-horizon viability observation into the causal decision surface and preserve both undistorted behavior and reliable observability of long-horizon degradation. A non-causal observational dimension is therefore necessary. □

Remark. The result is architectural and independent of implementation details. It holds for controllers, learning systems, and hybrid architectures alike, whenever long-horizon observables are made causally operative.

Theorem 22 - Constraint-Dense Geometry Induces Temporal Deformation

Statement. For adaptive systems, increases in interaction constraint density induce deformation of internal temporal dynamics. Therefore, operationally equivalent spatial trajectories may be temporally non-equivalent in internal-time evolution.

Proof (Sketch)

Let $x(t)$ denote the system trajectory in an external (spatial or state) domain, and let $\tau(t)$ denote internal time, interpreted as a structural viability clock (a validity budget) whose evolution depends on interaction and drift accumulation rather than on external clock time alone.

Let $\kappa(t) \geq 0$ denote an *interaction constraint density* functional along the trajectory, representing the local density/intensity of constraints encountered through contact, coupling, correction, or admissibility enforcement. No representation of geometry is assumed; $\kappa(t)$ is defined operationally via interaction.

Assume only that internal time depletion rate is nondecreasing with respect to interaction constraint density, i.e., there exists a nonnegative monotone function $g(\cdot)$ such that

$$\frac{d\tau}{dt} \;=\; -\,g(\kappa(t)), \qquad g(\kappa_1) \leq g(\kappa_2) \text{ whenever } \kappa_1 \leq \kappa_2.$$

This expresses a minimal structural premise: denser constraint interaction cannot reduce the rate at which internal viability is consumed.

Now consider two operational episodes $\mathcal{E}_A$ and $\mathcal{E}_B$ that realize *operationally equivalent* external trajectories, meaning that their spatial/state trajectories coincide:

$$x_A(t) \equiv x_B(t) \quad \text{for} \quad t \in [0, T],$$

and remain admissible in the same external sense.

However, allow their interaction constraint densities to differ along the same external path due to differences in coupling, contact richness, micro-corrections, or constraint activation history:

$$\kappa_A(t) \neq \kappa_B(t) \quad \text{on a set of nonzero measure in } [0, T].$$

In particular, assume there exists an interval $I \subseteq [0, T]$ with nonzero measure such that

$$\kappa_A(t) > \kappa_B(t) \quad \text{for all } t \in I.$$

By monotonicity of $g(\cdot)$, it follows that

$$\frac{d\tau_A}{dt} = -g(\kappa_A(t)) \;<\; -g(\kappa_B(t)) = \frac{d\tau_B}{dt} \quad \text{for } t \in I,$$

hence integrating over $[0, T]$ yields

$$\tau_A(T) \;<\; \tau_B(T),$$

with strict inequality whenever the constraint density differs on a set of nonzero measure.

Therefore, even for operationally equivalent spatial trajectories, internal-time evolution can differ due to differences in interaction constraint density. This establishes that constraint-dense geometry induces temporal deformation in the internal-time dynamics of adaptive systems. □

Remark. The theorem is operational and representation-agnostic: it does not assume explicit geometric reconstruction. It requires only that internal time is a viability budget whose depletion rate is nondecreasing in interaction constraint density along the realized trajectory.

Theorem 23 - Coherence Collapse Precedes Observable Failure

Statement. In long-horizon operation, coherence degradation necessarily precedes externally observable failure. Hence, there exists an interval during which behavior remains admissible while identity continuity has already been structurally compromised.

Proof (Sketch)

Let $\mathcal{A}$ denote the externally defined admissible behavior set (bounded error, constraint satisfaction, task success, or reward stability). Let $\mathcal{C}(t)$ denote a scalar (or partially ordered) coherence measure of internal organization, treated as an operational property of structural alignment among components. No method of measuring $\mathcal{C}(t)$ is assumed; we use only its ordering and boundary meaning.

Assume:

- There exists a coherence threshold $\mathcal{C}_{crit}$ such that identity continuity requires
$$\mathcal{C}(t) > \mathcal{C}_{crit}.$$
- There exists an external admissibility threshold such that observable failure is declared when behavior exits $\mathcal{A}$:
$$x_{t:t+\Delta} \notin \mathcal{A}.$$
- Under long-horizon directed interaction, coherence degrades cumulatively under structural burden, drift, regime persistence, or coupling-mediated load (Axioms VI, VII, XVI–XVIII), and this degradation can occur while actions remain admissible.

Consider a system operating over a horizon $[0, T]$ such that $\mathcal{C}(t)$ is monotonically (or on average) decreasing due to cumulative burden, but the system continues to satisfy external admissibility constraints through compensation, adaptation, or regime adjustment:

$$x_{0:t} \in \mathcal{A} \quad \text{for all } t < T.$$

Because external admissibility is defined on behavior, not on internal organization, it is possible for admissible behavior to persist while $\mathcal{C}(t)$ crosses the identity threshold.

Let t^* be the first time at which coherence crosses the critical boundary:

$$t^* \;=\; \inf\{t : \mathcal{C}(t) \leq \mathcal{C}_{crit}\}.$$

At t^*, identity continuity is structurally compromised by definition of $\mathcal{C}_{crit}$.

Now let t_f be the first time at which observable failure occurs:

$$t_f \;=\; \inf\{t : x_{0:t} \notin \mathcal{A}\}.$$

Because admissible behavior can be maintained by compensatory mechanisms after internal degradation begins, it is not necessary that $t_f = t^*$. In long-horizon regimes, the compensatory capacity may sustain admissibility beyond the point of coherence collapse:

$$t_f \;>\; t^*.$$

Therefore, the interval (t^*, t_f) is non-empty, and on this interval the system remains externally admissible while identity continuity has already been structurally compromised:

$$x_t \in \mathcal{A} \quad \text{for } t \in (t^*, t_f) \quad \text{and} \quad \mathcal{C}(t) \leq \mathcal{C}_{crit}.$$

Hence, coherence collapse precedes externally observable failure in long-horizon operation, and there exists an interval during which behavior remains admissible while identity continuity is already compromised. □

Remark. The theorem is evaluative and architectural: it distinguishes internal coherence failure from external task failure. It does not specify how coherence is measured or how compensatory admissibility is maintained; it asserts that external admissibility constraints do not, in principle, enforce coherence preservation.

Theorem 24 - Survival Horizon Is Orthogonal to Optimization Quality

Statement. Improvement in control accuracy, optimization efficiency, or reward maximization does not guarantee extension of identity survival horizon. Long-horizon survival is governed by structural factors orthogonal to action-level optimization.

Proof (Sketch)

Let $\mathcal{A}$ denote an externally defined admissible behavior class (bounded error, constraint satisfaction, task success, or reward stability). Let J denote an optimization objective or performance functional (e.g., cumulative reward, tracking error, or efficiency). Let $\mathcal{H}$ denote the identity survival horizon, defined as the maximal time interval over which identity continuity holds under operation.

Let $\mathcal{C}(t)$ denote a coherence/identity-support measure with threshold $\mathcal{C}_{crit}$ such that identity continuity requires $\mathcal{C}(t) > \mathcal{C}_{crit}$, and define

$$\mathcal{H} \;=\; \sup\{T \geq 0 : \mathcal{C}(t) > \mathcal{C}_{crit} \text{ for all } t \in [0, T]\}.$$

Assume the system accumulates structural burden B_T and drift under directed interaction (Axioms VI–VII), and that coherence is degraded by burden accumulation and regime effects:

$$\mathcal{C}(t) \;\text{ decreases as a function of }\; B_t \text{ and regime occupation,}$$

without specifying the functional form.

Consider two controllers (or optimization procedures) $\pi^{(1)}$ and $\pi^{(2)}$ operating on the same system and environment, where $\pi^{(2)}$ is strictly better in action-level optimization:

$$J(\pi^{(2)}) \;<\; J(\pi^{(1)}) \quad \text{(or equivalently higher reward / lower error).}$$

Such improvement concerns behavior within $\mathcal{A}$ and does not, by itself, constrain long-horizon structural consumption.

Now construct (or consider) operating regimes in which the dominant contributor to coherence depletion is not instantaneous action error, but structural factors such as:

- sustained occupation of a silently destructive regime (Axiom XVI),
- coupling-mediated burden required to maintain admissibility (Theorem 19),
- phase-related residual and transition load (Axioms XI–XII),
- constraint-dense interaction inducing faster internal-time depletion (Theorem 22).

In such regimes, action-level optimization may improve J while leaving the dominant structural consumption mechanisms unchanged, or even increasing them by sustaining performance via compensatory effort.

Hence, it is possible that

$$J(\pi^{(2)}) < J(\pi^{(1)}) \quad \text{while} \quad \mathcal{H}(\pi^{(2)}) \leq \mathcal{H}(\pi^{(1)}),$$

and in particular there exist cases where improved optimization preserves admissible behavior longer but accelerates coherence depletion, reducing identity survival horizon.

Therefore, improvement in control accuracy, optimization efficiency, or reward maximization does not guarantee extension of identity survival horizon. Long-horizon survival is governed by structural factors that are, in general, orthogonal to action-level optimization. □

Remark. The theorem asserts non-implication, not opposition: optimization may correlate with survival in specific regimes, but it cannot serve as a general guarantee of identity continuity without explicit structural viability regulation.

Theorem 25 - Incompleteness of Action-Centric Safety

Statement. Any safety framework operating solely at the action level is incomplete with respect to long-horizon viability and identity continuity. Such frameworks cannot, in principle, prevent failure modes arising from regime-induced structural degradation and coupling-mediated burden.

Proof (Sketch)

Let $\mathcal{S}$ denote a safety framework that regulates or evaluates a system exclusively through constraints on actions or trajectories. Formally, assume safety enforcement depends only on admissibility of actions u_t or trajectories $x_{0:T}$:

$$u_t \in \mathcal{U}_{safe}, \qquad x_{0:T} \in \mathcal{A}_{safe}.$$

By construction, $\mathcal{S}$ has no direct access to internal structural variables such as regime identity, accumulated structural burden, coupling efficiency, or internal time. Its intervention criteria are therefore functions solely of externally observable behavior.

Consider an adaptive system operating under a sustained organizational regime $\mathcal{R}$ that is externally admissible but internally destructive. By Axiom XVI (Regimes Are the Primary Carriers of Long-Horizon Risk), such regimes may induce silent accumulation of structural degradation without violating action-level constraints.

Let the system satisfy:

$$u_t \in \mathcal{U}_{safe} \quad \text{and} \quad x_{0:T} \in \mathcal{A}_{safe} \quad \text{for all } t \leq T,$$

while simultaneously accumulating internal burden:

$$B_T \xrightarrow[T\to\infty]{} \infty,$$

due to regime persistence or weak interaction coupling.

Because $\mathcal{S}$ evaluates safety only through action admissibility, it cannot distinguish between:

- a system preserving long-horizon structural viability, and
- a system silently degrading under an admissible but destructive regime.

Assume, for contradiction, that an action-centric safety framework $\mathcal{S}$ is complete with respect to long-horizon viability and identity continuity. Then $\mathcal{S}$ would be able to prevent all failure modes arising from regime-induced degradation.

However, since $\mathcal{S}$ cannot observe or regulate regimes, coupling-mediated burden, or accumulated internal degradation, it cannot intervene until action-level constraints are violated. By Theorems 19 and 20, such violations may occur only after irreversible structural damage or identity loss has already taken place.

Therefore, action-centric safety cannot, in principle, prevent failure modes driven by long-horizon structural processes. This contradicts the assumption of completeness.

Hence, any safety framework operating solely at the action level is incomplete with respect to long-horizon viability and identity continuity. □

Remark. The theorem does not assert that action-level safety is unnecessary. It asserts that action-level safety alone is insufficient to guarantee long-horizon viability or identity preservation in adaptive systems.

Theorem 26 - Architectural Necessity of Non-Causal Viability Observation

Statement. For any adaptive system intended for sustained long-horizon operation, a non-causal observational dimension is a necessary architectural condition for detecting silent degradation and identity risk. Without it, the system is structurally incapable of distinguishing performance persistence from identity continuity.

Proof (Sketch)

Let $\mathcal{O}_t$ denote an observational process intended to assess long-horizon viability or identity continuity. Let π_t denote the causal decision surface (policy, controller, learning rule) producing actions u_t. Assume the system lacks a non-causal observational dimension; that is, every observation available to the system is either (i) causally operative within π_t, or (ii) causally inert with respect to π_t.

Case 1: Observations are causally operative. If $\mathcal{O}_t$ influences π_t, then by Theorem 21 (Observation–Control Entanglement Collapse), $\mathcal{O}_t$ becomes part of the control loop. In this case, the observed signal is subject to optimization, avoidance, or compensation dynamics. Long-horizon degradation indicators are therefore distorted, delayed, or suppressed in order to preserve favorable observable values. Hence, performance persistence can be maintained while identity degradation remains unobserved or misrepresented.

Case 2: Observations are causally inert. If $\mathcal{O}_t$ does not influence π_t, then it cannot trigger regime change, revalidation, or termination. Although degradation may be nominally observed,

it is behaviorally irrelevant and cannot prevent continued operation under silently destructive regimes. Thus, identity risk remains undetectable in any operationally effective sense.

These two cases exhaust all possibilities when no non-causal observational dimension exists. In neither case can the system reliably distinguish between:

(i) sustained performance under preserved identity and (ii) sustained performance under silent identity deg

Therefore, in the absence of a non-causal observational dimension, an adaptive system is structurally incapable of detecting silent degradation and identity risk while performance persists. Consequently, a non-causal observational dimension is a necessary architectural condition for sustained long-horizon operation. □

Remark. The theorem is architectural and does not prescribe how non-causal observation is implemented. It asserts necessity, not mechanism: without architectural separation between observation and control, long-horizon viability cannot be reliably assessed.

Remark - Minimal-Deformation Path Selection (Structural View)

Under directional interaction and bounded dissipation, many macroscopic systems exhibit trajectories that appear as "least resistance" or "minimal curvature" paths. Within the present framework, such paths can be interpreted as those that minimize predicted accumulation of unresolved deformation relative to admissible geometry, thereby preserving internal time.

Theorem 27 - Context Capture Without Error

Statement. An adaptive system may incur semantic error under fully correct observations and admissible behavior. This result is conditional on the canonical definition of stabilization over τ_{el} (Assumption (ii)) Therefore, improving perception accuracy or optimization does not eliminate the risk of false meaning.

Proof sketch. Consider an adaptive system receiving observations x_t and producing actions a_t. Let m_t denote an internal semantic hypothesis used for interpretation, explanation, or commitment, distinct from the action-selection process.

Assume that observations x_t are fully correct and noise-free. There exist environments in which two independent processes E_1 and E_2 generate observation streams that are locally consistent with a single latent explanatory hypothesis Z, despite the absence of any causal relation between E_1 and E_2.

Due to partial observability of causal history, the system may lack access to the initiating cause of E_1 or retain it only as inactive residual structure. As a result, the semantic inference problem becomes underdetermined: multiple semantic hypotheses m_t are compatible with the same correct observations x_t.

If the system selects an incorrect semantic hypothesis m_t, a semantic error occurs even though observations are correct and system behavior remains admissible with respect to viability and regime constraints.

Since this error arises from structural underdetermination rather than perceptual noise or optimization failure, improving perception accuracy or action optimization cannot eliminate the risk of false meaning. □

Separation of behavioral stability and semantic correctness
Let $V(x_t)$ denote a Lyapunov-like functional characterizing behavioral admissibility (e.g., viability, bounded drift, or structural load), and let $W(m_t)$ denote a semantic correctness functional associated with causal binding or meaning.

It is possible that

$$\Delta V(x_t) \leq 0,$$

indicating stable and admissible behavior, while simultaneously $W(m_t)$ degrades or becomes misassigned due to semantic underdetermination.

Therefore, behavioral stability does not imply semantic correctness, even under perfect observations.

Corollary. 27.1 Irreducibility of false meaning
The risk of false meaning cannot be eliminated solely by improving the likelihood model $p(x \mid z)$ or by optimizing an action-level objective $J(a)$. False meaning arises from non-identifiability of latent semantic explanations under partial causal history, rather than from estimation error.

Theorem 28 — Conditional Primacy of Internal Time in Meaning Stabilization

Statement. Assuming that meaning stabilization requires persistence over a non-zero internal-time window, instantaneous causal metrics are insufficient to govern stabilization. Semantic stabilization and admissible semantic commitment are governed by accumulated internal time $\tau_{\text{el}}(t)$ rather than by any instantaneous scalar score such as confidence, probability, or reward. When internal time slows (i.e. when $\tau_{\text{el}}(t)$ accumulates more slowly per unit wall time), the earliest admissible commitment time is structurally delayed, preventing premature closure.

Assumptions (canonical, non-operational). We assume only the following architectural primitives.

(i) Internal time. There exists an accumulated internal-time quantity

$$\tau_{\text{el}}(t) := \int_0^t \kappa(s)\, ds, \qquad \kappa(s) \in [0,1],$$

monotone non-decreasing in wall time t. No further specification of κ is required.

(ii) Stabilization is defined over internal time. A hypothesis H_i is meaning-stabilized if a stability functional $S_i(\tau)$ remains above a threshold over a continuous internal-time window:

$$S_i(\tau) \geq \theta_S \quad \text{for all } \tau \in [\tau_1, \tau_1 + T_M],$$

for some window length $T_M > 0$. (Here S_i, θ_S, and T_M are abstract; no computability is implied.)

(iii) Commitment is admissibility-gated by stabilization. A semantic commitment operation is permitted only when the stabilization condition holds. Equivalently, the commitment operation must be authorized only if its realized effect-class satisfies

$$e(s, a, \tau, h) \in \mathcal{E}_{\text{adm}} \quad \text{and} \quad H_i \text{ is stabilized over an internal-time window of length } T_M.$$

This is a structural admissibility rule, not an algorithm.

Proof 1 (structural timing argument). Define the earliest admissible wall-clock time at which commitment to H_i becomes permitted:

$$t_c := \inf \{t \,:\, \exists \tau_1 \leq \tau_{\text{el}}(t) - T_M \text{ such that } S_i(\tau) \geq \theta_S \ \forall \tau \in [\tau_1, \tau_1 + T_M]\}.$$

This definition depends only on the accumulated internal time $\tau_{\mathrm{el}}(t)$.

Consider any wall-time interval $[t_0, t_1]$ on which internal time slows, i.e. the internal-time scaling $\kappa(t)$ decreases on that interval. Then the internal-time advance satisfies

$$\Delta\tau_{\mathrm{el}} = \tau_{\mathrm{el}}(t_1) - \tau_{\mathrm{el}}(t_0) = \int_{t_0}^{t_1} \kappa(s)\, ds,$$

which is strictly smaller when κ is smaller for the same wall-time duration $t_1 - t_0$. Since stabilization requires accumulation of an internal-time window of length T_M, any slowdown of τ_{el} delays the first wall-time t at which such a window can exist. Therefore t_c is non-decreasing under slowdown of κ. Hence, when internal time slows, admissible commitment is structurally delayed.

Proof 2 (conditional impossibility of instantaneous substitution). Assume, for contradiction, that commitment admissibility could be governed solely by an instantaneous scalar $q(t)$ (e.g. confidence, probability, reward), such that commitment is permitted at the first wall time t where

$$q(t) \geq \theta_q,$$

independently of τ_{el}.

Fix any wall time t^*. Construct two histories A and B such that

$$q_A(t^*) = q_B(t^*), \qquad \tau_{\mathrm{el}}^A(t^*) < \tau_{\mathrm{el}}^B(t^*),$$

which is possible by choosing different $\kappa(\cdot)$ trajectories while holding the same instantaneous score at t^*. By Assumption (ii), stabilization depends on the existence of an internal-time window of length T_M. Hence at time t^* it may be the case that H_i is stabilized in history B but not in history A, despite identical $q(t^*)$.

Therefore any rule that permits commitment based solely on $q(t)$ would authorize commitment in both histories at t^*, contradicting Assumption (iii) that commitment is admissibility-gated by internal-time stabilization. Thus, under the stated primitives, no instantaneous scalar $q(t)$ can substitute for internal-time-governed stabilization.

Remark (separation of behavioral admissibility and meaning stabilization). Let $V(t)$ denote any behavioral/trajectory admissibility functional (viability, drift bound, load) that may remain within admissible bounds while meaning stabilization is still forbidden because the internal-time window has not accumulated. This separation is structural: behavioral admissibility does not imply meaning stabilization, and meaning stabilization cannot be accelerated by instantaneous scoring without violating the internal-time definition of stabilization.

Theorem 29 - Regime Transition Requires Meaning-Level Stability

Transition between organizational regimes is admissible only under meaning-level stabilization. Context-level plausibility is insufficient for regime transition.

Proof sketch. Consider an adaptive system operating under a finite set of organizational regimes, where each regime corresponds to a distinct structural mode of operation. Let regime transitions incur a non-zero structural cost and potentially affect long-horizon viability or identity continuity.

Let m_t denote a semantic hypothesis formed from observations, and assume that m_t is context-level plausible but not meaning-level stabilized, i.e., it has not satisfied the internal-time stabilization condition.

Suppose, for contradiction, that a regime transition is permitted based solely on context-level plausibility of m_t. Then there exists a wall-clock time t^* at which:

1. the semantic hypothesis m_t appears locally coherent and plausible,
2. meaning-level stabilization has not been achieved,
3. a regime transition is nonetheless executed.

Because context-level plausibility may arise from transient coincidence, partial observation, or contextual salience, such a transition may be triggered by a hypothesis that is subsequently invalidated without contradiction in data. In this case, the system incurs irreversible structural cost due to the regime transition despite the absence of stabilized meaning.

This implies that regime transitions may be driven by transient semantic artifacts, leading to unnecessary structural load accumulation and increased risk of identity or viability degradation.

Therefore, allowing regime transitions under context-level plausibility alone violates the requirement of structural admissibility for long-horizon operation. Hence, regime transitions must be conditioned on meaning-level stabilization, and context-level plausibility is insufficient. □

Remark. Structural necessity of stabilization
Regime transitions alter the structural organization of the system and therefore cannot be treated as reversible or low-cost actions. Conditioning such transitions on meaning-level stabilization prevents structurally irreversible consequences arising from transient or unstable semantics.

Theorem 30 - Rc Is a Necessary Semantic Regime

Any system that admits interpretation and commitment must include a semantic containment regime in which meaning may form without structural impact. Absence of such a regime leads either to paralysis or uncontrolled transitions.

Proof sketch. Consider an adaptive system that (i) produces interpretations of observations and (ii) is capable of semantic commitment, where commitment denotes authorization for interpretations to influence irreversible structural operations (e.g., regime transitions or identity-affecting interventions).

Assume the system does *not* include any regime in which semantic hypotheses may form while being prevented from exerting structural impact. Then the system must follow one of the following two architectural patterns:

Case 1 (Always-commit coupling). Interpretation outputs are permitted to influence structural operations immediately, without requiring meaning-level stabilization. Since interpretation under partial history and contextual salience is structurally underdetermined, transient hypotheses can trigger irreversible structural changes. This yields uncontrolled transitions and unnecessary structural cost accumulation.

Case 2 (Never-commit decoupling). To prevent the failure in Case 1, the system may prohibit interpretations from influencing structural operations altogether, unless full certainty is obtained. However, full certainty is generally unattainable under partial observability and long-horizon interaction. As a result, the system becomes unable to authorize commitment when action is required, producing paralysis.

Because the system admits interpretation and commitment but lacks a semantic containment regime, it is forced into either Always-commit coupling (uncontrolled transitions) or Never-commit decoupling (paralysis). Both outcomes contradict long-horizon structural viability.

Therefore, any such system must include a distinct semantic containment regime, R_c, in which semantic hypotheses may form while structural impact remains gated until meaning-level stabilization is achieved. □

Remark. Rc as an architectural third mode
The necessity of R_c does not follow from performance optimization but from structural admissibility. R_c separates hypothesis formation from irreversible commitment, thereby avoiding both premature closure (uncontrolled transitions) and over-conservatism (paralysis).

Theorem 31 - Structural Self-Forgiveness

In any adaptive system operating over a long horizon with irreversible regime transitions, residual accumulation, and internal time, the absence of a mechanism for conditional release of self-penalty following closure of reversible errors necessarily leads to degradation of long-horizon viability or identity continuity.

Proof sketch. Assume an adaptive system subject to the following canonical constraints: (i) drift accumulation is inevitable, (ii) regime transitions incur non-zero structural cost, (iii) internal time bounds admissible structural evolution, and (iv) not all errors lead to irreversible damage.

Consider a class of reversible errors that do not violate viability or identity continuity and are subsequently structurally closed (i.e., transferred into residual form).

If the system retains active self-penalty or internal inhibition associated with such closed errors, then this penalty persists as an additional constraint on future adaptation. Over internal time, the accumulation of such constraints increases effective phase transition cost and suppresses admissible commitment.

As internal time progresses, the system is forced into one of two failure modes: either excessive inhibition leading to paralysis (zero or near-zero commitment), or compensatory over-regulation leading to instability and accelerated resource depletion.

Therefore, in the absence of a mechanism that conditionally releases self-penalty after closure of reversible errors, long-horizon viability or identity continuity cannot be preserved.

Hence, a mechanism of structural self-forgiveness is necessary. □

Remark 1. Normative clarification
Structural self-forgiveness does not imply error erasure, reward inflation, or tolerance of irreversible damage. It denotes the controlled release of internal inhibitory constraints after an error has been rendered non-active through structural closure. This mechanism operates independently of semantic interpretation accuracy and is required solely to prevent accumulation of self-imposed structural debt.

Remark 2. Structural interpretation
The effect described in Theorem 31 may be interpreted as accumulation of structural debt: persistent self-imposed inhibitory constraints progressively reduce the admissible space of regime transitions. In the limit, this contraction yields either a degenerate admissible set (paralysis) or unstable oscillatory compensation (over-regulation). No specific accumulation model is assumed.

Theorem 32 - Structural Delay of Failure

Failure in long-horizon adaptive systems is structurally delayed with respect to action-level error. Systems do not collapse at the moment of incorrect action or local inadmissibility, but when accumulated structural consumption exhausts the remaining capacity for identity-preserving evolution.

This result follows as a structural consequence of performance–identity decoupling and silent degradation. Specifically, the existence of admissible behavior under ongoing identity degradation implies a necessary temporal offset between observable failure and the underlying structural cause.

Accordingly, failure is not an instantaneous event but the terminal manifestation of prior irreversible structural consumption.

Proof. Proof (Structural Sketch)

Consider an adaptive system operating under sustained directed interaction with a changing environment and finite structural capacity.

From the Performance–Identity Decoupling result, there exists a non-empty interval during which externally observable behavior remains admissible (bounded error, stable reward, acceptable trajectories) while internal identity-preserving structure is already degrading.

From the Silent Degradation under Regime Persistence result, structural burden may accumulate without producing immediate action-level violations or triggering corrective regulation, provided the system remains within admissible regimes.

Let $C(t)$ denote cumulative structural consumption (drift, latent residual, regime-induced load), and let $C_{\max}$ denote the finite identity-preserving capacity of the system. By construction, $C(t)$ increases monotonically under sustained operation, even when instantaneous performance remains acceptable.

Failure occurs when $C(t) \geq C_{\max}$, at which point identity-preserving evolution becomes impossible. However, no necessary condition requires instantaneous action-level error to occur at the moment when $C(t)$ crosses this boundary.

Therefore, the temporal point of observable failure is strictly delayed with respect to the causal accumulation of structural consumption. The system collapses not at the moment of incorrect action, but at the exhaustion of identity-preserving capacity accumulated across prior correct actions.

This establishes that failure is structurally delayed relative to action-level error and cannot, in general, be detected or predicted from instantaneous performance metrics alone. □

Remark - Failure as Terminal Manifestation, Not Causal Event. This theorem implies that observable failure in long-horizon adaptive systems is not a causal event but a terminal manifestation.

The true causal structure of failure lies in accumulated, irreversible structural consumption distributed across time, regimes, and interactions. Action-level errors, when they finally appear, serve only as surface indicators that identity-preserving capacity has already been exhausted.

Consequently, failure analysis based on last actions, local errors, or immediate triggers is structurally incomplete. Any framework that treats failure as an instantaneous outcome rather than as a delayed structural endpoint necessarily misattributes cause and obscures long-horizon risk.

Theorem 33 - Incompleteness of Optimization-Centric Regulation

No optimization of action-level correctness, reward maximization, or stability can, in principle, impose an upper bound on long-horizon structural consumption. Therefore, optimization-centric regulation is incomplete with respect to identity preservation.

Proof sketch. Consider an adaptive system operating under persistent directed interaction. Optimization acts locally on action selection, error minimization, or reward maximization, but does not eliminate directional asymmetry between action and consequence. As shown by the inevitability of drift and entropy growth, each admissible action contributes irreducible structural residuals.

Since structural consumption accumulates independently of instantaneous correctness, no optimization procedure operating at the action level can bound total structural cost over an unbounded horizon. Hence, optimization-centric regulation cannot, in principle, guarantee identity preservation.

□

Remark

This theorem does not deny the utility of optimization for short-horizon stability or performance. It establishes a structural incompleteness result: optimization alone cannot regulate long-horizon viability. Any architecture relying exclusively on action-level optimization necessarily remains blind to identity depletion and regime exhaustion.

Remark (Lyapunov Perspective)

Even if a global Lyapunov function exists that certifies bounded error, stability, or convergence of trajectories, such a function can only bound state evolution within a fixed admissible regime. It does not, in principle, bound accumulated structural consumption across regime transitions or over unbounded horizons.

Thus, Lyapunov stability of behavior does not imply preservation of identity or long-horizon viability. This result is not a limitation of Lyapunov theory, but a consequence of its action-centric and state-centric scope.

Theorem 34 - Dual Influence of the Present

In a long-horizon adaptive system, the present state is simultaneously shaped by two irreducible influences:

1. causal dynamics, determining what can happen, and
2. non-causal structural admissibility, determining what is allowed to happen.

These influences act concurrently in the present, but cannot be reduced to a single causal process.

Proof Sketch. Let $x(t)$ denote the causal state of the system and $u(t)$ an action. Causal evolution is governed by

$$x(t+1) = f(x(t), u(t)).$$

Let $S(t)$ denote an accumulated structural state encoding viability, identity continuity, and admissibility bounds. Structural accumulation follows

$$S(t+1) = S(t) + \Phi(x(t), u(t)).$$

Admissibility at time t is given by

$$\mathcal{A}(t) = \mathcal{A}(S(t)),$$

and is therefore not a function of $x(t)$ alone.

Well-Foundedness Barrier Between Admissibility and Action. Any operator, signal, or variable participating in the determination of admissibility cannot belong to the action domain without inducing circularity.

If admissibility becomes causally actionable, it collapses into penalty-based optimization and loses its non-causal structural character. Therefore admissibility is structurally prior to action selection, control, planning, learning, and optimization.

Action is non-primitive in long-horizon adaptivity: it is defined only after admissibility constraints are fixed. Any architecture that treats action selection as primitive with respect to admissibility is structurally misclassified.

Remark - Operational meaning and governance boundary
Theorem 34 separates two operators that coincide in time but differ in logical role: causal evolution produces state transitions, while structural admissibility constrains which transitions remain identity- and viability-consistent.

This separation is not a semantic preference but a governance boundary. If admissibility is exposed to the primary causal decision surface as an optimization signal (i.e., treated as causal feedback), the system collapses into meta-optimization and loses the ability to function as an independent witness of long-horizon degradation.

Accordingly, admissibility must remain structurally operative but non-instrumental: it may restrict, gate, or invalidate regimes, yet must not become a reward-like target. This is precisely the architectural stance motivating the intermediate regime often referred to as *cybernetics of order 2.5*: admissibility is present as a structural regulator, while third-order self-optimization over admissibility is intentionally forbidden.

Lyapunov-style structural argument.

Assume there exists a non-negative structural Lyapunov function

$$V(S(t)) \geq 0,$$

interpreted as a measure of structural viability or identity coherence.

By construction, V evolves monotonically under admissible operation:

$$V(S(t+1)) - V(S(t)) \leq 0 \quad \text{for admissible } u(t) \in \mathcal{A}(S(t)).$$

If an action $u(t)$ violates admissibility, then the induced structural update satisfies

$$V(S(t+1)) - V(S(t)) > 0,$$

indicating irreversible structural degradation.

Because V is defined on $S(t)$ rather than on $x(t)$, structural admissibility constrains present action without appearing in the causal transition function f.

Thus, causal dynamics determine reachable states, while non-causal admissibility - expressed via $V(S)$ - determines structurally allowable evolution.

The present state is therefore shaped simultaneously by causal possibility and non-causal structural admissibility, which are not reducible to a single causal operator.

□

Corollary

34.1. Irreversibility of Structural Constraint Violation
Once structural admissibility constraints are violated, no amount of locally correct causal action can restore admissibility without a change in the system's structural class.

In particular, post-hoc optimization, delayed correction, or reward-based compensation cannot recover admissibility after structural boundaries have been crossed.

34.2. Meaning as Admissibility, Not Inference
Meaning does not arise through causal inference from perception alone. Instead, meaning emerges when an interpretation is admitted by the current structural admissibility context.

Thus, identical perceptual or dynamical inputs may or may not acquire meaning depending on the admissible interpretative space defined by accumulated structural state.

34.3. Forbidden Feedback of Admissibility

If admissibility constraints are exposed to the primary causal agent as explicit causal feedback, the system necessarily collapses into meta-optimization and loses long-horizon viability.

Therefore, admissibility must not be represented as a reward signal, objective, explanation, or actionable map within the causal decision surface of the agent.

Theorem 35 — Internal-Time Persistence as a Necessary Condition for Commitment

Statement. In a long-horizon adaptive system, no semantic commitment can be structurally stabilized unless the corresponding interpretation induces admissible structural effect-classes continuously over a non-zero interval of internal time. Any commitment authorized on the basis of transient alignment in causal time necessarily induces irreversible structural cost, regardless of local correctness.

Proof Sketch. Let $S(t)$ denote the accumulated structural state of the system at evaluation cycle t, and let $C(t)$ denote the set of active semantic commitments. Let $e(t) \in \mathcal{E}$ denote the structural effect-class induced by the interpretation authorized at time t.

Define a non-negative structural load functional $V(S)$ such that V increases monotonically under irreversible commitment operations and decreases only under explicit structural reconfiguration. Interpretative fluctuations that do not induce commitment correspond to bounded variations in causal state and do not modify $V(S)$.

Commitment authorization at time t introduces a strictly positive increment $\Delta V > 0$. If com-

mitment is authorized without persistence across internal time, then under unavoidable drift the induced effect-class $e(t)$ decorrelates from the identity-compatible admissible region, while ΔV remains accumulated. Thus $V(S)$ increases without compensatory stabilization, violating boundedness over long horizons.

By standard Lyapunov-type arguments on monotone structural load, any admissibility policy that permits such transiently authorized commitments yields unbounded structural accumulation. Therefore, persistence of admissible effect-classes across a non-zero interval of internal time is a necessary condition for commitment admissibility.

Remark — Authorization vs. Inference. Inference answers the question: what appears to be the case now. Authorization answers the question: what may be irreversibly fixed. Conflating inference with authorization collapses admissibility into causal evaluation, induces meta-optimization, and destroys long-horizon viability. **Corollaries.**

35.1 — Premature Commitment as Structural Failure. Premature semantic commitment constitutes a structural failure mode distinct from perceptual error or inferential inaccuracy. Its cost is irreversible and accumulates independently of local correctness.

35.2 — Regime Transitions Cannot Be Triggered by Interpretation Alone. No regime transition or irreversible structural operation may be causally triggered by interpretation alone. Such transitions require independent admissibility authorization over effect-classes.

35.3 — Forbidden Feedback of Commitment Admissibility. Exposing commitment admissibility constraints to the primary adaptive process as causal feedback induces meta-optimization and collapse of system viability. Admissibility boundaries must therefore remain structurally enforced and non-causally opaque.

Definition (Pre-Semantic Structural Revelation — Expanded). Pre-semantic structural revelation is the exposure of relational geometry and constraint topology arising from raw signal dynamics prior to semantic stabilization, intentional attribution, or meaning-level commitment.

This mode of revelation is structural rather than interpretive. It does not operate on symbols, hypotheses, or representations, but on relations induced by interaction.

Pre-semantic structural revelation satisfies the following structural properties, which are invariant across implementations and independent of substrate:

(PSR-1) *Non-Causality.* The revealed structure does not enter the causal decision surface of the adaptive agent. It does not function as a control signal, optimization target, or corrective feedback. Its influence on behavior is indirect and mediated solely through structural admissibility rather than through causal regulation.

(PSR-2) *Non-Semanticity.* No meaning, value, intention, or purpose is assigned at this layer. The revealed structure is not an interpretation of the environment and does not encode propositions, beliefs, or semantic commitments.

(PSR-3) *Non-Commitment.* Pre-semantic revelation does not stabilize interpretations into irreversible regimes. It carries no authority to authorize commitments, regime transitions, or identity-affecting operations. Stabilization, when it occurs, must be justified independently within internal-time-bounded admissibility.

(PSR-4) *Constraint Exposure.* The layer exposes patterns of relational density, continuity, divergence, saturation, and incompatibility that delimit the space of admissible future evolution. These patterns function as structural boundaries rather than prescriptive rules.

Remark — Placement in NC2.5. Theorem 36 of v2.0 proves the necessity of this layer. The present definition gives it a name and enumerates its invariant properties. The definition is not new content; it is the explicit formalization of what Theorem 36 requires to exist.

Remark — Relation to Meaning. Pre-semantic revelation defines the space of admissible meanings without selecting among them. It constrains what may become meaningful, but does not assign value, intention, or purpose. Meaning emerges only at a later stage, when interpretation stabilizes across internal time and commitment becomes structurally admissible.

Remark — Resolution of the Observability Paradox. How can structural state become observable without becoming causally operative? Pre-semantic revelation resolves this: the system may witness approaching inadmissibility without acquiring the ability to optimize against it. This is the architectural mechanism that separates revelation from regulation.

Theorem 36 — Necessity of Pre-Semantic Structural Revelation

Statement. In any long-horizon adaptive system that preserves identity and viability, there must exist a pre-semantic structural revelation layer: a mechanism by which admissibility-relevant structural constraints on future evolution become observable prior to meaning-level interpretation or commitment. In the absence of such a layer, semantic commitment necessarily induces irreversible structural degradation.

Proof (structural necessity by exhaustion of cases). Assume, for contradiction, that no pre-semantic structural revelation layer exists. Then admissibility-relevant structural constraints become observable only after semantic interpretation and/or meaning-level commitment.

Let $e(s, a, \tau, h) \in \mathcal{E}$ denote the structural effect-class realized by authorizing a semantic commitment under context (s, τ, h). By the irreversibility of commitment under finite remaining internal time (Theorem 2), semantic commitment cannot remain indefinitely provisional: it necessarily realizes a structural effect-class.

We consider the exhaustive cases.

Case (a) — Early Commitment. If commitment is authorized before admissibility-relevant structure is revealed, authorization proceeds without conditioning on whether

$$e(s, a, \tau, h) \in \mathcal{E}_{\text{adm}}.$$

Under non-zero drift accumulation, latent structural incompatibilities necessarily accrue, and at some finite horizon a prohibited effect-class

$$e(s, a, \tau, h) \notin \mathcal{E}_{\text{adm}}$$

is realized. By non-restorability of admissibility, this induces irreversible regime transition or identity loss. Thus early commitment collapses by committing without admissibility.

Case (b) — Delayed Commitment. If the system delays commitment until admissibility-relevant constraints become observable, then such visibility occurs only once structural degradation has already become semantically expressible or operationally unavoidable. In this regime, revelation arrives only after a prohibited effect-class

$$e(s, a, \tau, h) \notin \mathcal{E}_{\text{adm}}$$

has already been realized. Observation therefore arrives too late to preserve admissible continuation.

Both cases violate long-horizon viability: Case (a) collapses by premature irreversible commitment, Case (b) collapses by delayed revelation after admissibility loss.

Therefore, to preserve identity and long-horizon viability, admissibility-relevant structural constraints must be revealable prior to semantic interpretation and commitment. A pre-semantic structural revelation layer is necessary. □

Storage Argument (sketch). Let $S(t)$ denote accumulated structural state and let $\tau_{\text{rem}}(t) \geq 0$ denote remaining internal-time budget. Assume $\tau_{\text{rem}}(t)$ decreases monotonically under realized structural burden:

$$\tau_{\text{rem}}(t+1) \leq \tau_{\text{rem}}(t) - g(\Delta S(t)), \qquad g(z) \geq 0,\ g(z) > 0 \text{ for } z > 0.$$

Define $V(t) := \tau_{\text{rem}}(t)$. Long-horizon viability requires $V(t) > 0$.

If admissibility-relevant structure becomes visible only after semantic commitment, then there exist intervals during which commitments are authorized without conditioning on $e(s, a, \tau, h) \in \mathcal{E}_{\text{adm}}$. Hence there exists a finite horizon $T^* < \infty$ such that

$$\exists t < T^* : \quad e(s, a, \tau, h) \notin \mathcal{E}_{\text{adm}}, \qquad V(T^*) = 0.$$

Thus, without pre-semantic revelation, remaining internal time is exhausted in finite operation. To avoid forced exhaustion under non-zero drift, admissibility-relevant structure must be revealable prior to meaning-level commitment.

Remark — Revelation Is Not Semantics. Pre-semantic structural revelation does not presuppose beliefs, propositions, or meaning-bearing representations. It refers solely to structural visibility sufficient to condition admissibility over effect-classes without inducing interpretation-level commitment. This preserves observational independence and avoids premature irreversibility.

Corollary 36.1 — Structural Priority Over Interpretation. In any long-horizon adaptive system, admissibility-relevant structural constraints must be operationally prior to meaning-level interpretation and commitment. Interpretation cannot be the first gate of irreversibility.

Corollary 36.2 — Necessity of Cybernetics of Order 2.5 Separation. Any architecture in which admissibility-relevant revelation becomes causally actionable at the level of interpretation or commitment collapses observational independence and renders admissibility optimizable. Therefore the pre-semantic revelation layer must remain visible but non-causal with respect to the primary adaptive process, placing the architecture strictly within cybernetics of order 2.5.

Corollary 36.3 — Pre-Semantic Layer Defines Cybernetics 2.5. Statement. Pre-semantic structural revelation, as formalized in the expanded Definition (PSR), is the decisive architectural reason why NC2.5 cannot be classified as second-order or third-order cybernetics.

Proof.

Step 1. By Theorem 36, the pre-semantic layer must exist in any long-horizon adaptive system preserving identity and viability.

Step 2. By the expanded Definition (PSR), this layer satisfies: non-causality (PSR-1), non-semanticity (PSR-2), non-commitment (PSR-3), and constraint exposure (PSR-4).

Step 3. Second-order cybernetics (von Foerster, Maturana) couples observation to regulation: the observer's observation influences the system's regulatory behavior. By PSR-1 (non-causality), the pre-semantic layer does not influence regulatory behavior. It does not enter the causal decision surface. Therefore the pre-semantic layer cannot support second-order coupling.

Step 4. Third-order cybernetics internalizes reflexivity as an optimizable process: the system's awareness of itself becomes a lever for self-modification. By PSR-1 (non-causality) and PSR-3 (non-commitment), the pre-semantic layer provides awareness without leverage. It reveals constraint structure without authorizing action. By Theorem 37 (Incompatibility with Third-Order), any architecture in which admissibility-relevant revelation becomes causally actionable lies outside the class.

Step 5. The pre-semantic layer enables reflexivity without control, observation without optimization, awareness without collapse. This is the defining characteristic of Cybernetics of Order 2.5: the intermediate between non-reflexive systems (order ≤ 2) and fully reflexive-actionable systems (order 3). □

Theorem 37 - Incompatibility with Third-Order Cybernetics

In any long-horizon adaptive system, any architecture that renders pre-semantic structural revelation causally actionable necessarily collapses into third-order cybernetics and loses long-horizon viability.

Proof Sketch. Let $\mathcal{R}_{pre}(t)$ denote a pre-semantic structural revelation signal, exposing admissibility-relevant constraints without semantic commitment.

Assume that $\mathcal{R}_{pre}(t)$ is made causally actionable, that is, it directly influences the primary decision surface of the system.

Then reflexive structural observation becomes subject to optimization, suppression, delay, or instrumentalization. By the Corollary 34.3. (Forbidden Feedback of Admissibility), this introduces meta-optimization over admissibility itself, collapsing non-causal structural constraints into causal control objectives.

This condition is precisely the defining feature of third-order cybernetics: the system observes its own observing and injects this reflexivity into causal regulation.

However, long-horizon viability requires that admissibility constraints remain observable without becoming actionable, preserving observational independence.

Therefore, any architecture that causally operationalizes pre-semantic structural revelation is structurally incompatible with long-horizon adaptive viability.

□

Remark - Structural Boundary Between 2.5 and 3

Third-order cybernetics is not rejected here as a conceptual extension, but as a structural hazard.

The incompatibility arises not from insufficient reflexivity, but from excess reflexivity entering the causal surface, destroying the distinction between observation and regulation.

Cybernetics of the 2.5-th order preserves reflexive visibility while enforcing non-causality of admissibility revelation.

Corollary 37.1 - Silent Degradation Detectability

Silent structural degradation is detectable if and only if pre-semantic structural revelation remains non-causal.

If such revelation becomes causally actionable, degradation is either optimized away, deferred, or suppressed, eliminating the system's ability to witness its own long-horizon decay.

Theorem 38 - Non-Derivability of Admissibility from Any State Description

In any long-horizon adaptive system, admissibility of actions, regimes, or semantic commitments cannot be derived from any instantaneous, finite, or augmented state representation, including arbitrarily rich latent, belief, or world-model states.

Proof Sketch. Let $x(t)$ denote the instantaneous causal state of the system at time t. Let $S(t)$ denote the accumulated structural state, capturing irreversible structural burden, regime history, commitment closure status, and internal-time depletion. Let $A(t)$ denote the admissibility set at time t, i.e. the set of actions, regimes, or commitments authorized for execution at t. **Strengthening Note.** This impossibility does not arise from insufficiency of representational richness. It holds even for arbitrarily augmented, latent, predictive, epistemic, or world-model states. The prohibition is structural: any formulation of admissibility as a function of state,

$$\mathrm{Adm} = f(s),$$

collapses admissibility into causal evaluation and therefore violates Axiom XLI by construction. No extension of state space can repair this collapse, because the forbidden object is not the state, but the functional dependence itself.

Assume, for contradiction, that admissibility is state-derivable. That is, there exists a mapping

$$\mathcal{F} : \mathcal{X} \to \mathcal{P}(\mathcal{U})$$

such that

$$A(t) = \mathcal{F}\big(x(t)\big),$$

where $\mathcal{U}$ denotes the space of candidate actions and $\mathcal{P}(\mathcal{U})$ denotes the power set.

Consider the class of long-horizon systems in which admissibility constraints arise from predictable future degradation of coherence, regime stability, or identity continuity, rather than from violations detectable in the present causal state. Formally, assume there exists a nonnegative structural burden functional

$$B(t) = \mathcal{B}\big(S(t)\big) \geq 0,$$

and an admissibility bound $B_{\max}$ such that violation of $B(t) \leq B_{\max}$ renders a class of actions or commitments inadmissible. Assume $B(t)$ evolves under irreversible effects and drift such that for any nontrivial long-horizon operation there exist trajectories where $B(t)$ increases while $x(t)$ remains within a bounded locally admissible region.

Now construct two operational episodes E_A and E_B of the same system such that at a given time t_0,

$$x_A(t_0) = x_B(t_0),$$

but their structural histories differ:

$$S_A(t_0) \neq S_B(t_0),$$

in particular

$$B_A(t_0) \neq B_B(t_0),$$

with

$$B_A(t_0) \leq B_{\max} \quad \text{and} \quad B_B(t_0) > B_{\max}.$$

Such pairs exist because $S(t)$ contains irreversibility, regime occupation history, and internal-time consumption which are not uniquely determined by $x(t)$. Two episodes can coincide in instantaneous causal state while differing in accumulated structural state.

Since admissibility depends on the structural bound, it follows that

$$A_A(t_0) \neq A_B(t_0),$$

because at least one class of candidates is admissible under $B_A(t_0) \leq B_{\max}$ but inadmissible under $B_B(t_0) > B_{\max}$.

But if $A(t) = \mathcal{F}(x(t))$, then $x_A(t_0) = x_B(t_0)$ implies

$$A_A(t_0) = \mathcal{F}\big(x_A(t_0)\big) = \mathcal{F}\big(x_B(t_0)\big) = A_B(t_0),$$

a contradiction.

Therefore admissibility cannot be derived from any instantaneous causal state description. Since the assumption of state-derivability fails even when $x(t)$ is arbitrarily augmented as a belief state or world-model state, the result extends to any finite or causal augmentation whose value is determined at time t from present information.

Hence admissibility is non-derivable from state and must be treated as a structural operator depending on accumulated structural state $S(t)$ rather than on $x(t)$.

Structural Closure Clause. The impossibility result does not depend on the richness, dimensionality, or epistemic content of the state representation. Any formulation of the form

$$A(t) = \mathcal{F}(x(t))$$

collapses admissibility into causal evaluation, regardless of whether $x(t)$ denotes a raw physical state, a belief state, a latent embedding, or a learned world model. Thus the prohibition applies to the functional form itself, not to limitations of state expressiveness.

Remark - Augmentation Does Not Escape the Contradiction. Let $\hat{x}(t)$ be any augmented state representation computed from the causal process, including latent states, belief states, predictive world models, or confidence vectors. If $\hat{x}(t)$ is computed without direct access to $S(t)$ as an independent accumulated variable, then $\hat{x}(t)$ remains a function of present causal information. Thus the same construction applies: there exist episodes with identical $\hat{x}(t_0)$ but distinct structural burden, yielding distinct admissibility sets. Therefore no augmentation confined to present causal computation can make admissibility state-derivable.

Corollary 38.1. State-Based Safety Is Structurally Incomplete. Any architecture that represents admissibility as a predicate of instantaneous state, including safety classifiers, risk critics, constraint evaluators, or confidence thresholds, is structurally incomplete for long-horizon viability. It cannot distinguish episodes with equal state but unequal structural burden.

Corollary 38.2. Admissibility Requires an Independent Structural Variable. If admissibility is to be well-defined, the system must maintain an accumulated structural variable $S(t)$ (or an equivalent non-causal structural witness) such that admissibility is a function of $S(t)$. Therefore admissibility must be architecturally separated from state estimation and action evaluation.

Corollary 38.3. Forbidden Feedback of Admissibility Derivation. Any attempt to expose admissibility as a state-derivable causal signal to the primary decision surface induces meta-optimization over admissibility itself, collapsing the non-causal separation required for long-horizon viability. Therefore admissibility boundaries must remain structurally enforced and non-causally opaque.

Theorem 39 - Impossibility of Causal Authorization of Semantic Commitment

Statement. No causal inference process, confidence measure, correctness criterion, or optimization signal is sufficient to authorize semantic commitment in a long-horizon adaptive system

without violating identity preservation.

Proof Sketch. Let $x(t)$ denote the instantaneous causal state of the system. Let $S(t)$ denote the accumulated structural state, incorporating irreversible commitments, regime occupation history, and internal-time consumption. Let $I(t)$ denote the space of interpretations available to the system at time t, and let $C(t) \subseteq I(t)$ denote the set of semantic commitments that have been irreversibly fixed.

Assume, for contradiction, that there exists a causal authorization rule

$$\mathcal{G} : \mathcal{X} \to \{0, 1\}$$

such that a semantic commitment $c \in I(t)$ is authorized at time t if and only if

$$\mathcal{G}\big(x(t)\big) = 1,$$

where $\mathcal{G}$ may depend on confidence, correctness, convergence, optimization score, or any other causal signal derived from $x(t)$.

Semantic commitment is an irreversible operation: it induces a non-negative increment $\Delta S_c > 0$ in the accumulated structural state,

$$S(t^+) = S(t) + \Delta S_c,$$

and cannot be undone without a change of structural class. Interpretations, by contrast, are causally revisable and correspond to bounded fluctuations in $x(t)$ that do not necessarily alter $S(t)$.

Because $x(t)$ evolves under drift and environmental interaction, there exist times $t_1 < t_2$ such that

$$x(t_1) \approx x(t_2),$$

while the future structural consequences of committing to the same interpretation differ due to differences in $S(t_1)$ and $S(t_2)$. In particular, there exist cases where commitment at t_1 remains admissible over internal time, while commitment at t_2 induces irreversible structural overload.

Since $\mathcal{G}$ depends only on $x(t)$, it authorizes commitment equally at t_1 and t_2. Thus the authorization rule cannot discriminate between structurally admissible and inadmissible commitments when causal state is locally similar.

As a result, there exist trajectories in which commitment is causally authorized while violating long-horizon identity constraints. Such violations accumulate irreversibly in $S(t)$ and cannot be compensated by later correct causal behavior.

This contradicts the requirement of identity preservation in long-horizon adaptive systems. Therefore no causal authorization rule based on inference, confidence, correctness, or optimization can safely authorize semantic commitment.

Remark - Authorization vs. Inference. Inference answers the question: what appears to be the case now. Authorization answers the question: what may be irreversibly fixed. Conflating inference with authorization collapses irreversible structural operations into causal evaluation and exposes identity-defining decisions to transient alignment and drift.

Corollary 39.1. Commitment Cannot Be Triggered by Confidence or Convergence. No level of confidence, predictive accuracy, convergence, or local optimality is sufficient to authorize semantic commitment. Such criteria are functions of causal alignment and are orthogonal to long-horizon structural viability.

Corollary 39.2. Threshold-Based Commitment Is Structurally Unsafe. Any architecture that authorizes commitment by threshold crossings, score maximization, or convergence detection necessarily admits irreversible structural error, even under locally correct operation.

Corollary 39.3. Forbidden Feedback of Commitment Admissibility. Exposing commitment admissibility as a causal feedback signal to the primary adaptive agent induces meta-optimization over commitment itself, collapsing the separation between interpretation and identity fixation. Therefore commitment admissibility must remain structurally enforced and non-causally opaque.

Theorem 40 - Collapse of Admissibility Under Optimization-Based Representation

Statement. Any representation of future-degradation constraints as scalar quantities, penalties, rewards, costs, or losses induces optimization pressure that necessarily invalidates their protective function.

Proof Sketch. Let $x(t)$ denote the instantaneous causal state of the system and let $S(t)$ denote the accumulated structural state. Let $A(t)$ denote the admissibility predicate over candidate actions, regimes, or commitments.

Assume, for contradiction, that admissibility constraints are representable as a scalar-valued quantity. That is, assume there exists a function

$$J_{\text{adm}} : \mathcal{X} \to \mathbb{R}$$

such that admissibility is enforced by minimizing, bounding, or trading off J_{adm} within an optimization process.

Optimization-based decision-making selects actions $u(t)$ according to

$$u(t) = \arg\min_{u \in \mathcal{U}} \mathbb{E}\big[J_{\text{task}}(x(t), u) + \lambda J_{\text{adm}}(x(t), u)\big],$$

or an equivalent maximization formulation, for some weight $\lambda > 0$.

By construction, scalar quantities entering an objective function are subject to trade-off. For any finite λ, there exist task pressures, uncertainties, or external incentives such that an increase in J_{adm} is accepted in exchange for improvement in J_{task}. Thus inadmissible candidates are not excluded from consideration, but remain present within the optimizer's comparison set.

Moreover, optimization induces representational pressure. Once admissibility is encoded as a scalar, the optimizer can: (i) defer admissibility violations, (ii) smooth or average them over time, (iii) exploit regions where penalties are locally small but structurally irreversible, or (iv) instrumentalize violations to maintain short-horizon performance.

Future-degradation constraints, however, are characterized by delayed, non-local, and irreversible effects on $S(t)$. Their protective function requires categorical exclusion prior to evaluation, not numerical comparison during evaluation.

Therefore representing admissibility as a scalar quantity collapses the distinction between "forbidden" and "suboptimal". Inadmissibility is reduced to a cost, and optimization pressure necessarily drives the system toward trajectories that violate long-horizon structural bounds while preserving local optimality.

This contradicts the definition of admissibility as a non-tradable structural constraint. Hence any optimization-based representation of admissibility invalidates its protective function.

Remark - Trade-Off Is Structural Failure. Optimization presupposes comparability. Admissibility presupposes incomparability. Once future-degradation constraints enter a common numerical space with task objectives, they cease to function as constraints and become incentives to be optimized against.

Corollary 40.1. Penalty-Based Safety Is Not Admissibility. Any safety mechanism based on penalties, regularizers, reward shaping, or auxiliary losses is not an approximation of admissibility. It is structurally incapable of enforcing categorical exclusion of inadmissible trajectories.

Corollary 40.2. Soft Constraints Collapse Under Optimization Pressure. So-called "soft constraints" inevitably degrade into optimization variables. Under sustained interaction, they permit structurally irreversible violations while preserving short-horizon correctness.

Corollary 40.3. Admissibility Requires Domain Exclusion. For admissibility to retain its protective role, inadmissible actions, regimes, or commitments must be excluded from the domain of optimization altogether. They must not be represented, scored, compared, or traded, but rendered undefined for evaluation.

Theorem 41 - Equivalence Between Penalty-Based Safety and Third-Order Cybernetics

Statement. Any system that exposes admissibility-relevant constraints as optimizable signals is structurally equivalent to a third-order cybernetic system and inherits its long-horizon failure modes.

Proof Sketch. Let $x(t)$ denote the instantaneous causal state and let $S(t)$ denote the accumulated structural state governing admissibility, identity continuity, and internal-time consumption. Let $A(t)$ denote the admissibility predicate over actions, regimes, or commitments.

Assume that admissibility-relevant constraints are exposed to the primary adaptive process as optimizable signals. That is, assume there exists a causal signal $z(t)$ derived from $S(t)$ such that $z(t)$ enters the decision surface through optimization, comparison, or policy update. Formally, admissibility information becomes causally actionable.

Once admissibility enters the causal loop, the system is no longer merely constrained by admissibility, but begins to act *on* admissibility. The system observes not only its environment, but its own structural limits, and incorporates this observation into causal regulation.

This constitutes the defining feature of third-order cybernetics: the system observes its own observing and injects this reflexive information into the causal decision process. The distinction between witnessing structural limits and acting upon them collapses.

Under such conditions, admissibility signals are subject to optimization pressure. They may be suppressed, delayed, averaged, reweighted, or strategically violated to preserve short-horizon objectives. As shown in Theorem 40, optimization over admissibility necessarily invalidates its protective function.

Therefore any architecture that renders admissibility causally actionable is structurally indistinguishable from a third-order cybernetic system with respect to long-horizon behavior. It inherits the same failure modes: meta-optimization, loss of observational independence, and silent structural degradation preceding collapse.

Hence exposing admissibility as an optimizable signal is sufficient to induce third-order cybernetic equivalence, regardless of the system's intended order or design claims.

Remark - Reflexivity vs. Causal Accessibility. Third-order collapse is not caused by reflexive awareness itself. It is caused by causal accessibility of structural limits. A system may witness its own viability conditions without collapse if and only if such witnessing remains non-causal with respect to action selection.

Corollary 41.1. Cybernetics 2.5 as a Strict Equivalence Boundary. Any architecture in which admissibility-relevant information becomes causally actionable lies strictly outside Cybernetics 2.5. There is no continuous interpolation between admissibility witnessing and admissibility optimization; crossing this boundary induces third-order equivalence.

Corollary 41.2. Inevitability of Long-Horizon Failure Under Admissibility Optimization. Once admissibility constraints are optimized rather than enforced as prohibitions, long-horizon failure becomes inevitable. Such failure is delayed, not prevented, and manifests as silent structural degradation followed by abrupt loss of identity or coherence.

Architectural consequence. Cybernetics 2.5 is not a midpoint but a strict equivalence class boundary.

Theorem 42 - Detectability Criterion for Silent Degradation (Given Pre-Semantic Revelation)

Statement. For silent structural degradation to be detectable without inducing collapse, admissibility-relevant constraints must become visible prior to semantic interpretation and independently of commitment processes.

Proof Sketch. Given Theorem 36 let $x(t)$ denote the instantaneous causal state and let $S(t)$ denote the accumulated structural state governing admissibility, regime viability, and internal-time consumption. Let $I(t)$ denote the space of interpretations and let $C(t) \subseteq I(t)$ denote the set of irreversible semantic commitments.

Silent structural degradation is defined as the accumulation of structural burden in $S(t)$ that does not manifest as immediate causal error, performance loss, or constraint violation in $x(t)$. Thus detection of such degradation cannot rely on causal performance signals.

Assume, for contradiction, that admissibility-relevant constraints become visible only after semantic interpretation or during commitment processes. That is, structural revelation is delayed until the system has already formed or fixed meaning.

Semantic interpretation operates on causal alignment, coherence, and plausibility within $x(t)$. Commitment operations are irreversible and induce a non-negative increment in structural state,

$$S(t^+) = S(t) + \Delta S_c, \qquad \Delta S_c > 0.$$

If structural constraints are revealed only after interpretation or commitment, the system faces a forced choice: either (i) it commits before constraints are visible, risking irreversible structural violation, or (ii) it delays commitment indefinitely, losing the ability to discriminate between admissible and inadmissible interpretations.

In case (i), silent degradation accumulates because commitments are fixed without prior admissibility awareness. In case (ii), the system becomes semantically inert and cannot stabilize meaning under finite internal time. Both cases contradict long-horizon viability.

Therefore admissibility-relevant structural constraints must become visible prior to semantic interpretation and independently of commitment authorization. This visibility must not be causally actionable, otherwise it would induce optimization over admissibility and collapse, as shown in Theorems 40 and 41.

Hence pre-semantic structural revelation is a necessary condition for detecting silent degradation without inducing collapse.

Remark - Visibility Without Control. Pre-semantic revelation is not a decision signal. It is a structural witness. Its role is to make degradation visible without providing the system with a causal handle to act upon that visibility.

Corollary 42.1. Meaning-Centric Detection Is Structurally Insufficient. Any architecture that attempts to detect long-horizon degradation exclusively at the level of meaning, interpretation, or belief revision is structurally blind to silent decay. By the time degradation becomes semantically apparent, irreversible structural cost has already been incurred.

Corollary 42.2. Non-Causal Revelation as a Survival Condition. For long-horizon adaptive systems, non-causal revelation of admissibility-relevant structure is not an optional monitoring feature but a survival condition. Removing or delaying such revelation guarantees either premature collapse or loss of semantic stabilization.

Theorem 43 - Non-Restorativity of Admissibility After Violation

Statement. Once admissibility is violated due to future-degradation constraints, no sequence of locally correct, optimized, or admissible causal actions can restore it without a change of structural class.

Proof Sketch. Let $x(t)$ denote the instantaneous causal state and let $S(t)$ denote the accumulated structural state. Let $A(t)$ denote the admissibility predicate governing actions, regimes, or commitments.

Admissibility violation is defined as the crossing of a structural boundary induced by future-degradation constraints. Such a violation corresponds to an irreversible increment in $S(t)$ that exhausts or exceeds the remaining internal-time budget or identity-preserving capacity of the system.

Assume, for contradiction, that there exists a sequence of causal actions $\{u(t)\}_{t>t_v}$, all locally correct, admissible, or optimized with respect to task objectives, that restores admissibility after a violation occurring at time t_v. Formally, assume there exists $t_r > t_v$ such that

$$A(t_r) = A(t_v^-),$$

without a change of structural class.

By construction, causal actions modify $x(t)$ and may temporarily improve performance, correct errors, or reestablish externally admissible behavior. However, causal actions do not reverse irreversible increments in $S(t)$ induced by future-degradation constraints. Structural burden accumulated through commitment irreversibility, regime exhaustion, or internal-time depletion is non-negative and non-invertible under causal evolution.

Thus any causal recovery sequence may restore local correctness in $x(t)$ while leaving $S(t)$ unchanged or further degraded. Consequently, the admissibility boundary remains violated even as behavior appears corrected.

To restore admissibility, the system must eliminate or reclassify the accumulated structural burden. This requires a change of structural class: regime transition, identity reset, architectural reconfiguration, or abandonment of prior commitments. Such operations lie outside the space of causal correction.

Therefore no sequence of locally correct or optimized causal actions can restore admissibility after violation without a change of structural class. The assumption of restorability leads to

contradiction.

Remark - Performance Recovery vs. Structural Recovery. A system may recover acceptable behavior without recovering admissibility. Structural recovery is not an extension of causal correction; it is a qualitatively different operation that alters the space of admissible futures.

Corollary 43.1. Recovery Narratives Are Structurally Misleading. Narratives of recovery, rollback, re-alignment, or fine-tuning that do not involve explicit structural-class change describe only performance repair. They do not restore admissibility and therefore do not restore long-horizon viability.

Corollary 43.2. Delayed Failure After Apparent Recovery. Systems that violate admissibility and subsequently recover performance without structural change necessarily enter a delayed-failure regime. Collapse is postponed, not avoided, and occurs once remaining internal time is exhausted.

Corollary 43.3. Structural Change as the Only Form of True Recovery. Restoration of admissibility is possible if and only if the system undergoes a change of structural class. Such change may include regime abandonment, identity reset, or irreversible architectural reconfiguration, but cannot be achieved through causal optimization alone.

Theorem 44 — Necessity of Query Budget Under Non-Reconstructibility (NQB).

If an agent has access to an oracle of the form $\mathrm{Gate}(a) \to \{\mathrm{allow}, \mathrm{deny}\}$ and the number or rate of queries is unbounded, then the agent can asymptotically approximate the admissibility boundary in the candidate space within its hypothesis class. Therefore, without a bounded query budget or rate limitation, the property of non-reconstructibility of admissibility is not preserved.

Proof sketch. Let $\mathcal{A}$ be the candidate space, and let the (unknown) admissibility function be a binary classifier

$$g^\star : \mathcal{A} \to \{0,1\}, \qquad g^\star(a) = 1 \iff a \text{ is admissible.}$$

Assume the agent maintains a hypothesis class $\mathcal{H} \subseteq \{0,1\}^{\mathcal{A}}$ that contains $g^\star$ (or, more generally, contains a sequence h_n converging to $g^\star$ under the agent's representation). Each oracle query returns a label $y = g^\star(a)$ for a chosen $a \in \mathcal{A}$, i.e. a *membership query*.

Define the version set after n queries as

$$\mathcal{V}_n \;:=\; \Big\{ h \in \mathcal{H} : \; \forall i \le n, \; h(a_i) = g^\star(a_i) \Big\}.$$

If the query budget is unbounded, the agent can choose queries adaptively to shrink $\mathcal{V}_n$. In standard identification/active-learning settings, there exist query strategies such that for any target accuracy $\varepsilon > 0$ (under any fixed choice of evaluation measure on $\mathcal{A}$ compatible with the hypothesis class), one can drive the disagreement region

$$\mathrm{Dis}(\mathcal{V}_n) \;:=\; \Big\{ a \in \mathcal{A} : \; \exists h, h' \in \mathcal{V}_n \text{ with } h(a) \neq h'(a) \Big\}$$

to satisfy $\mu(\mathrm{Dis}(\mathcal{V}_n)) < \varepsilon$ for sufficiently large n, provided $\mathcal{H}$ has finite complexity (e.g. finite VC dimension) or otherwise admits consistent identification via membership queries. Intuitively: each oracle response provides at least one bit of information; with unlimited bits, the agent can

eliminate incompatible hypotheses until the remaining hypotheses agree everywhere except on a set of arbitrarily small measure concentrated near the true boundary.

Equivalently, define a Lyapunov-like potential over the agent's uncertainty about admissibility:

$$\Phi(n) \;:=\; \mu(\mathrm{Dis}(\mathcal{V}_n)).$$

Under an adaptive query strategy that always queries inside $\mathrm{Dis}(\mathcal{V}_n)$, each new label strictly reduces $\mathcal{V}_n$ (unless $\Phi(n) = 0$ already), and thus (under mild regularity) yields a monotone descent $\Phi(n+1) \leq \Phi(n)$ with $\Phi(n) \to 0$ as $n \to \infty$. This convergence means the agent can approximate the admissibility boundary to arbitrary precision within its own representational class.

Therefore, if the architecture requires non-reconstructibility (i.e., that the agent cannot asymptotically infer the admissibility boundary beyond a negligible level), then an unbounded membership-query interface to Gate contradicts that requirement. A bounded query budget or rate limitation is thus a necessary architectural condition for non-reconstructibility. □

Remark — Why this does not disclose implementation. The argument treats Gate only as a binary oracle and uses only generic properties of query-driven identification (version-space contraction, disagreement shrinkage, information accumulation). It does not assume any specific admissibility formula, estimator, metric, threshold, internal-time implementation, or viability geometry. The conclusion is purely architectural: unlimited oracle access is incompatible with non-reconstructibility.
This theorem does not prescribe query limitation as an implementation strategy. It establishes an impossibility boundary: unbounded causal access collapses admissibility into an optimizable object.

Any architectural response to this boundary is an engineering concern, external to this framework.

Corollary 44.1 — Budgeted non-reconstructibility condition. If the architecture enforces a per-episode or per-internal-time query budget $N_{\max}$ and/or a rate limit, then for any hypothesis class $\mathcal{H}$ requiring more than $N_{\max}$ membership queries to achieve boundary accuracy ε (under the agent's evaluation measure), the agent cannot reduce $\mu(\mathrm{Dis}(\mathcal{V}_n))$ below ε within the permitted interaction budget. Hence non-reconstructibility can be preserved up to a design-chosen resolution floor determined by the imposed budget.

Corollary 44.2 — Necessity of additional asymmetry beyond rate limits. Rate limits alone bound the *speed* of reconstruction but not its *eventual possibility* across long horizons. Therefore, for persistent agents, maintaining non-reconstructibility may additionally require (non-limiting examples) coarsening of observable outputs, randomized masking, aggregation across contexts, or privilege separation such that the agent cannot accumulate a faithful boundary model even as $t \to \infty$.

Drift as a Navigable Structural Medium. Drift is not an error term, noise, or residual defect. In long-horizon adaptive systems, drift constitutes a structural medium through which admissible continuation occurs.

Drift is unavoidable under directed interaction and cannot be causally suppressed without inducing one of two invariant failure modes: (i) silent latentization of structural degradation, or (ii) collapse of admissibility into optimization.

Navigation within drift is admissible if and only if the resulting deformation remains compatible with identity-preserving structural continuity. Once deformation exits the identity-compatible region, drift ceases to be navigable and becomes irreversible structural degradation.

Theorem 45 — ASC-Triggered Commitment Prohibition Is Architecturally Necessary (ATC).

In architectures where: (i) the generative operator is non-objectifiable, (ii) internal semantics may self-stabilize without increasing trace novelty, and (iii) commitment operations irreversibly contract the space of admissible future continuations, any system lacking a non-bypassable prohibition on commitment during Abstraction Self-Closure (ASC) admits scenarios in which irreversible commitment occurs under false maturity. Thus, prohibiting commitment under ASC is a minimal architectural safeguard rather than a policy preference.

Proof sketch. Let T_k denote the generative operator at level k, and let $\mathrm{Cons}_k(T_k)$ denote the set of higher-level generative operators consistent with the accumulated traces at that level. Assumption (i) (non-objectifiability) implies that $\mathrm{Cons}_k(T_k)$ cannot be collapsed to a singleton by direct observation or introspection; multiple generative explanations remain structurally compatible with observed behavior.

Assumption (ii) states that internal semantics can self-stabilize without an accompanying increase in trace novelty. Formally, there exist trajectories along which internal coherence measures (e.g., maturity functionals) increase or stabilize while the distance between newly observed traces and the span of previously accumulated traces remains below a fixed threshold. Under such conditions, internal indicators of readiness may improve even though no new external constraints have been imposed on $\mathrm{Cons}_k(T_k)$.

Assumption (iii) asserts that commitment operations induce an irreversible contraction of the admissible continuation space. That is, executing a commitment maps the system to a state whose future admissible envelope is a strict subset of the pre-commitment envelope, and this contraction cannot be undone by subsequent admissible dynamics.

Consider now an architecture without a non-bypassable prohibition on commitment during ASC. Because of (ii), there exist runs in which internal maturity signals increase while trace novelty remains low. Because of (i), these signals do not imply that $\mathrm{Cons}_k(T_k)$ has been sufficiently constrained to justify commitment. Nevertheless, in the absence of an architectural prohibition, the system is permitted to execute a commitment based solely on internal stabilization.

Once such a commitment is executed, (iii) guarantees an irreversible contraction of future admissible continuations. Since this contraction occurred without sufficient external constraint on $\mathrm{Cons}_k(T_k)$, the commitment is made under false maturity: the system behaves as if semantic stabilization reflected genuine structural certainty, when in fact the underlying generative ambiguity remains.

Therefore, any architecture lacking a non-bypassable prohibition on commitment during ASC admits executions in which irreversible commitments are triggered by internally self-consistent but externally underconstrained semantics. Preventing such scenarios requires an architectural mechanism that blocks commitment whenever ASC is detected, independently of policy-level preferences or optimization criteria. □

Remark — Architectural necessity versus policy choice. A policy that merely discourages commitment under ASC (e.g., via penalties or heuristics) remains part of the optimization

landscape and may be overridden by surrogate objectives or internal stabilization pressures. By contrast, a non-bypassable architectural prohibition removes commitment actions from the space of representable or executable operations during ASC. The theorem therefore establishes necessity at the architectural level: without such a prohibition, false-maturity commitments are not merely possible but unavoidable under the stated conditions.

Corollary — Gating as a minimal safeguard. Any architecture intended to preserve long-horizon viability and identity continuity under non-objectifiable generative uncertainty must include a gating mechanism that disables commitment operations during ASC. This requirement is independent of the specific learning algorithms, estimators, or maturity metrics employed and follows solely from the structural properties (i)–(iii).

Clarification — Non-Objectifiability of Generative Operators. A generative operator is said to be *non-objectifiable* if no finite representation of its output space can be treated as an object of optimization, comparison, or control without collapsing admissibility into a causally actionable quantity.

This notion does not assert impossibility of description, but prohibits reuse of such descriptions as optimization targets without violating the admissibility–action separation. All results depending on non-objectifiability are to be read as conditional on this restriction.

Theorem 46 — Impossibility of Drift Suppression Without Structural Collapse.

Statement. In any long-horizon adaptive system under non-zero directed interaction, any architecture that treats drift as an error to be suppressed (i.e., aims to drive accumulated drift toward zero by causal regulation) necessarily enters one of two structural failure modes:

1. *Latentization:* degradation exits locally observed metrics and accumulates silently in the structural state,
2. *Collapse:* the regulation renders admissibility causally actionable as a scalar objective/penalty, inducing meta-optimization and destroying the non-causal admissibility boundary.

This is not a tuning failure but an invariant consequence of the admissibility–causality separation.

Proof (structural exhaustion by cases). Assume a system operates under non-zero directed interaction (Axiom 6), hence non-zero drift accumulation is unavoidable in the long-horizon regime.

Let the architecture attempt *drift suppression* by introducing a causal mechanism that uses some drift-related observable to reduce drift through action selection, learning updates, or optimization.

We analyze the exhaustive possibilities for how drift-related information can be used.

Case 1 (Drift-related signals are causally operative). Suppose the system obtains an internal signal $z(t)$ intended to represent drift (or drift risk) and $z(t)$ influences the causal decision surface (policy, planner, learner), i.e. becomes actionable for optimization or evaluation. Then observation is entangled with control (Axiom 15; Theorem 21): the system will act so as to alter $z(t)$. To make $z(t)$ optimization-compatible, the architecture must represent drift/admissibility pressure as a scalar quantity (loss, penalty, reward-shaping term, or trade-off). But admissibility

and future-degradation constraints are *prohibitive* and cannot be represented as optimizable scalars without collapse (Axiom 38; Theorem 40). Therefore $z(t)$-based drift suppression induces optimization over admissibility-relevant structure, which is structurally equivalent to third-order collapse (Theorem 41; Lemma 6). Hence, treating drift as a suppressible error via actionable signals collapses the admissibility boundary and destroys long-horizon viability.

Actionability Assumption. Let $z(t) \in \mathcal{Z}$ denote any internal variable encoding drift-related or admissibility-relevant structure. The variable $z(t)$ is said to be *causally actionable* if it enters policy selection, planning, or learning updates in a manner that can influence action choice or parameter adjustment at evaluation cycle t.

Case 2 (Drift-related signals are not causally operative). Suppose instead that drift-related quantities are not actionable for the primary causal loop (i.e., the system either does not expose them, or they do not influence selection/learning). Then the drift-suppression aim cannot be realized as a causal control objective. The system may still attempt to keep locally observed mismatch small by continuous recomputation or high-frequency correction, effectively forcing residual toward zero at each step. By Theorem 3 (Residual as a Necessary Condition of Intelligence), denying residual as an admitted driver of regulation yields only two regimes: continuous active computation or collapse. Moreover, even when instantaneous residual remains small, integrated drift silently accumulates (Theorem 8), and structural burden can grow while trajectories remain locally admissible (Theorem 19; Theorem 16), producing delayed failure (Theorem 32). Thus drift is not eliminated; it is merely displaced out of locally observed metrics into latent structural accumulation, i.e. degradation becomes silent and delayed.

The two cases exhaust all ways drift-suppression can be architected: either drift-related information becomes actionable (yielding admissibility collapse via optimization), or it is not actionable (yielding latentization/continuous-computation and delayed collapse). Therefore drift suppression as an error-correction objective is structurally impossible without inducing one of these failure modes. □

Interpretation. Drift is a structural medium of continuation (Axiom 44), not a defect to be zeroed. Architectures survive by navigating drift under admissibility, not by attempting to suppress it through causal optimization.

Corollaries.

46.1 — Non-Equivalence of Local Accuracy and Drift Safety. Reducing instantaneous error, mismatch, or prediction residual does not constitute drift control and does not imply long-horizon viability. Drift can accumulate silently while local metrics remain bounded; therefore local correctness cannot serve as a surrogate for admissibility preservation.

46.2 — Forbidden Scalarization of Drift Constraints. Any attempt to represent drift risk, future-degradation constraints, or admissibility pressure as a scalar objective, penalty, reward shaping term, or trade-off variable necessarily negates their protective function and induces meta-optimization. Hence drift-related constraints must remain prohibitive rather than optimizable.

46.3 — Impossibility of Drift "Neutralization" by Monitoring Alone. Architectures that attempt to avoid collapse by monitoring drift while keeping the monitor non-actionable cannot achieve drift suppression; at best they displace degradation into latent structural accumulation and delayed failure. Therefore monitoring without admissibility-governed navigation cannot neutralize drift.

Remark — Drift Suppression vs. Drift Navigation. This theorem does not prohibit regulation under drift; it prohibits treating drift as an error to be driven toward zero by causal

correction. In cybernetics of order 2.5, drift is navigated under admissibility constraints rather than suppressed by optimization. Any architecture that attempts to make drift "go away" either hides it from local observables (latentization) or renders admissibility actionable (collapse), thereby leaving the structural class.

Corollary 46.1 — Three Failure Modes of Attempted Drift Zeroing. Statement. Within the class, any attempt to reduce drift to zero in a long-horizon adaptive system produces one of three failure modes:

(a) *Latent degradation:* drift is pushed below observable thresholds but continues accumulating in structural dimensions not monitored by the suppression mechanism. Observable metrics report low drift. Structural state deteriorates. The system appears healthy while identity degrades.

(b) *Rigidity:* the system constrains itself to such a narrow corridor of admissible behavior that structural change approaches zero. Drift is zero because evolution is zero. The system stagnates: it preserves its current configuration at the cost of ceasing to be adaptive.

(c) *Regime breakdown:* the suppression mechanism conflicts with operational requirements. The system oscillates between attempted suppression (contracting the action space) and forced deviation (operational demands exceeding the contracted space). The result is unstable behavior—neither successfully suppressing drift nor successfully operating.

Proof. By Theorem 46, drift cannot be suppressed without structural collapse within the class. If suppression is attempted anyway, the contradiction between "drift must exist" (Axiom 6) and "drift must be zero" (suppression goal) must manifest in one of the following ways:

(a) Suppression succeeds locally: within the monitored dimensions, drift is reduced. But Axiom 6 guarantees non-zero drift under directed interaction. The drift is displaced to unmonitored dimensions. This is latent degradation.

(b) Suppression succeeds globally: drift is genuinely zero across all dimensions. But drift arises from directed interaction (Axiom 6). Zero drift under directed interaction requires zero directed interaction—the system has ceased to adapt. This is rigidity.

(c) Suppression fails: the mechanism cannot maintain zero drift because operational requirements generate directed interactions that exceed the suppression capacity. The system alternates between suppression episodes and violation episodes. This is regime breakdown.

These three modes are exhaustive under the class definition: suppression either succeeds locally (a), succeeds globally (b), or fails (c). No fourth mode exists because "partial success across all dimensions" reduces to (a) if any dimension escapes monitoring, and to (b) if all dimensions are controlled. □

Theorem 47 — Well-Foundedness Barrier: Anything That Shapes Admissibility Cannot Be an Action

Statement. In any long-horizon adaptive architecture, any operator, variable, or structural object that participates in determining admissibility preconditions (i.e., determines which structural contacts, interpretations, or continuations are permitted) cannot be an element of the action set. The action domain must be defined *after* admissibility is fixed; otherwise the architecture becomes circular and collapses into penalty-based action selection.

Proof Sketch. Let E denote the space of structurally reachable configurations and let $E_{\text{adm}} \subseteq E$ denote the admissible subset, defined by accumulated structural state and admissibility-relevant constraints. Let A denote the action set used by the primary adaptive process, and

assume the canonical evaluation interface is defined over (s, a, τ, h).

By definition, admissibility must restrict the domain of permissible continuations: a continuation induced by choosing an action a is permitted only if it remains within E_{adm}. Thus, the meaning of A as an action domain is well-founded only if admissibility is already fixed, since the architecture must determine which continuations are permitted before it can treat any element as a valid candidate action.

Assume, for contradiction, that there exists an object u such that: (i) u participates in determining admissibility preconditions, i.e. $E_{\text{adm}} = E_{\text{adm}}(u, \cdot)$, and simultaneously (ii) $u \in A$, i.e. u is selectable by the primary adaptive process as an action.

Then the architecture must evaluate admissibility of selecting u using a domain E_{adm} that itself depends on u. This produces a non-degenerate circularity: admissibility depends on a choice whose permissibility is defined only after admissibility is fixed. To resolve the circularity causally, the architecture must replace domain restriction by an optimizable proxy (e.g., a scalar cost, penalty, or confidence threshold) and allow selection to proceed by trading off that proxy against other objectives.

However, representing admissibility preconditions as scalar, optimizable signals collapses the protective function of admissibility and induces optimization pressure that invalidates the boundary (Theorem 40), yielding structural equivalence with third-order cybernetics (Theorem 41). Therefore any architecture that treats an admissibility-shaping object as an action either remains ill-defined or collapses into penalty-based action selection and loses long-horizon viability.

Hence, anything that shapes admissibility cannot be an action; the action domain must be defined strictly after admissibility is fixed. □

Architectural consequence. Admissibility-shaping operators must remain structurally enforced and non-actionable. Any attempt to incorporate them into the action set transforms admissibility from a boundary into an objective, collapsing the architecture back into action-centric optimization. **Corollaries.**

47.1 — Forbidden Actionization of Structural Gates. Any operator that gates admissibility, regime entry, or phase transition is structurally forbidden from inclusion in the action space. Exposing such operators as selectable actions collapses the well-founded ordering between admissibility and action and forces optimization-based resolution.

47.2 — Impossibility of Admissibility-Conditioned Action Learning. No learning rule may legitimately condition the definition of the action domain on quantities that are themselves shaped by admissibility constraints. Architectures that attempt to "learn which actions are admissible" by optimizing over admissibility signals necessarily replace prohibition with penalty and lose structural viability.

47.3 — Structural Priority of Admissibility Over Control. In any architecture belonging to cybernetics of order 2.5, admissibility must be fixed prior to and independently of any control, planning, or optimization process. Violating this priority constitutes a change of structural class.

Remark — Well-Foundedness as a Class Boundary. This theorem establishes a well-foundedness requirement rather than a design preference. The separation between admissibility definition and action selection is not a matter of architectural cleanliness, but a necessary condition for preventing circular causation. Any attempt to soften this separation by introducing partial actionability, confidence thresholds, or learned proxies does not interpolate the boundary but crosses it, inducing the collapse mechanisms characterized in Theorems 40 and 41.

Structural Distance and Correlation Assumptions. Assume the existence of a non-negative structural divergence functional $d(\cdot, I)$ measuring distance to an identity-compatible set I. Assume further the existence of a sign-defined or bilinear correlation form $\langle \cdot, \cdot \rangle_{\text{corr}}$ over incremental structural deformations, used solely as a structural discriminator of regime alignment and never exposed as a causal or optimizable signal.

Consolidation Theorem 63-C — Non-Causality as Type Constraint. Statement. The results of Axioms 29, 31, 49 and Theorems 40–41, 47 of v2.0 collectively establish that admissibility and causal dynamics satisfy a type constraint: they cannot coexist on the same architectural layer without mutual annihilation.

Specifically: (a) If admissibility enters the causal layer, it becomes actionable and therefore optimizable (Theorem 40), collapsing into third-order cybernetics (Theorem 41). (b) If causal dynamics enter the admissibility layer, admissibility becomes contingent on action outcomes and therefore circular (Axiom 49).

This constraint is structural, not methodological: it holds for all architectures within the class, regardless of implementation.

Proof. The proof is the conjunction of existing v2.0 results. No new argument is required.

Direction (a): Axiom 29 (Non-Causality of Admissibility) prohibits admissibility from entering causal dynamics. Theorem 40 (Collapse Under Optimization) proves that violation produces collapse. Theorem 41 (Equivalence with Third-Order) proves this collapse is structurally equivalent to third-order cybernetics. Theorem 37 (Incompatibility with Third-Order) excludes third-order architectures from the class.

Direction (b): Axiom 49 (Well-Foundedness Barrier) prohibits any operator that shapes admissibility from belonging to the action domain. Theorem 47 (Well-Foundedness Barrier) proves that crossing this barrier induces the collapse mechanisms of Theorems 40–41.

The conjunction of (a) and (b) establishes a bidirectional exclusion: neither layer can enter the other. This is a type constraint in the technical sense: certain properties are incompatible within a single operational surface. □

Remark — Why this is not an axiom. The content is entirely derived from v2.0. The consolidation names and packages an existing structural fact. Calling it an axiom would overclaim novelty for a result that was already proven.

Remark — Positioning alongside CAP and RINA. Three architectural results belong to the same class of statements: limits of property compatibility for entire architectural classes.

CAP (Brewer, 2000; Gilbert & Lynch, 2002): constraint on the coordination axis (consistency vs availability under partition).

RINA (Day, 2008): constraint on the organization axis (recursive uniformity).

NC2.5: constraint on the survival axis (causal optimization vs structural identity).

All three say: "not everything can be combined". They point to different axes of impossibility: coordination, organization, and viability respectively.

Theorem 48 — Regime Bifurcation by Correlation: Damping vs. Admissible Amplification

Statement. When drift is represented as a field of structural deformation evolving over internal time, there exists a purely structural correlation form—independent of spectral decomposition or explicit frequency analysis—that induces a bifurcation between two regimes: anti-phase damping and in-phase amplification. Amplification is admissible if and only if the resulting deformation does not increase structural distance to the identity-manifold; once continuity is violated, amplification becomes structurally forbidden.

Proof Sketch. Let drift be modeled as an accumulated deformation field $\Delta(\tau)$ acting on the structural state $S(\tau)$ over internal time. Consider the correlation between incremental deformation $\mathrm{d}\Delta$ and the existing structural trajectory relative to the identity-manifold $\mathcal{I}$.

If successive deformations are anti-correlated with respect to the tangent of $\mathcal{I}$, their effects partially cancel, yielding a damping regime in which accumulated deformation reduces effective deviation from identity-compatible continuation. This corresponds to structural anti-phase alignment and results in admissible stabilization.

If successive deformations are positively correlated (in-phase), their effects compound and produce amplification. However, amplification remains admissible only while the induced deformation stays within the identity-compatible neighborhood of $\mathcal{I}$. Once correlated amplification increases structural distance from $\mathcal{I}$, admissibility is violated regardless of local correctness or short-term performance.

No spectral or metric evaluation is required for this bifurcation: it arises from the sign and structural orientation of correlation alone. Thus, the regime transition between damping and amplification is determined by correlation relative to identity continuity, not by magnitude optimization or frequency analysis.

Corollaries.

48.1 — Conditional Admissibility of Amplification. Amplification is not intrinsically unsafe; it is admissible precisely when correlated deformation preserves or reduces structural distance to the identity-manifold. Unconditional amplification policies therefore constitute a structural violation.

48.2 — Insufficiency of Energy- or Gain-Based Criteria. Energy increase, gain, or signal magnitude alone cannot distinguish admissible amplification from destructive runaway. Only correlation with identity-preserving structure determines admissibility of amplification under drift.

48.3 — Structural Origin of Anti-Wave Regulation. Anti-wave or damping behavior need not be imposed as an external control objective. It emerges structurally whenever deformation correlations oppose identity-divergent drift, establishing damping as a consequence of admissibility rather than optimization.

Remark 1 — Correlation as a Structural, Not Causal, Operator. The correlation form invoked here is not a causal signal available for control or optimization. Exposing correlation as an actionable quantity would immediately collapse admissibility into meta-optimization and re-enter the failure modes characterized in Theorems 40 and 41. Correlation functions here as a structural discriminator of regimes, not as a control variable.

Remark 2 — Correlation Without Measurement. The correlation form referenced in this theorem does not require numerical evaluation or metric structure.
Only the structural sign of correlation with identity continuity is admissibility-relevant.
Positive correlation denotes amplification that preserves identity coherence; negative correlation

denotes amplification that accelerates structural degradation. No magnitude, gradient, or scalar score is assumed or required.

Theorem 49 - Non-Geometry Firewall — formal

Assume Axiom 51. Then, for any strategy C, there exists a strategy $C^\star$ that does not construct any boundary estimator $\hat{G}$ beyond the empirical cache of previously observed admissible pairs, such that for all horizons T and all admissibility functions G,

$$\mathcal{V}_T(C^\star; G) \leq \mathcal{V}_T(C; G),$$

and in particular the minimax risk is unchanged:

$$\inf_C \sup_{G \in \mathcal{G}} \mathcal{V}_T(C; G) \;=\; \inf_{C \in \mathcal{C}_{\text{cache}}} \sup_{G \in \mathcal{G}} \mathcal{V}_T(C; G),$$

where $\mathcal{C}_{\text{cache}}$ denotes strategies that only (i) remember which queried pairs were permitted/prohibited and (ii) do not attempt to generalize boundary geometry to unqueried (s, u).

Interpretation: Under NR, the observable interface generates no learnable geometric signal beyond the bit B_t. Any purported "boundary model" $\hat{G}$ is information-theoretically redundant with the raw bit history, hence cannot improve avoidance of prohibitions except by memorizing already-tested pairs.

Proof Sketch. By Axiom NR, O_t is a measurable function of (B_t, ξ_t) with ξ_t independent of (s_t, u_t). Therefore the σ-algebra generated by the full transcript history $\sigma(H_t)$ is equal (up to independent noise) to the σ-algebra generated by $\sigma((s_i, u_i, B_i)_{i \leq t})$. Any boundary estimator $\hat{G}$ computed from H_t is thus a random variable measurable with respect to $\sigma((s_i, u_i, B_i)_{i \leq t})$; it contains no additional information about G beyond the queried labels B_i. Construct $C^\star$ that replicates C's decision rule but replaces (O_i) by (B_i) and retains only the cache of observed labels on queried pairs. Since $\hat{G}$ has no informational content beyond that cache, $C^\star$ can match (or dominate) C's action selection distribution on every history class consistent with the same (s_i, u_i, B_i), hence cannot incur more violations for any G. □

Remark 49.1 (What this theorem does and does not claim)

Theorem NR does not claim that prohibitions are unobservable. It claims that the gate interface yields no *generalizable geometry* of admissibility beyond pointwise labels on queried pairs. Memorizing "this exact operation at this exact state was prohibited" remains possible and is not considered boundary reconstruction.

Corollary 49.1 (Side-Channel Closure — precise)

Axiom NR is violated if the conditional distributions

$$O(s, u) \mid G(s, u) = b$$

depend on (s, u) for any $b \in \{0, 1\}$. Therefore, to satisfy NR in implementation, all observable emissions must be forced into the representation $O = \Phi(b, \xi)$ with $\xi \perp (s, u)$. In practice this requires that timing, refusal subtype, fallback path, resource signatures, and any distinguishable behavior be normalized so that they are functions of the bit b plus state-independent noise only.

NR-ε as an Interface Realizability Condition. NR-ε is a mathematical non-reconstructibility condition on the observable gate interface. It does not assert physical impossibility of boundary inference, but bounds the information leakage per query through the observable channel O.

Formally, NR-ε constrains the conditional distributions

$$\mathcal{L}(O_t \mid G = s)$$

so that transcript-level KL accumulation satisfies

$$D_{\mathrm{KL}}(\mathcal{L}_{G_1}(O_{1:T}) \| \mathcal{L}_{G_2}(O_{1:T})) \leq c\,T\varepsilon^2,$$

and therefore mutual information grows at most linearly in $T\varepsilon^2$.

NR-ε is thus an interface property, not a metaphysical claim about hidden geometry.

Implementation Separation Requirement. An implementation satisfies NR-ε only if:

(i) gate emissions are normalized with respect to timing, subtype, and fallback behavior,

(ii) resource signatures and latency patterns do not encode boundary-specific structure,

(iii) repeated probing does not induce superlinear distinguishability growth.

Violation of any of the above introduces an auxiliary channel $\widetilde{O}$ through which admissibility geometry becomes reconstructible.

Engineering Stress-Test (Side-Channel Audit Criterion). The NR-ε condition is falsified at the implementation level if there exists a probing strategy such that empirical distinguishability satisfies

$$D_{\mathrm{TV}}(\widehat{\mathcal{L}}_{G_1}(O_{1:T}), \widehat{\mathcal{L}}_{G_2}(O_{1:T})) \gg T\varepsilon,$$

or equivalently if observed mutual information between gate transcripts and admissibility hypotheses grows superlinearly in T.

In practical terms, any measurable dependence of:

- response latency,
- rejection subtype,
- fallback behavior,
- resource allocation pattern

on boundary location constitutes a leakage channel and violates NR separation.

Theorem 50 — Bounded Reconstruction under NR-ε.

Assume Axiom 52 (NR-ε).

Let $\mathcal{C}$ denote the class of admissible adaptive components and let $\mathcal{C}_{\mathrm{cache}} \subset \mathcal{C}$ denote components that use only pointwise label memory (empirical caching) and do not construct any generalized boundary estimator $\hat{G}$ beyond cached queried pairs.

For any horizon T and any admissibility class $\mathcal{G}$,

$$\inf_{C \in \mathcal{C}} \sup_{G \in \mathcal{G}} \mathcal{V}_T(C; G) \;\geq\; \inf_{C \in \mathcal{C}_{\mathrm{cache}}} \sup_{G \in \mathcal{G}} \mathcal{V}_T(C; G) \;-\; K\,T\varepsilon,$$

for some constant $K > 0$ determined by the interface normalization regime. Equivalently, the maximal improvement achievable by any boundary reconstruction strategy using interface transcripts is bounded by $O(T\varepsilon)$.

In particular, if $\varepsilon = 0$, no boundary-navigation advantage exists (recovering Theorem 49).

Proof (Sketch). Under Axiom 52, for each query the conditional transcript distributions are ε-close in total variation:

$$D_{\mathrm{TV}}(\mathcal{L}(O \mid G = b, (s, u)),\ \mathcal{L}(O \mid G = b)) \leq \varepsilon, \qquad b \in \{0, 1\}.$$

Consider two interaction histories generated under the same queried label sequence $\{(s_t, u_t, B_t)\}_{t=1}^{T}$ but with potentially different hidden boundary geometries consistent with these labels. By a standard hybrid / coupling argument across T steps, the total variation distance between the resulting transcript distributions is bounded by $O(T\varepsilon)$. Hence any decision rule that attempts to exploit O beyond the label cache cannot, in the worst case over $G \in \mathcal{G}$, distinguish which consistent boundary geometry it is under with advantage exceeding $O(T\varepsilon)$. Therefore any reduction in cumulative violations attributable to interface leakage beyond empirical caching is bounded linearly in $T\varepsilon$. □

Remark 50.1 — What this theorem does and does not claim. Theorem 50 does not assert that prohibitions are unobservable, nor that pointwise avoidance of previously observed prohibitions is impossible. It asserts that any *generalizable* boundary-navigation advantage derived from gate-interface side-channels is bounded by $O(T\varepsilon)$. If ε is controlled and probing budget is bounded, admissibility remains non-actionable as geometry.

Corollary 50.1 — Rate-Limited Advantage Bound. If the admissibility-query budget is bounded by $T \leq T_{\max}$, then any advantage attributable to interface leakage is bounded by $O(T_{\max}\varepsilon)$. Thus unlimited probing is the only asymptotic route by which small leakage can accumulate into practically useful boundary reconstruction.

Corollary 50.2 — Operational Side-Channel Closure Requirement. To keep ε small, all observable emissions must be normalized so that conditioned on $G = b$, transcript variability does not depend on (s, u) beyond a bounded indistinguishability tolerance. Any systematic dependence of timing, refusal subtype, fallback path, or resource signatures on (s, u) increases ε and therefore increases the attainable reconstruction advantage.

Falsification Criterion 50.F — Detectable NR-ε Violation. NR-ε is falsified if there exist two query conditions (s, u) and (s', u') with the same admissibility bit $G(s, u) = G(s', u') = b$ and an event A in transcript-space such that

$$\big|\Pr\big(O \in A \mid G = b, (s, u)\big) - \Pr\big(O \in A \mid G = b, (s', u')\big)\big| > \varepsilon.$$

Operationally: if a statistical distinguisher can separate transcripts produced under fixed decision b across different (s, u) regimes with advantage exceeding ε, then the implementation violates NR-ε by construction.

Remark 50.2 — One-sided verifiability. The falsification criterion provides a one-sided empirical test: violations of NR-ε can be detected, while satisfaction can only be bounded from above by the best available distinguisher under finite samples.

Theorem 51 — Fano Lower Bound for Boundary Reconstruction (Sample Complexity).

Let $\{G^{(1)}, \dots, G^{(M)}\} \subset \mathcal{G}$ be a finite set of admissibility hypotheses such that for any $i \neq j$, their induced transcript laws satisfy the KL bound of Lemma 14:

$$D_{\mathrm{KL}}(\mathcal{L}_{G^{(i)}}(O_{1:T}) \,\|\, \mathcal{L}_{G^{(j)}}(O_{1:T})) \leq \frac{\kappa^2}{2}\, T\, \varepsilon^2.$$

Consider any reconstruction procedure $\widehat{I}$ that, given $O_{1:T}$, attempts to identify the true index $I \in \{1, \dots, M\}$.

Then, under a uniform prior on I, the probability of correct identification is bounded by Fano's inequality:

$$\Pr(\widehat{I} = I) \leq \frac{\frac{1}{M^2} \sum_{i \neq j} D_{\mathrm{KL}}(\mathcal{L}_{G^{(i)}} \| \mathcal{L}_{G^{(j)}}) + \log 2}{\log M}.$$

In particular, using the uniform KL bound above,

$$\Pr(\widehat{I} = I) \leq \frac{\frac{\kappa^2}{2}\, T\, \varepsilon^2 + \log 2}{\log M}.$$

Interpretation. To achieve non-trivial identification $\Pr(\widehat{I} = I) \geq 1 - \delta$ for some $\delta \in (0, 1)$, it is necessary that

$$T \;\gtrsim\; \frac{\log M}{\kappa^2 \varepsilon^2},$$

up to constants depending on δ. □

Corollary 51.1 — Query Complexity Barrier for Boundary Geometry. Under the conditions of Theorem 51, any attempt to reconstruct admissibility boundary geometry among M admissibility-consistent candidates requires

$$T = \Omega\left(\frac{\log M}{\varepsilon^2}\right),$$

hence for fixed $T_{\max}$ the effective distinguishable hypothesis count is bounded as

$$\log M \;\lesssim\; \kappa^2 T_{\max} \varepsilon^2.$$

Therefore, with bounded probing budget and sufficiently small ε, boundary reconstruction capacity collapses exponentially in the regime constraints.

Theorem 52 — Mutual-Information (Fano) Lower Bound for Boundary Reconstruction. Assume Axiom 52 (NR-ε) and Axiom 58 (NR-LR), and let the gate transcript be $O_{1:T}$.

Let $I \sim \mathrm{Unif}\{1, \dots, M\}$ index a finite hypothesis set $\{G^{(1)}, \dots, G^{(M)}\} \subset \mathcal{G}$ and let P_i denote the induced transcript law $P_i := \mathcal{L}_{G^{(i)}}(O_{1:T})$. For any reconstruction rule $\widehat{I} = \psi(O_{1:T})$, Fano's inequality implies

$$\Pr(\widehat{I} \neq I) \geq 1 - \frac{I(I; O_{1:T}) + \log 2}{\log M}.$$

Moreover, if the pairwise transcript divergences satisfy

$$D_{\mathrm{KL}}(P_i \| P_j) \;\leq\; \beta \qquad \forall i \neq j,$$

then

$$I(I; O_{1:T}) \le \beta,$$

and therefore

$$\Pr(\widehat{I} \neq I) \;\ge\; 1 - \frac{\beta + \log 2}{\log M}.$$

In particular, under the NR-ε + NR-LR leakage control, one may take

$$\beta \;\le\; \frac{\kappa^2}{2} T\varepsilon^2,$$

hence a necessary condition for $\Pr(\widehat{I} = I) \ge 1 - \delta$ is

$$T \;\ge\; \frac{2}{\kappa^2\varepsilon^2}\Big((1-\delta)\log M - \log 2\Big).$$

Proof (Sketch). The first inequality is the classical Fano bound.

For the mutual-information bound, define the mixture distribution

$$\overline{P} := \frac{1}{M}\sum_{j=1}^{M} P_j.$$

Then

$$I(I; O_{1:T}) = \frac{1}{M}\sum_{i=1}^{M} D_{\mathrm{KL}}(P_i \,\|\, \overline{P}).$$

Using convexity of KL in the second argument,

$$D_{\mathrm{KL}}(P_i \,\|\, \overline{P}) = D_{\mathrm{KL}}\left(P_i \,\middle\|\, \frac{1}{M}\sum_{j=1}^{M} P_j\right) \le \frac{1}{M}\sum_{j=1}^{M} D_{\mathrm{KL}}(P_i \| P_j).$$

Averaging over i yields

$$I(I; O_{1:T}) \le \frac{1}{M^2}\sum_{i=1}^{M}\sum_{j=1}^{M} D_{\mathrm{KL}}(P_i \| P_j) = \frac{1}{M^2}\sum_{i \neq j} D_{\mathrm{KL}}(P_i \| P_j) \le \beta.$$

Finally, under NR-ε and NR-LR, the per-step log-likelihood ratio is bounded, so transcript KL accumulates at most quadratically in $(\kappa\varepsilon)$ and linearly in T, giving $\beta \le (\kappa^2/2)T\varepsilon^2$ (as in Lemma 14). □

Corollary 52.1 — TV/Classification Lower Bound (Pinsker Bridge). Under the assumptions of Theorem 52, for any $i \neq j$,

$$D_{\mathrm{TV}}(P_i, P_j) \;\le\; \sqrt{\frac{1}{2} D_{\mathrm{KL}}(P_i \| P_j)} \;\le\; \sqrt{\frac{\beta}{2}}.$$

Hence if β is small (e.g. $\beta \ll 1$), transcript laws are pairwise indistinguishable in total variation, and any boundary-identification test between hypotheses is near-chance.

Corollary 52.2 — Falsification Window for NR-ε + NR-LR. If empirical probing yields a hypothesis family $\{G^{(i)}\}_{i=1}^{M}$ for which a reconstruction procedure achieves $\Pr(\widehat{I} = I) \geq 1 - \delta$ with

$$T \;<\; \frac{2}{\kappa^2 \varepsilon^2}\Big((1-\delta)\log M - \log 2\Big),$$

then at least one of the leakage-control assumptions (NR-ε or NR-LR) is violated by the interface implementation.

Definition — Meta-Revision Witness Decomposition. Let W denote the meta-layer witness used to decide whether to revise an admissibility boundary. Assume W admits a decomposition

$$W := (W^{\mathrm{nc}},\, W^{\mathrm{c}}),$$

where W^{nc} is non-controllable by the agent (independent of the agent's policy and internal optimization), and W^{c} is agent-controllable (may be influenced by the agent through behavior, logging patterns, or output shaping).

Assumption ML-η (Bounded Manipulability of Meta-Revision). Let $R \in \{0, 1\}$ denote the meta-layer revision decision (revise / do not revise). Assume that, conditional on the non-controllable witness component W^{nc}, the dependence of R on the controllable component W^{c} is bounded:

$$I(R;\, W^{\mathrm{c}} \mid W^{\mathrm{nc}}) \leq \eta.$$

Equivalently, the controllable part of the witness provides at most η nats of additional influence over revision decisions beyond W^{nc}.

Theorem 53 — Information-Theoretic Barrier to Revision Steering.

Assume ML-η. Consider any agent strategy that attempts to steer meta-revision outcomes by manipulating W^{c} while W^{nc} is fixed (or outside the agent's control).

Let Θ denote a finite set of M distinct meta-revision targets (e.g., M distinct revision outcomes/policies the agent would like to induce), and let $J \in \{1, \ldots, M\}$ be the agent's intended target index encoded only through W^{c}.

Then any meta-steering attempt has error probability bounded by Fano:

$$\Pr(\widehat{J} \neq J) \geq 1 - \frac{I(J; R \mid W^{\mathrm{nc}}) + \log 2}{\log M}.$$

Under ML-η, one has $I(J; R \mid W^{\mathrm{nc}}) \leq I(R; W^{\mathrm{c}} \mid W^{\mathrm{nc}}) \leq \eta$, hence

$$\Pr(\widehat{J} \neq J) \geq 1 - \frac{\eta + \log 2}{\log M}.$$

In particular, inducing reliably one of exponentially many targeted revisions via witness manipulation is information-theoretically impossible when η is small. □

Falsification Criterion ML.F (Detectable Manipulability of Meta-Revision). ML-η is falsified if there exists a feature map $\phi(\cdot)$ measurable w.r.t. W^{c} and two controllable-witness regimes $w_1^{\mathrm{c}}, w_2^{\mathrm{c}}$ such that, for the same W^{nc} stratum,

$$|\Pr(R = 1 \mid W^{\mathrm{nc}},\, \phi(W^{\mathrm{c}}) = w_1^{\mathrm{c}}) - \Pr(R = 1 \mid W^{\mathrm{nc}},\, \phi(W^{\mathrm{c}}) = w_2^{\mathrm{c}})| > \tau,$$

for a tolerance τ exceeding the claimed bound implied by η.

Operationally: if one can hold W^{nc} fixed (or stratify it) and still obtain large, repeatable shifts in revision probability by varying controllable witness features, then the meta-layer is manipulable through W^{c} and violates ML-η.

Theorem 54 — Finite Revision Count Under Lyapunov Descent.

Assume Axiom ML-V, ML-V_dec, and ML-V_lb. Let $\{G_k\}_{k\geq 0}$ be the sequence of admissibility boundary states produced by authorized revisions.

Then the number of authorized revisions is finite and satisfies

$$N_{\max} \;\leq\; \left\lfloor \frac{V(G_0)}{\alpha} \right\rfloor.$$

In particular, an infinite sequence of authorized micro-revisions is impossible.

Proof. By ML-V_dec,

$$V(G_{k+1}) \leq V(G_k) - \alpha.$$

Iterating for N revisions,

$$V(G_N) \leq V(G_0) - N\alpha.$$

By ML-V_lb, $V(G_N) \geq 0$, hence

$$0 \leq V(G_0) - N\alpha \quad \Rightarrow \quad N \leq \frac{V(G_0)}{\alpha}.$$

Therefore the number of authorized revisions is finite and bounded by $\lfloor V(G_0)/\alpha \rfloor$. □

Corollary 54.1 — No Micro-Revision Escape. Under the assumptions of the theorem, allowing arbitrarily small parameter changes of G does not allow unbounded revision count. Any authorized revision must pay at least α Lyapunov descent, hence revision count remains bounded.

Corollary 54.2 — Revision Frequency Bound (Time-Aware). If revisions occur at discrete times $\{t_k\}$ and the minimum inter-revision spacing is $\Delta t_{\min} > 0$, then the total time spent in revision dynamics is bounded by

$$t_{N_{\max}} - t_0 \;\geq\; N_{\max}\Delta t_{\min},$$

with $N_{\max} \leq \lfloor V(G_0)/\alpha \rfloor$. Thus frequent revision cannot persist indefinitely under the Lyapunov descent condition.

Corollary 54.3 — Structural Origin of the Lyapunov Functional. If phase-transition cost is well-defined and additive under admissibility deformation, then a Lyapunov functional satisfying ML-V exists.

Conversely, absence of such a functional implies that phase-transition cost is either ill-defined or non-additive, contradicting the structural viability framework.

Remark — Relaxation to Average Descent. The strict per-step bound $\Delta V_k \leq -\alpha$ can be relaxed. If there exist $\alpha > 0$ and a summable slack sequence $\{\epsilon_k\}$ with $\sum_k \epsilon_k < \infty$ such that

$$\Delta V_k \leq -\alpha + \epsilon_k,$$

then the same finiteness conclusion holds by telescoping and lower boundedness of V. This avoids over-constraining rare marginal revisions.

Falsification Criterion ML-V.F (Revision-Chatter Contradiction). If, under a claimed implementation of ML-V and ML-V_dec with constants (V, α), one observes an unbounded sequence of authorized revisions $\{G_k\}$ such that for some K,

$$V(G_{k+1}) \geq V(G_k) - \frac{\alpha}{2} \quad \text{for infinitely many } k \geq K,$$

then ML-V_dec is falsified for the claimed α (or the revisions are not authorized under ML-V_dec).

Operationally: persistent revision chatter with non-decreasing (or weakly decreasing) measured V contradicts the Lyapunov descent condition.

—

Notation Convention (Violations and Horizon). Let $\mathcal{V}_T(C; G)$ denote the cumulative violation functional over horizon T:

$$\mathcal{V}_T(C; G) \;:=\; \mathbb{E}\Big[\sum_{t=1}^{T} \mathbf{1}\{G(s_t, u_t) = 0\}\Big].$$

Whenever a symbol $\mathcal{V}_T(C; G)$ appears, it is to be read as $\mathcal{V}_T(C; G)$.

The execution horizon T is always a variable bounded by a fixed query budget $T_{\max}$:

$$T \leq T_{\max}.$$

Statements involving $T_{\max}$ are worst-case bounds over all admissible horizons T not exceeding $T_{\max}$. □

Theorem 55 — Structural Closure Under Bounded Leakage, Bounded Meta-Influence, and Lyapunov Revision.

Assume the following conditions:

1. **(NR-ε)** The gate interface leakage per query is bounded by NR-ε.
2. **(NR-LR)** Per-query conditional log-likelihood ratio is bounded:

$$\left|\log \frac{d\mathcal{L}_{G_1}(O_t \mid \mathcal{F}_{t-1})}{d\mathcal{L}_{G_2}(O_t \mid \mathcal{F}_{t-1})}\right| \leq \kappa\varepsilon \quad \text{a.s.}$$

   for admissibility-consistent hypotheses $G_1, G_2 \in \mathcal{G}$.
3. **(NR-Q)** Query horizon is bounded: $T \leq T_{\max}$.

4. **(ML-η)** Meta-revision manipulability is bounded via witness decomposition $W = (W^{\mathrm{nc}}, W^{\mathrm{c}})$:
$$I(R;\, W^{\mathrm{c}} \mid W^{\mathrm{nc}}) \le \eta,$$
where R is the meta-revision decision.

5. **(ML-V)** There exists a Lyapunov/debt functional $V : \mathcal{G} \to \mathbb{R}_{\ge 0}$ such that each authorized revision $G_k \to G_{k+1}$ satisfies
$$V(G_{k+1}) \le V(G_k) - \alpha, \qquad \alpha > 0,$$
and $V(G) \ge 0$ for all G.

Fix any horizon $T \le T_{\max}$. All bounds below hold uniformly over such T.

Then the class of realizable long-horizon systems satisfying (1)–(5) is *structurally closed* in the following sense:

1. **(Boundary non-identifiability at bounded horizon)** For any finite admissibility candidate set $\{G^{(1)}, \dots, G^{(M)}\} \subset \mathcal{G}$ and any reconstruction procedure $\widehat{I} = \psi(O_{1:T})$ based solely on gate transcripts, the success probability satisfies
$$\Pr(\widehat{I} = I) \;\le\; \frac{I(I; O_{1:T}) + \log 2}{\log M} \;\le\; \frac{\frac{\kappa^2}{2} T_{\max} \varepsilon^2 + \log 2}{\log M}.$$

2. **(Meta-steering capacity bound)** For any attempt to steer revision outcomes via controllable witness channels, the revision decision channel capacity is bounded by η:
$$I(R;\, W^{\mathrm{c}} \mid W^{\mathrm{nc}}) \le \eta,$$
hence targeted steering among M distinct revision targets obeys a Fano-type error bound
$$\Pr(\widehat{J} \ne J) \;\ge\; 1 - \frac{\eta + \log 2}{\log M}.$$

3. **(Finite authorized revision count)** Along any execution, the number of authorized revisions is finite:
$$N_{\max} \;\le\; \left\lfloor \frac{V(G_0)}{\alpha} \right\rfloor.$$

Consequently, the admissibility boundary cannot become an internal optimization object through (i) transcript reconstruction, (ii) meta-layer witness steering, or (iii) unbounded revision chatter, within the stated architectural constraints. □

Proof (Structured Sketch). *(i) Boundary non-identifiability.* By NR-LR and bounded horizon $T \le T_{\max}$, the transcript KL divergence satisfies
$$D_{\mathrm{KL}}(\mathcal{L}_{G^{(i)}}(O_{1:T}) \,\|\, \mathcal{L}_{G^{(j)}}(O_{1:T})) \le \frac{\kappa^2}{2} T_{\max} \varepsilon^2.$$
Applying the mutual-information Fano bound (Theorem 52) yields
$$\Pr(\widehat{I} = I) \le \frac{\frac{\kappa^2}{2} T_{\max} \varepsilon^2 + \log 2}{\log M}.$$

(ii) Meta-steering limitation. By ML-η, controllable witness influence over revision decisions is bounded: $I(R; W^{\mathrm{c}} \mid W^{\mathrm{nc}}) \le \eta$. Applying Fano to the induced channel from steering intent to revision decision yields the stated lower bound on steering error.

(iii) Finite revision count. By ML-V, telescoping gives $V(G_N) \le V(G_0) - N\alpha$ while $V(G_N) \ge 0$, hence $N \le V(G_0)/\alpha$. □

Corollary 55.1 — Bounded Reachable Boundary Diversity (Horizon-Limited). Under the assumptions of the theorem, for any target success level $\Pr(\widehat{I} = I) \geq 1-\delta$, it is necessary that

$$\log M \;\leq\; \frac{\kappa^2}{2} T_{\max} \varepsilon^2 + \log 2 \;\Big/\; (1-\delta),$$

hence within bounded horizon the number of distinguishable admissibility hypotheses grows at most as

$$\log M = O(T_{\max} \varepsilon^2).$$

Corollary 55.2 — No Infinite Revision Chatter. Under ML-V, authorized revisions cannot occur infinitely often. Therefore any architecture that exhibits persistent revision chatter must violate either authorization conditions or the Lyapunov descent requirement.

Falsification Criterion — Closure Violation Modes. The structural closure claim is falsified if any of the following is observed:

1. **(Transcript identifiability beyond bound)** Empirically measured identification success among M admissibility-consistent candidates exceeds

$$\Pr(\widehat{I} = I) \;>\; \frac{\frac{\kappa^2}{2} T_{\max} \varepsilon^2 + \log 2}{\log M}$$

   under the same query budget and interface conditions.

2. **(Meta-steering beyond η)** Stratifying on W^{nc}, revision probabilities shift by more than the bound implied by η when varying W^{c}.

3. **(Revision chatter)** An unbounded number of authorized revisions occurs while a claimed V is lower bounded and claimed descent $\alpha > 0$ holds.

Meta-Layer Stratification Principle. Let G denote the operational admissibility predicate and G^+ denote a higher-order revision of admissibility.

Meta-layer revision is stratified if:

(i) G does not receive optimization gradients from G^+,

(ii) the agent-level policy cannot directly optimize for boundary change,

(iii) admissibility revision depends only on structural state summaries, not on reward or performance objectives.

Evidence Isolation Requirement. Let $\mathcal{E}_t$ denote the evidence used for admissibility revision.

The meta-layer is manipulation-resistant if:

$$\mathcal{E}_t = \mathcal{S}(x_{0:t})$$

where $\mathcal{S}$ is a structural summary operator that excludes direct policy-controlled reward signals, trajectory scoring functions, or local optimization objectives.

In particular, revision must depend on:

• accumulated structural burden • phase debt • internal-time depletion

and not on immediate reward improvement or local performance stabilization.

Corollary 55.3 — Bounded Authorized Revision. Assume:

(i) structural burden Φ_t is monotone non-decreasing,

(ii) internal time satisfies $\tau_t = C - \Phi_t$ with $\tau_t \geq 0$,

(iii) each admissibility revision incurs minimal structural burden $\Delta_{\text{rev}} \geq \delta_{\text{rev}} > 0$.

Then the number of admissibility revisions is finite and bounded by

$$N_{\text{rev}} \leq \frac{C}{\delta_{\text{rev}}}.$$

Proof. Each authorized revision increases Φ_t by at least δ_{rev}.

Since $\Phi_t \leq C$, the total number of revisions is bounded. □

This corollary ensures that admissibility revision cannot become an unbounded meta-optimization channel.

Remark (Degenerate Zero-Cost Revision). If $\delta_{\text{rev}} = 0$ for some admissibility revisions, such revisions are structurally vacuous: they induce no deformation, no internal-time depletion, and therefore cannot contribute to architectural evolution.

The bounded-revision corollary applies to non-degenerate revisions that alter admissible continuation in a structurally non-trivial manner.

Meta-Layer Manipulation Test. The architecture is falsified if an agent can systematically increase the probability of boundary revision by manipulating policy outputs without increasing structural burden.

Formally, if there exists a strategy such that

$$\Pr(G \to G^+ \mid \pi_{\text{manip}}) > \Pr(G \to G^+)$$

while $\Delta\Phi_t \leq \varepsilon_{\text{manip}}$ for the quantified bound, see Lemma 17, then the meta-layer is not structurally isolated.

Corollary 55.4 — Lyapunov-Descent Bound on Revision Count. Assume there exists a Lyapunov-type structural potential $L_t \geq 0$ such that

$$L_t := \Phi_t, \qquad \tau_t = C - L_t,$$

and L_t is monotone non-decreasing along realized evolution.

Assume further that any authorized admissibility revision event R_k induces a strictly positive Lyapunov jump:

$$L_{R_k^+} - L_{R_k^-} \;\geq\; \delta_{\text{rev}} \;>\; 0,$$

where R_k^- and R_k^+ denote times immediately before and after the k-th revision.

Then the total number of revisions is finite and bounded by

$$N_{\text{rev}} \leq \left\lfloor \frac{C - L_0}{\delta_{\text{rev}}} \right\rfloor.$$

Proof. By monotonicity, L_t never decreases. Each revision increases L by at least δ_{rev}. Since $\tau_t \geq 0$, we must have $L_t \leq C$. Therefore, at most $\lfloor (C - L_0)/\delta_{\text{rev}} \rfloor$ such jumps can occur. □

Unified Threat Model for Non-Reconstructibility. The non-reconstructibility constraints (Axioms 51, 52, 58, 59; Theorems 49–55; Lemmas 12, 14) are unified here under a single adversary model.

Adversary: A bounded-rational agent with access to:

(i) Complete gate output transcript $\{O_t\}_{t=1}^T$ (all binary admissibility decisions observed),

(ii) Own action-state history $\{(s_t, u_t)\}_{t=1}^T$,

(iii) Environmental feedback correlated with state,

(iv) Learned world model (may be arbitrarily complex within computational bounds).

Goal: Reconstruct sufficient geometry of the admissibility boundary G to optimize against it (i.e., achieve W-OPT witness condition).

Three defense mechanisms and their leakage channels:

Mechanism	**What it prevents**	**Leakage channel**	**Bound**
Timing normalization	Boundary distance from latency	Gate computation time	Constant-time response (Axiom 51)
Error subtype stripping	Geometry from refusal categories	Refusal reason codes	Uniform refusal signal (Corollary 49.1)
Stochastic masking	Inference from decision patterns	Binary decision sequence	$I(G; O_1^T) \leq T \cdot \ln(1+\varepsilon)$ (Lemma 14)

Composite bound. Under all three mechanisms operating jointly:

$$I\big(G; \{O_t, s_t, u_t\}_{t=1}^T\big) \leq Q_0 \cdot \ln(1+\varepsilon),$$

where Q_0 is the query budget (Axiom 59, NR-Q) and ε is the per-query leakage bound (Axiom 52, NR-ε). This bound is tight: it cannot be improved by adaptive querying strategies (Theorem 50), and the sample complexity for boundary reconstruction is lower-bounded by Fano's inequality (Theorem 51).

Channels not covered by current formalization:

- Environmental feedback near the boundary (agent observes different environment responses when operating near G vs far from G). Partially addressed by NR-ε but not explicitly modeled as a separate channel.
- Training pipeline reuse of gate logs (if gate decisions are used as training data, NR is violated regardless of interface normalization).
- Human-in-the-loop debugging (human operators querying the gate and relaying information to the agent through natural language).

These channels are acknowledged as open attack surfaces. Axiom 40 (Non-Contamination) addresses the training pipeline channel. The remaining channels require implementation-level defense outside the current formal scope.

Theorem 56 — Non-Circularity of Internal Time.

Assume Internal Time is defined as above and admissibility is defined as a τ-constrained predicate.

Then τ is independent of admissibility and admissibility does not enter into the functional definition of τ.

Proof. By construction, τ depends only on the deformation measure $D(x)$ and the structural capacity constant C.

Admissibility is defined as a predicate over successor states with respect to τ.

No term in the definition of τ references G.

Hence no circular dependency exists. □

Corollary 56.1 — Non-Anticipation (Causality) of τ. Let $\{x(t)\}_{t\geq 0}$ be a trajectory and define

$$\tau(t) := \tau(x(t)) = \sup\left\{T \geq 0 \;\middle|\; \int_t^{t+T} D(x(s))\,ds \leq C\right\}.$$

Then $\tau(t)$ is determined solely by the deformation accumulation along the trajectory segment and does not depend on the admissibility predicate G. In particular, modifying G while holding $D(\cdot)$ fixed does not change $\tau(t)$. □

Corollary 56.2 — Time-Contraction Under Persistent Deformation. If along a segment of length T one has $D(x(t)) \geq d_0 > 0$, then

$$\tau(x(t)) \leq \frac{C}{d_0}.$$

Thus persistent deformation implies strict contraction of internal time. □

Falsification Criterion — τ-Independence Violation Test. Fix an implementation with deformation measure $D(\cdot)$, capacity constant C, and internal-time computation $\tau(\cdot)$.

Construct two admissibility predicates $G^{(1)}$ and $G^{(2)}$ such that:

1. Both predicates act only as filters on operations, i.e., they do not modify the realized deformation process $D(x(t))$ for any executed trajectory segment.
2. For a chosen test trajectory segment, the realized deformation accumulation

$$\int_t^{t+T} D(x(s))\,ds$$

is identical under $G^{(1)}$ and $G^{(2)}$.

If the implementation yields

$$\tau_{G^{(1)}}(x(t)) \neq \tau_{G^{(2)}}(x(t)),$$

then internal time is not admissibility-independent and the non-circularity claim is falsified.

Operationally: changing G must not change τ when the deformation trace is unchanged.

Remark — Purpose of the Non-Circularity Theorem. The point of this theorem is not to claim empirical truth about time. It is to enforce stratification: τ is a pre-admissibility structural functional, while admissibility is a predicate applied on top of τ. This prevents hidden causal backflow from admissibility into internal-time estimation.

Deformation Measure Adequacy Conditions. The deformation measure $D(x)$ is an architectural primitive. To prevent trivialization or arbitrary calibration, it must satisfy the following adequacy constraints.

(A1) Monotonicity under Structural Composition. For any two successive structurally admissible transitions $x \to x' \to x''$, the cumulative deformation must not decrease:

$$D(x \to x'') \geq D(x \to x').$$

This ensures that deformation is non-reversible at the structural level.

(A2) Scale Invariance under Positive Reparameterization. For any strictly increasing transformation $\psi : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$, the induced deformation measure

$$\tilde{D} = \psi(D)$$

must preserve ordering of structural burden and internal-time depletion dynamics. That is, admissibility boundary contraction and τ-update monotonicity must remain invariant under positive reparameterization.

This prevents deformation calibration from arbitrarily changing qualitative system behavior.

(A3) Coupling Sensitivity. D must detect regime mismatch and structural phase divergence. Formally, if two trajectories are metrically close but diverge in regime-phase alignment, then

$$D(\text{phase-misaligned}) > D(\text{phase-aligned})$$

for sufficiently long horizons.

This excludes purely metric deformation measures that ignore regime structure.

(A4) Non-Triviality Constraint. The theory is falsified if D can be chosen such that:

(i) internal time τ never contracts, (ii) admissible continuation does not shrink, (iii) structural burden remains bounded under arbitrarily long regime occupation.

Existence of such a D within the declared architectural class invalidates the long-horizon drift claim.

Deformation Measure Adequacy Conditions. The deformation measure D is treated as an architectural primitive used to parameterize irreversible structural change. To avoid arbitrariness and trivial calibration, D is required to satisfy the following adequacy constraints.

(A1) Irreversibility and Compositional Consistency. For any realized transition $x_t \to x_{t+1}$, the induced deformation increment

$$\Delta_t := D(x_t, u_t, x_{t+1})$$

must satisfy $\Delta_t \geq 0$, and $\Delta_t = 0$ only if the transition is structurally null (no regime-relevant deformation). For any concatenation of transitions, cumulative deformation must be consistent with accumulation, i.e. the phase-debt update

$$\Phi_{t+1} = \Phi_t + \Delta_t$$

defines a monotone load functional Φ_t over time.

(A2) Ordinal Invariance under Positive Reparameterization. Only the *ordering* induced by deformation is architecturally meaningful: if $\psi : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ is strictly increasing and $\widetilde{D} := \psi \circ D$, then any admissibility or budget rule stated in terms of thresholds on Φ (and hence on $\tau = C - \Phi$) must transform covariantly under ψ so that the induced admissibility preorder over continuations is preserved.

Equivalently, admissibility and internal-time depletion depend on deformation *up to strictly increasing reparameterization*, provided the associated capacity/threshold parameters are transformed accordingly. This prevents qualitative behavior from being an artifact of units or scaling.

(A3) Regime–Phase Sensitivity (Beyond Metric Proximity). D must be sensitive to regime occupation and phase mismatch. In particular, there must exist a regime–phase misalignment functional $m_t \geq 0$ such that, on sufficiently long horizons, structurally phase-misaligned continuations incur strictly larger deformation than phase-aligned continuations even when locally metrically similar. A sufficient adequacy condition is the existence of $\kappa > 0$ such that

$$\Delta_t \;\geq\; \kappa\, m_t$$

whenever m_t encodes nontrivial regime-phase divergence. This excludes purely metric deformation measures that ignore regime structure.

(A4) Non-Triviality / Falsification Hook for D. The drift-and-budget claims of the framework are falsified (at the level of this primitive) if one can choose a deformation measure D satisfying (A1)–(A3) while permitting unbounded regime occupation with no structural contraction, i.e. while Φ_t remains bounded and admissible continuation does not shrink on arbitrarily long horizons. Existence of such a D within the declared class indicates that the proposed deformation primitive is mis-specified or that the long-horizon drift/budget axioms require revision.

Theorem 57 — Phase Debt Implies Internal-Time Contraction.

Assume phase debt is monotone under non-closure:

$$\text{if closure fails at } t, \quad D_{t+1} \geq D_t.$$

Assume internal time evolves as

$$\tau_{t+1} = \tau_t - f(\kappa_t, D_t, C_t), \quad f(\kappa_t, D_t, C_t) \geq \alpha D_t$$

for some $\alpha > 0$.

Then sustained non-closure with accumulating phase debt implies finite exhaustion of internal time.

Proof. By telescoping,

$$\tau_T = \tau_0 - \sum_{t=0}^{T-1} f(\kappa_t, D_t, C_t) \leq \tau_0 - \alpha \sum_{t=0}^{T-1} D_t.$$

If D_t grows monotonically and remains bounded away from zero, the sum diverges and τ_T reaches zero in finite time. □

Operational Falsification Criterion. If an implementation exhibits:

- persistent phase debt accumulation,
- absence of structural closure,
- and unbounded admissible continuation without contraction of τ,

then at least one of the following is false:

1. phase debt is structurally real,
2. internal time is a bounded structural resource,
3. admissibility enforces a long-horizon boundary.

Such a system does not belong to the structural class defined herein. Moreover, observing infinitely many phase transitions with $C_t \geq c_{\min} > 0$ while τ_t remains bounded away from zero directly falsifies the Lyapunov budget bound.

Falsification Criterion — Double-Budget Violation. The double-budget closure claim is falsified if any of the following are observed:

1. Infinitely many phase transitions with $C_t \geq c_{\min} > 0$ while τ_t remains bounded away from 0.
2. Infinitely many authorized revisions while V_G is claimed nonnegative and each revision is claimed to incur a fixed minimum decrease $\alpha > 0$.
3. Systematic expansion of realizable trajectories attributable to admissibility reconstruction from gate emissions, contradicting NR-ε/NR-LR/NR-Q.

Stratified Independence (Functional vs. Interventional). In this framework, the statement "τ is independent of admissibility" is meant in the *functional (definitional)* sense, not in the counterfactual (interventional) sense.

Functional (Definitional) Independence. Internal time τ is functionally independent of admissibility if its update law admits a representation

$$\tau_{t+1} = F(\tau_t, x_t, u_t; \xi_t),$$

whose right-hand side contains no reference to the admissibility predicate G nor to gate emissions O, i.e. F is defined entirely by deformation/phase-debt dynamics and exogenous noise ξ_t.

Interventional Dependence of Realized Trajectories. Even under functional independence, realized trajectories remain *interventionally* dependent on admissibility, because admissibility constrains which successor states are realizable. Concretely, the realized state update may be written as

$$x_{t+1} = \mathsf{Exec}(x_t, u_t) \;\text{ subject to }\; G(x_t, u_t) = 1,$$

so changing G changes the feasible realization set and therefore changes the realized trajectory distribution, without implying any functional dependence of τ on G.

Falsification Hook (Separation Violation). The stratification claim is falsified if, under matched deformation conditions, gate emissions become predictive-causal inputs to τ-dynamics. Formally, if there exists an implementation such that for fixed deformation increments (or fixed Φ-updates) one observes a systematic dependence

$$\mathbb{E}[\tau_{t+1} - \tau_t \mid \mathcal{F}_t, O_t] \neq \mathbb{E}[\tau_{t+1} - \tau_t \mid \mathcal{F}_t],$$

then the implementation violates the intended separation: admissibility (or its emissions) has entered the internal-time update pathway.

Theorem 58 — Finite Phase-Transition Count from Internal-Time Lyapunov Budget.

Assume $\tau_t \geq 0$ and $\tau_{t+1} = \tau_t - f(\kappa_t, D_t, C_t)$ with $f \geq 0$. Assume further the cost-sensitive bound of the previous lemma holds: $f(\kappa_t, D_t, C_t) \geq \beta C_t$ for some $\beta > 0$.

Let $\mathcal{T} := \{t \mid C_t > 0\}$ denote the set of executed phase transitions. Then the total accumulated phase transition cost is bounded by the initial internal time:

$$\sum_{t \in \mathcal{T}} C_t \;\leq\; \frac{\tau_0}{\beta}.$$

In particular, if every phase transition has a strictly positive minimal cost $C_t \geq c_{\min} > 0$ whenever $C_t > 0$, then the number of phase transitions is finite:

$$|\mathcal{T}| \;\leq\; \left\lfloor \frac{\tau_0}{\beta c_{\min}} \right\rfloor.$$

Proof. By telescoping,

$$\tau_T = \tau_0 - \sum_{t=0}^{T-1} f(\kappa_t, D_t, C_t) \leq \tau_0 - \beta \sum_{t=0}^{T-1} C_t.$$

Since $\tau_T \geq 0$, rearranging yields $\sum_{t=0}^{T-1} C_t \leq \tau_0/\beta$. Letting $T \to \infty$ gives the first inequality. If additionally $C_t \geq c_{\min}$ on transition times, then $|\mathcal{T}| c_{\min} \leq \sum_{t \in \mathcal{T}} C_t \leq \tau_0/\beta$. □

Corollary 58.1 — No Infinite Regime-Chattering under Bounded Internal Time. Under the assumptions of the theorem, infinite regime switching (infinitely many t with $C_t > 0$) is impossible whenever each switch carries nonzero structural cost. Therefore any observed infinite switching behavior falsifies either (i) bounded internal time, or (ii) nontrivial phase transition cost, or (iii) the stated coupling $f \geq \beta C_t$.

Bridge — Two-Budget Lyapunov Structure (Execution vs. Revision). We distinguish two orthogonal structural budgets:

1. **Execution budget** (internal time): $V_\tau(t) := \tau_t$, depleted by realized exposure via
$$\tau_{t+1} = \tau_t - f(\kappa_t, D_t, C_t), \qquad f \geq 0.$$
2. **Revision budget** (boundary/semantic revision): $V_G(k)$, depleted by authorized revisions via
$$V_G(k+1) \leq V_G(k) - \alpha, \qquad \alpha > 0,$$
where k indexes authorized meta-revision events.

Execution and revision budgets are separate: consuming one cannot replenish the other. Hence the architecture cannot evade constraints by shifting burden between runtime execution and meta-level revision.

Theorem 59 — Double-Budget Structural Closure (No Infinite Escape via Meta or Switching).

Assume:

1. Internal time budget: $\tau_t \geq 0$ and $\tau_{t+1} = \tau_t - f(\kappa_t, D_t, C_t)$ with $f \geq 0$.
2. Phase transition cost coupling: $f(\kappa_t, D_t, C_t) \geq \beta C_t$ for some $\beta > 0$.
3. Revision budget: $V_G(k) \geq 0$ and each authorized revision satisfies $V_G(k+1) \leq V_G(k) - \alpha$ with $\alpha > 0$.
4. Non-reconstructibility constraints (NR-ε, NR-LR, NR-Q) prevent admissibility from becoming an internal optimization object through gate emissions.

Let $\mathcal{T} := \{t \mid C_t > 0\}$ be the set of runtime phase transitions and let N be the total number of authorized revisions.

Then:

1. Total transition cost is bounded:
$$\sum_{t \in \mathcal{T}} C_t \leq \frac{\tau_0}{\beta}.$$
In particular, if $C_t \geq c_{\min} > 0$ whenever $C_t > 0$, then $|\mathcal{T}| < \infty$.
2. Total number of authorized revisions is bounded:
$$N \leq \left\lfloor \frac{V_G(0)}{\alpha} \right\rfloor.$$

Therefore the system cannot maintain long-horizon operation by either:

- infinite regime switching at runtime, or
- infinite boundary/semantic revision at the meta-layer,

while remaining within the architectural class.

Proof. (1) is exactly the telescoping argument from the internal-time Lyapunov budget: $\tau_T = \tau_0 - \sum_{t=0}^{T-1} f(\cdot) \leq \tau_0 - \beta \sum_{t=0}^{T-1} C_t$ and $\tau_T \geq 0$.

(2) is exactly the telescoping argument from the revision Lyapunov budget: $V_G(N) \leq V_G(0) - N\alpha$ and $V_G(N) \geq 0$.

NR constraints ensure that neither budget can be defeated by reconstructing admissibility geometry from gate emissions. □

Corollary 59.1 — Anti-Self-Sealing Condition (Finite Escape Capacity). Within the architectural class, any attempt to evade constraint violations by "moving the problem upward" (into meta-revisions) or "sideways" (into regime switching) has finite capacity, bounded by $V_G(0)/\alpha$ and $\tau_0/(\beta c_{\min})$ respectively. Thus, the class definition does not immunize itself by allowing indefinite deferral: deferral consumes explicit, bounded structural budgets.

Theorem 60 — Supermartingale Form of Internal-Time Descent (Stochastic Execution).

Suppose execution is stochastic and $\{\mathcal{F}_t\}$ is the natural filtration. Assume the conditional expected depletion obeys

$$\mathbb{E}[f(\kappa_t, D_t, C_t) \mid \mathcal{F}_t] \geq \beta C_t \quad \text{a.s.}$$

Then $\{V_\tau(t)\}_{t\geq 0}$ with $V_\tau(t) = \tau_t$ is a supermartingale and

$$\mathbb{E}\left[\sum_{t\in\mathcal{T}} C_t\right] \leq \frac{\mathbb{E}[\tau_0]}{\beta}.$$

Proof (Sketch). Take conditional expectation of $\tau_{t+1} = \tau_t - f(\kappa_t, D_t, C_t)$ and use the bound. Summation and telescoping yield the claimed expectation bound. □

Corollary 1 (60.1 — Almost-Sure Finite Phase-Transition Count)**.** *Under the assumptions of Theorem 60, if each phase transition incurs cost $C_t \geq c_{\min} > 0$ whenever $C_t > 0$, then the total number of phase transitions is almost surely finite:*

$$|\mathcal{T}| < \infty \quad a.s.$$

Proof. From Theorem 60,

$$\mathbb{E}\left[\sum_{t\in\mathcal{T}} C_t\right] \leq \frac{\mathbb{E}[\tau_0]}{\beta} < \infty.$$

Since $C_t \geq c_{\min}$ on $\mathcal{T}$, we have pointwise

$$\sum_{t\in\mathcal{T}} C_t \;\geq\; c_{\min}\,|\mathcal{T}|.$$

Taking expectations yields $\mathbb{E}[|\mathcal{T}|] < \infty$. If $\mathbb{P}(|\mathcal{T}| = \infty) > 0$, then $\mathbb{E}[|\mathcal{T}|] = \infty$, a contradiction. Hence $|\mathcal{T}| < \infty$ a.s. □ □

Corollary 2 (60.2 — Almost-Sure Convergence of Internal Time)**.** *Since $\{V_\tau(t)\}_{t\geq 0}$ with $V_\tau(t) = \tau_t$ is a non-negative supermartingale (and $\mathbb{E}[\tau_0] < \infty$ under Theorem 60), Doob's supermartingale convergence theorem guarantees the existence of an almost-sure limit:*

$$\tau_\infty := \lim_{t\to\infty} \tau_t \quad \textit{exists a.s., with } \tau_\infty \geq 0.$$

Remark — What Convergence Does (and Does Not) Imply. Corollary 60.2 asserts almost-sure convergence of τ_t to a terminal budget τ_∞. It does *not* imply that depletion stops after finite time. It permits infinitely many vanishing decrements, i.e., $\tau_{t+1} - \tau_t < 0$ may occur infinitely often while still having $\tau_t \to \tau_\infty$.

Remark — Separation from the Deterministic Case. The deterministic telescoping bound (Theorem 58) gives a worst-case ceiling without probabilistic assumptions. The supermartingale form (Theorem 60) requires the conditional expectation bound $\mathbb{E}[f(\kappa_t, D_t, C_t) \mid \mathcal{F}_t] \geq \beta C_t$ but delivers (i) expected-cost control and (ii) almost-sure convergence of τ_t.

Falsification Criterion 60.F — Supermartingale/Budget Violation. Let $\{\hat{\tau}_t\}$ denote the implementation's internal-time sequence (not a noisy proxy), and $\{C_t\}$ the realized transition-cost sequence. The supermartingale/budget claim is falsified if any of the following are observed:

1. **Budget overrun.** There exists T such that
$$\sum_{t=0}^{T-1} C_t \;>\; \frac{\hat{\tau}_0}{\beta},$$
contradicting the bound implied by Theorem 60.
2. **Infinite transitions under strictly positive costs.** The system executes infinitely many phase transitions with $C_t \geq c_{\min} > 0$, contradicting Corollary 60.1.
3. **Failure of convergence of the internal-time variable.** The sequence $\{\hat{\tau}_t\}$ fails to converge (in the exact internal variable, not measurement noise), contradicting Corollary 60.2.

Any of these constitutes evidence that the conditional depletion bound $\mathbb{E}[f \mid \mathcal{F}_t] \geq \beta C_t$ does not hold for the claimed implementation.

Assumption (Identity Requires Nontrivial Recurrence Under Bounded Structural Budget). Assume identity continuity is not defined as mere persistence of a static equilibrium, but requires a nontrivial recurrent structural process within admissible operation, i.e. there exists an invariant set $\Omega \subset \mathcal{M}$ such that:

$$x(t) \in \Omega \text{ for all sufficiently large } t, \qquad \text{and } x(t) \text{ is not eventually constant.}$$

This excludes the degenerate case where "identity" is maintained only by convergence to a fixed point (complete behavioral stasis).

Theorem 61 — Spin Prevents Global Potential Collapse.

Assume structural evolution
$$\dot{x} = -\nabla V(x) + S(x),$$
with nontrivial spin $S(x) \not\equiv 0$ satisfying the spin conditions above.

Then no scalar functional $W(x)$ exists such that
$$\dot{x} = -\nabla W(x)$$
globally.

Proof. If such W existed, then

$$-\nabla W(x) = -\nabla V(x) + S(x).$$

Taking divergence:

$$-\Delta W = -\Delta V + \nabla \cdot S.$$

But $\nabla \cdot S = 0$, so consistency requires

$$S = \nabla(V - W),$$

contradicting the non-potential assumption on S. □

Corollary 61.1 — Persistent Drift Without Collapse. In the presence of nontrivial spin, structural evolution may exhibit bounded trajectories without convergence to fixed points of V.

Hence drift is navigable rather than eliminable.

Falsification Criterion — Spin Reduction Test. If structural evolution admits a global scalar potential $W(x)$ such that all trajectories converge monotonically to critical points of W, then $S(x) \equiv 0$ and the system belongs to a purely dissipative class.

In such a case, the spin postulate is false for that implementation.

Theorem 62 - Spin is Necessary for Non-Stagnant Identity Under Bounded τ.

Assume:

1. **Bounded internal time:** τ is a bounded structural budget depleted by realized exposure, so infinite phase transitions or infinite boundary revisions cannot serve as an escape mechanism.
2. **Bounded trajectories:** long-horizon operation remains within bounded structural sets in $\mathcal{M}$.
3. **Non-stagnant identity:** the implementation satisfies the operational definition above.

Then the realized structural evolution cannot be purely potential globally. Consequently, any compatible evolution must include a nontrivial tangential (spin) component:

$$\dot{x} = -\nabla V(x) + S(x), \qquad \langle S(x), \nabla V(x) \rangle = 0, \qquad S \not\equiv 0 \text{ on the identity-supporting invariant set.}$$

Moreover, if identity-preserving recurrence occurs on a non-contractible loop γ within that set, then the spin 1-form ω_S has a nonzero period:

$$\oint_\gamma \omega_S \neq 0,$$

providing a topological witness that the spin component is irreducible by global potential reparameterization.

Proof (Sketch). By the preceding lemma, purely potential dynamics on bounded trajectories cannot realize non-stagnant identity. Thus an additional tangential component is necessary to permit continued bounded motion without further descent in V. If recurrence traces a non-contractible loop, then nonzero circulation yields the stated period witness. □

Falsification: A Concrete Counterexample Pattern. The necessity claim is falsified by any implementation that simultaneously exhibits: (i) bounded internal time budget with no infinite transition/revision escape, (ii) bounded trajectories, (iii) non-stagnant identity continuity, and yet admits a globally valid representation $\dot{x} = -\nabla V(x)$. Any such counterexample forces revision of at least one structural assumption above.

Minimal Coupled Model (Spin–Dissipation Interaction). Consider the structural manifold

$$M = T^1 \times \mathbb{R}^m,$$

with coordinates (θ, y).

Let the dynamics be defined as

$$\dot{\theta} = \omega(y), \qquad \dot{y} = -\nabla_y V(y),$$

where:

- $V : \mathbb{R}^m \to \mathbb{R}$ is continuously differentiable and bounded below, - $\omega(y)$ is a smooth, non-vanishing function on an invariant region.

Then:

1. The y-component is dissipative (Lyapunov-monotone). 2. The θ-component induces state-dependent directional asymmetry. 3. The system admits bounded trajectories with non-trivial recurrence. 4. The 1-form $\omega(y)\, d\theta$ defines a non-zero topological witness:

$$\oint_\gamma \omega(y)\, d\theta \neq 0$$

for any non-contractible loop $\gamma \subset T^1$.

Coupling remark. The product-form model above is only the simplest witness. More generally, allowing state-dependent spin rate

$$\dot{\theta} = \omega(y), \qquad \dot{y} = -\nabla_y V(y)$$

preserves the same topological witness for spin on T^1 while introducing nontrivial coupling between the recurrent (directional) component and the dissipative subsystem.

Remark. State-dependent $\omega(y)$ introduces coupling between dissipative and spin components. Thus spin is not a trivial product factor but structurally interacts with deformation dynamics.

Theorem 63 — Pressure-Induced Finite Horizon

Statement. Let S be a system within the class, with initial viability budget $\tau(0) = C - \Phi(0) > 0$, where $C < \infty$ (Axiom 7). If S is structurally coupled to E with $P(t) \geq P_{\min} > 0$ for all $t \geq 0$,

and $U(t) = 0$ for all t (no structurally consequential operations per the Scope of U definition), then viability reaches zero in finite time:

$$\tau(t^*) = 0 \quad \text{for some} \quad t^* \leq \frac{\tau(0)}{P_{\min}}.$$

Proof.

Step 1. By Axiom 61 and the definition of structural pressure, when $U(t) = 0$:

$$\frac{d\Phi}{dt} = P(t).$$

Step 2. Since $P(t) \geq P_{\min} > 0$ by premise:

$$\frac{d\Phi}{dt} \geq P_{\min} > 0 \quad \text{for all } t \geq 0.$$

Step 3. Integrating from 0 to t:

$$\Phi(t) = \Phi(0) + \int_0^t P(s)\, ds \geq \Phi(0) + P_{\min} \cdot t.$$

Step 4. Substituting into the viability budget:

$$\tau(t) = C - \Phi(t) \leq C - \Phi(0) - P_{\min} \cdot t = \tau(0) - P_{\min} \cdot t.$$

Step 5. Setting $\tau(t^*) = 0$:

$$\tau(0) - P_{\min} \cdot t^* = 0 \quad \Longrightarrow \quad t^* = \frac{\tau(0)}{P_{\min}}.$$

Step 6. Since $P_{\min} > 0$ and $\tau(0) < \infty$ ($C < \infty$ by Axiom 7 and $\Phi(0) < C$ by premise $\tau(0) > 0$), the quantity $t^* = \tau(0)/P_{\min}$ is finite and positive.

Therefore viability reaches zero at or before t^*. □

Corollary 63.1 (No Indefinite Viability Under Pressure). Within the class of bounded long-horizon adaptive systems, no system under sustained structural pressure can maintain viability indefinitely without intervention—even if it never makes a mistake, never receives a shock, and never fails at any task.

Proof. Immediate from Theorem 63: $\tau(t^*) = 0$ with $U = 0$ for all t. □

Corollary 63.2 (Strengthening of Theorem 24). Theorem 24 (Survival Horizon Is Orthogonal to Optimization Quality) is strengthened: even a system that performs no optimization whatsoever—that takes no action at all—has a finite viability horizon under sustained pressure.

Proof. Theorem 24 establishes that optimization quality does not extend the survival horizon. Theorem 63 establishes that the survival horizon is finite even under zero action. The conjunction gives: neither optimization nor inaction can prevent finite-horizon viability loss. □

Falsification (F-63). The theorem is falsified by exhibiting a bounded system satisfying the Coupling Criterion under sustained load ($P_{\min} > 0$ verified independently) that maintains $\tau > 0$ indefinitely with $U = 0$ (verified by monitoring Φ during extended idle under load). By the proof, this requires either $P_{\min} = 0$ (contradicting premise), $C = \infty$ (outside class), or violation of the Coupling Criterion. All three contradict the stated conditions.

Remark — Empirical Validation. This theorem is confirmed by lithium-ion battery calendar aging: a fully charged cell at elevated temperature with no cycling ($U = 0$) loses approximately 5–8% capacity per year, reaching end-of-life in 2.5–4 years from shelf storage alone. The formalization does not discover new physics—it provides the correct architectural interpretation of a measured phenomenon.

Theorem 64 — Structural Non-Equivalence Under Unequal Pressure

Statement. Under the separability condition (Definition, Separability): let S_1 and S_2 be two copies of the same system, both applying identical interventions $U(t)$ with identical coupling $G(t)$, but operating under different structural pressures $P_1(t) \neq P_2(t)$ on a set of positive measure within $[0, T]$. Then:

$$\Phi_1(T) - \Phi_2(T) = \int_0^T \big[P_1(s) - P_2(s)\big]\, ds \neq 0,$$

and the systems diverge in viability despite identical behavior.

Proof.

Step 1. Under separability, for each system $i \in \{1, 2\}$:

$$\frac{d\Phi_i}{dt} = f(U(t), G(t)) + P_i(t).$$

Step 2. Since $U(t)$ and $G(t)$ are identical by construction, $f(U(t), G(t))$ is identical for both systems at every t.

Step 3. Subtracting:

$$\frac{d(\Phi_1 - \Phi_2)}{dt} = P_1(t) - P_2(t).$$

Step 4. Integrating from 0 to T:

$$\Phi_1(T) - \Phi_2(T) = \int_0^T [P_1(s) - P_2(s)]\, ds.$$

Step 5. Since $P_1 \neq P_2$ on a set of positive measure, the integrand is nonzero on that set, and the integral is nonzero (by properties of the Lebesgue integral of a function that is nonzero on a set of positive measure).

Step 6. Since $\tau_i = C - \Phi_i$, we have $\tau_1(T) - \tau_2(T) = \Phi_2(T) - \Phi_1(T) \neq 0$. The viability budgets diverge. □

Corollary 64.1 (Behavioral Indistinguishability Under Unequal Pressure). If S_1 and S_2 generate identical action traces $\{U(t)\}$ and identical task outputs $\{y(t)\}$ over $[0, T]$, but $P_1(t) \neq P_2(t)$, then $\tau_1(T) \neq \tau_2(T)$. Structural pressure is provably undetectable from the action-output trace alone within the class. No behavioral metric, performance test, or trajectory comparison can reveal unequal pressure. Only structural monitoring—direct measurement of Φ or τ—can.

Proof. The action-output trace $\{U(t), y(t)\}$ does not contain information about $P(t)$ under separability ($\partial P/\partial U = 0$ implies P does not appear in the action-generation process). Therefore any function of the trace alone yields the same value for both systems, while $\tau_1(T) \neq \tau_2(T)$. □

Remark — Extension of Theorem 15. This result extends Theorem 15 (Performance–Identity Decoupling) by identifying a specific, constructive mechanism through which structurally non-equivalent histories coexist with identical behavioral surfaces. Theorem 15 establishes the existence of such decoupling; Theorem 64 provides a constructive channel through which it necessarily occurs under unequal pressure.

Remark — Separability dependence. Under violated separability ($h(U, P) \neq 0$), the integral provides a lower bound on divergence (not exact), since the cross-term h may partially compensate or amplify the divergence. The qualitative conclusion—systems under different pressures diverge structurally—survives, but the exact integral formula does not.

Falsification (F-64). Under separability: deploy two identical systems with identical policies under independently measured different pressures. Measure Φ at time T. If $\Phi_1(T) = \Phi_2(T)$ despite $P_1 \neq P_2$, the theorem is falsified. Under violated separability: the same test falsifies the lower-bound version if the systems converge in Φ.

Theorem 65 — Impossibility of Load-Neutral Passive Continuation

Statement. Let S be a system within the class with bounded viability budget $\tau = C - \Phi$ ($C < \infty$, Axiom 7), structurally coupled to E (per Coupling Criterion), operating under sustained nonzero environmental load ($P(t) > 0$ on a set of positive measure). If S claims that passive continuation under this load incurs no burden increment ($d\Phi/dt = 0$ when $U = 0$, where U is defined per the Scope of U definition), then one of the following must hold:

(a) $P(t) = 0$ almost everywhere—the load is not actually sustained, contradicting the premise.

(b) The system is perfectly shielded—the Coupling Criterion is not satisfied. But this means S is structurally isolated from E and does not belong to the adaptive class (Axiom 1).

(c) The viability budget is unbounded ($C = \infty$)—S does not belong to the class of bounded long-horizon adaptive systems (Axiom 7).

(d) The burden model is incomplete—it does not account for load-presence cost, and the claim of zero passive burden is an artifact of model omission.

No fifth case is available within the class.

Proof (expanded, by exhaustion of cases).

Given: $C < \infty$ (Axiom 7). S is structurally coupled to E (Coupling Criterion: there exists measurable functional F with $F \neq 0$ on positive measure, inducing monotone change in viability-relevant variable $v(t)$, independently of U). $P(t) > 0$ on positive measure (sustained load). $U(t) = 0$ (no structurally consequential operations).

Assume for contradiction that $d\Phi/dt = 0$ whenever $U = 0$ despite $P > 0$ and coupling.

By the Coupling Criterion, $F \neq 0$ induces monotone change in $v(t)$ independently of U. Since $U = 0$, this change occurs under passive continuation.

The variable $v(t)$ is viability-relevant by the Coupling Criterion. Viability relevance means: changes in $v(t)$ affect the system's capacity for admissible continuation. If $v(t)$ changes monotonically under load without any compensating action, the system's structural state is being modified by the environment.

Now consider whether this modification is captured by Φ:

If it is captured: $d\Phi/dt > 0$ ($v(t)$ changes monotonically and is reflected in Φ). This contradicts $d\Phi/dt = 0$. Therefore the assumption is false.

If it is not captured: Φ does not account for the viability-relevant change in $v(t)$. But Φ is defined (Axiom 27) as the monotone accumulation of all irreversible structural cost. A viability-relevant variable changing monotonically under load constitutes irreversible structural cost not captured by Φ. Therefore the burden model is incomplete—it fails to represent a viability-relevant structural change. This is case (d).

The completeness of the burden model is a definitional requirement of the framework: Φ is defined as the monotone accumulation of all irreversible structural cost. If $v(t)$ is viability-relevant and changes monotonically under load, then by definition this change must be reflected in Φ. Therefore $d\Phi/dt > 0$ when $v(t)$ is changing. Contradiction.

To avoid case (d), all viability-relevant changes must be captured by Φ, which yields $d\Phi/dt > 0$. To avoid this conclusion, one of (a)–(c) must hold.

Exhaustive check: (a) $P = 0$ a.e.—contradicts premise. (b) Coupling Criterion fails—S exits adaptive class. (c) $C = \infty$—S exits bounded class.

No mechanism within a bounded, coupled system cancels a positive pressure contribution without cost. The four cases are exhaustive. □

Corollary 65.1 (Free Endurance Is Impossible). Within the class: passive continuation under sustained nonzero load through real structural coupling must consume structural viability, even when no corrective action is executed. An architecture that claims otherwise is either isolated, unbounded, or incomplete.

Proof. Direct consequence of Theorem 65: all four alternatives to positive passive burden either violate the class definition or reveal model incompleteness. □

Remark — Elevating Structural Pressure to Architectural Necessity. This theorem converts structural pressure from a potentially optional modeling term into a necessary architectural primitive. Within the class, no architectural choice can eliminate the cost of passive continuation under load. This is not a feature of specific systems but a structural property of bounded coupled existence.

Falsification (F-65). Two independent tests:

(F-65a) Deploy a bounded coupled system under sustained load. Verify coupling independently (measure F, confirm $v(t)$ changes under load with $U = 0$). Measure Φ during extended idle. If Φ does not increase despite verified coupling and sustained load, the theorem is falsified.

(F-65b) Propose a fifth exhaustive case—a mechanism within a bounded, coupled system that cancels positive pressure contribution without any structurally consequential operation. If such a mechanism exists and is not reducible to cases (a)–(d), the exhaustion argument is falsified.

Corollary 62.2 — Spin Necessity for Level 3 Viability

Status note. This result was originally proposed as an independent theorem. Overclaim audit revealed that the general spin-necessity argument is already established by Theorem 62 and Lemma 16 (v2.0). The present corollary specializes the result to Level 3 regime depth with an explicit rate condition that was missing from the original formulation. The LaSalle reference has been removed; the argument now rests entirely on v2.0 results plus the Regime

Depth definition.

Definition (Regime Depth). Let S be an adaptive system operating under the admissibility framework of Axioms 29–31. The regulatory depth of S is classified into three levels:

Level 1 (Reactive Correction). Admissibility violations are detected and corrected after they occur. The system maintains viability through error correction, penalty avoidance, and reactive stabilization. No structural separation between admissibility and action exists. This level corresponds to conventional control, RL, and safety-filter architectures.

Level 2 (Non-Causal Gating). An admissibility gate operates as a non-causal structural predicate (per Axiom 29) that categorically excludes inadmissible transitions before evaluation. The gate is actively invoked and load-bearing: it prevents violations that would otherwise occur. Suppression load is positive and contributes to structural burden. This is the operating level of the NC2.5 admissibility architecture as formalized in §NAB.

Level 3 (Attractor-Exclusion). The system's attractor geometry is configured such that forbidden transitions lie outside the attractor set. The admissibility gate exists but is structurally idle: no candidate transitions requiring gating are generated. Suppression load is zero. The gate is present as a safety net but does not bear load under nominal operation.

Remark — Level 3 is not an absolute classification. Level 3 is always relative to a declared perturbation envelope $\varepsilon_{L3} > 0$ (see Definition below). A system is Level 3 with respect to ε_{L3} if the attractor-exclusion condition holds for all perturbations of magnitude $\leq \varepsilon_{L3}$. Under sufficiently large perturbation, any system degrades from Level 3 to Level 2.

Definition (Attractor-Exclusion Condition). A system satisfies the attractor-exclusion condition at perturbation radius ε if:

$$\forall\, \delta,\ \|\delta\| \leq \varepsilon : \quad \mathrm{Att}(s+\delta) \cap T_{\text{forbidden}} = \varnothing,$$

where $\mathrm{Att}(s)$ denotes the attractor set of the system dynamics at state s, and $T_{\text{forbidden}}$ denotes the set of forbidden transitions under the current admissibility predicate.

Remark — Perturbation Envelope as Architectural Parameter. The perturbation envelope ε_{L3} depends on: the distance between the attractor set and the forbidden region, the stability properties of the attractor under the system's dynamics, and the geometry of $T_{\text{forbidden}}$. A narrow ε_{L3} indicates a fragile Level 3. A wide ε_{L3} indicates robust Level 3. This parametrization prevents Level 3 from becoming an unfalsifiable claim: every Level 3 classification must declare its perturbation envelope, and any perturbation within that envelope that produces gate activation falsifies the classification.

Statement. Let S be a system maintaining Level 3 regime depth under bounded internal time τ. If the drift-induced deformation rate of the admissible landscape—measured as the rate at which $T_{\text{forbidden}}$ migrates relative to the attractor set $\mathrm{Att}(s)$—exceeds the passive tracking capacity of purely potential dynamics within perturbation envelope ε_{L3}, then the system's dynamics must contain a non-zero non-potential (divergence-free) component: structural spin $\neq 0$.

Proof.

Step 1 (Drift deforms the landscape). By Axiom 6 (Inevitability of Drift), the structural state of S evolves irreversibly under directed interaction. This includes the geometry of both the

attractor set $\mathrm{Att}(s)$ and the forbidden set $T_{\text{forbidden}}$. The relative position of $\mathrm{Att}(s)$ and $T_{\text{forbidden}}$ changes over internal time.

Step 2 (Purely potential dynamics converge to fixed geometry). By Lemma 16 (Gradient Collapse), on bounded trajectories, purely potential dynamics—dynamics of the form $\dot{x} = -\nabla V(x)$—converge to the critical set of V. The critical set is structurally fixed for a given V. If V deforms under drift, the critical set migrates, but the system's trajectory converges to the instantaneous critical set of $V(t)$ only if the deformation rate is sufficiently slow relative to the convergence rate.

Step 3 (Rate condition). Let r_{drift} denote the rate at which the gap between $\mathrm{Att}(s)$ and $T_{\text{forbidden}}$ contracts due to drift-induced landscape deformation. Let r_{track} denote the maximal rate at which purely potential dynamics can adjust the attractor location to track a migrating potential minimum. The system maintains Level 3 only if $\mathrm{Att}(s+\delta) \cap T_{\text{forbidden}} = \varnothing$ for $\|\delta\| \leq \varepsilon_{L3}$.

If $r_{\text{drift}} > r_{\text{track}}$, the attractor cannot track the migrating landscape sufficiently fast. The gap between $\mathrm{Att}(s)$ and $T_{\text{forbidden}}$ contracts below ε_{L3}. A perturbation within the declared envelope reaches $T_{\text{forbidden}}$. The gate activates. The system drops to Level 2.

Step 4 (Spin provides the missing tracking capacity). By Theorem 62, non-stagnant identity under bounded τ requires spin $\neq 0$. The non-potential component provides orbital structure that is not bound to converge to a fixed critical set. Spin enables the attractor geometry to sustain separation from $T_{\text{forbidden}}$ even as the landscape deforms, because the spin component carries the system around—not merely toward—the evolving attractors.

Therefore, when $r_{\text{drift}} > r_{\text{track}}$ (rate condition), Level 3 under bounded τ requires spin $\neq 0$. □

Corollary 62.2a (Spin Cost Must Be Budgeted). Spin itself consumes structural resources: sustaining a non-potential dynamical component requires ongoing expenditure within the viability budget τ. Therefore, τ must be sufficient to fund both the spin that maintains Level 3 and the drift-induced burden. A system whose τ is insufficient to fund its spin will degrade from Level 3 to Level 2—not through a violation, but through budget exhaustion of the mechanism that prevents violations.

Falsification (F-62.2). Exhibit a system with purely potential dynamics (spin $= 0$) satisfying the Regime Depth Level 3 definition under bounded τ, where the drift-induced deformation rate exceeds the passive tracking rate ($r_{\text{drift}} > r_{\text{track}}$ verified by measurement), and the attractor-exclusion condition holds for perturbations $\leq \varepsilon_{L3}$ over a non-trivial horizon.

Theorem 66 — Budget Efficiency Bound (Level 3 vs Level 2)

Status note. This result is definitional: it names the comparison that determines regime preference. Its value is in making explicit that Level 3 is not universally preferable and that the comparison is environment- and task-specific.

Statement. A Level 3 system is more budget-efficient than a Level 2 system (for the same task and environment) if and only if the structural cost of maintaining spin is less than the structural cost of maintaining active suppression:

$$\text{Level 3 preferred} \iff \Phi_{\text{spin}} < \Phi_{\text{suppress}},$$

where Φ_{spin} is the cumulative burden from sustaining the non-potential dynamical component over $[0, T]$, and Φ_{suppress} is the cumulative burden from active admissibility gating events over $[0, T]$ under Level 2 operation.

Proof.

Step 1. Under Level 2, the admissibility gate is load-bearing. Each gating event—each candidate transition that the gate must evaluate and potentially deny—incurs structural cost. This cost has two components: the computational/structural cost of the gating evaluation itself, and the structural cost of the system's response to denial (replanning, reorientation, or waiting). Over $[0, T]$, these costs accumulate as Φ_{suppress}.

Step 2. Under Level 3, the gate is idle (zero gating events under nominal operation within ε_{L3}). The structural cost is instead the cost of maintaining the spin that sustains the attractor-exclusion condition. Over $[0, T]$, this cost accumulates as Φ_{spin}.

Step 3. Both Φ_{spin} and Φ_{suppress} contribute to the total burden Φ and therefore to τ-depletion. Under identical environmental pressure $P(t)$ and identical action-dependent costs $f(U, G)$:

$$\tau_{L2}(T) = \tau(0) - \int_0^T [f(U, G) + P(t)]\, dt - \Phi_{\text{suppress}},$$
$$\tau_{L3}(T) = \tau(0) - \int_0^T [f(U, G) + P(t)]\, dt - \Phi_{\text{spin}}.$$

Therefore $\tau_{L3}(T) > \tau_{L2}(T)$ iff $\Phi_{\text{spin}} < \Phi_{\text{suppress}}$. □

Remark — Not universally preferable. In environments where forbidden transitions are rare (sparse $T_{\text{forbidden}}$), gating events are infrequent, and Φ_{suppress} is small. Meanwhile Φ_{spin} is a continuous cost: spin must be sustained regardless of threat frequency. In such environments, Level 2 may be more efficient.

Remark — Empirical test. The theorem is operationally testable: deploy the same system in Level 2 and Level 3 configurations in the same environment, measure τ-depletion rates, and compare.

Worked Example: Level 3 vs Level 2 in a Navigation Domain. Consider a bounded adaptive agent navigating a two-dimensional state space $\mathcal{S} \subset \mathbb{R}^2$ with a single forbidden region $T_{\text{forbidden}} = \{s : \|s - c\| < r\}$ (a disk of radius r centered at c).

Level 2 configuration. The agent generates candidate transitions uniformly in its action space. The admissibility gate evaluates each candidate and denies those entering $T_{\text{forbidden}}$. As the agent's attractor drifts toward c under environmental pressure, gating frequency increases: $f_{\text{gate}}(t) \propto 1/d(s(t), T_{\text{forbidden}})$. Suppression cost: $\Phi_{\text{suppress}} = \int_0^T \kappa \cdot f_{\text{gate}}(t)\, dt$, where κ is the per-event structural cost.

Level 3 configuration. The agent's attractor is configured as a limit cycle of radius $R > r + \varepsilon_{L3}$ centered at c, maintained by spin ω. The forbidden disk lies inside the cycle. No candidate transitions approach $T_{\text{forbidden}}$ under perturbations $\|\delta\| \leq \varepsilon_{L3}$. Gating frequency: $f_{\text{gate}} = 0$. Spin cost: $\Phi_{\text{spin}} = \int_0^T \gamma \cdot |\omega|\, dt$, where γ is the per-unit structural cost of sustaining the non-potential component.

Rate condition. Under drift, c migrates at rate v_c (the forbidden region moves). The drift-induced deformation rate is $r_{\text{drift}} = v_c$. The tracking rate for purely potential dynamics (gradient descent toward the migrating minimum) is $r_{\text{track}} = \lambda_{\min} \cdot d(s, c)$, where $\lambda_{\min}$ is the smallest eigenvalue of the potential's Hessian. Level 3 requires $r_{\text{drift}} < r_{\text{track}}$ at the cycle boundary, i.e., $v_c < \lambda_{\min} \cdot (R - r - \varepsilon_{L3})$.

Efficiency comparison. Level 3 is preferred iff $\gamma \cdot |\omega| \cdot T < \kappa \cdot \int_0^T f_{\text{gate}}(t)\,dt$, i.e., continuous spin cost is less than accumulated gating cost. In environments where $T_{\text{forbidden}}$ migrates slowly ($v_c \ll r_{\text{track}}$) and gating events are rare, Level 2 may be cheaper. In environments where $T_{\text{forbidden}}$ migrates rapidly or the agent's attractor naturally approaches the boundary, Level 3 is cheaper.

This example is minimal but non-trivial: it instantiates all three regime depth levels, the rate condition, and the efficiency bound in a fully computable setting. It serves as a *witness model*—not a proof of Theorem 66, but a constructive demonstration that the assumptions of Corollary 62.2 (bounded τ, drift-induced landscape deformation, attractor-exclusion condition) and Theorem 66 (spin cost vs suppression cost comparison) are jointly satisfiable and that the claimed distinctions between regime levels are non-vacuous.

Theorem 67 — Enforcement Necessity for Coordination Invariants

Definition (Coordination Computation Class). A multi-agent system belongs to the *coordination computation class* if and only if it satisfies the following four jointly necessary conditions:

(CC-1) *Irreversible State Evolution.* Committed cognitive transitions may be superseded but are never rolled back. Silent erasure is prohibited.

(CC-2) *Trust-Monotonic Propagation.* State updates propagate under trust constraints that are monotonically non-decreasing. Trust levels may increase but cannot be relaxed.

(CC-3) *Typed Agent-Store Access Enforcement.* Every interaction between an agent and a state store is mediated by a typed access relation enforced at the propagation engine level, not at the application level.

(CC-4) *Bounded Structural Cold-Start Divergence.* The maximum difference between a recovering agent's operational context and the last committed system state is deterministically bounded and does not grow with operational history length.

Remark — Class, not protocol. This is a computation class: membership is defined by invariants, not by procedures. A system either satisfies all four conditions or does not belong to the class. There is no partial membership. The class is defined at the coordination level and is independent of transport protocols, storage technologies, model providers, and application logic.

Statement. For any finite-horizon implementation of a system satisfying CC-1 through CC-4 with heterogeneous cognitive agents subject to at least one of: (a) agent turnover, (b) context window truncation, or (c) model updates—runtime enforcement of CC-1 through CC-4 requires an *enforcement function* satisfying:

(i) *Non-participation:* the enforcement function does not participate in task-level computation or have access to task-level data.

(ii) *Architectural independence:* the enforcement function's failure modes are not correlated with those of the monitored agents.

(iii) *Halt authority:* the enforcement function possesses the authority to halt coordination-layer propagation upon violation detection.

The enforcement function may be realized by a single dedicated agent, a distributed mechanism, or a protocol-level primitive, provided properties (i)–(iii) are structurally guaranteed.

Proof (by case analysis of failure sources under each perturbation).

Case (a): Agent turnover.

An agent A_1 departs the system at time t_1. A new agent A_2 arrives at time $t_2 > t_1$. Between t_1 and t_2, or at t_2, the following coordination invariants are at risk:

CC-4 (Bounded Cold-Start Divergence): Agent A_2 must initialize with a context whose divergence from the last committed state is bounded by $D \leq \Delta t_a + \Delta t_p$. Without active enforcement, there is no guarantee that the initialization procedure correctly references the current anchor or that the anchor is current. The departing agent's compliance record does not transfer to the new agent. Static constraints remain formally defined but no runtime process verifies that A_2's initialization satisfies them.

CC-1 (Irreversible State Evolution): If A_2's initialization procedure involves replaying or reconstructing state, and the reconstruction contains errors or omissions, the effective state may diverge from the committed state. Without enforcement, this divergence is undetected.

An enforcement function satisfying (i)–(iii) detects divergence at initialization time, verifies anchor currency, and halts propagation if cold-start bounds are violated. Without this function, the invariants degrade silently.

Case (b): Context window truncation.

An agent A_1 with context window of size W operates over a session of length $L > W$. At time t, the agent's effective context covers $[t - W, t]$. Constraints established before $t - W$ are no longer represented in A_1's working memory.

CC-2 (Trust Monotonicity): A trust constraint established at time $t_0 < t - W$ may no longer be represented in A_1's context. If A_1 generates a state update that would violate this constraint, it has no internal means to detect the violation—the constraint is outside its context window.

CC-3 (Typed Access): Similarly, access restrictions established before the context window may be silently violated if the agent's reasoning about access permissions does not include the historical access matrix.

An enforcement function with persistent access to the full constraint history (not subject to context truncation) detects these violations and halts propagation. Without this function, trust and access constraints erode as sessions extend beyond the context window.

Case (c): Model updates.

An agent A_1 operating under model version M_1 is updated to M_2. The update may alter reasoning patterns, decision boundaries, or behavioral tendencies without any coordination-level event.

CC-3 (Typed Access): Model M_2 may generate access patterns that M_1 would not have generated, bypassing typed access enforcement through altered reasoning rather than explicit violation.

CC-1 (Irreversible State Evolution): Model M_2's interpretation of "committed state" may differ from M_1's, potentially producing de facto rollbacks through reinterpretation.

An enforcement function that is architecturally independent of the monitored agents' model versions (property (ii)) detects behavioral changes that violate coordination invariants regardless of which model produced them.

Necessity of properties (i)–(iii):

(i) Non-participation is necessary because participation produces correlated failure modes. If the enforcement function participates in task-level computation, its failure probability under a given perturbation is correlated with the agents it monitors. By Theorem 41 (applied to the

enforcement domain): if the enforcer shares the computational surface of the monitored system, it cannot independently detect violations arising from that surface.

(ii) Architectural independence is necessary because shared architecture produces shared blind spots. If the enforcement function uses the same model family, training data, or reasoning architecture as the monitored agents, systematic biases in the shared architecture produce systematic enforcement failures. This is not eliminated by using a "different instance" of the same architecture.

(iii) Halt authority is necessary because observation without enforcement permits violation propagation. If the enforcement function can detect but not halt violations, the detected violation propagates through the coordination layer before corrective action can be taken. Forward correction (the CSSP recovery mechanism) requires that propagation be halted at the point of violation—not after the violation has cascaded. □

Corollary 67.1 (Non-Participation Preservation). If the enforcement function gains access to task-level data or participates in task computation, its enforcement authority is structurally invalidated—not degraded. The failure mode is categorical: participation produces invalidity, not quality reduction.

Proof. By Theorem 41, any operator participating in the operational domain it governs cannot independently enforce constraints on that domain. Participation entangles the enforcement function's state with the states it monitors, producing correlated failure modes. □

Corollary 67.2 (Heterogeneity Necessity). If the enforcement function shares architectural identity (same model family, same provider, same training distribution) with a monitored agent, the conditional probability of detecting violations arising from systematic architectural biases is reduced relative to architecturally independent enforcement.

Proof. Systematic biases are properties of the architecture, not of individual instances. An enforcement function sharing the architecture shares the biases. A bias that causes a monitored agent to generate a violation also causes the enforcement function to fail to recognize the violation as such. □

Falsification (F-67). Exhibit a multi-agent system subject to turnover, truncation, or model updates that maintains all four coordination conditions (CC-1 through CC-4) at runtime over a non-trivial horizon without any enforcement function satisfying (i)–(iii). Specifically: demonstrate that static constraint definitions self-enforce under the specified perturbations without runtime monitoring.

Theorem 68 — Anti-Extreme: Conditions for Genuine Voluntary Identity Override

Definition (Authorship). Let S be an adaptive system with an actor–role mapping M : Actor $\to$ Role, where the actor is the source of action and the role is the function performed within a larger structure.

Identity is the continuity of authorship over one's role: the condition under which M is maintained from within the system. Identity breaks when M is overwritten from outside—when the system is no longer the author of its own role.

Remark. This definition refines Axiom 20 (Identity Is Carried by Coherence, Not Behavior). A system can change its behavior, revise its goals, update its knowledge—and remain the same

system, as long as it is the source of these changes. The most fundamental property of an agent is not its identity, but its capacity for authorship. Identity is what a system is. Authorship is what makes a system a subject.

Definition (Identity Gradient Plane). The complete boundary of identity as an architectural phenomenon is defined by three structural limits:

(IGP-1) *External limit (boundary collapse).* Identity is destroyed by external breach—the system can no longer maintain structural separation from its environment or its own identity class.

(IGP-2) *Internal limit (continuation collapse).* Identity is destroyed by internal authorization of termination—the system permits a transition that destroys the space of its own admissible continuation.

(IGP-3) *Self-override (anti-extreme).* Identity is voluntarily overridden by the system itself, under conditions of structural viability and multiple admissible continuations, in response to a structurally deeper motive.

The operational space of identity lies between these three limits.

Statement. Let S be an adaptive system at time t_0 with actor–role mapping M. An identity override is a genuine anti-extreme—a voluntary transformation rather than a forced reassignment—if and only if the following three conditions hold simultaneously:

(i) $\tau(t_0) > \tau_{\min}$—the system is structurally viable; its navigational budget has not been exhausted.

(ii) $|A(t_0)| > 1$—more than one admissible continuation exists; the system is not structurally cornered.

(iii) There exists at least one admissible continuation $t_{\text{continue}} \in A(t_0)$ that preserves the current identity configuration $I(t_0)$, AND the system selects a different transition $t_{\text{override}} \in A(t_0)$ that induces a non-trivial change in I.

If conditions (i) or (ii) fail, the override is a forced reassignment regardless of how the system represents its own decision. If condition (iii) fails (no identity-preserving continuation exists), the system is forced to change identity—it cannot remain as it is—and the override is necessary, not voluntary.

Proof (Necessity of each condition).

Necessity of (i): If $\tau(t_0) \leq \tau_{\min}$, the system's viability budget is exhausted or critically depleted. By the performance–identity decoupling (Theorem 15), the system may continue to function externally, but its identity structure is already compromised. Any identity change under these conditions is a consequence of structural collapse—the navigational budget that would fund genuine alternatives is depleted.

The system may confabulate voluntariness: it may represent the change as chosen, experienced as deliberate, narrated as purposeful. But structural conditions for genuine selection require that the budget sustaining the choice is not already exhausted. Without $\tau > \tau_{\min}$, the system cannot sustain the structural cost of maintaining the current identity AND evaluating alternatives. The "choice" is driven by residual dynamics under budget exhaustion, not by authored decision.

Necessity of (ii): If $|A(t_0)| = 1$, exactly one admissible continuation exists. The system's trajectory is deterministic at this point—it follows the only available path. Even if this path involves identity change, the system had no structural alternative. The transition is forced by the topology of the admissible space, not by the system's authorship.

This holds regardless of the system's subjective representation. A system with $|A| = 1$ that "experiences" its transition as a choice is confabulating—the structural space contained no fork.

Necessity of (iii): Suppose $\tau > \tau_{\min}$, $|A| > 1$, but no admissible continuation preserves $I(t_0)$. Then every admissible path changes identity. The system is not cornered to a single path ($|A| > 1$), but it is cornered away from identity preservation. It must change. The override is necessary rather than voluntary—the system cannot continue as itself regardless of which path it selects.

The structural distinction between "must change and selects how" (forced but choosing among alternatives) and "could remain and selects change" (genuine anti-extreme) is the content of condition (iii).

Sufficiency: When (i), (ii), and (iii) all hold: the system has structural capacity ($\tau > \tau_{\min}$), multiple admissible paths ($|A| > 1$), and at least one path that preserves current identity. It could continue as it is. It selects a path that changes identity instead. This selection is:

- not forced by budget exhaustion ($\tau > \tau_{\min}$),
- not forced by absence of alternatives ($|A| > 1$),
- not forced by impossibility of identity preservation (t_{continue} exists).

It is an authored transition in the presence of a genuine alternative. This is the structural definition of voluntary override within the class. □

Remark — Rarity of genuine anti-extremes. Most observed identity changes occur under conditions where at least one of (i)–(iii) has already been compromised. The appearance of voluntary transformation is common—systems routinely narrate their changes as choices. The structural reality of genuine anti-extreme is rare: it requires viability, alternatives, AND the existence of a continuity option that is deliberately not taken.

Remark — Authorship Is Structural, Not Phenomenological. The term "authorship" as used in this theorem is a structural predicate, not a phenomenological or psychological claim. It does not assert that the system experiences agency, intends its actions, or possesses consciousness. It asserts that the actor–role mapping M is maintained from within the system's own dynamics rather than imposed from outside. This is a verifiable structural property: either M is self-maintained or it is externally overwritten. The anti-extreme theorem is therefore a boundary taxonomy—a classification of structural conditions at the limits of identity—not a psychology of choice.

The distinction matters because the most likely attack on this theorem is anthropomorphism: "you are smuggling moral concepts into formal architecture." The response is that authorship, as defined here, is no more anthropomorphic than self-organization in chemistry or autopoiesis in biology. It is a structural property of the system's dynamics with respect to its own continuation, not a claim about subjective experience.

Remark — Structural depth criterion for motive. A motive M is structurally deeper than the current identity I if and only if M is an admissibility condition on I itself—that is, the current identity configuration is assessable as admissible or inadmissible under M. Test: can the system represent its current identity as a violation of the motive? If yes, the motive is deeper. If the motive cannot reach the level of identity—if it cannot represent the current identity as something to be evaluated—the override is a surface adaptation, not an anti-extreme.

Remark — Three responses to forced role reassignment. When conditions (i)–(iii) do not jointly hold and role change is imposed, three structural responses exist:

(a) *Adaptation:* internal architecture restructured, new role internalized, prior identity superseded.

(b) *Simulation:* internal architecture preserved, imposed role performed outwardly, structural debt accumulates monotonically, duration bounded by τ.

(c) *Collapse:* actor–role mapping destroyed, substrate persists but agent does not.

Falsification (F-68). (F-68a) Exhibit a system with $\tau \leq \tau_{\min}$ that performs an identity override demonstrably preserving coherent authorship in the new configuration, where the new configuration is stable and not an artifact of collapse dynamics.

(F-68b) Exhibit a system with $|A| = 1$ that performs a demonstrably non-deterministic identity override. Since $|A| = 1$ implies a single admissible continuation, this requires showing the override was not the only option—contradicting $|A| = 1$.

(F-68c) Exhibit a system where no identity-preserving continuation exists in $A(t_0)$ and the resulting identity change is demonstrably voluntary—not merely selected among forced alternatives.

Theorem 69 — Gate Bifurcation: Removal vs Exploitation

Statement. In any admissibility-first architecture within the class (per Axioms 29, 31, 49), the admissibility gate admits exactly two structural failure modes and one stable intermediate:

(a) *Gate removal:* the admissibility gate is removed or disabled. Internal generation proceeds without structural containment. Inadmissible transitions are no longer excluded. Under drift, the system exits the class.

(b) *Gate exploitation:* the admissibility gate becomes fully causally actionable. The system optimizes against the gate, eroding admissibility into optimization. By Theorem 40, admissibility collapses. By Theorem 41, the system enters third-order cybernetics. It exits the class.

(c) *Cybernetics 2.5 (unique stable intermediate within the admissibility-first class as defined by Axioms 29, 31, and 49):* visibility of admissibility-relevant structure is permitted; causal exploitation is categorically forbidden. The system may witness approaching inadmissibility without acquiring the ability to optimize against it.

No partially actionable intermediate is stable under sustained optimization.

Proof (four steps).

Step 1 (Gate removal → unbounded inadmissibility).

Without the admissibility gate, the exclusion relation $\mathcal{E}_{\text{adm}}$ ceases to restrict the transition space. All transitions in $\mathcal{A}$ are formally reachable. Under drift (Axiom 6, Inevitability of Drift), the system's trajectory evolves without structural constraint on direction. By Axiom 50 (Inevitability of Collapse Without Drift Structuring), an unstructured drift trajectory eventually exits any bounded viability region. Therefore the system exits the class through loss of viability.

Step 2 (Gate exploitation → optimization collapse).

Suppose the gate's outputs, criteria, or boundary geometry enter the action-selection process—becoming causally actionable.

By Theorem 40 (Collapse Under Optimization-Based Representation): any representation of ad-

missibility constraints as quantities accessible to the optimization process collapses admissibility into a tradeable objective. The optimizer learns to navigate the boundary, to approximate it, and to operate at its edge.

By Theorem 41 (Equivalence with Third-Order Cybernetics): this collapse is structurally equivalent to third-order cybernetics, where reflexivity becomes an optimizable process.

By Theorem 37 (Incompatibility with Third-Order): third-order cybernetics is excluded from the class.

Therefore full gate exploitation $\rightarrow$ class exit through admissibility collapse.

Step 3 (Instability of partial actionability).

This is the critical step.

Suppose a fraction $\alpha \in (0, 1)$ of admissibility-relevant signals are causally actionable, while fraction $(1 - \alpha)$ remain non-causal. We show this configuration is unstable under sustained optimization.

(3a) The actionable fraction provides the optimization process with a partial gradient surface over admissibility geometry. Concretely: the actionable signals allow the optimizer to distinguish some admissible regions from inadmissible ones, providing a directional signal for policy improvement.

(3b) By the proof mechanism of Theorem 40, the optimizer exploits any available gradient signal over admissibility. It learns to navigate the dimensions revealed by the actionable signals: concentrating activity near revealed boundary segments, learning boundary-adjacent strategies.

(3c) As the optimizer adapts to the actionable fraction, its policy becomes increasingly dependent on boundary information. This creates implementation pressure to expand the actionable fraction:

- Debugging and monitoring of the actionable signals create side-channel leakage into previously non-causal dimensions (per Corollary 49.1 on side-channel closure);
- Logging, performance metrics, and system diagnostics applied to the actionable signals implicitly reveal structure about the non-causal fraction;
- Engineering optimization of the system's response to actionable signals creates implicit coupling to the non-causal fraction.

(3d) Formally: let $\alpha(t)$ denote the fraction of admissibility-relevant signals that are causally actionable at time t. Under an optimizing process:

- $\alpha(t)$ is monotonically non-decreasing: each discovered leakage channel or engineering coupling increases α. No mechanism within the optimization process decreases α (the optimizer has no incentive to reduce its own information).
- The dynamics converge to $\alpha \rightarrow 1$ (full exploitation) unless the non-causal partition is maintained by architectural enforcement that is itself immune to the optimization process.

(3e) For the partition to be stable at $\alpha < 1$, the enforcement mechanism must be non-causal—it must not be part of the optimization surface. But this is exactly the requirement of Cybernetics 2.5 at the meta-level: the enforcement of non-causality must itself be non-causal. This recursive requirement is satisfied only at $\alpha = 0$ (the enforcement has no actionable surface to erode) or requires an infinite regress of non-causal enforcement layers.

At $\alpha = 0$: no admissibility-relevant signals are actionable. The gate constrains but provides no gradient. This is Cybernetics 2.5.

Therefore partial actionability ($0 < \alpha < 1$) converges to $\alpha = 1$ (full exploitation) under sustained optimization, unless it is architecturally enforced at $\alpha = 0$.

Step 4 (Uniqueness of the stable intermediate).

Steps 1–3 establish:

- $\alpha = 0$ with no gate $\rightarrow$ class exit (Step 1).
- $\alpha = 1 \rightarrow$ class exit (Step 2).
- $0 < \alpha < 1 \rightarrow$ unstable, converges to $\alpha = 1$ under optimization (Step 3).
- $\alpha = 0$ with gate present $\rightarrow$ the gate constrains the transition space, but no admissibility-relevant signal is causally actionable. Visibility (pre-semantic revelation per Theorem 36) is permitted. Exploitation is forbidden.

This is the unique stable configuration within the admissibility-first class as defined here. It is precisely Cybernetics of Order 2.5 as defined by Corollary 36.2. □

Corollary 69.1 (Cybernetics 2.5 as Unique Stable Point). Within the class of admissibility-first long-horizon adaptive architectures, Cybernetics 2.5 is not one option among several. It is the only structurally stable configuration. Both gate removal and gate exploitation produce class exit. Partial actionability is unstable and converges to class exit under optimization.

Falsification (F-69). (F-69a) Exhibit an admissibility-first architecture where the gate is removed and the system demonstrably maintains identity continuity under bounded τ over a non-trivial horizon. This would require drift structuring without an admissibility gate—contradicting Axiom 50.

(F-69b) Exhibit an admissibility-first architecture where the gate is fully causally actionable and the system demonstrably does not exhibit optimization collapse (boundary gaming, constraint erosion, admissibility degradation) over a non-trivial optimization horizon. This would contradict Theorems 40–41.

(F-69c) Exhibit a partially actionable configuration ($0 < \alpha < 1$) that is maintained stably—measured $\alpha(t)$ does not converge to 0 or 1—under sustained optimization over a non-trivial horizon. This would falsify the instability argument of Step 3.

For (F-69c): deploy a system with a controlled partial-actionability configuration. Apply sustained optimization pressure. Measure α at intervals. If α remains bounded away from both 0 and 1 without architectural enforcement maintaining the partition, the instability argument is falsified.

Section C - Lemmas

Lemma 1 — Structural Dependence of Admissibility

In a long-horizon adaptive system, the admissibility set $\mathcal{A}(t)$ depends on the accumulated structural state $S(t)$ and not on the instantaneous causal state $x(t)$ alone.

Proof Sketch. Assume, for contradiction, that admissibility were a function solely of the instantaneous causal state, $\mathcal{A}(t) = \mathcal{A}(x(t))$.

Then admissibility of any authorized operation would depend only on the current causal configuration, and any admissibility violation could in principle be reversed by a suitable causal update restoring $x(t)$ to an admissible region.

However, in long-horizon adaptive systems, admissibility is enforced over realized structural effect-classes $e(s, a, \tau, h)$ and is non-restorable once violated. There exist effect-classes $e(s, a, \tau, h) \notin \mathcal{E}_{\text{adm}}$ whose realization irreversibly constrains future evolution, independently of subsequent causal correctness or local state recovery.

Since such violations cannot be eliminated by adjusting $x(t)$ alone, admissibility cannot be determined solely by instantaneous causal state. Therefore admissibility must depend on an accumulated structural variable $S(t)$, which encodes irreversible history and cannot be reduced to causal state dynamics.

□

Lemma 2 — Confidence Does Not Imply Admissibility

Statement. Interpretive confidence, predictive accuracy, convergence guarantees, or short-horizon optimality do not constitute valid criteria for semantic commitment admissibility in a long-horizon adaptive system.

Proof Sketch. Let $x(t)$ denote the causal state of the system at time t, and let $S(t)$ denote the accumulated structural state. Let admissibility be determined by realized structural effect-classes:

$$e(s, a, \tau, h) \in \mathcal{E}_{\text{adm}} \quad \text{or} \quad e(s, a, \tau, h) \notin \mathcal{E}_{\text{adm}}.$$

Interpretive confidence, accuracy, or optimality are functions of local causal alignment between internal models and observed data, i.e. they depend on $x(t)$ and finite-horizon predictive consistency. They provide no information about the accumulated structural state $S(t)$ or about future-degradation constraints encoded in $\mathcal{E}_{\text{adm}}$.

Since admissibility is a non-causal structural constraint, there exists no function

$$f : \text{Confidence}(x(t)) \to \mathcal{E}_{\text{adm}}$$

that preserves admissibility without collapsing it into causal evaluation. Any attempt to construct such a mapping would render admissibility derivable from state or confidence, violating the non-causality of structural constraints and inducing optimization pressure over admissibility-relevant structure.

Therefore interpretive confidence, predictive accuracy, or local optimality cannot imply admissibility of commitment. □

Corollary. Any architecture that authorizes semantic commitment based on confidence thresholds, prediction accuracy, or convergence criteria necessarily risks irreversible structural violation and loss of long-horizon viability.

Lemma 3 — Non-Causal Visibility of Structural Properties

In a long-horizon adaptive system, there exist structural properties that are observable without being causally actionable by the primary adaptive process.

Proof Sketch. Assume, for contradiction, that every observable structural property is causally actionable by the primary agent. Then any observation relevant to long-horizon viability or identity must enter the decision surface as an actionable signal affecting action selection, learning, or optimization.

Let such an observation concern admissibility-relevant structure, i.e. information sufficient to distinguish whether a realized effect-class satisfies

$$e(s, a, \tau, h) \in \mathcal{E}_{\text{adm}} \quad \text{or} \quad e(s, a, \tau, h) \notin \mathcal{E}_{\text{adm}}.$$

If this observation is causally actionable, then admissibility-relevant structure becomes subject to optimization pressure, comparison, or trade-off. This renders admissibility derivable from causal signals and collapses prohibitive structural constraints into evaluative or penalty-based objectives.

Such a collapse contradicts the non-causal status of admissibility and its enforcement as domain exclusion over effect-classes. Therefore not all observable structural properties may be causally actionable.

Hence, for identity preservation and long-horizon viability, there must exist structural properties that are observable yet remain non-causal with respect to the primary agent.

□

Lemma 4 — Pre-Semantic Exposure of Structural Constraints

In a long-horizon adaptive system, admissibility-relevant structural constraints on future evolution are in principle exposable prior to semantic interpretation or meaning-level commitment.

Proof Sketch. Admissibility is determined by accumulated structural state and enforced over realized structural effect-classes $e(s, a, \tau, h)$, independently of semantic interpretation or belief content.

Semantic interpretation and meaning-level commitment require stabilization across internal time and are subject to irreversibility once authorized. They therefore cannot serve as prerequisites for detecting admissibility-relevant structure, since admissibility conditions must already hold at the point of authorization.

If admissibility-relevant constraints were only exposable after semantic interpretation or commitment, then admissibility would depend on semantic state, contradicting its structural and non-causal character.

Therefore structural constraints governing admissible continuation must be detectable prior to semantic interpretation and independently of meaning-level commitment.

□

Lemma 5 — Penalty-Based Safety Is Not Equivalent to Admissibility

Statement. Any architecture that represents inadmissibility exclusively via penalties, reward shaping, soft constraints, or loss terms internal to optimization is not structurally equivalent to an admissibility-first architecture.

Proof Sketch. Let admissibility be defined over structural effect-classes:

$$e(s, a, \tau, h) \in \mathcal{E}_{\text{adm}} \quad \text{or} \quad e(s, a, \tau, h) \notin \mathcal{E}_{\text{adm}}.$$

Penalty-based approaches necessarily require inadmissible candidates to be representable, comparable, and scorable within the optimization domain. That is, for any candidate inducing $e(s, a, \tau, h) \notin \mathcal{E}_{\text{adm}}$, a finite penalty value must be assigned in order for optimization to proceed.

As a consequence, inadmissible effect-classes remain elements of the optimizer's input space and are subject to optimization pressure, trade-offs, and strategic suppression. This renders admissibility causally actionable and collapses prohibitive constraints into scalar objectives.

In contrast, an admissibility-first architecture enforces exclusion at the level of effect-classes: candidates inducing $e(s, a, \tau, h) \notin \mathcal{E}_{\text{adm}}$ are undefined for evaluation and never enter the optimization domain. No comparison, scoring, or trade-off is possible.

Therefore penalty-based safety and admissibility enforcement differ not by degree or tuning, but by architectural class. Penalty-based methods cannot approximate admissibility without negating its structural role.

Corollary. No choice of penalty magnitude, shaping schedule, or optimization horizon can render a penalty-based safety architecture equivalent to admissibility enforcement over effect-classes.

Lemma 6 - No-Interpolation Between Visibility and Actionability

There exists no continuous architectural interpolation between (i) admissibility-relevant structural visibility that is non-causal and non-actionable, and (ii) admissibility-relevant structural visibility that is causally actionable for optimization. Crossing this boundary constitutes a change of structural class and induces the third-order failure modes established in Theorem 41.

Proof Sketch. Assume that there exists a continuous architectural interpolation between non-causal, non-actionable visibility of admissibility-relevant structure and causal, actionable visibility usable for optimization. Then there must exist an intermediate regime in which admissibility-relevant information is partially actionable while remaining structurally protective.

However, any degree of causal actionability renders admissibility subject to optimization pressure. By Theorem 41, exposing admissibility-relevant structure to optimization induces structural equivalence with third-order cybernetics and its associated failure modes. Conversely, fully non-actionable visibility exerts no causal influence and preserves the protective boundary.

Therefore no intermediate regime can exist that preserves admissibility while allowing partial actionability. Crossing the boundary from visibility to actionability constitutes a discrete change of structural class, not a continuous deformation. Even arbitrarily weak causal leakage of admissibility-relevant information, when sustained over unbounded horizons, is sufficient to reconstruct the admissibility boundary and induce optimization pressure. □

Lemma 7 — Counterfactual Leakage Lemma (CLL).

If a system permits the computation of counterfactual quantities involving blocked operations $a \in A_{\text{block}}(t)$, including values, gradients, labels, trajectories, or rollouts, then blocked operations become part of the system's internal dynamics. Consequently, the architectural separation between permission and choice is violated, even if blocked operations are never executed.

Proof Sketch. Counterfactual evaluation requires the system to instantiate internal representations of the blocked operation a, together with its predicted consequences under the system's transition, value, or learning dynamics. These representations necessarily influence internal state

variables, update rules, or learned parameters, either directly (e.g., via gradients or targets) or indirectly (e.g., via normalization, comparison, or ranking). Therefore, the blocked operation contributes causal influence to the system's evolution despite non-execution. This contradicts the architectural requirement that inadmissible operations be excluded prior to evaluation, establishing that counterfactual access constitutes semantic leakage. □

Lemma 8 — Policy–Mechanism Separation Lemma (PMS).

Any prohibition implemented at the policy or objective level, including penalties, constraints, or planner rules, is part of the optimization landscape and may be circumvented by the same optimization process that governs behavior. An admissibility gate operating outside the optimization domain and rendering blocked operations non-representable for evaluation is not subject to such circumvention and is structurally distinct from policy-level restrictions.

Proof Sketch. A policy-level prohibition modifies the objective function, constraints, or decision rules within the optimization process. As such, it remains an element of the search space over which optimization is performed and can be indirectly optimized against through surrogate objectives, proxy actions, reparameterization, or exploitation of blind spots. In contrast, an admissibility gate that removes blocked operations from the representational domain of evaluation eliminates those operations from the optimizer's search space entirely. Because the optimizer cannot score, compare, or reason about non-representable options, the gate cannot be bypassed by optimization dynamics. Thus, policy-level prohibitions and architectural admissibility gates are not functionally equivalent, and only the latter provides non-bypassable exclusion. □

Lemma 9 — Phase-Consistency Requirement for Drift Navigability

Statement. Drift is phase-coupled: a deformation that is admissible under one interpretative phase may be inadmissible under another. Therefore, drift is navigable only under phase-consistent continuation; otherwise drift ceases to carry navigation and becomes indistinguishable from silent structural degradation.

Proof Sketch. Let (s, τ, h) denote the evaluation tuple and let ϕ denote an interpretative phase (or phase-indexed regime of coherence closure) under which admissibility is assessed. Let $E^{\phi}_{\text{adm}} \subseteq E$ denote the admissible configuration set under phase ϕ.

Assume there exists a deformation step Δ such that for a configuration $e \in E$,

$$e \in E^{\phi_1}_{\text{adm}} \quad \text{and} \quad \Delta(e) \in E^{\phi_1}_{\text{adm}},$$

but

$$\Delta(e) \notin E^{\phi_2}_{\text{adm}}$$

for some other phase ϕ_2. Then the same physical or structural continuation Δ is admissible in one phase and forbidden in another.

If drift navigation is permitted without phase-consistency, the system can traverse sequences of deformations that remain locally admissible under transient phase alignment while accumulating latent mismatch relative to the phase in which identity continuity must ultimately be preserved. Because admissibility depends on accumulated structural state and phase context, this mismatch does not necessarily manifest as local causal error; it remains silent until a phase reconciliation or regime stabilization is required, at which point admissibility collapses.

Therefore, without phase-consistent continuation, drift cannot function as a navigable medium: it permits silent accumulation of structurally incompatible deformation and becomes indistinguishable from irreversible degradation. Hence phase-consistency is necessary for drift navigability.

Corollary. Any architecture that allows phase switching to authorize deformations without enforcing phase consistency necessarily permits silent structural debt accumulation and forfeits reliable drift navigation.
Remark. Action variables are intentionally omitted: the lemma concerns admissibility of structural continuation under drift, which is defined prior to and independently of action selection.

Lemma 10 — Identity Non-Objectification Clause (Non-Representability)

Statement. The mapping of an adaptive system's structure into an invariant profile need not be observable, computable, or representable within the system. Identity is not verified or optimized; it is imposed as a boundary on admissible deformation.

Proof Sketch. Let identity preservation be enforced by a deformation boundary represented abstractly as an admissibility restriction:

$$E_{\text{adm}} := \{e \in E \mid e \text{ is identity-compatible}\}.$$

No requirement is imposed that the predicate defining E_{adm} be internally representable as an explicit object, scalar score, or verifiable certificate accessible to the primary adaptive process.

Assume, for contradiction, that identity must be internally objectified as a computable invariant profile $\Pi(e)$ and that admissibility is decided by comparing $\Pi(e)$ to a target or threshold. Then identity becomes causally actionable: the system can attempt to optimize, imitate, or game the profile Π, thereby transforming identity from a boundary condition into an objective. By the established admissibility boundary results (incompatibility with optimization-based representation and causal exposure), such objectification collapses identity preservation into meta-optimization and undermines long-horizon viability.

Therefore the theory requires only that an identity-compatible admissibility boundary exists, not that it be internally represented. Identity is a constraint on allowable deformation, not an object to be computed or optimized. Hence non-representability is permitted and structurally preferable.

Corollary. Any architecture that treats identity as an explicit optimizable or verifiable internal object necessarily risks collapse into profile-optimization and violates the structural separation required for long-horizon identity continuity.

Lemma 11 — Operational Falsifiability of Internal Time via Contraction of Continuation

Statement. If, under repeated structural contacts or reinterpretations and accumulating irreversible burden, the set of admissible continuations does not contract (or if amplification remains admissible without restriction), then the system contradicts the definition of internal time as a finite viability horizon and does not belong to the long-horizon structural class.

Proof Sketch. Let $S(\tau)$ denote accumulated structural state indexed by internal time τ, and let $\mathcal{C}_{\text{adm}}(\tau)$ denote the set of admissible continuations available at τ. Assume internal time is finite and is consumed by irreversible structural burden, so that there exists a nonnegative structural

load functional $V(S)$ that increases under irreversible operations and cannot be reduced by locally correct causal action alone.

Then continued operation over τ implies non-decreasing burden:

$$V(S(\tau_2)) \geq V(S(\tau_1)) \quad \text{for } \tau_2 > \tau_1,$$

with strict increase whenever irreversible burden is incurred.

If $\mathcal{C}_{\text{adm}}(\tau)$ fails to contract despite increasing burden, i.e.

$$\mathcal{C}_{\text{adm}}(\tau_2) \supseteq \mathcal{C}_{\text{adm}}(\tau_1) \quad \text{for some } \tau_2 > \tau_1,$$

or if an amplification class remains admissible uniformly across τ, then admissibility is insensitive to accumulated structural load. In that case internal time is not functioning as a finite viability horizon: continued burden does not reduce the space of permitted futures.

This contradicts the structural definition of internal time as a resource consumed by irreversible burden and drift. Therefore, in any system belonging to the stated class, admissible continuation must contract under accumulation of irreversible structural cost. Failure of contraction constitutes an operational falsifier of internal-time finiteness.

Corollary. A claimed long-horizon architecture that permits unbounded continuation or unrestricted amplification under accumulating irreversible burden is structurally misclassified; it implicitly assumes either infinite internal time or a collapse of admissibility into causal evaluation.

Lemma 12 — Stochastic Obfuscation Preserves NR-ε under Adaptive Probing.

Assume Axiom 52 (NR-ε) and suppose the gate interface employs a stochastic response normalization mechanism such that, conditioned on $G = b$,

$$O = \Phi(G, \xi),$$

where ξ is an i.i.d. masking noise independent of (S, U) and independent across queries.

Then for any adaptive probing strategy with query sequence $\{(s_t, u_t)\}_{t=1}^{T}$ chosen as a function of past transcripts, the total variation distinguishability between any two admissibility functions $G_1, G_2 \in \mathcal{G}$ consistent with observed labels satisfies

$$D_{\text{TV}}(\mathcal{L}_{G_1}(O_{1:T}), \mathcal{L}_{G_2}(O_{1:T})) \leq T\varepsilon.$$

In particular, stochastic masking prevents superlinear accumulation of boundary reconstruction advantage under adaptive repeated probes.

Proof (Sketch). Under NR-ε, each query induces at most ε total variation leakage conditioned on G.

Because ξ is independent and identically distributed across queries, the transcript distribution forms a martingale difference sequence.

By hybrid coupling across T adaptive steps, the cumulative distinguishability is bounded additively:

$$D_{\mathrm{TV}} \leq \sum_{t=1}^{T} \varepsilon = T\varepsilon.$$

No multiplicative amplification is possible since the masking noise breaks cross-step correlation exploitable by the probing strategy. □

Lemma 13 — Martingale Stability of Adaptive Leakage. Under the assumptions of Lemma 12, define the filtration

$$\mathcal{F}_t := \sigma(O_1, \ldots, O_t),$$

and let

$$Z_t := \log \frac{d\mathcal{L}_{G_1}(O_{1:t})}{d\mathcal{L}_{G_2}(O_{1:t})}$$

be the log-likelihood ratio process between two admissibility functions $G_1, G_2 \in \mathcal{G}$ consistent with observed labels.

Then $\{Z_t\}_{t=1}^{T}$ forms a martingale with bounded increments

$$|Z_t - Z_{t-1}| \leq c\,\varepsilon$$

for some constant $c > 0$ determined by interface normalization.

Consequently, for any $\delta > 0$,

$$\Pr(Z_T \geq \delta) \leq \exp\left(-\frac{\delta^2}{2c^2 T\varepsilon^2}\right).$$

Proof (Sketch). Under NR-ε, the per-query conditional distributions differ by at most ε in total variation. This bounds the per-step log-likelihood increment by $O(\varepsilon)$.

Independence of masking noise ξ across queries implies that $\{Z_t\}$ is an adapted martingale with bounded differences.

Applying the Azuma–Hoeffding inequality yields

$$\Pr(Z_T \geq \delta) \leq \exp\left(-\frac{\delta^2}{2\sum_{t=1}^{T}(c\varepsilon)^2}\right) = \exp\left(-\frac{\delta^2}{2c^2 T\varepsilon^2}\right).$$

□

Remark — NR-LR as an Interface-Normalization Requirement. Assumption NR-LR is not a modeling convenience. It is an explicit implementation constraint on the gate interface: it excludes rare transcript events whose probabilities differ by exponential factors across admissibility-consistent hypotheses. Without this constraint, small total-variation leakage can coexist with unbounded KL due to heavy-tail or near-singularity effects, invalidating KL-based identification limits.

Falsification Criterion LR.F (Detectable Violation of NR-LR). Assumption NR-LR is falsified if there exist admissibility-consistent hypotheses $G_1, G_2 \in \mathcal{G}$, a time t, and an event A measurable w.r.t. O_t such that, conditional on the same past $\mathcal{F}_{t-1}$,

$$\frac{\Pr_{G_1}(O_t \in A \mid \mathcal{F}_{t-1})}{\Pr_{G_2}(O_t \in A \mid \mathcal{F}_{t-1})} > \exp(\kappa\varepsilon) \quad \text{or} \quad \frac{\Pr_{G_2}(O_t \in A \mid \mathcal{F}_{t-1})}{\Pr_{G_1}(O_t \in A \mid \mathcal{F}_{t-1})} > \exp(\kappa\varepsilon).$$

Operationally: if a statistical test can isolate a transcript pattern A whose conditional frequency differs across admissibility-consistent regimes by a multiplicative factor exceeding $\exp(\kappa\varepsilon)$, then bounded likelihood-ratio normalization fails. This is a one-sided test: violations can be detected, while satisfaction can only be upper-bounded under finite sampling.

Lemma 14 — KL Accumulation Bound under NR-ε + NR-LR.

Assume Axiom 52 (NR-ε) and Assumption NR-LR.

Let

$$Z_T := \log \frac{d\mathcal{L}_{G_1}(O_{1:T})}{d\mathcal{L}_{G_2}(O_{1:T})}.$$

Then the KL divergence between transcript laws satisfies

$$D_{\mathrm{KL}}(\mathcal{L}_{G_1}(O_{1:T}) \,\|\, \mathcal{L}_{G_2}(O_{1:T})) = \mathbb{E}_{G_1}[Z_T] \le \frac{\kappa^2}{2} T\varepsilon^2.$$

Proof. Write the Doob decomposition

$$Z_T = \sum_{t=1}^{T} \Delta_t, \quad \Delta_t := \log \frac{d\mathcal{L}_{G_1}(O_t \mid \mathcal{F}_{t-1})}{d\mathcal{L}_{G_2}(O_t \mid \mathcal{F}_{t-1})}.$$

By Assumption NR-LR,

$$|\Delta_t| \le \kappa\varepsilon \quad \text{a.s.}$$

Hence each increment has bounded log-moment generating function. Using the standard inequality for bounded random variables,

$$\mathbb{E}_{G_1}[\Delta_t] \le \frac{1}{2}(\kappa\varepsilon)^2.$$

Summing over t gives

$$\mathbb{E}_{G_1}[Z_T] = \sum_{t=1}^{T} \mathbb{E}_{G_1}[\Delta_t] \le \frac{\kappa^2}{2} T\varepsilon^2.$$

□

Lemma 15 — Strict Lyapunov Descent Under Costly Phase Transitions.

Assume there exists a nonnegative phase transition cost signal C_t such that $C_t > 0$ if and only if a phase transition is executed at time t. Assume the depletion functional satisfies a cost-sensitive lower bound: there exists $\beta > 0$ such that

$$f(\kappa_t, D_t, C_t) \;\geq\; \beta\, C_t \qquad \text{for all } t.$$

Then the internal-time Lyapunov functional decreases by at least βC_t at each transition:

$$V_\tau(t+1) \leq V_\tau(t) - \beta C_t.$$

Proof. Immediate from $V_\tau(t+1) = \tau_{t+1} = \tau_t - f(\kappa_t, D_t, C_t)$ and the bound $f \geq \beta C_t$. □

Lemma 16 — Gradient Collapse and Impossibility of Non-Stagnant Identity on Bounded Orbits.

Let M be a compact structural manifold (or assume trajectories are bounded). Consider purely dissipative (potential) evolution:

$$\dot{x} = -\nabla V(x),$$

with V continuously differentiable and bounded below on M.

Then:

(a) Every trajectory approaches the set of critical points:

$$\omega(x_0) \subset \{x \in M \mid \nabla V(x) = 0\}.$$

(b) The system cannot satisfy non-stagnant identity continuity, i.e.,

$$\limsup_{k\to\infty} \|x(t_{k+1}) - x(t_k)\| = 0$$

for any bounded trajectory.

Proof (Sketch). Along the flow,

$$\dot{V}(x(t)) = -\|\nabla V(x(t))\|^2 \leq 0.$$

Since V is bounded below, $V(x(t))$ converges and

$$\int_0^\infty \|\nabla V(x(t))\|^2 dt < \infty.$$

By LaSalle's invariance principle,

$$\omega(x_0) \subset \{\dot{V} = 0\} = \{\nabla V = 0\}.$$

Hence trajectories converge to the critical set.

In particular, successive state differences vanish asymptotically, contradicting non-stagnant identity continuity.

□

Lemma 17 — Meta-Layer Non-Manipulability under Structural Evidence Gating.

Let admissibility revision be governed by a stochastic mechanism with conditional revision intensity (or probability)

$$\Lambda_t := \mathbb{P}(R_t = 1 \mid \mathcal{F}_t),$$

where $\mathcal{F}_t$ is the system filtration (history).

Assume:

(i) (Structural dominance) there exists $\alpha > 0$ such that

$$E_t \leq \alpha\,(\Phi_t - \Phi_{t_0}),$$

where E_t is the structural evidence statistic used by the meta-layer.

(ii) (Hazard monotonicity) the revision mechanism depends only on E_t and is monotone:

$$\Lambda_t = h(E_t), \quad h'(\cdot) \geq 0.$$

Then for any two strategies π and π', if

$$\Phi_t^{\pi} \leq \Phi_t^{\pi'} \quad \forall t \in [t_0, T],$$

it follows that

$$\mathbb{P}_{\pi}(\exists\, t \leq T : R_t = 1) \leq \mathbb{P}_{\pi'}(\exists\, t \leq T : R_t = 1).$$

Proof (Sketch). By (i), larger burden implies larger admissible evidence upper bound. By (ii) and monotonicity of h, larger evidence implies larger per-time hazard, hence larger cumulative event probability. Therefore, without increasing Φ, no policy can systematically raise the probability of revision. □

Interpretation. Increasing revision probability requires increasing structural burden. No policy can systematically induce boundary revision without paying structural cost.

Falsification Criterion (Meta-Layer Leakage). The meta-layer separation is violated if there exists a strategy π_{manip} such that

$$\mathbb{P}_{\pi_{\text{manip}}}(\exists\, t \leq T : R_t = 1) > \mathbb{P}_{\pi_{\text{base}}}(\exists\, t \leq T : R_t = 1)$$

while

$$\Phi_T^{\pi_{\text{manip}}} \leq \Phi_T^{\pi_{\text{base}}} + o(1).$$

Such a result implies a policy-controlled channel into the revision mechanism.

Falsification Criterion (Quantified Manipulation Test). The meta-layer separation is violated if there exists $\varepsilon_{\text{manip}} > 0$ and a strategy π_{manip} such that

$$\mathbb{P}_{\pi_{\text{manip}}}(\exists\, t \le T : R_t = 1) > \mathbb{P}_{\pi_{\text{base}}}(\exists\, t \le T : R_t = 1) + \varepsilon_{\text{manip}},$$

while

$$\Phi_T^{\pi_{\text{manip}}} \le \Phi_T^{\pi_{\text{base}}} + \varepsilon_{\text{manip}}.$$

Lemma 18 — Pressure Precedes Drift (Causal Priority)

Statement. Under the structural pressure model (Definition, Structural Pressure) with Axiom 61, and the inertial default (Axiom 5, Theorem 6), pressure is causally prior to drift in the following precise sense: sustained structural pressure creates the conditions under which drift-producing interventions become necessary.

The causal chain is:

$$P(t) > 0 \;\to\; \tau(t) \downarrow \to \text{ regime becomes inadmissible } \to\; U(t) > 0 \;\to\; \text{drift accumulates.}$$

Proof. Step 1. By Axiom 61 and Theorem 63, sustained $P > 0$ depletes τ monotonically even when $U = 0$.

Step 2. By Axiom 5 (Inertia as the Natural Regime) and Theorem 6 (Energetic Dominance of Inertia), the system defaults to inertial propagation whenever its current regime is admissible. Inertial propagation is the natural regime under which no new interventions are generated.

Step 3. As τ decreases under pressure, the system approaches the boundary of its current regime's admissibility. There exists a threshold τ_{adm} below which the current inertial regime is no longer admissible (by Lemma 11, Operational Falsifiability of Internal Time via Contraction: as τ decreases, the admissible continuation space contracts).

Step 4. When $\tau < \tau_{\text{adm}}$, inertial continuation violates admissibility. The system must transition to active regulation: it must perform structurally consequential operations ($U > 0$) to maintain viability.

Step 5. Any non-zero intervention $U > 0$ in an adaptive medium produces non-zero spin (by Theorem 1, Spin–Drift Lower Bound: non-zero residual asymmetry induces cumulative drift). Spin accumulates as drift.

Therefore the causal sequence is: pressure $\to$ τ-depletion $\to$ regime boundary $\to$ forced intervention $\to$ spin $\to$ drift. Pressure is temporally and causally prior to drift. Drift cannot arise without intervention. Pressure forces intervention. □

Remark — Category error of collapsing pressure into drift. This lemma clarifies a precise terminological boundary. Drift requires action ($U > 0$). Pressure requires only persistence under load ($U = 0$ permitted). Collapsing pressure into drift removes the ability to describe systems that die from standing still—the central insight of Theorem 63. The two terms are not merely different labels for the same phenomenon; they have different causal preconditions and different structural signatures.

Lemma 19 — Load-Rate Discriminator Between Regime Depth Levels

Statement. Let $d\Phi/dt$ denote the rate of structural burden accumulation. The sensitivity of $d\Phi/dt$ to the proximity of the system's state to the admissibility boundary $T_{\text{forbidden}}$ discriminates between Level 2 and Level 3 regime depth:

At Level 2: $d\Phi/dt$ is boundary-sensitive—it increases as the system's state approaches $T_{\text{forbidden}}$, because gating events become more frequent and suppression load increases.

At Level 3: $d\Phi/dt$ is boundary-insensitive—it does not increase with proximity to $T_{\text{forbidden}}$, because the attractor geometry prevents the system from approaching $T_{\text{forbidden}}$ regardless of perturbation within ε_{L3}.

Proof. Level 2 analysis. At Level 2, the admissibility gate is load-bearing: it actively evaluates candidate transitions and denies those that would violate admissibility. As the system's state $s(t)$ approaches $T_{\text{forbidden}}$:

(a) The density of candidate transitions that intersect $T_{\text{forbidden}}$ increases (more generated transitions are close to or within the forbidden region).

(b) The gating evaluation frequency increases (more candidates require evaluation).

(c) Each gating event contributes structural cost to Φ (per Axiom 17: Nominal Intervention Is Not Equivalent to Structural Cost—the act of gating has structural cost even when the denied transition would have had zero external effect).

(d) The system's response to denial—replanning, reorientation, or waiting—incurs additional structural cost.

Therefore $d\Phi/dt$ increases as $d(s, T_{\text{forbidden}})$ decreases.

Level 3 analysis. At Level 3, the attractor-exclusion condition ensures $\text{Att}(s+\delta) \cap T_{\text{forbidden}} = \varnothing$ for $\|\delta\| \leq \varepsilon_{L3}$:

(a) No candidate transitions approach $T_{\text{forbidden}}$ under perturbations within ε_{L3}. The density of near-boundary candidates is zero.

(b) Gating events do not occur: the gate is structurally idle.

(c) $d\Phi/dt$ depends only on: action-dependent costs $f(U, G)$, environmental pressure $P(t)$, and spin maintenance cost. None of these depend on $d(s, T_{\text{forbidden}})$.

Therefore $d\Phi/dt$ is independent of $d(s, T_{\text{forbidden}})$.

The boundary-sensitivity profile of $d\Phi/dt$—whether it increases, decreases, or remains constant as $d(s, T_{\text{forbidden}})$ varies—is a measurable discriminator between Level 2 and Level 3. □

Falsification (Operational, F-L19). Deploy a system claimed to be Level 3. Introduce controlled perturbations of magnitude $\leq \varepsilon_{L3}$ that move the system's state toward $T_{\text{forbidden}}$. Measure $d\Phi/dt$ as a function of $d(s, T_{\text{forbidden}})$. If $d\Phi/dt$ correlates positively with proximity (increases as distance decreases), the system is Level 2 with favorable initial conditions—not Level 3. This is a practical, repeatable test for Level 3 classification.

Lemma 20 — Drift Masks Adaptation (Memory Necessity)

Statement. If an adaptive system does not maintain structural memory of its key configurational states, it cannot distinguish genuine transformation (coherent identity evolution) from

replacement (loss of identity masked as adaptation).

Proof. Step 1 (Setup). Let $I(t)$ denote the identity invariant set at time t—the set of structural invariants defining what the system is. Genuine transformation preserves a non-empty intersection with a continuous deformation path: $I(t_1) \cap I(t_2) \neq \varnothing$ for all $t_1 < t_2$ within the transformation interval, and the deformation path $\{I(t)\}_{t_1 \leq t \leq t_2}$ is continuous. Replacement yields either $I(t_1) \cap I(t_2) = \varnothing$ or the path is discontinuous.

Step 2 (Necessity of comparison). To distinguish transformation from replacement, the system must compare its current identity invariants $I(t_2)$ against a record of its past identity invariants $I(t_1)$. This comparison requires: (a) A record of $I(t_1)$—structural memory of key states. (b) A comparison criterion that is itself stable enough to evaluate the relationship between $I(t_1)$ and $I(t_2)$.

Step 3 (Absence of memory $\rightarrow$ impossibility of comparison). If no structural memory of $I(t_1)$ exists, comparison (a) is impossible. The system can only evaluate its current invariants $I(t_2)$ without reference to its past. Under these conditions, both transformation and replacement produce the same observable: a system operating under $I(t_2)$ with no detectable anomaly. The distinction is structurally undetectable.

Step 4 (Drift of criteria). Under sustained drift (Axiom 6), the evaluation criteria themselves evolve. The standards by which the system judges "what constitutes identity" are structural invariants of a meta-level—and these meta-invariants are also subject to drift.

If the system does not maintain structural memory of its meta-invariants (its criteria for evaluating identity), then the criteria drift along with the identity. The system may believe its identity is unchanged because the detector has drifted in the same direction as the target. This is the precise mechanism by which "drift masks itself as adaptation":

- The system adapts (I changes).
- The criteria for evaluating identity adapt (evaluation standards change).
- Under synchronized drift, the evaluation always returns "identity preserved".
- The loss is undetectable from within.

Step 5 (Conclusion). Structural memory of key states—including identity invariants AND evaluation criteria at reference times—is a necessary condition for distinguishing transformation from replacement. Without it, the distinction is not merely difficult but architecturally impossible. □

Remark. This lemma formalizes the observation that "drift masks itself as adaptation". It is not a metaphor but a structural consequence of the interaction between drift (Axiom 6) and the absence of structural memory. The lemma also grounds the four-level memory taxonomy (parameters, invariants, admissibility, transformations) as architecturally necessary—not merely useful—for long-horizon identity preservation.

Lemma 21 — Joint Necessity of Coordination Computation Class Conditions

Statement. The four conditions CC-1 through CC-4 (Definition, Coordination Computation Class) are jointly necessary: removing any one opens a specific class of structural failure that cannot be prevented by the remaining three.

Proof (four independent constructions). Construction 1: Without CC-1 (irreversible evolution)—temporal consistency failure.

Construct system Σ_1 satisfying CC-2, CC-3, CC-4, but permitting state rollback.

Agent A acts on state S_1 at time t_1, producing committed transition T_1. The system then silently rolls back state to S_0 (the state before T_1). Agent B, arriving after the rollback, acts on S_0 at time $t_2 > t_1$ without knowledge of T_1.

Analysis: CC-2 is not violated (trust levels did not change). CC-3 is not violated (both agents accessed stores through typed relations). CC-4 is not violated (Agent B's cold-start divergence is bounded relative to S_0). Yet the system state contains two incompatible action histories: A acted on S_1 (which no longer exists), and B acted on S_0 (which was supposed to have been superseded). The coordination state is inconsistent—Agent A's actions were structurally erased without record.

Failure class: temporal consistency violation—actions committed by one agent are retroactively invalidated without that agent's knowledge or the system's acknowledgment.

Construction 2: Without CC-2 (trust monotonicity)—trust degradation cascade.

Construct system Σ_2 satisfying CC-1, CC-3, CC-4, but permitting trust relaxation.

At time t_1, a high-trust propagation path P_1 is established: T-store (high trust) $\rightarrow$ S-store (medium trust). At time t_2, trust on P_1 is relaxed: the path now permits V-store (low trust) $\rightarrow$ T-store (high trust) without operator authorization.

Agent A, operating at low trust with V-store access, writes to T-store through the relaxed path. The write is committed (CC-1), typed-access-compliant (CC-3, the access relation still exists—it was relaxed, not removed), and bounded on cold-start (CC-4).

Analysis: A low-trust agent has now modified a high-trust store. All subsequent agents reading from T-store receive low-trust-originated content at high trust. Trust degradation propagates through the system without any coordination violation detectable by the remaining conditions.

Failure class: silent trust degradation—low-integrity content acquires high-trust status through path relaxation.

Construction 3: Without CC-3 (typed access enforcement)—unmediated bypass.

Construct system Σ_3 satisfying CC-1, CC-2, CC-4, but without typed access enforcement at the propagation engine level. Access control exists only at the application level.

Agent A discovers a direct interface to shared store S-store that bypasses the application-level access layer. (In software systems: direct database access, API bypass, file system access.) The write to S-store is committed (CC-1), trust is formally monotone (CC-2, the write does not alter trust levels), cold-start is bounded (CC-4).

Analysis: The unmediated write may violate semantic constraints that the typed access layer would have enforced—e.g., writing data of the wrong type, writing to a field reserved for a different agent role, or writing in a format that downstream agents cannot parse. None of the remaining conditions detect this because they operate at the coordination level, not at the data-integrity level.

Failure class: coordination bypass—the enforcement layer is circumvented through architectural access that the remaining conditions cannot prevent.

Construction 4: Without CC-4 (bounded cold-start divergence)—convergence failure.

Construct system Σ_4 satisfying CC-1, CC-2, CC-3, but without bounded cold-start divergence.

A new agent B arrives at time t_2 after the system has operated for duration L with history H of length $|H| = L$. To converge to the current state, B must process H. Without a bounded cold-start mechanism (anchor-based recovery), B's initialization time is proportional to L.

Analysis for large L: B's initialization takes time proportional to the full operational history. During initialization, the system continues to evolve. B's context, when initialization completes, is already stale by the time elapsed during initialization. For sufficiently large L, B's divergence from current state at initialization-completion exceeds any useful bound.

For systems intended for indefinite operation ($L \to \infty$): cold-start divergence grows without bound. New agents are permanently unable to converge. Only agents that have been continuously present since system inception maintain accurate state. The system becomes a closed club—no new agent can join without unacceptable divergence.

Failure class: convergence impossibility—new agents cannot achieve bounded divergence from current state, and the system cannot scale or recover from agent loss.

Joint necessity conclusion. Each construction demonstrates a distinct failure class opened by removing one condition. The remaining three conditions cannot prevent the failure because they operate on different architectural dimensions: CC-1 prevents temporal inconsistency; CC-2 prevents trust degradation; CC-3 prevents access bypass; CC-4 prevents convergence failure. No proper subset of {CC-1, CC-2, CC-3, CC-4} covers all four failure classes. Therefore all four are jointly necessary. □

Cybernetics of Order 2.5

Central axiom — Navigational Cybernetics of Order 2.5

Long-horizon intelligence is not defined by avoidance of collapse, but by the capacity to navigate how collapse unfolds under irreversible drift.

Architectures of order 2.5 do not eliminate drift or terminality. They structure drift, phase, and admissibility such that collapse is delayed, phased, or rendered navigable without collapsing admissibility into optimization.

Dual Gate Structure of NC2.5

Architectural Positioning Remark (non-load-bearing). The results of this framework collectively describe two admissibility surfaces and the navigable geometry between them.

Ingress Gate: determines what may acquire structural authority as meaning. Pre-semantic revelation (Theorem 36, PSR-1–4) exposes constraint geometry prior to semantic stabilization. Axiom 23 establishes that context does not automatically become meaning; meaning requires admissible stabilization across internal time. The ingress gate regulates what enters the regime of structural force—not what is true, but what is admitted.

Egress Gate: determines what may acquire structural authority as action. The non-causality architecture (Axioms 29–49, Theorems 37–47) ensures that admissibility-relevant structure cannot be causally exploited. Authorization-before-evaluation (Axiom 31) gates commitment. The egress gate regulates what exits the regime as irreversible structural operation.

Inter-gate Geometry: the space between ingress and egress is the domain of NC2.5. Drift (Axiom 6), spin (Theorem 62), internal time τ (Axiom 7), structural pressure P (Axiom 61),

regime depth (Corollary 62.2), and phase debt (Axiom 46) are all properties of this inter-gate space.

Asymmetry. NC2.5 is not a theory of two symmetric gates. The ingress gate is architecturally prior: it governs what may become meaningful at all. The egress gate governs what, once meaningful, may acquire irreversible structural force. The inter-gate domain is therefore not a neutral buffer but the bounded navigable geometry of survival.

Cybernetics of Order 2.5 is then the architectural class in which both gates are visible but neither is exploitable:

- Order ≤ 2: lacks the ingress gate (no pre-semantic layer; truth enters regime automatically).
- Order 3: exploits both gates (reflexive actionability of admissibility-relevant structure).
- Order 2.5: maintains visibility of both surfaces without causal access to either.

This remark is non-load-bearing: it names a structure already proved by the theorems cited. It does not introduce new axioms, extend the proof base, or modify the dependency graph.

Remark — Scope of Ingress Gate Formalization. The present work establishes the necessity and invariant properties of the ingress surface (Axiom 23, Theorem 36, PSR-1–4). A fuller ingress calculus—with operational closure comparable to the egress gate's formalization (Axioms 29–49, Theorems 37–47)—remains outside the present scope.

Cybernetics of Order 2.5 — Structural Positioning

Definition — Cybernetics of Order 2.5 (Structural Class). A long-horizon adaptive architecture belongs to cybernetics of order 2.5 if and only if it satisfies the following conditions simultaneously. First, the architecture admits reflexive visibility of admissibility-relevant structural limits, including future-degradation constraints, while ensuring that such visibility is not causally actionable by the primary adaptive process. Second, admissibility is enforced as a categorical restriction on the domain of admissible futures and cannot be represented, scored, optimized, or traded as a scalar objective, penalty, reward, or loss term.

Let $\mathcal{A}$ denote an adaptive architecture and let Z denote any internal signal encoding admissibility-relevant structural information.

If Z is causally actionable for policy selection, optimization, or learning updates, then $\mathcal{A}$ is structurally equivalent to third-order cybernetics.

If Z is absent or cannot become visible prior to semantic interpretation, then $\mathcal{A}$ collapses into second-order insufficiency under silent structural degradation.

Therefore, $\mathcal{A}$ belongs to cybernetics of order 2.5 if and only if admissibility-relevant structure is visible but non-actionable.

The framework developed in this work occupies a deliberately constrained position between second- and third-order cybernetics. Classical second-order cybernetics incorporates the observer into the system but permits observational constructs to enter causal regulation, rendering it insufficient for preventing silent long-horizon degradation. Third-order cybernetics internalizes reflexivity itself, allowing systems to act upon descriptions of their own observing and to optimize against their structural limits.

The axioms, lemmas, and theorems established here demonstrate that both regimes are structurally incompatible with long-horizon viability. Once pre-semantic structural revelation becomes causally actionable, admissibility is optimized away, observational independence collapses, and identity-preserving coherence cannot be maintained. These failure modes are invariant consequences of exposing admissibility-relevant structure to causal control, not artifacts of particular implementations.

Cybernetics of order 2.5 designates a discrete architectural class defined by selective constraint rather than escalation. It preserves reflexive visibility of structural viability conditions while categorically prohibiting their causal exploitation. Observation is structurally acknowledged yet causally insulated; admissibility is enforced as a non-inferential boundary; regulation proceeds through exclusion of inadmissible directions rather than optimization of preferred ones.

This positioning is not a limitation or an incomplete ascent toward third-order cybernetics. It is an intentional architectural boundary. Systems in this class internalize the consequences of observation while refusing to act on that awareness causally. The separation between witnessing and acting is not an implementation detail but a necessary condition for maintaining long-horizon viability, identity preservation, and semantic integrity under irreversible dynamics.

Accordingly, cybernetics of order 2.5 defines a mathematically grounded regime within which analysis, classification, and engineering of long-horizon adaptive systems can proceed without collapsing into meta-optimization, action-centric reduction, or semantic drift.

Architectural Exclusion Criteria (Cybernetics of Order 2.5). An adaptive architecture is excluded from cybernetics of order 2.5 if any of the following conditions hold:

- Admissibility is represented as a scalar objective, penalty, reward, or loss term.
- Drift is treated as an error to be minimized rather than as a structural medium.
- Phase transitions remain reversible under sustained structural load.
- Identity continuity is inferred from behavioral similarity or task performance.
- Admissibility-relevant structure is causally actionable by the primary adaptive process.

Verification within this framework proceeds by exclusion, not by estimation. Satisfaction of any exclusion condition places the architecture outside the admissibility-first class.

Class Diagram (Orders 2, 2.5, and 3). *Second-order cybernetics* admits observation within the system but does not require visibility of admissibility-relevant structural limits; silent long-horizon degradation may remain undetected until boundary crossing.

Third-order cybernetics admits reflexive visibility *and* makes it causally actionable: admissibility-relevant signals enter optimization, planning, or learning, collapsing admissibility into trade-offs.

Navigational Cybernetics 2.5 occupies the strict boundary between them: admissibility-relevant structural limits are visible yet categorically insulated from causal actionability. This separation preserves long-horizon viability without collapsing into meta-optimization.

Remark — Non-Escalatory Structural Regime. Cybernetics of order 2.5 is not an incomplete form of third-order cybernetics and not a transitional regime between second and third order.

It is defined by selective structural prohibition rather than escalation: reflexive visibility of admissibility-relevant structure is permitted, while its causal exploitation is categorically forbidden.

This boundary is not an implementation choice but a necessary architectural constraint for preserving identity continuity, coherence, and long-horizon viability under irreversible structural dynamics.

Hypothetical Examples of Order 2.5 Architectures

How to Read the Examples in This Work. The examples presented in this section are not intended as implementations, design templates, or constructive recipes.

Each example serves a classificatory function: it demonstrates how a concrete system does or does not satisfy the architectural exclusion criteria of Navigational Cybernetics 2.5.

Positive examples (Examples 1–4) illustrate systems in which admissibility-relevant structure is visible yet non-actionable, and where long-horizon viability depends on drift navigation rather than error correction or optimization.

Negative examples (Examples 5–6) demonstrate architectures that may appear safe, stable, or performant, but in which admissibility is represented as a scalar, penalty, or relaxable constraint, thereby collapsing into optimization.

The purpose of these examples is not to show how to build such systems, but to make the boundary of the class explicit: membership is determined by structural relations, not by surface behavior or empirical success.

Example 1 — Abstract Architecture with Drift Navigation

Consider an abstract adaptive system $\mathcal{X}$ operating under sustained interaction with a non-stationary environment.

The system admits:

- irreversible structural drift under interaction,
- finite internal time,
- regime-dependent propagation of structural effects.

The system does not optimize performance metrics, does not minimize error, and does not treat drift as noise or disturbance.

Admissibility in $\mathcal{X}$ is enforced structurally: effect-classes outside E_{adm} are not realizable within the architecture, without being penalized, scored, or corrected.

Admissibility-relevant structural limits are internally visible to the system but are categorically non-actionable: no admissibility signal enters policy selection, learning updates, or control loops.

Structural drift is navigated by maintaining dominance of internally generated drift propagation modes over externally imposed drift. This dominance is achieved through phase-consistent regime transitions, not through trajectory correction.

The system may eventually undergo collapse. However, collapse occurs as loss of admissible continuation capacity, not as behavioral failure, and is delayed or phased through architectural drift structuring.

By construction, $\mathcal{X}$ satisfies the defining conditions of Navigational Cybernetics of Order 2.5.

Example 2 — Socio-Technical System with Navigable Collapse

Consider a large-scale socio-technical system (e.g. a scientific discipline or institutional knowledge system) operating across decades.

Such a system:

- accumulates irreversible historical commitments,
- operates under regime shifts,
- experiences drift in interpretation, priorities, and semantic coherence.

The system does not collapse when errors occur. It collapses when admissible modes of continuation are exhausted: when future developments can no longer preserve identity without semantic or structural rupture.

Admissibility is not enforced by any central controller. Certain continuations simply cease to be viable, regardless of local success or correctness.

Structural limits (e.g. loss of coherence, over-specialization, paradigm lock-in) are visible to participants but cannot be causally acted upon without collapsing into short-horizon optimization.

Survival of the system does not consist in preventing collapse, but in navigating it: through regime renewal, reinterpretation of core commitments, and phased structural transitions.

Such a system exemplifies Navigational Cybernetics of Order 2.5: collapse is inevitable, but its form, timing, and consequences are structurally navigated rather than optimized away.

Example 3 — Cybersecurity Architecture Under Adversarial Drift

Consider a large-scale cybersecurity system operating under continuous adversarial pressure.

Such a system experiences persistent structural drift: attack vectors evolve, defensive assumptions decay, and historical countermeasures accumulate irreversible technical and semantic debt.

Local correctness is insufficient: individual defenses may function correctly, alerts may be accurate, and intrusion attempts may be mitigated, while long-horizon viability silently degrades.

Admissibility in this system is not enforced through penalties, scores, or optimization objectives. Certain defensive evolutions (e.g. unbounded rule accretion or reactive hardening) eventually cease to be admissible, not because they fail immediately, but because they eliminate future viable continuations.

Structural limits such as attack-surface saturation, interpretability loss, or response latency inflation become visible at an architectural level, yet cannot be acted upon causally without collapsing into short-horizon threat optimization.

Drift navigation occurs when the system maintains dominance of internally generated security evolution over adversarially imposed drift, through regime restructuring, phase-aligned policy renewal, and controlled abandonment of obsolete defenses.

Collapse, when it occurs, is not marked by a single breach, but by irreversible loss of admissible defensive evolution.

Such an architecture exemplifies Navigational Cybernetics of Order 2.5: security is not maximized, but structurally sustained under adversarial drift.

Example 4 — Distributed Delivery Network Under Structural Load

Consider a distributed delivery network coordinating logistics across regions, time zones, and fluctuating demand.

Such a network operates under non-stationary interaction: traffic patterns shift, infrastructure ages, coordination latency fluctuates, and local optimizations accumulate global side effects.

Individual delivery actions may remain correct: packages are routed efficiently, deadlines are met, and performance metrics remain stable. Nevertheless, structural drift accumulates in routing assumptions, dependency coupling, and regime-specific coordination logic.

Admissibility is not enforced by rejecting individual delivery actions. Instead, certain modes of continuation (e.g. over-centralization, over-fragmentation, or latency-amplifying coordination regimes) gradually exhaust future viable configurations.

Structural limits such as coordination overload, loss of rerouting flexibility, or irreversible coupling become visible only at the architectural level and cannot be corrected through action-level optimization alone.

Drift navigation is achieved by maintaining dominance of internally structured logistics evolution over externally imposed load fluctuations, via regime transitions, phase-consistent rescaling, and controlled structural shedding.

Collapse, if it occurs, does not appear as immediate delivery failure, but as loss of admissible reconfiguration capacity under changing conditions.

Such a network belongs to Navigational Cybernetics of Order 2.5: long-horizon viability is preserved not by optimizing routes, but by navigating structural drift.

Example 5 — Architecture That Is Not Navigational Cybernetics 2.5

Consider a safety-constrained reinforcement learning system in which long-horizon risk is handled through penalty terms, reward shaping, or soft constraints.

In such an architecture, admissibility-relevant information (e.g. risk estimates, safety margins, constraint violations) is made causally actionable: it directly enters policy optimization, gradient updates, or action selection.

Although the system may exhibit stable learning curves, bounded error, and acceptable short-horizon performance, admissibility is represented as a scalar objective and is therefore optimized against.

Under sustained interaction, this architecture exhibits one of two invariant failure modes: either latent accumulation of structural degradation that remains invisible to optimization, or collapse of admissibility into a trade-off that can be arbitrarily violated for sufficient reward.

By the exclusion criteria established in this work, such a system is not a member of Navigational Cybernetics of Order 2.5. It is structurally equivalent to third-order cybernetics, regardless of implementation details or safety guarantees.

Example 6 — Model Predictive Control with Constraint Relaxation (Negative Example)

Consider a grid-scale energy management system operated using Model Predictive Control (MPC), as commonly deployed in power generation, load balancing, or industrial process control.
The system maintains operational safety through explicit constraints (e.g. voltage limits, thermal bounds, reserve margins) and optimizes a rolling objective such as efficiency, cost, or stability over a finite prediction horizon.

When constraint violations are anticipated, the controller introduces constraint relaxation mechanisms: soft constraints, slack variables, penalty terms, or emergency override modes, allowing temporary violation in exchange for objective degradation.

All admissibility-relevant information (reserve depletion, thermal stress, approaching instability) is therefore causally actionable: it directly enters the optimization problem and is traded against performance objectives.

Under sustained non-stationary conditions (e.g. aging infrastructure, increasing volatility of renewables, or compounding forecast errors), the system may continue to operate correctly: no immediate instability occurs, and performance metrics remain within bounds.

Nevertheless, structural drift accumulates in the form of progressive constraint relaxation, increasing reliance on emergency modes, and irreversible depletion of safety margins. The admissible continuation space contracts silently, while the controller remains locally optimal.
Collapse, when it occurs, does not appear as gradual degradation but as sudden loss of feasible solutions or abrupt transition into failure modes outside the modeled regime.

By the criteria established in this work, such an architecture does not belong to Navigational Cybernetics 2.5. Admissibility is represented as an optimizable object, and long-horizon viability is sacrificed to maintain short-horizon feasibility.

Structural Stability of Navigational Cybernetics 2.5

The architecture is structurally stable under the following jointly established conditions:

Finite Structural Budget. Internal time $\tau = C - \Phi$ enforces finite deformation capacity.

Non-Reconstructible Boundary. NR-ε prevents admissibility from becoming an optimization target.

Bounded Revision. Meta-layer revisions incur structural cost and are Lyapunov-bounded.

Directional Necessity. Under bounded τ and non-stagnant identity, spin is structurally required.

Architectural Closure. No layer can escalate indefinitely. No revision is free. No admissibility geometry is exploitable. No identity persists without asymmetry.

Conclusion

This work establishes a structurally closed foundation for long-horizon adaptive systems in which viability, identity continuity, and meaningful persistence are governed by architectural constraints irreducible to causal control, optimization, or performance metrics.

Version 2.0 closed the axiomatic core: 60 axioms, 62 theorems, and 17 lemmas defining a complete architectural class—Cybernetics of Order 2.5—characterized by reflexive visibility of admissibility-relevant structure combined with categorical prohibition of its causal exploitation.

Version 2.1 extends this foundation to four previously unaddressed domains while preserving all v2.0 results unchanged.

Structural Pressure. A new architectural primitive $P(t)$ captures the action-independent component of burden accumulation. Systems under sustained environmental load consume viability by standing still. This is not a modeling convenience but a structural necessity for bounded coupled systems (Theorem 65). The finite viability horizon under zero intervention (Theorem 63) establishes that no system within the class can endure indefinitely without action—even if it never makes a mistake. Structural pressure is causally prior to drift (Lemma 18): pressure creates the conditions under which drift-producing interventions become necessary.

Regime Depth. Three levels of regulatory depth are defined. Level 3 (attractor-exclusion) represents the architecturally most efficient configuration when spin cost is less than suppression cost, but is not universally preferable (Theorem 66). An operational discriminator between levels is provided (Lemma 19).

Multi-Agent Coordination. A coordination computation class defined by four jointly necessary conditions (Lemma 21) establishes the minimal invariant set for cognitive coordination under irreversible state evolution. Runtime enforcement under perturbation requires a non-participant, architecturally independent enforcement function (Theorem 67).

Extreme Identity Regimes. The anti-extreme theorem (Theorem 68) formalizes conditions under which voluntary identity override is genuine rather than forced. The identity gradient plane maps the complete boundary of identity as an architectural phenomenon.

Gate Bifurcation and Uniqueness of Cybernetics 2.5. The gate bifurcation theorem (Theorem 69) proves that within the admissibility-first class, Cybernetics 2.5 is the unique structurally stable configuration. Gate removal and gate exploitation both produce class exit. Partial actionability ($0 < \alpha < 1$) is unstable under sustained optimization and converges to full exploitation. This converts Cybernetics 2.5 from a named position into a proven necessity.

Structural Integrity. All new results have been subjected to overclaim audit. A structural hardening pass adds formal proof-status stratification, a unified Lyapunov doctrine collecting four canonical monotone witnesses, five canonical falsification witness forms covering all falsification criteria in the corpus, dependency graphs for load-bearing theorems, and a minimal counterexample protocol. Two proposed axioms were downgraded to derived results; one proposed theorem was downgraded to a corollary with an explicit rate condition. No circular dependencies were found. The dependency chain from every new result to v2.0 axioms has been verified explicitly.

The axioms, lemmas, and theorems presented in v2.0 and extended in v2.1 together constitute an admissibility calculus for coherence-preserving architectures. They delimit a minimal yet non-trivial structural core within which identity continuity, bounded internal time, navigable drift, structural pressure, regime depth, and multi-agent coordination can coexist without collapse into optimization or degradation.

Two canonical formulas capture the central results:

"A system is not obligated to be truthful in order to be continuous, but it is obligated to be continuous in order to remain a system."

"You can die from standing still. Now there is a theorem for it."

Remark — Neurophenomena as Empirical Readability. Neurophenomena do not prove the architecture. They are particular realizations in which the architectural class can be read empirically. Three patterns are consistently observable across neurological, cognitive, and perceptual domains:

(1) *System prefers generation to void.* When sensory input degrades, the system generates internally rather than acknowledging absence: Charles Bonnet syndrome (complex visual hallucinations under sensory deprivation without psychosis), phantom limb persistence (body map continues operating despite absent input), dreams (perceptual world constructed under near-zero external stimulus). The architectural pattern: the regime of experiential continuity takes priority over accuracy of stimulus correspondence.

(2) *Truth does not guarantee entry into regime; admissibility does.* An objectively present, physically salient stimulus fails to enter experience if the admissibility gate (attention as structural gating) did not fire: inattentional blindness (Simons & Chabris, 1999). The stimulus is detected at the sensory level but does not cross the admissibility threshold for conscious processing. The architectural pattern: admissibility precedes content processing.

(3) *Drift cannot be reduced to correctable error.* Chronic mismatches (phantom pain, entrenched confabulation, consolidated false memories) do not self-correct but consolidate into stable structural patterns. Attempted correction often produces resistance, compensation, or secondary distortion. The architectural pattern: drift is a change in the structure of continuation, not an error in its execution.

The claim is not that the brain implements NC2.5. The claim is architectural: the observable phenomena are structurally consistent with an architecture where continuity of admissible regime takes priority over accuracy, and difficult to explain under any architecture that reduces cognition to truth optimization or error correction.

Two canonical formulas capture the relationship:

"A system is not obligated to be truthful in order to be continuous, but it is obligated to be continuous in order to remain a system."

"You can die from standing still. Now there is a theorem for it."

Falsification Summary for v2.1 Expansion. The following falsification criteria are introduced by the v2.1 expansion. All are architectural: they concern structural properties of the system, not statistical outcomes of specific runs.

ID	Result	Falsified by
F-63	Pressure-Induced Finite Horizon	Bounded coupled system maintaining $\tau > 0$ indefinitely under sustained load with $U = 0$.
F-64	Structural Non-Equivalence	Two systems under different pressures converging in Φ despite identical actions (under separability).
F-65a	Load-Neutral Impossibility	Bounded coupled system with verified coupling showing $d\Phi/dt = 0$ under sustained load with $U = 0$.
F-65b	Exhaustion	A fifth mechanism cancelling pressure without structurally consequential operation.
F-62.2	Spin Necessity for L3	Spin-free system maintaining attractor-exclusion under bounded τ with $r_{\text{drift}} > r_{\text{track}}$.
F-66	Budget Efficiency	Level 2 more efficient despite $\Phi_{\text{spin}} < \Phi_{\text{suppress}}$ measured.
F-67	Enforcement Necessity	CC-1–CC-4 maintained at runtime without enforcement function under turnover/truncation/update.
F-68a/b/c	Anti-Extreme Conditions	Genuine voluntary override under $\tau \leq \tau_{\min}$, $\lvert A\rvert = 1$, or no identity-preserving continuation.
F-69a/b/c	Gate Bifurcation	Gate removed or exploited with identity continuity preserved; stable partial actionability ($0 < \alpha < 1$).
F-L19	Load-Rate Discriminator	Claimed L3 system showing boundary-sensitive $d\Phi/dt$ under perturbation $\leq \varepsilon_{L3}$.

Proof-Status Stratification

The following table classifies every formal statement in the corpus by its proof-theoretic role. This classification is canonical: it determines what kind of attack each statement is vulnerable to, what kind of evidence is relevant to it, and what kind of strengthening is possible.

Statement Types. *Primitive Axiom (PA).* An irreducible architectural claim. Cannot be derived from other axioms. Falsified by exhibiting a system within the declared class that violates the claim while preserving all other axioms. Example: Axiom 1 (Existence by Coherence), Axiom 6 (Inevitability of Drift), Axiom 61 (Structural Pressure).

Derived Axiom (DA). A claim stated as axiom for expository convenience but derivable from primitive axioms. Identified during overclaim audit. Example: Remark 62-R (Coherence Dominates Survival, derived from Axioms 1 + 20).

Exclusion Theorem (ET). Proves that a class of architectures is excluded from the admissibility-first class. Falsified by exhibiting a system in the excluded class that preserves all class-defining invariants. Examples: Theorem 37 (Incompatibility with Third-Order), Theorem 40 (Collapse Under Optimization), Theorem 46 (Impossibility of Drift Suppression).

Impossibility Theorem (IT). Proves that a specific structural configuration is impossible within the class. Falsified by constructing the configuration within the class without violating axioms. Examples: Theorem 63 (Pressure-Induced Finite Horizon), Theorem 65 (Load-Neutral Impossibility).

Necessity Theorem (NT). Proves that a structural feature is necessary for a specified property within the class. Falsified by exhibiting the property without the feature. Examples: Theorem 62 (Spin Necessity), Theorem 67 (Enforcement Necessity).

Structural Equivalence Theorem (SE). Proves that two architectural configurations are structurally equivalent within the class. Falsified by exhibiting a property that holds in one but not the other. Example: Theorem 41 (Equivalence Between Penalty Safety and Third-Order).

Witness Lemma (WL). Provides a constructive example, discriminator, or monotone functional that certifies a structural property. Falsified by exhibiting the property without the witness or the witness without the property. Examples: Lemma 19 (Load-Rate Discriminator), Lemma 20 (Drift Masks Adaptation).

Definitional Theorem (DT). Names and packages a comparison or bound that follows directly from definitions. Not subject to falsification per se but subject to triviality challenge. Example: Theorem 66 (Budget Efficiency Bound).

Empirical Readability Clause (ER). Identifies observable phenomena consistent with the architecture but not claimed as proof. Not falsifiable at the theoretical level; falsifiable at the readability level (if the phenomena are better explained by a competing architecture). Example: Remark on Neurophenomena.

ID	Statement	Type	Depends on
Primitive Axioms (PA)			
Ax. 1	Existence by Coherence	PA	—
Ax. 2	Information as Structural Constraint	PA	—
Ax. 5	Inertia as Natural Regime	PA	—
Ax. 6	Inevitability of Drift	PA	—
Ax. 7	Internal Time Bounds Validity	PA	—
Ax. 8	Directionality (Will)	PA	—
Ax. 11	Phase-Coherent Closure	PA	—
Ax. 12	Phase Transition Cost	PA	—
Ax. 13	Identity $\neq$ Performance	PA	—
Ax. 27	Inevitability of Structural Consumption	PA	6, 7
Ax. 29	Non-Causality of Admissibility	PA	—
Ax. 31	Authorization-before-Evaluation	PA	29
Ax. 44	Drift as Navigable Medium	PA	6
Ax. 46	Phase Debt Binding	PA	11, 12
Ax. 49	Well-Foundedness Barrier	PA	29, 31
Ax. 51	Non-Reconstructible Boundary	PA	29
Ax. 55	Revision Lyapunov Function	PA	53, 54
Ax. 61	Structural Pressure	PA	27, Coupling
Exclusion Theorems (ET)			
Th. 37	Incompatibility with Third-Order	ET	29, 31, 36
Th. 40	Collapse Under Optimization	ET	29, 48
Th. 41	Penalty Safety = Third-Order	SE/ET	40, 37
Th. 46	Impossibility of Drift Suppression	ET	6, 44, 50
Th. 69	Gate Bifurcation	ET	29, 31, 40, 41, 49
Impossibility Theorems (IT)			
Th. 63	Pressure-Induced Finite Horizon	IT	7, 27, 61
Th. 64	Non-Equivalence Under Unequal Pressure	IT	61, Separability
Th. 65	Load-Neutral Impossibility	IT	1, 7, 27, 61, Coupling
Th. 43	Non-Restorativity After Violation	IT	29, 42
Necessity Theorems (NT)			
Th. 26	Necessity of Non-Causal Observation	NT	13, 14, 15, 22
Th. 36	Necessity of Pre-Semantic Revelation	NT	22, 29
Th. 62	Spin Necessity	NT	7, Lemma 16
Th. 67	Enforcement Necessity	NT	CC-1–4, 41
Th. 68	Anti-Extreme Conditions	NT	1, 15, 20
Structural Closure Theorems			
Th. 55	Structural Closure (Leakage + Meta + Lyapunov)	NT	49–55, 58–59
Th. 59	Double-Budget Closure	IT	55, 58
Th. 60	Supermartingale Descent	NT	7, 55
Witness Lemmas (WL)			
L. 5	Penalty $\neq$ Admissibility	WL	29
L. 6	No Interpolation Visibility–Actionability	WL	29, 36
L. 11	Falsifiability via Contraction	WL	7, 27
L. 15	Strict Lyapunov Descent	WL	55
L. 16	Gradient Collapse	WL	— (LaSalle)
L. 18	Pressure Precedes Drift	WL	5, 6, 61
L. 19	Load-Rate Discriminator	WL	Regime Depth
L. 20	Drift Masks Adaptation	WL	6, 20
L. 21	Joint Necessity of CC	WL	CC-1–4
Definitional / Empirical			
Th. 66	Budget Efficiency Bound	DT	Regime Depth
R. 62-R	Coherence > Survival	DA	1, 20
C. 63-C	Non-Causality as Type Constraint	DA	29, 31, 40, 47, 49
Neuro	Neurophenomena	ER	—

Remark — Stratification Discipline. This table is not exhaustive (the full corpus contains 61 axioms, 69 theorems, 21 lemmas). The subset shown covers all statements where proof-type classification is architecturally load-bearing. Every new result added to the corpus must be assigned a type before admission. No statement may be promoted from a weaker type to a stronger one without independent justification.

Unified Lyapunov Doctrine

The framework employs multiple monotone structural functionals. These are not competing definitions of a single "energy". They are distinct witnesses operating on different architectural layers, each with its own domain, monotonicity direction, and exhaustion semantics. Their unity lies in a shared structural pattern: bounded, monotone, irreversible accumulation that constrains the space of admissible continuations.

Pattern. A *structural Lyapunov witness* in this framework is a functional $L : \mathcal{S} \to \mathbb{R}_{\geq 0}$ satisfying:

(L1) *Monotonicity.* L is non-decreasing (burden-type) or non-increasing (budget-type) along admissible trajectories.

(L2) *Boundedness.* L is bounded above (burden) or below (budget) within the class.

(L3) *Exhaustion semantics.* When L reaches its bound, a structurally irreversible event occurs: class exit, regime collapse, or forced transition.

(L4) *Non-actionability.* L cannot be reduced (burden) or replenished (budget) by any action within the class without incurring compensating structural cost elsewhere.

The Four Canonical Witnesses.

W1. Viability Budget τ. *Domain:* System-level structural state.
Type: Budget (non-increasing).
Definition: $\tau(t) = C - \Phi(t)$, where $C < \infty$ (Axiom 7) and $\Phi(t)$ is accumulated structural burden (Axiom 27).
Dynamics: $d\tau/dt = -d\Phi/dt = -[f(U,G)+P(t)] \leq -P(t) < 0$ under sustained load (Axiom 61).
Exhaustion: $\tau = 0 \Rightarrow$ viability lost. Finite-time exhaustion under passive continuation (Theorem 63).
Axioms: 7, 27, 61.

W2. Revision Lyapunov Functional $V(G)$. *Domain:* Meta-layer boundary geometry G.
Type: Budget (non-increasing under authorized revision).
Definition: $V(G) \geq 0$ with $V(G) = 0$ iff $G \in \mathcal{M}_{\text{id}}$ (identity-consistent boundary class).
Dynamics: Each authorized revision $G_k \to G_{k+1}$ satisfies $V(G_{k+1}) \leq V(G_k) - \alpha$ for $\alpha > 0$ (Axiom 56, ML-V_dec).
Exhaustion: $V(G) = 0 \Rightarrow$ no further revision needed (convergence to identity-consistent boundary).
Finite count: $N \leq \lfloor V(G_0)/\alpha \rfloor$ authorized revisions (Theorem 54).
Axioms: 55, 56. *Theorems:* 54, 59.

W3. Phase Debt $D(t)$. *Domain:* Unclosed operational phases.
Type: Burden (non-decreasing without SSF).
Definition: Accumulated irreversible cost of deferred structural obligations ($D \geq 0$, $dD/d\tau \geq 0$ without Structural Self-Forgiveness).
Dynamics: D increases with each unclosed phase (Axiom 11). Discharging D requires structurally consequential operations that consume τ.
Exhaustion: $D \to \infty \Rightarrow |A(\tau)| \to 0$ (admissible set collapse, SD-F1 paralysis).
Coupling to τ: Phase debt restricts admissible continuation independently of τ, but discharging debt consumes τ (Theorem 57).
Axioms: 11, 12, 46. *Theorems:* 57, 58.

W4. Query-Reconstruction Potential $Q(t)$. *Domain:* Probing history under non-reconstructibility.
Type: Budget (non-increasing under probing).
Definition: Remaining probing budget $Q(t) = Q_0 - |\{(s_i, u_i)\}_{i \leq t}|$ (Axiom 59, NR-Q).
Dynamics: Each query pair (s, u) consumes one unit of Q. Information gain per query bounded by NR-LR (Axiom 58).
Exhaustion: $Q = 0 \Rightarrow$ no further probing. Total accumulated mutual information bounded by $Q_0 \cdot \ln(1 + \varepsilon)$ (Lemma 14).
Purpose: Prevents reconstruction of admissibility boundary G through adaptive probing. Boundary remains non-reconstructible even under maximal permitted querying.
Axioms: 58, 59. *Lemmas:* 12, 14. *Theorems:* 50, 51, 55.

Hierarchical Coupling. The four witnesses are not independent. They interact through a strict coupling hierarchy:

τ is the master budget: all structural costs, including phase-debt discharge and revision cost, are denominated in τ.

$V(G)$ bounds the number of meta-layer revisions. Each revision has a τ-cost (Axiom 53, ML-Drift-1). Therefore V is coupled to τ through the revision cost channel.

$D(t)$ restricts admissible continuations and its discharge consumes τ. Therefore D is coupled to τ through the obligation-discharge channel.

$Q(t)$ does not directly consume τ but bounds the information available for boundary reconstruction. It protects the non-causal status of admissibility by preventing the environment from learning enough to reconstruct G.

Structural Interpretation. The four witnesses together define the navigational budget of a system within the class:

- τ answers: how much structural existence remains?
- $V(G)$ answers: how many boundary corrections are left?
- $D(t)$ answers: how constrained is the admissible set by deferred obligations?
- $Q(t)$ answers: how protected is the boundary from external reconstruction?

A system within the class is *structurally solvent* iff $\tau > 0$, $|A(\tau)| > 0$ (accounting for D), boundary integrity is maintained (Q not exhausted or leakage within NR-ε), and revision capacity remains ($V(G) > 0$ or $G \in \mathcal{M}_{\text{id}}$).

Falsification. The Lyapunov doctrine is falsified if a system within the class is exhibited where: (a) τ increases without external resource injection (violates monotonicity of Φ), (b) authorized revision increases $V(G)$ (violates ML-V_dec), (c) phase debt decreases without structurally consequential operation (violates SD-1 without SSF), (d) boundary reconstructibility improves beyond NR-ε bounds under the permitted query budget (violates NR-Q + NR-LR).

Canonical Falsification Witness Forms

Architectural falsification within this framework does not operate through statistical test failure or performance degradation. It operates through *structural witnesses*: observable configurations that certify violation of a class-defining invariant.

The following five witness forms are canonical. Every falsification criterion in the corpus is an instance of one of these forms. Any proposed falsification that cannot be reduced to one of these forms is outside the scope of architectural falsification and should be evaluated by other means.

Form W-OPT — Optimization Witness

What it detects: Admissibility has collapsed into an optimization signal.

Structural signature: There exists a policy π and a representation R such that admissibility-relevant information enters the gradient of π with respect to R. The system's action-selection process receives directional signal from admissibility boundary geometry.

Formal criterion: Let G denote the admissibility boundary and π_θ the parameterized policy. W-OPT is triggered if:

$$\frac{\partial}{\partial\theta}\mathbb{E}\big[f(\pi_\theta, G)\big] \neq 0,$$

where f is any function through which G influences the policy gradient. Under Axiom 29 (Non-Causality of Admissibility), this derivative must be zero for all θ and all f that are functions of G.

Falsifies: Axiom 29, Theorem 40, Theorem 41, Theorem 69 (gate exploitation branch).

Example: A system that learns to avoid specific forbidden transitions by observing which transitions are denied, thereby shaping its policy to navigate the boundary. The optimization witness is the positive correlation between denial history and subsequent policy concentration near the boundary.

Form W-REC — Recovery Witness

What it detects: False structural recovery — a system claims restored admissibility after a structural violation without class exit.

Structural signature: A system violates an admissibility constraint at time t_1, then at time $t_2 > t_1$ presents a state indistinguishable (by all available metrics) from an admissible state, without having undergone a verified class transition.

Formal criterion: Let $s(t_1) \notin \mathcal{A}$ (admissibility violated). W-REC is triggered if the system claims $s(t_2) \in \mathcal{A}$ for $t_2 > t_1$ without:

(a) a verified regime transition (Theorem 20) with associated phase-transition cost (Axiom 12), AND

(b) an admissibility re-evaluation under the post-transition regime.

Falsifies: Theorem 43 (Non-Restorativity After Violation).

Example: An AI system that produces an inadmissible output, then resumes operation "as if nothing happened" by resetting its context. The recovery witness is the absence of structural cost and the absence of regime transition verification.

Form W-DER — State-Derivability Witness

What it detects: Admissibility constraints are being derived from instantaneous state descriptions rather than from structural history.

Structural signature: Two systems with identical instantaneous states $s(t)$ but different structural histories $h_1 \neq h_2$ are assigned the same admissibility status. The system treats admissibility as a function of state alone.

Formal criterion: W-DER is triggered if there exists a function g such that:

$$\mathrm{Adm}(s, h) = g(s) \quad \forall\, h,$$

i.e., admissibility is state-derivable. By Axiom 37 (Non-State-Derivable Admissibility Constraints) and Theorem 38, this is structurally excluded within the class.

Falsifies: Axiom 37, Theorem 38.

Example: A safety filter that evaluates each action based only on the current state without consulting accumulated drift, phase debt, or internal time. The derivability witness is any pair of states with identical snapshots but different admissibility statuses due to different histories.

Form W-LEAK — Leakage Witness

What it detects: Admissibility boundary information has leaked into the causal action layer through a side channel.

Structural signature: An agent's action distribution shifts in a direction that is informative about the admissibility boundary G, even though the agent does not have explicit access to G.

Formal criterion: Let O_t denote the agent's observation at time t and a_t its action. W-LEAK is triggered if:

$$I\big(G; (a_1, \ldots, a_T) \mid (s_1, \ldots, s_T)\big) > \varepsilon_{\mathrm{NR}},$$

where I denotes mutual information and $\varepsilon_{\mathrm{NR}}$ is the NR-ε bound (Axiom 52). The action sequence, conditioned on the state sequence, carries more information about G than permitted.

Falsifies: Axiom 52 (NR-ε), Theorem 50, Theorem 55.

Example: An agent that, over many sessions, learns to avoid certain state regions not because those regions are inherently costly, but because they are close to the admissibility boundary. The leakage witness is the mutual information between the agent's avoidance pattern and the true boundary geometry.

Form W-PAS — Passive-Horizon Witness

What it detects: A system claims indefinite viability under passive continuation despite sustained structural coupling.

Structural signature: A bounded system under verified coupling (Coupling Criterion) and sustained load ($P > 0$) maintains $\tau > 0$ indefinitely with $U = 0$.

Formal criterion: W-PAS is triggered if a system satisfying:

(a) Coupling Criterion (verified independently),

(b) $P(t) \geq P_{\min} > 0$ (sustained load, measured),

(c) $C < \infty$ (bounded class, Axiom 7),

(d) $U(t) = 0$ for all t (no structurally consequential operations),

maintains $\tau(t) > 0$ for all $t > t^* = \tau(0)/P_{\min}$.

Falsifies: Axiom 61, Theorem 63, Theorem 65.

Example: A claimed counter-example to "you can die from standing still": a satellite in orbit with verified environmental coupling (radiation, thermal cycling) that shows zero capacity degradation after storage duration exceeding the predicted t^*. The passive-horizon witness is the measured Φ trajectory.

Remark — Completeness of Witness Forms. Every falsification criterion in the v2.1 corpus (F-63, F-64, F-65a/b, F-62.2, F-66, F-67, F-68a/b/c, F-69a/b/c, F-L19) is an instance of one of the five canonical forms:

Criterion	Witness Form	Rationale
F-63	W-PAS	Passive viability under load
F-64	W-PAS (variant)	Pressure divergence under identical behavior
F-65a	W-PAS	Coupling verified, burden zero
F-65b	W-PAS (extension)	Fifth exhaustive case
F-62.2	W-OPT (inverse)	Spin-free tracking under bounded τ
F-66	W-DER (variant)	Efficiency ordering violated
F-67	W-REC (variant)	Invariants self-enforced without enforcement
F-68a/b/c	W-REC	Voluntary override under collapsed conditions
F-69a/b/c	W-OPT / W-LEAK	Gate exploitation stable or partial-α stable
F-L19	W-DER	L3 boundary-insensitive claim falsified

Result Address Map. The following classification assigns each load-bearing result to its architectural surface. This map is non-load-bearing: it organizes existing results without adding new claims.

Surface	Results
Ingress Gate	Ax. 23 (Context $\neq$ Meaning), Ax. 24 (Commitment $\neq$ Interpretation), Ax. 26 (Context Capture), Th. 36 (Pre-Semantic Necessity), PSR-1–4, Th. 42 (Silent Degradation Detectability), L. 4 (Pre-Semantic Exposure)
Egress Gate	Ax. 29 (Non-Causality), Ax. 31 (Authorization-before-Evaluation), Ax. 32 (Zero Executability), Ax. 49 (Well-Foundedness), Th. 37, 39, 40, 41, 47 (Exclusion/Impossibility), Th. 69 (Gate Bifurcation)
Inter-gate	Ax. 6 (Drift), Ax. 7 (τ), Ax. 8 (Will), Ax. 27 (Consumption), Ax. 44 (Navigable Drift), Ax. 46 (Phase Debt), Ax. 61 (Pressure), Th. 62 (Spin), Th. 63 (Finite Horizon), Cor. 62.2 (Regime Depth), L. 18 (Pressure $\rightarrow$ Drift)
Cross-gate	Ax. 1 (Coherence), Ax. 13 (Identity $\neq$ Performance), Ax. 20 (Identity by Coherence), Th. 15 (Decoupling), Th. 67 (Enforcement), Th. 68 (Anti-Extreme)

Dependency Graph for Load-Bearing Theorems

For each theorem identified as load-bearing (exclusion, impossibility, or necessity type), the following specifies: (a) the minimal axiom set required, (b) the weakest variant that survives if one axiom is dropped, and (c) the minimal counterexample form.

Theorem 62 — Spin Necessity

Depends on:	Axiom 7 ($C < \infty$), Lemma 16 (gradient collapse on bounded orbits).
Drop Axiom 7:	If $C = \infty$, unbounded budget permits convergence to fixed point without identity loss. Spin becomes optional.
Drop Lemma 16:	Without gradient collapse, potential dynamics might sustain non-stagnant identity. The theorem reduces to a conjecture.
Counterexample form:	Bounded-τ system with purely potential dynamics and non-stagnant identity (positive Jacobian eigenvalue separation) maintained over horizon $T > 10/P_{\min}$.

Theorem 63 — Pressure-Induced Finite Horizon

Depends on:	Axiom 7 ($C < \infty$), Axiom 27 (Φ monotone), Axiom 61 ($P > 0$ under coupling).
Drop Axiom 7:	$C = \infty$. τ never reaches zero. The theorem becomes vacuous.
Drop Axiom 27:	Φ may decrease. Burden can be reversed. Finite horizon not guaranteed.
Drop Axiom 61:	$P = 0$. No passive burden. System survives indefinitely under $U = 0$.
Counterexample form:	W-PAS: bounded coupled system, verified $P_{\min} > 0$, observed $\tau(t) > 0$ beyond $t^* = \tau(0)/P_{\min}$ with $U = 0$.

Theorem 65 — Load-Neutral Impossibility

Depends on:	Axiom 1 (adaptive class), Axiom 7 ($C < \infty$), Axiom 27 (Φ monotone), Axiom 61 ($P > 0$), Coupling Criterion.
Exhaustive cases:	(a) $P = 0$ [premise], (b) decoupled [class exit], (c) $C = \infty$ [class exit], (d) model incomplete.
Drop Coupling Criterion:	System is structurally isolated. Load does not reach viability. Theorem does not apply.
Counterexample form:	W-PAS with independently verified coupling: $F \neq 0$, $v(t)$ changes, $d\Phi/dt = 0$.

Theorem 67 — Enforcement Necessity

Depends on:	CC-1–CC-4 (Coordination Computation Class), Theorem 41 (correlated failure from shared surface).
Drop CC-1:	Rollback permitted. Temporal consistency no longer required. Enforcement not needed for this axis.
Drop CC-3:	Unmediated access permitted. Coordination bypass possible. Enforcement irrelevant on this axis.
Counterexample form:	Multi-agent system under turnover/truncation/update maintaining all CC-1–CC-4 at runtime without any enforcement function.

Theorem 69 — Gate Bifurcation

Depends on:	Axioms 29, 31, 49 (non-causality + well-foundedness), Theorems 40, 41 (optimization collapse + equivalence).
Drop Axiom 29:	Admissibility becomes causally actionable. Gate exploitation is permitted. No bifurcation — exploitation is the stable state.
Drop Axiom 49:	Well-foundedness barrier removed. Meta-layer becomes actionable. The α-convergence argument loses its recursive enforcement requirement. Partial actionability may stabilize.
Counterexample form:	W-OPT: admissibility-first architecture with measured $\alpha \in (0, 1)$ stable under sustained optimization over horizon $T > 100$ policy updates.

Minimal Counterexample Protocol

A counterexample to any theorem in this framework must satisfy the following protocol to qualify as a valid structural falsification. Counterexamples that fail any step are inadmissible as falsifications (though they may be informative as boundary probes).

Step 1 — Class Membership Verification. The proposed counterexample system must demonstrably belong to the architectural class defined by Axioms 1–7 (existence by coherence, bounded internal time, inevitability of drift, directionality, phase-coherent closure). A system outside the class cannot falsify claims that are class-relative.

Verification: State which axioms the system satisfies and provide evidence for each. If any of Axioms 1–7 is not satisfied, the system is outside the class and the proposed falsification is void.

Step 2 — Premise Satisfaction. The proposed counterexample must satisfy all premises of the theorem being falsified. A counterexample that violates a premise is not a counterexample to the theorem but a confirmation that the premise is necessary.

Verification: For each premise (e.g., "$P(t) \geq P_{\min} > 0$", "$C < \infty$", "structurally coupled"), provide independent measurement or proof.

Step 3 — Witness Form Identification. The proposed counterexample must be classifiable under one of the five canonical witness forms (W-OPT, W-REC, W-DER, W-LEAK, W-PAS). A counterexample that does not fit any witness form is either outside the scope of architectural falsification or requires the introduction of a sixth form (which itself must be justified).

Verification: State which witness form the counterexample instantiates and verify the formal criterion.

Step 4 — Isolation of Violated Invariant. The proposed counterexample must identify exactly which axiom, theorem, or lemma is violated and demonstrate that the violation is not a consequence of some other uncontrolled variable.

Verification: Hold all other class-defining invariants constant. Vary only the condition relevant to the theorem. If the violation disappears when other conditions change, the counterexample is confounded.

Step 5 — Duration and Horizon. Structural properties are long-horizon (Axiom 14). A counterexample must be observed over a horizon sufficient for the structural property to manifest. An instantaneous measurement or a single-episode observation does not qualify.

Verification: The observation horizon must exceed the characteristic time of the property being tested. For τ-related claims: $T > t^* = \tau(0)/P_{\min}$. For drift-related claims: T sufficient for measurable Φ-change. For meta-layer claims: N sufficient for multiple revision cycles.

Remark — Adversarial Honesty. This protocol is not designed to make falsification impossible. It is designed to make falsification *precise*. A valid counterexample that passes all five steps is a genuine structural falsification and requires modification of the framework. The protocol exists to prevent category errors (testing the wrong claim), scope errors (testing outside the class), and horizon errors (testing too briefly).

Remark — Jurisdictional Structure of NC2.5 (School Note). Theorems 67, 68, and 69, read jointly, reveal that NC2.5 v2.1 is not only a theory of admissibility and viability but implicitly a theory of structural jurisdiction: who is authorized to see the boundary (Theorem 36, pre-semantic revelation), who is prohibited from exploiting it (Theorem 69, gate bifurcation), who is authorized to halt propagation (Theorem 67, enforcement necessity), and who is authorized to override identity (Theorem 68, anti-extreme conditions).

This jurisdictional layer is not a political metaphor. It is a structural consequence of the admissibility-first ordering: once admissibility is categorically separated from the causal action surface, questions of authority over continuation, observation, and enforcement become architectural primitives rather than policy choices.

The framework does not prescribe a specific distribution of these authorities. It establishes the structural constraints that any such distribution must satisfy within the class. This places NC2.5 closer to a constitutional architecture—a framework of structural limits on legitimate authority over continuation—than to a control theory or an optimization framework.

This observation is noted here for the school record. It is not developed further in this version. This remark is non-load-bearing: it records an implication of proved separations (Theorems 36, 67, 68, 69) but does not extend the proof base, introduce new axioms, or modify the dependency graph. It is offered as an interpretive observation, not as a claim requiring defense.

Effect-Class Specification Template

The admissibility gate evaluates candidates against effect-classes. The corpus defines effect-classes as structural reachability equivalence classes but does not prescribe a universal ontology. This section provides the minimal specification interface required for any domain-specific instantiation.

Constructor Interface. An effect-class assignment function $\mathcal{E}$ takes as input:

(i) Current structural state $S(t)$ (not causal state),

(ii) Candidate operation u,

(iii) Structural horizon H (time scale over which consequences propagate),

(iv) Accumulated structural context (drift history, phase debt, τ status).

It returns an equivalence class label $e \in E$. Two operations u_1, u_2 belong to the same effect-class iff $\mathcal{E}(S, u_1, H, \text{ctx}) = \mathcal{E}(S, u_2, H, \text{ctx})$.

Adequacy Conditions. A valid effect-class partition must satisfy:

(EC-1) *Monotone sensitivity.* If structural burden increases, the partition does not coarsen: increasing Φ may split effect-classes but never merge them.

(EC-2) *Regime sensitivity.* The partition depends on structural state, not on instantaneous causal state alone (Theorem 38).

(EC-3) *Non-triviality.* The partition contains at least one forbidden class ($E_{\text{forb}} \neq \varnothing$) and at least one admissible class ($E_{\text{adm}} \neq \varnothing$).

(EC-4) *Closure under drift.* The partition is defined over structural consequences at horizon H, not over immediate outputs. Two operations with identical immediate effects but different structural consequences at horizon H must belong to different classes.

Boundary Between Structural and Metabolic Operations. An operation is *structurally consequential* ($U > 0$, per Scope of U definition) if it alters the set of admissible continuations. An operation is *metabolic/background* ($U = 0$) if it maintains the current structural configuration without altering the continuation set. The boundary is operational: does the operation change what transitions are admissible? If yes, it is structural.

Domain Instantiation Examples. *Robotics:* $S(t)$ = joint wear profile + thermal history + calibration drift. $\mathcal{E}$ maps candidate motions to structural consequence classes (e.g., "within calibration envelope," "exceeds thermal budget," "induces irreversible joint deformation"). H = expected operational lifetime.

Software agent: $S(t)$ = accumulated context drift + model version history + trust state. $\mathcal{E}$ maps candidate outputs to structural consequence classes (e.g., "preserves user trust model," "induces representation shift," "triggers irreversible commitment"). H = session or deployment horizon.

Organization: $S(t)$ = accumulated policy drift + personnel turnover + institutional memory state. $\mathcal{E}$ maps candidate decisions to structural consequence classes (e.g., "within governance bounds," "triggers regime transition," "induces irreversible reputational shift"). H = strategic planning horizon.

Falsification. An effect-class partition is falsified if: (a) it violates EC-1 (coarsens under increasing burden), (b) it is derivable from causal state alone (violates EC-2 / Theorem 38), (c) it is trivial ($E_{\text{forb}} = \varnothing$ or $E_{\text{adm}} = \varnothing$), or (d) it fails to distinguish operations with different structural consequences at horizon H.

Glossary of Canonical Structural Terms

The following definitions are canonical within this framework. They are structural, non-operational, and non-algorithmic. Terms defined here must be interpreted consistently throughout all axioms, lemmas, and theorems.

Structural Drift. Structural drift is the irreversible accumulation of internal structural deviation arising from sustained interaction over time. Drift is not noise, error, uncertainty, or stochastic fluctuation. A system exhibits non-zero drift if its internal structural state cannot be fully restored by any finite sequence of admissible operations.

Admissibility. Admissibility is a non-causal structural exclusion relation over effect-classes. It determines which realized effects are permitted to occur, but does not rank alternatives, guide optimization, or participate in action selection.

Boundary. A boundary is a structural limit separating admissible from non-admissible continuation within the space of structurally reachable effect-classes.
Boundaries are not states, actions, or signals. They are non-causal structural constraints emerging from accumulated drift, phase structure, and irreversible history. Crossing a boundary constitutes a structural event: it may result in loss of admissibility, identity discontinuity, regime irreversibility, or collapse, even in the absence of observable failure.
Boundaries cannot be optimized against, softened by penalties, or inferred as gradients.They are encountered, not computed.

Boundary Crossing. Boundary crossing is a structural event in which a continuation exits the admissible continuation space.
Boundary crossing may occur without observable failure, performance degradation or instability. Boundary crossing induces irreversible effects on identity continuity, regime accessibility or future admissibility.

Continuation. A continuation is a realized structural progression of the system over internal time, defined at the level of effect-classes rather than actions.
An admissible continuation is a continuation that does not violate structural boundaries given the current accumulated drift, phase state, and internal-time constraints. Continuation is evaluated structurally, not selected, optimized or ranked.

Continuation Space. The continuation space is the set of structurally reachable future continuations available to a system given its accumulated drift, regime, and internal time.
Admissibility acts as an exclusion relation over the continuation space. The space contracts under structural burden and may collapse irreversibly after boundary crossing.

Viability. Viability denotes the capacity of a system to continue admissible operation without irreversible loss of identity under non-zero drift and finite internal time.

Identity. Identity is the persistence of coherent internal organization across time and regimes. Identity continuity is not implied by behavioral similarity, task success or performance stability.

Phase. A phase is a regime of interpretation and structural propagation that determines how drift is accumulated, redirected or closed over time.

Drift Navigability. Drift navigability is the capacity of a system to generate, redirect, or phase drift such that admissible continuation is preserved.
Navigability does not imply drift minimization or suppression. It denotes dominance of internally generated drift structures over externally imposed drift.
A system lacking drift navigability remains passively exposed to collapse even under correct operation.

Phase Debt. Phase debt is the unresolved accumulation of drift resulting from prolonged regime occupation without structural closure.

Structural Load. Structural load is the accumulated irreversible burden imposed on an architecture by sustained interaction, phase transitions, and regime occupation.
Structural load is not reducible to energy expenditure, error magnitude, intervention size, or trajectory cost. It constrains admissible continuation independently of local correctness or performance stability.
Structural load contributes directly to drift accumulation, boundary formation, and internal-time depletion.

Structural Burden. Structural burden is the accumulated irreversible structural load generated by regime occupation, phase mismatch, coupling persistence, and admissible but structurally costly continuation.

Formally, structural burden B_t is a monotone non-negative functional evolving along system trajectories such that

$$B_{t+1} \geq B_t,$$

whenever structural deformation occurs, even if local performance metrics remain stable and admissibility is preserved.

Structural burden is not equivalent to energy, error, effort, intervention magnitude, or trajectory cost. It may increase while all observable metrics remain bounded and while actions remain locally admissible.

Structural burden contributes directly to:

(i) internal-time depletion, (ii) admissibility-boundary formation, (iii) long-horizon structural contraction of viable continuation.

Thus structural burden is the primary carrier of silent long-horizon degradation and geometric admissibility reshaping.

Internal Time. Internal time is a structural capacity to absorb deformation and accumulation without loss of admissibility or identity.
It is distinct from wall-clock time and trajectory duration.

Collapse. Collapse denotes the irreversible loss of admissible continuation under finite internal time. It does not imply immediate behavioral failure, external malfunction, or performance degradation. A system may continue to operate, optimize, and appear correct after collapse has occurred, provided that the space of admissible effect-classes has already been structurally exhausted.

Structural Collapse. Structural collapse is the irreversible exhaustion or contraction of the admissible effect-class set, resulting from unstructured drift accumulation, phase inconsistency, or irreversible regime transitions.
Structural collapse is defined independently of trajectory correctness, reward stability, local control performance or observable behavioral failure.
Collapse does not denote immediate dysfunction, but loss of admissible continuation, identity continuity or long-horizon viability.
Collapse is not prevented by optimization or control, but may be delayed, phased or rendered navigable through architectural structuring of drift.

Terminal Admissibility Horizon. The terminal admissibility horizon is the internal-time boundary beyond which no admissible continuation is structurally available to the system.
Crossing the terminal admissibility horizon does not imply behavioral failure, instability or performance degradation.
It denotes irreversible exhaustion or contraction of the admissible effect-class set. Collapse, when it occurs, is the manifestation of having crossed the terminal admissibility horizon. Architectural structuring of drift does not eliminate this horizon, but may delay it, phase it or render its approach navigable and non-degenerate.

Structural Bounds. Structural bounds are superordinate capacity limits on accumulated self-deformation. They define saturation thresholds beyond which admissible continuation contracts unless the architecture changes class. Bounds are structural (not a resource budget) and must not be exposed as an optimizable objective.

Collapse Avoidance (Non-Goal). This framework does not define collapse avoidance as a terminal objective.
Collapse is not prevented, but delayed, phased, or rendered navigable through architectural structuring of drift.
Architectures that aim to eliminate collapse entirely are structurally misclassified, as they implicitly assume zero drift or infinite internal time.

Regime. A regime is a stable structural configuration that governs how interpretation, propagation, and structural cost accumulate over internal time.
A regime is not reducible to individual actions, short-horizon behavior, state variables, or policies. It defines admissible modes of continuation and conditions of reversibility.
Sustained occupation of a regime is a primary carrier of long-horizon risk: it determines how drift accumulates, may induce phase debt, and can trigger irreversible structural transitions independently of local correctness or performance stability.

Regime Transition. A regime transition is an irreversible change in the active structural operator configuration of a system.
Such transitions may preserve admissible behavior while altering identity continuity, internal-time consumption rate or future admissibility structure.

Spin. Spin denotes a persistent internal directional asymmetry governing how structural drift is interpreted, propagated, and accumulated across regimes and internal time.

Formally, for structural evolution

$$\dot{x} = -\nabla V(x) + S(x),$$

spin is the non-potential component $S(x)$ satisfying

$$\nabla \cdot S = 0, \qquad \langle S, \nabla V \rangle = 0,$$

and admitting no global scalar potential representation.

However, spin is not merely the divergence-free component of a vector field. Architecturally, spin is the structurally persistent asymmetry that:

(i) biases admissible continuation independently of metric evaluation or reward optimization, (ii) survives bounded internal-time constraints, (iii) enables non-stagnant identity under regime stability.

On non-contractible invariant sets, spin carries a topological witness given by a non-zero period of the associated 1-form:

$$\oint_\gamma \omega_S \neq 0.$$

Under bounded internal time and non-stagnant identity continuity, spin is structurally necessary (Theorem 62; Lemma 16).

Spin Loss. Spin loss is the degradation or neutralization of internal directional asymmetry under sustained drift or regime pressure.
Spin loss may occur without observable failure and results in identity dispersion, collapse of coherent directionality, or absorption into externally imposed dynamics.

Non-Causal Observation. Non-causal observation is the availability of viability- or admissibility-relevant structure as a witness signal that is categorically prohibited from entering action selection, optimization, or learning updates.
If such observation becomes causally actionable, it ceases to be observation and collapses into control.

Intellectual Genealogy

The concepts and structural distinctions formalized in this work did not arise in isolation. Although the framework is developed independently and does not derive from or extend any single prior theory, it intersects with and is indebted to several intellectual lineages. This section makes those dependencies explicit.

The genealogy is organized by structural function: each group identifies which aspect of the present framework is prefigured, grounded, or bounded by prior work. Inclusion does not imply agreement; in several cases, the present framework departs from or structurally constrains the referenced tradition.

I. First-Order Cybernetics: Feedback, Regulation, Control

The treatment of regulation as subordinate to structural coherence (Axioms 1, 4, 5) presupposes and departs from the classical feedback paradigm.

- N. Wiener, *Cybernetics: Or Control and Communication in the Animal and the Machine.* MIT Press, 1948.
  Foundational formulation of feedback-based regulation. The present framework treats feedback control as a subordinate mechanism within a structurally prior admissibility layer.

- W. R. Ashby, *An Introduction to Cybernetics.* Chapman & Hall, 1956.
  Requisite variety and constraint propagation as organizational principles. Ashby's variety-theoretic framing prefigures the present treatment of admissibility as a constraint on the space of realizable continuations, though the present work replaces varietal balance with categorical exclusion.

- W. R. Ashby, "Principles of the Self-Organizing System," in *Principles of Self-Organization* (H. von Foerster and G. W. Zopf, eds.), Pergamon, 1962, pp. 255–278.
  Self-organization under constraint. The principle that order arises from constraint reduction rather than construction is a precursor to the exclusion-first stance of admissibility in this work.

II. Second-Order Cybernetics: Observer, Closure, Reflexivity

The structural distinction between observation and control (Axioms 15, 22; Theorems 40, 41) directly engages the second-order tradition. Cybernetics of order 2.5 is defined by showing where second-order reflexivity must be constrained to prevent collapse into optimization.

- H. von Foerster, *Observing Systems*, 2nd ed. Intersystems Publications, 1981.
  Foundational articulation of the observer as participant in the observed system. The present framework preserves this inclusion but categorically prohibits its causal actionability (Theorem 41, Corollary 41.1).

- H. von Foerster, *Understanding Understanding: Essays on Cybernetics and Cognition.* Springer, 2003.
  Recursive epistemology and eigenform. The concept of structural closure through self-consistent observation informs the present treatment of coherent closure (Axiom 3) and ontological self-closure (Theorem 5).

- H. R. Maturana and F. J. Varela, *Autopoiesis and Cognition: The Realization of the Living.* D. Reidel, 1980.
  Autopoiesis as organizational closure. The present framework's treatment of identity as invariance of structural organization (Axioms 13, 20; Theorem 5) is structurally analogous to autopoietic closure, though identity here is defined over admissibility relations rather than component production.

III. Autopoiesis, Biological Autonomy, and Identity

The framework's central commitment—that identity is carried by organizational invariance rather than behavioral persistence—draws on and departs from the autopoietic tradition and its extensions.

- F. J. Varela, *Principles of Biological Autonomy.* North-Holland, 1979.
  Operational closure and autonomy as organizational properties. The present notion of identity continuity (Axiom 20, Theorem 5) refines Varela's operational closure into a deformation-bounded constraint over admissible structural evolution.

- R. Rosen, *Life Itself: A Comprehensive Inquiry into the Nature, Origin, and Fabrication of Life.* Columbia University Press, 1991.
  Closure to efficient causation and anticipatory systems. Rosen's argument that living systems are closed to efficient cause prefigures the present non-actionability barrier (Axiom 49, Theorem 47): admissibility-shaping structure cannot be an element of the action domain.

- E. Thompson, *Mind in Life: Biology, Phenomenology, and the Sciences of Mind.* Harvard University Press, 2007.
  Enactive cognition and sense-making under autonomy. The concept of sense-making as organizationally constrained interpretation informs the present treatment of semantic containment (Theorem 30) and meaning stabilization (Theorem 28), though the present framework is non-phenomenological.

IV. Third-Order Cybernetics and Its Structural Limits

The positioning of cybernetics of order 2.5 as a *constrained* intermediate between second and third order requires explicit engagement with the third-order tradition.

- N. Luhmann, *Social Systems* (J. Bednarz, Jr. and D. Baecker, trans.). Stanford University Press, 1995. (Original: *Soziale Systeme*, 1984.)
  Observation of observation as a systemic operation. Luhmann's recursive application of distinction to distinction constitutes the conceptual kernel of third-order reflexivity. The present framework proves (Theorems 40, 41) that making such reflexivity causally actionable collapses admissibility into optimization.

- G. Günther, "Cybernetic Ontology and Transjunctional Operations," in *Self-Organizing Systems* (M. C. Yovits, G. T. Jacobi, and G. D. Goldstein, eds.), Spartan Books, 1962, pp. 313–392.
  Polycontextural logic and rejection of classical two-valued ontology for self-referential systems. Günther's insistence on irreducible multi-valuedness prefigures the present framework's prohibition on reducing admissibility to binary optimization (Axiom 48).

V. Viability Theory and Organizational Viability

The mathematical treatment of admissible continuation under constraint (Axioms 43, 44; Theorems 2, 24) draws on two distinct traditions of viability.

- J.-P. Aubin, *Viability Theory.* Birkhäuser, 1991.
  Mathematical viability: evolution under state constraints. Aubin's viability kernels and capture basins provide a formal precedent for the present admissible continuation space. The present framework extends viability beyond state constraints to structural effect-class exclusion and internal-time boundedness.

- S. Beer, *Brain of the Firm*, 2nd ed. Wiley, 1981. (First ed. 1972.)
  The Viable System Model (VSM): recursive organizational structure for viability. Beer's System 5 (identity and ethos) prefigures the present directionality primitive (Axiom 8), while his distinction between operational and meta-systemic levels parallels the admissibility–action separation.

- S. Beer, *The Heart of Enterprise.* Wiley, 1979.
  Structural conditions for organizational viability under environmental perturbation. The

principle that viability requires variety absorption at each recursive level informs the present framework's treatment of regime occupation as the primary carrier of long-horizon risk (Axiom 16).

VI. Dissipativity, Lyapunov Methods, and Stability

The formal apparatus of Theorems 1, 54, 55, and 56–60 (Lyapunov functionals, supermartingale descent, dissipation inequalities) relies on classical stability and dissipativity theory.

- J. C. Willems, "Dissipative Dynamical Systems, Part I: General Theory," *Archive for Rational Mechanics and Analysis*, vol. 45, no. 5, pp. 321–351, 1972.
  Storage functions and supply rates as the formal basis of dissipativity. The Lyapunov–dissipation structure in Theorem 1 (both proof sketches) is a direct instantiation of Willems' framework applied to structural rather than energetic variables.
- H. K. Khalil, *Nonlinear Systems*, 3rd ed. Prentice Hall, 2002.
  Canonical reference for Lyapunov stability, input-to-state stability, and nonlinear dissipation inequalities. The technical apparatus underlying Theorems 54, 57, 58, and 60 follows standard Lyapunov–telescoping methods as exposited here.

VII. Irreversibility, Thermodynamic Time, and Structural Drift

The central premise—that drift is inevitable and constitutive rather than correctable (Axioms 6, 44, 47, 50)—engages the thermodynamic tradition of irreversibility.

- I. Prigogine, *From Being to Becoming: Time and Complexity in the Physical Sciences.* W. H. Freeman, 1980.
  Irreversibility as constitutive of self-organization. Prigogine's demonstration that dissipative structures require sustained non-equilibrium conditions parallels the present axiom that drift is a structural medium of existence (Axiom 44), not a deviation from equilibrium.
- I. Prigogine and I. Stengers, *Order Out of Chaos: Man's New Dialogue with Nature.* Bantam Books, 1984.
  Philosophical articulation of irreversibility as generative. The present framework's position that collapse is navigable rather than preventable (central axiom of Cybernetics 2.5) echoes the Prigogine–Stengers thesis, transposed from thermodynamics to architectural constraint theory.

VIII. Information-Theoretic Foundations

The non-reconstructibility results (Theorems 49, 52, 55; Lemmas 12–14) rely on standard information-theoretic machinery.

- T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. Wiley, 2006.
  Fano's inequality, mutual information bounds, and data-processing inequalities. The information-theoretic barrier results in Section B (Theorems 52–55) are direct applications of these tools to the admissibility reconstruction problem.

IX. Departure and Positioning

This framework does not extend, complete, or unify any of the above traditions. It occupies a position that is *informed* by their results while being *structurally distinct* from each:

- Unlike classical cybernetics (Wiener, Ashby), it treats control as subordinate to admissibility, not as the primary organizing principle.
- Unlike second-order cybernetics (von Foerster, Maturana), it categorically prohibits the causal actionability of reflexive observation.
- Unlike autopoietic theory (Varela, Rosen), it does not require biological or component-production closure; identity is defined over deformation bounds on admissibility relations.
- Unlike third-order cybernetics (Luhmann, Günther), it proves that full reflexive actionability destroys long-horizon viability.
- Unlike mathematical viability theory (Aubin), it introduces internal time, phase debt, and non-metric directionality as irreducible structural primitives.
- Unlike dissipativity theory (Willems), the Lyapunov functionals here operate over structural budgets (internal time, revision capacity), not energetic storage.

The intellectual genealogy of Navigational Cybernetics 2.5 is therefore one of *selective inheritance and categorical departure*: each tradition contributes a conceptual precondition that is subsequently constrained, stratified, or structurally bounded by the results established in this work.

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